

CXR

Interpretation

<http://www.radiologyassistant.nl>

Collected By

Ahmad Mokhtar Abodahab

Sohag University

MAR 2017

Chest X-Ray

1

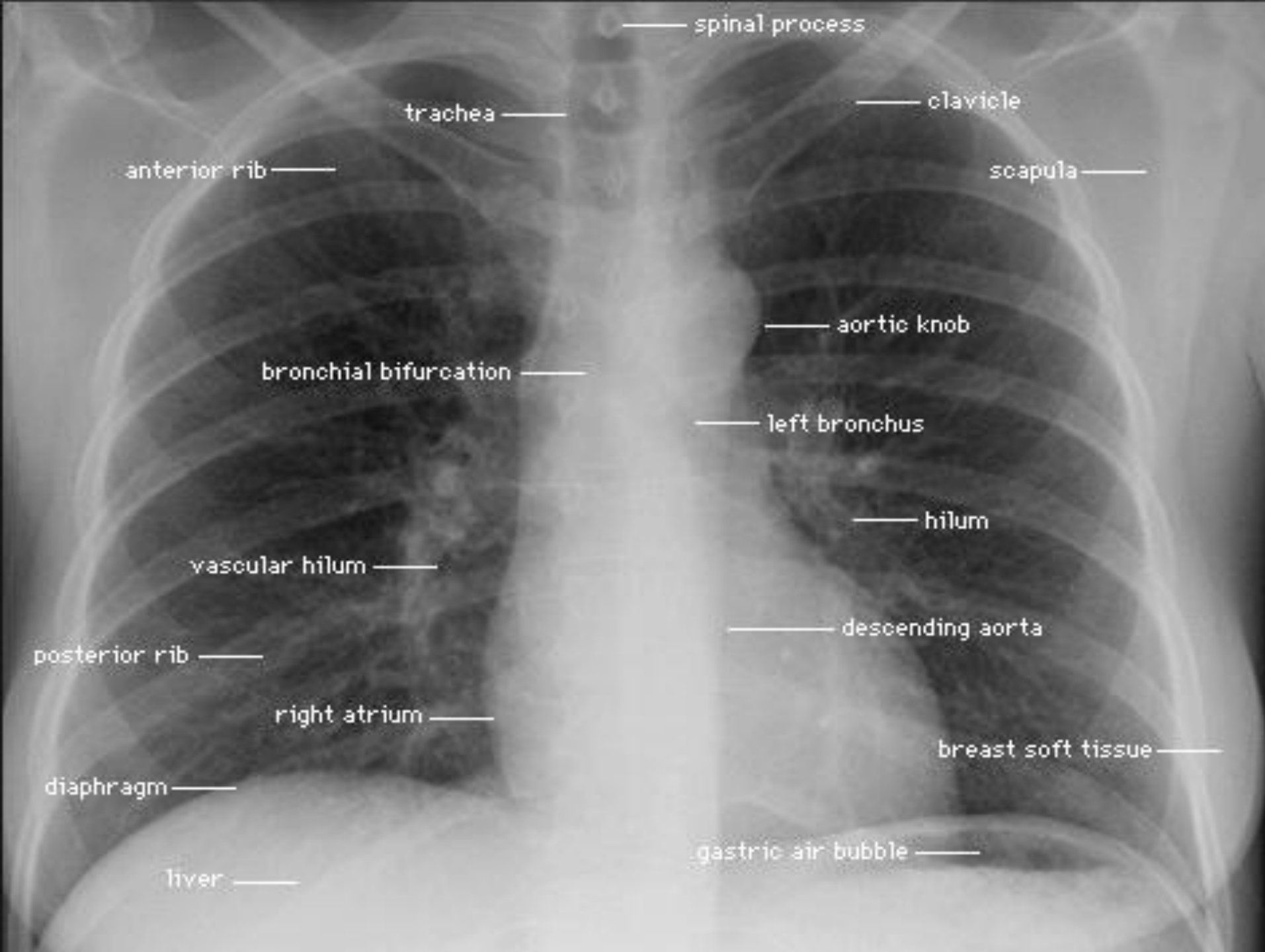
Basic Interpretation

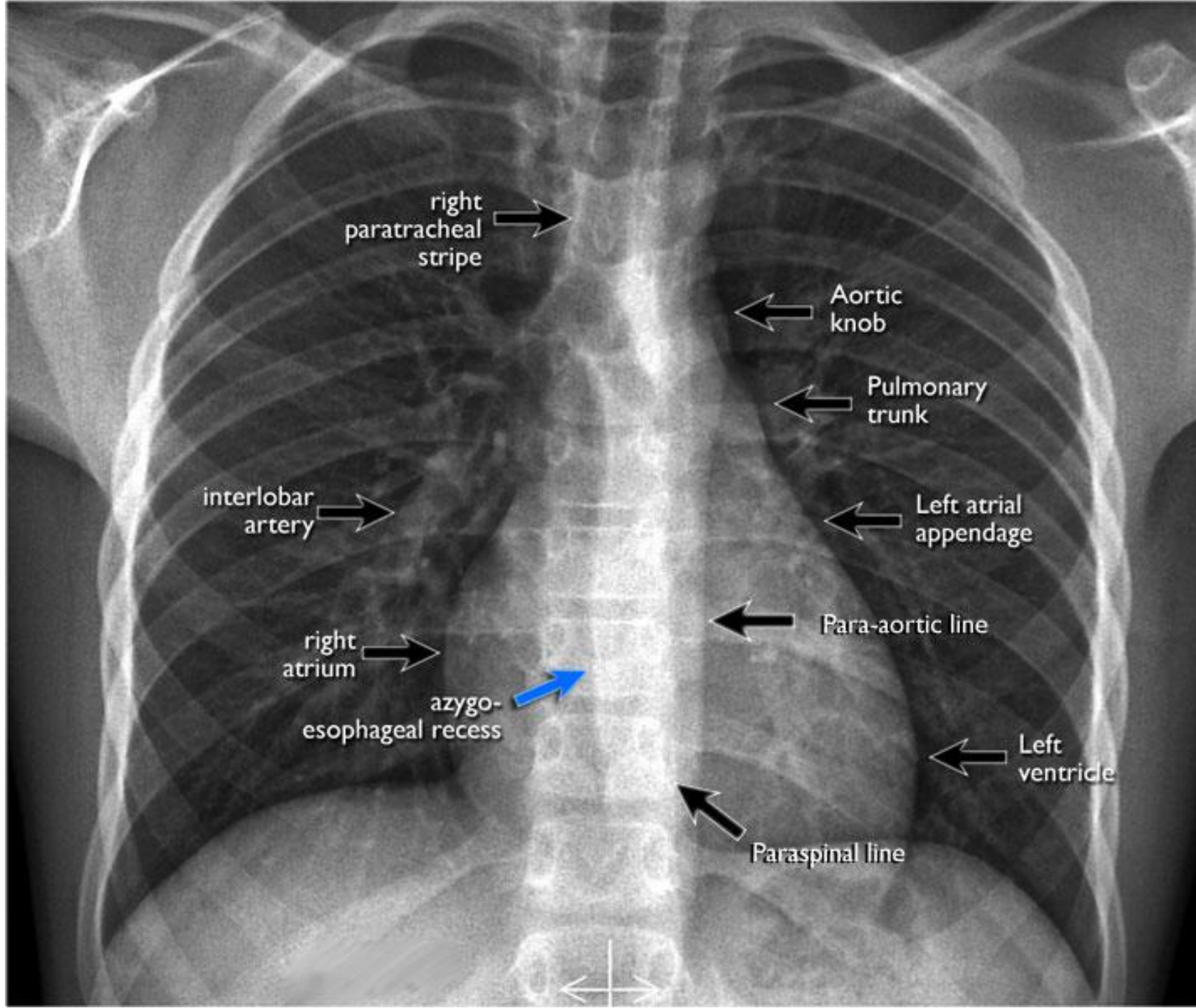
Robin Smithuis and Otto van Delden

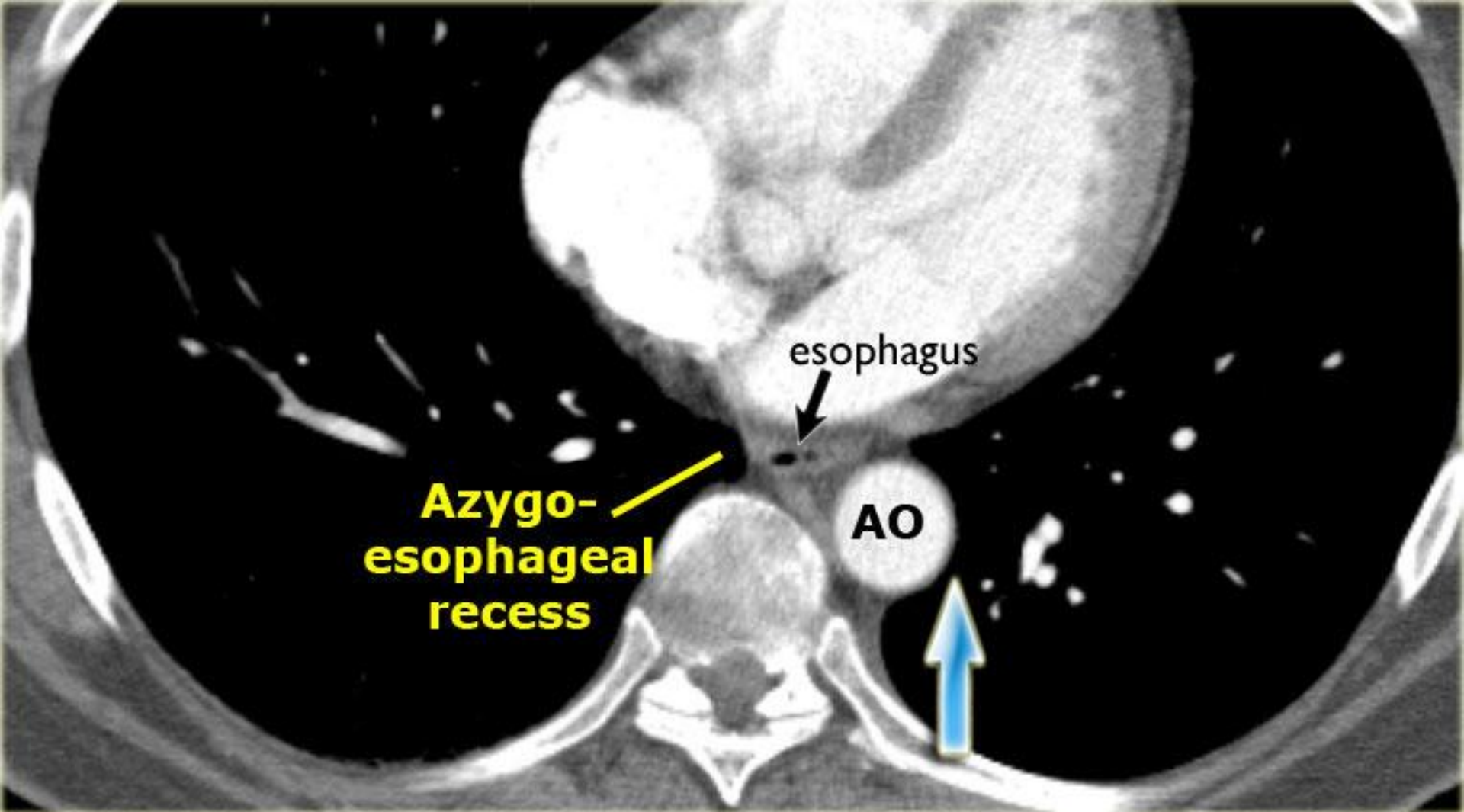
Radiology Department of the Rijnland Hospital,
Leiderdorp and the Academical Medical
Centre, Amsterdam, the Netherlands

[http://www.radiologyassistant.nl/en/
p497b2a265d96d/chest-x-ray-basic-
interpretation.html](http://www.radiologyassistant.nl/en/p497b2a265d96d/chest-x-ray-basic-interpretation.html)

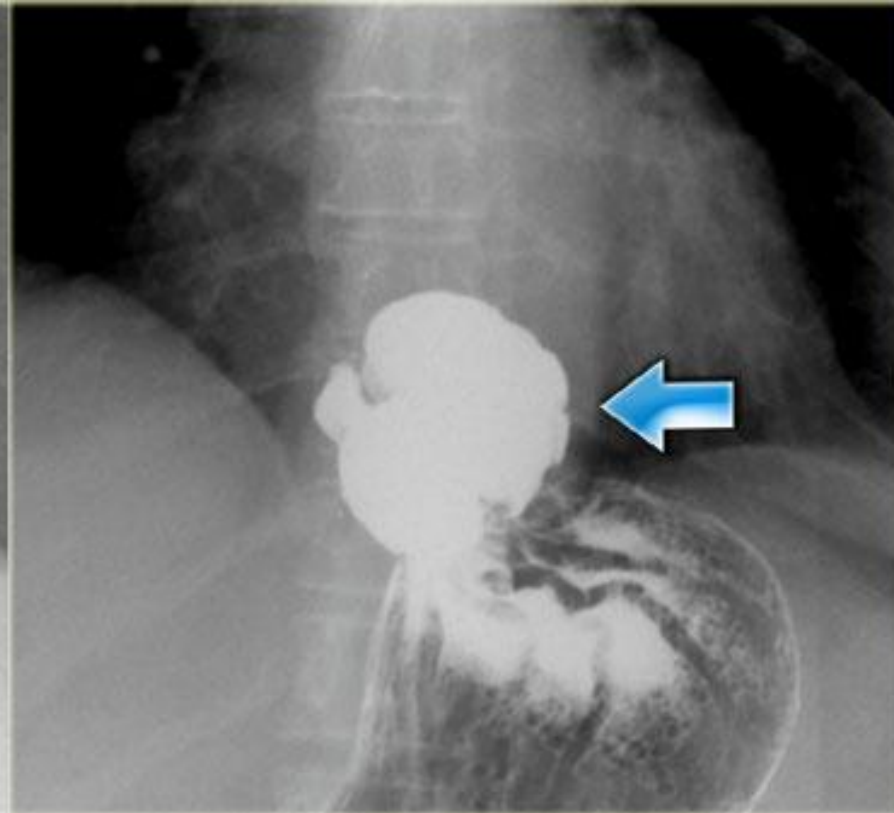
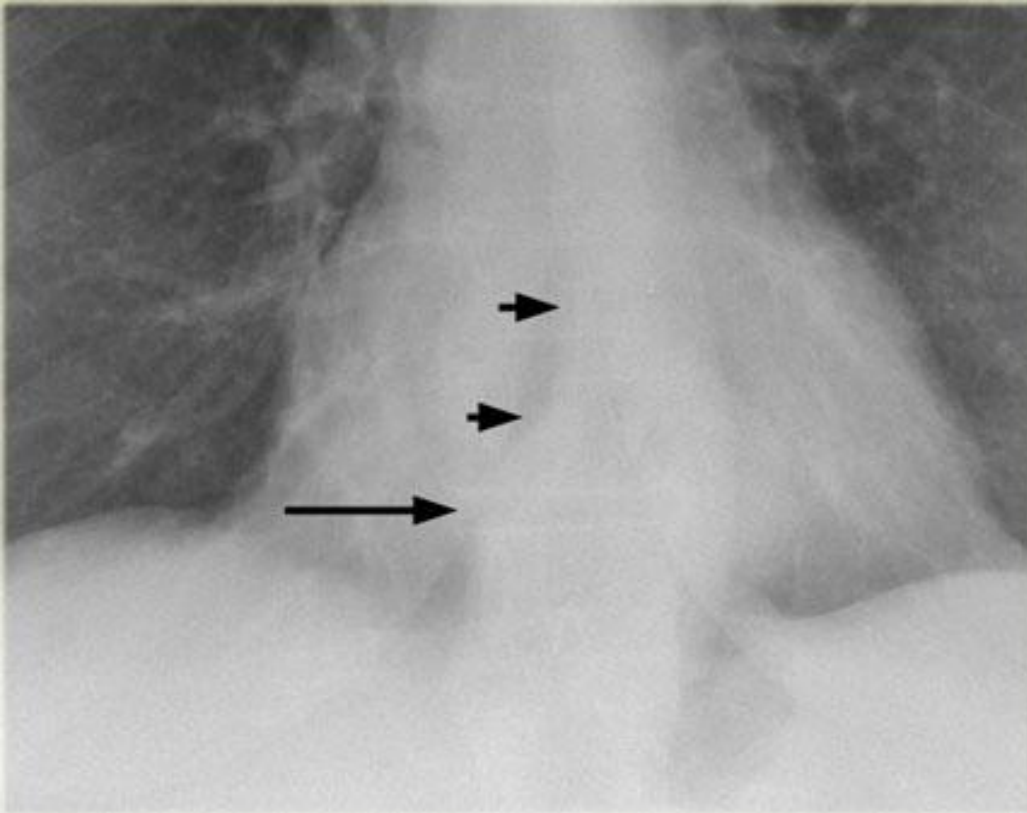
Collected by
A.M. Abodahab Mar 2017



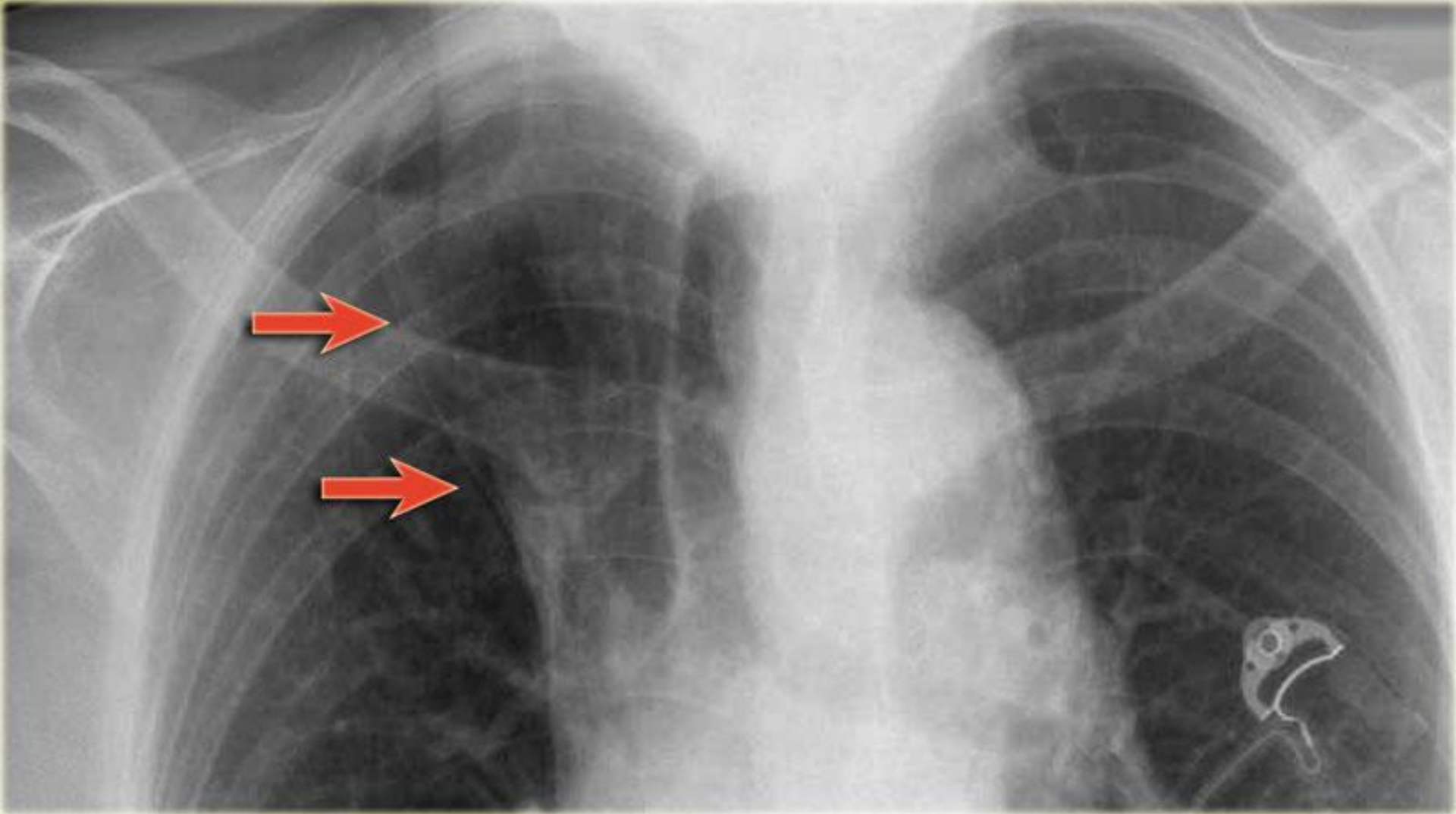




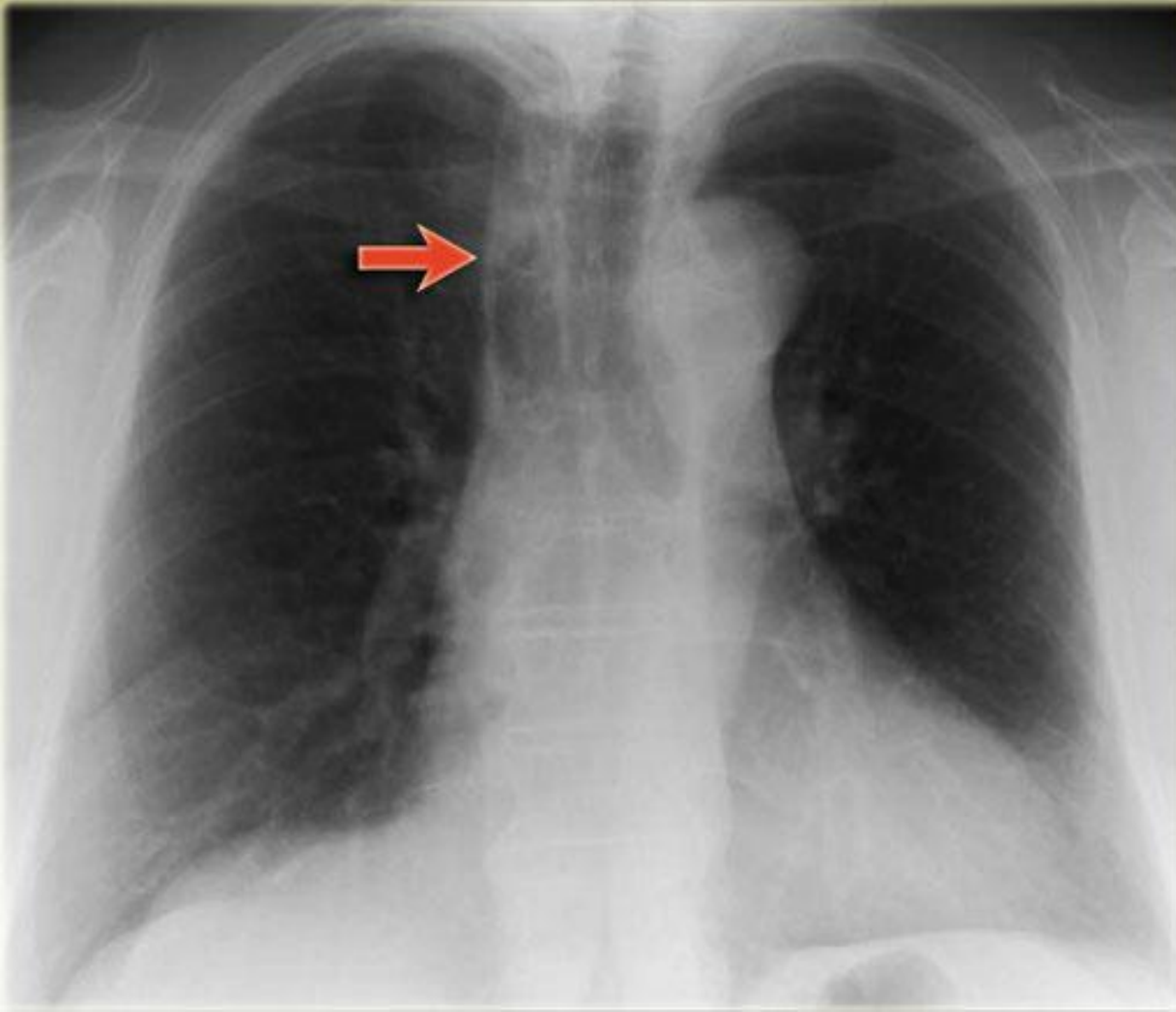
- Deviation of the azygoesophageal line is caused by (5):
 - Hiatal hernia
 - Esophageal disease
 - Left atrial enlargement
 - Subcarinal lymphadenopathy
 - Bronchogenic cyst



- Notice the deviation of the azygoesophageal line on the PA-film.
- It is caused by a hiatal hernia.

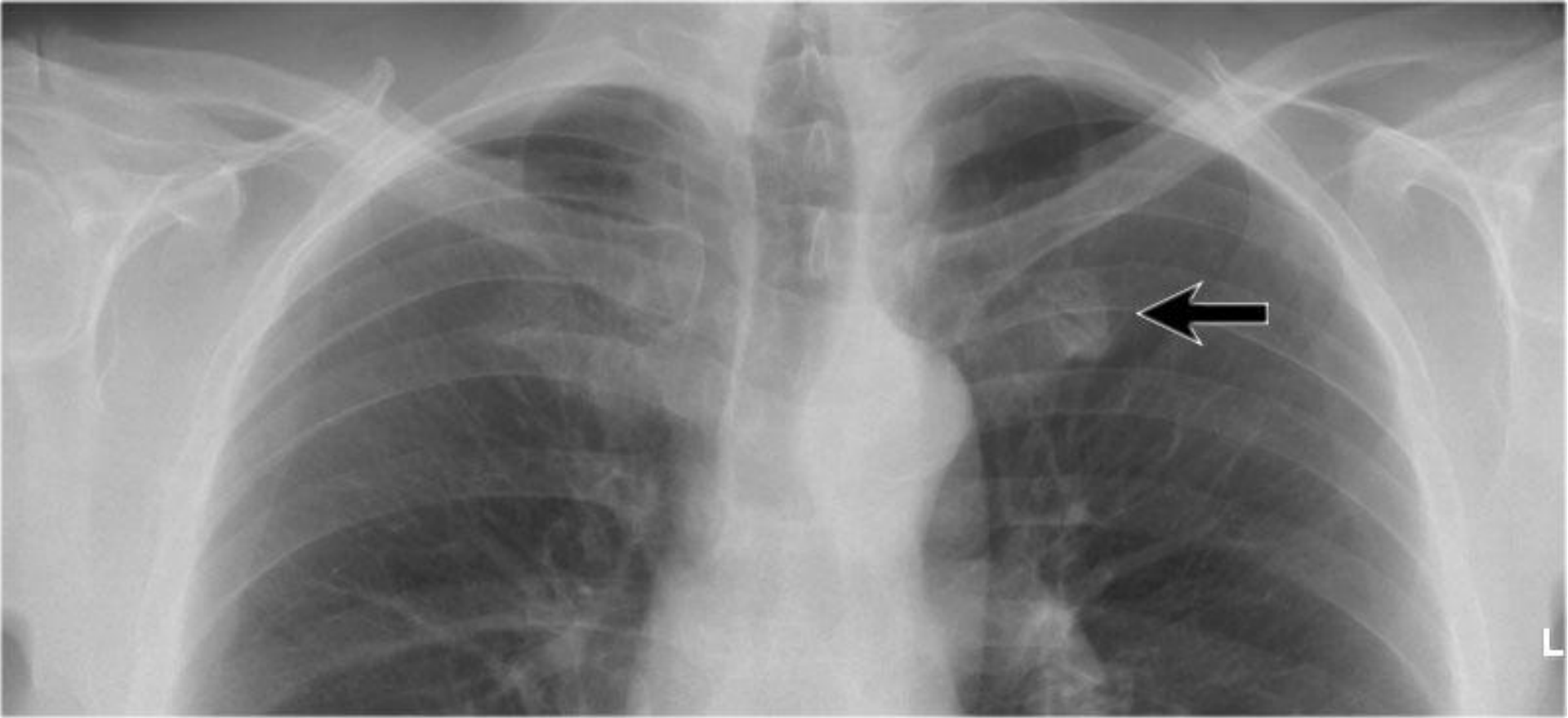


- **Vena azygos lobe**
- A common normal variant is the azygos lobe.
- The azygos lobe is created when a laterally displaced azygos vein makes a deep fissure in the upper part of the lung.
- On a chest film it is seen as a fine line that crosses the apex of the right lung.

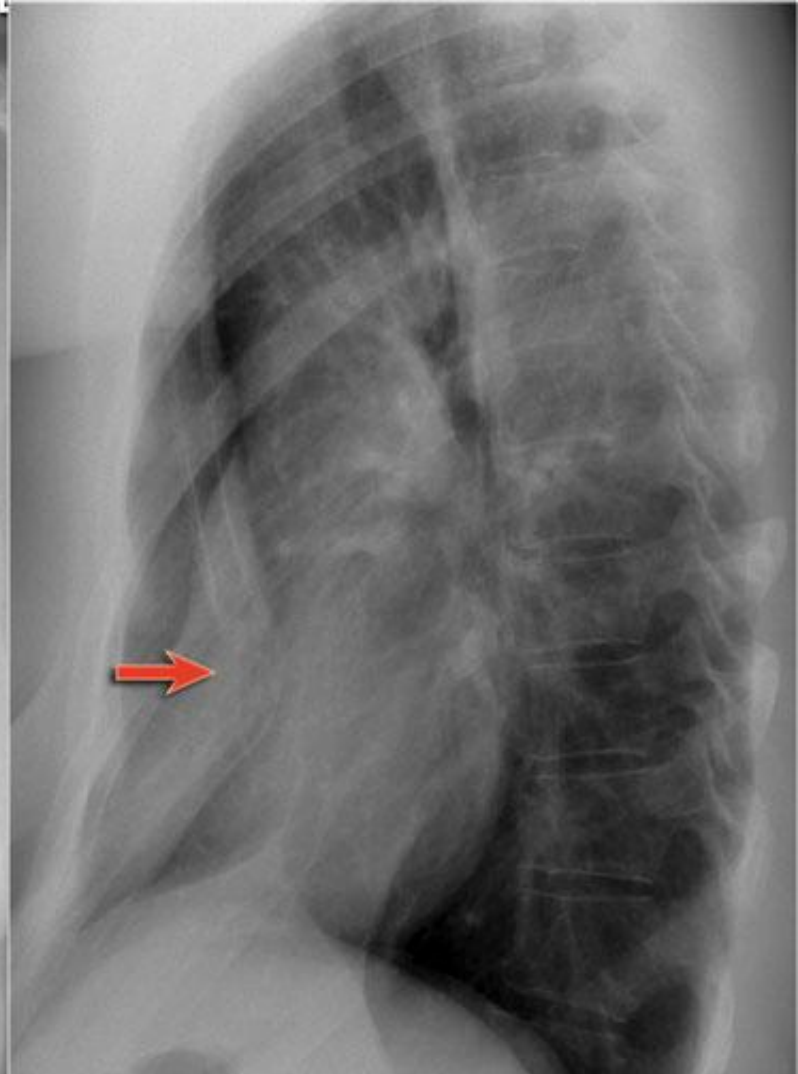
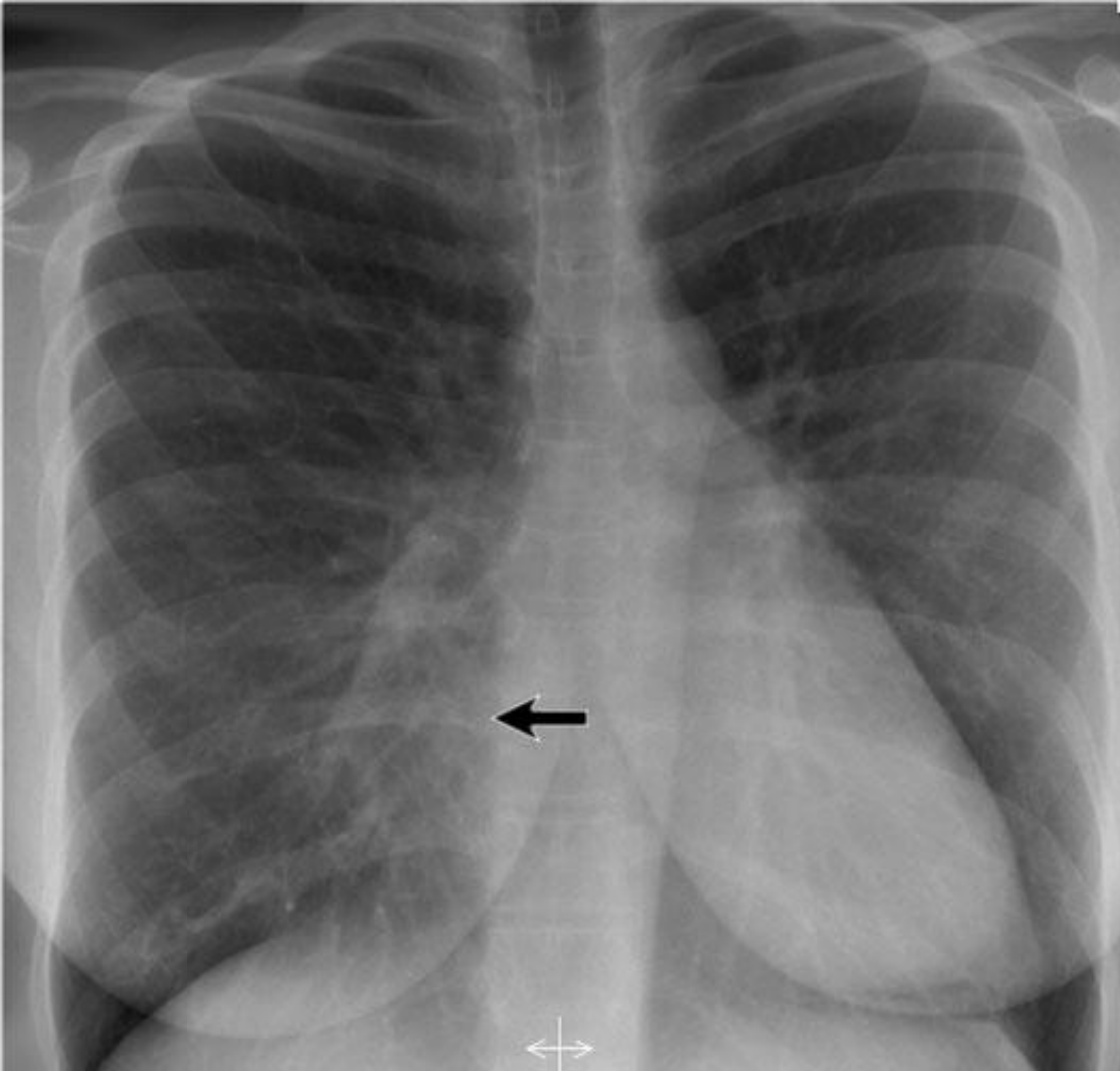


- ***Azygos lobe.***

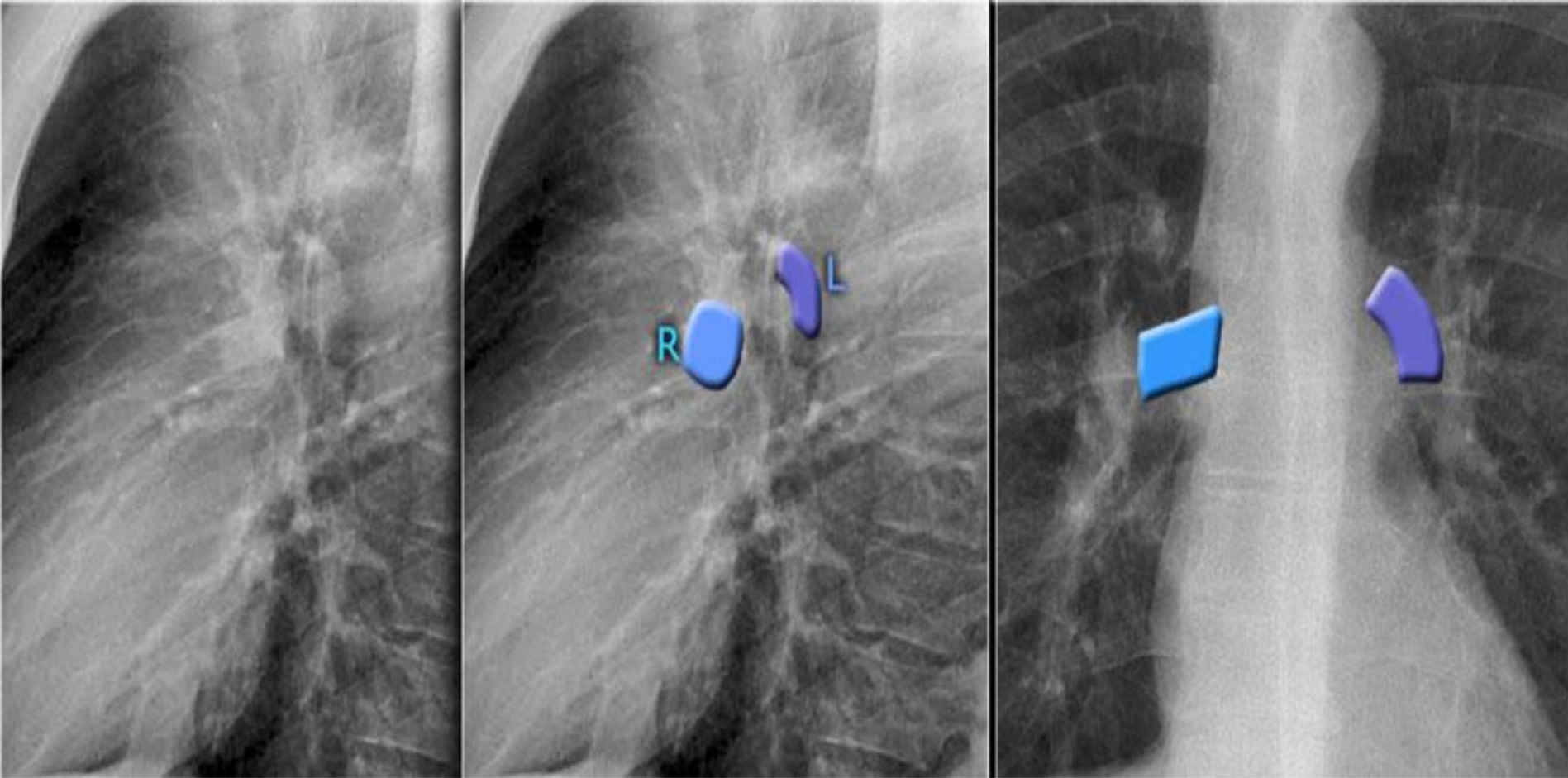
The azygos vein is seen as a thick structure within the azygos fissure.



- In some patients an extra joint is seen in the anterior part of the first rib → at the point where the bone meets the calcified cartilagenous part (arrow).
- This may simulate a lung mass.

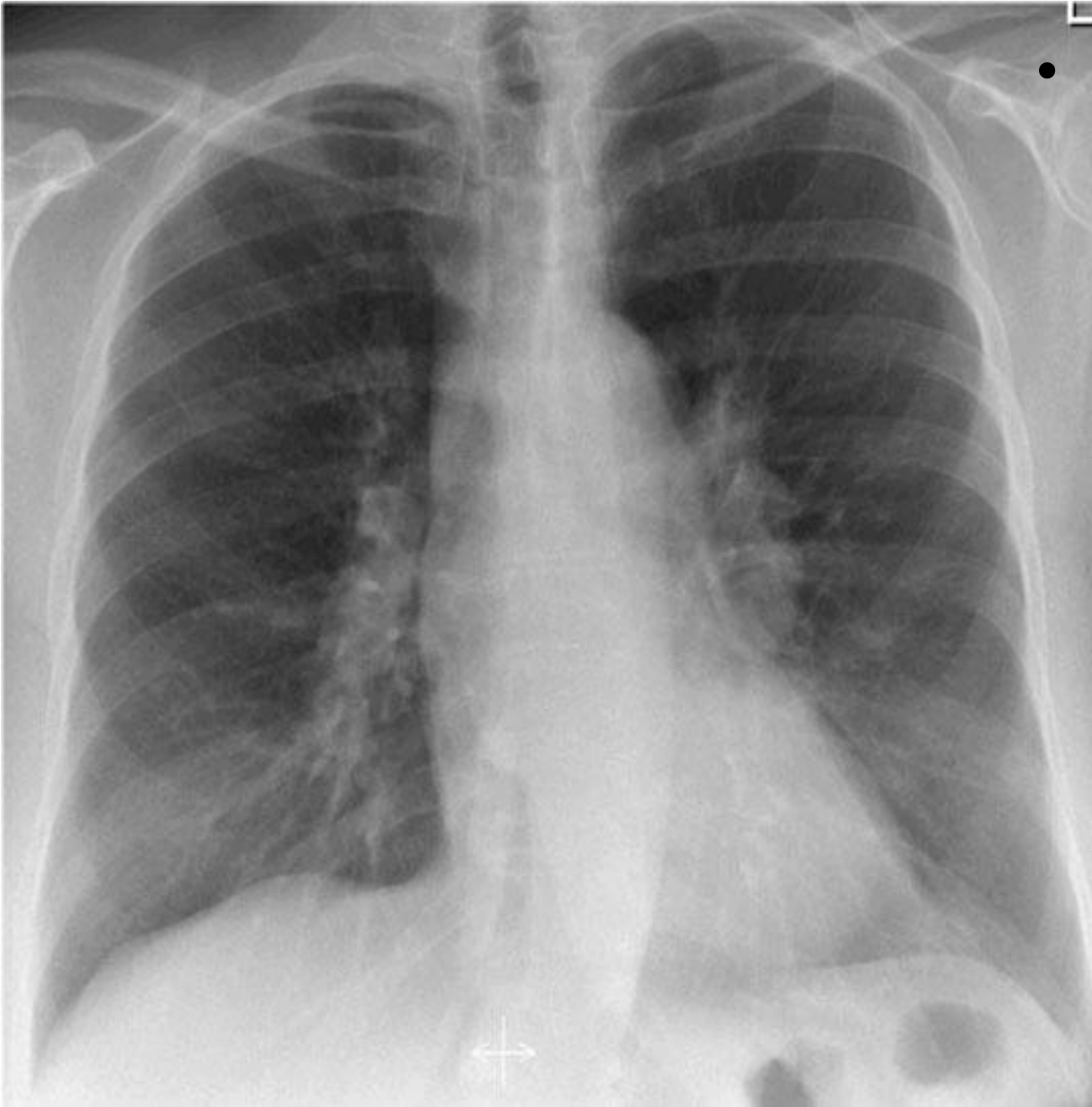


- **Pectus excavatum** : is a congenital deformity of the ribs and the sternum → a concave appearance of the anterior chest wall.
- In which right heart border can be **ill-defined**, but this is normal. It produces a [silhouette sign](#) → simulating a consolidation or atelectasis of the right middle lobe.
- The lateral view is helpful in such cases.



- The left main pulmonary artery (in purple) passes over the left main bronchus and is higher than the right pulmonary artery (in blue) which passes in front of the right main bronchus.

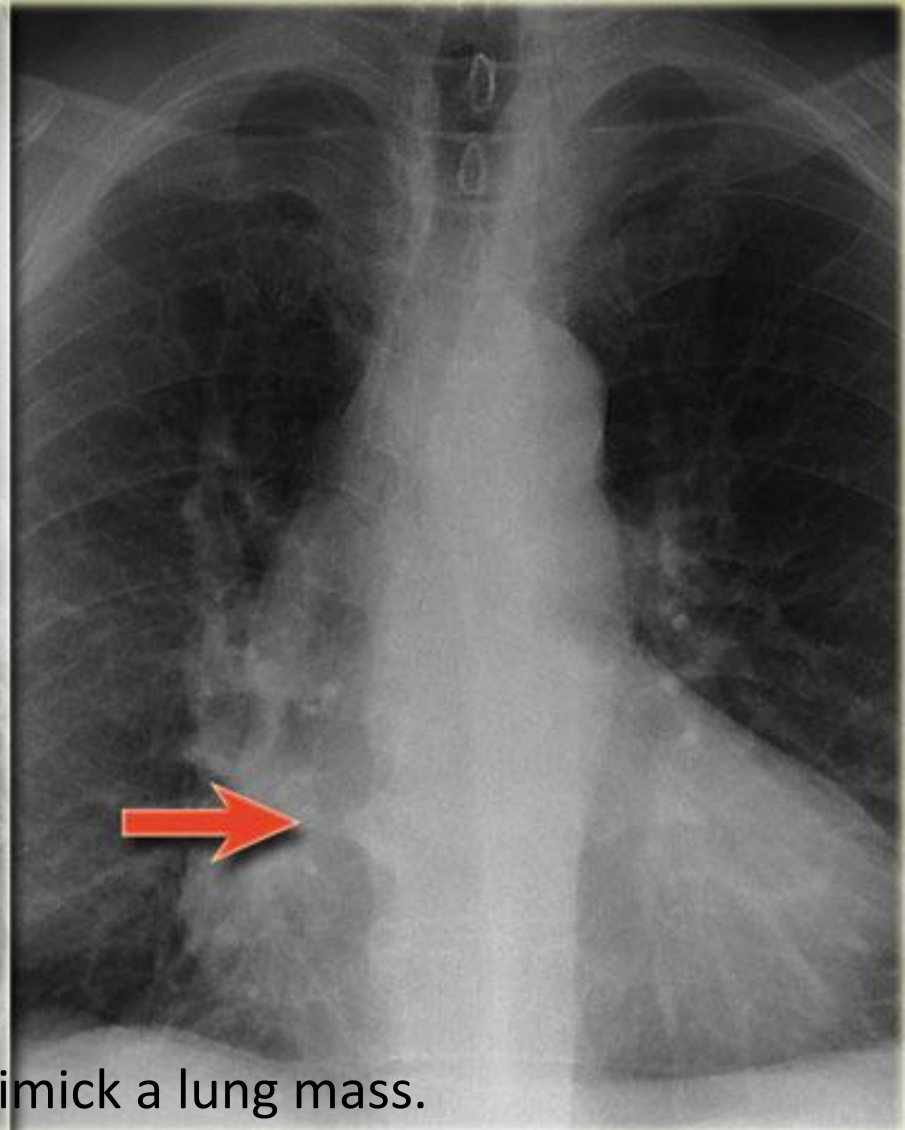
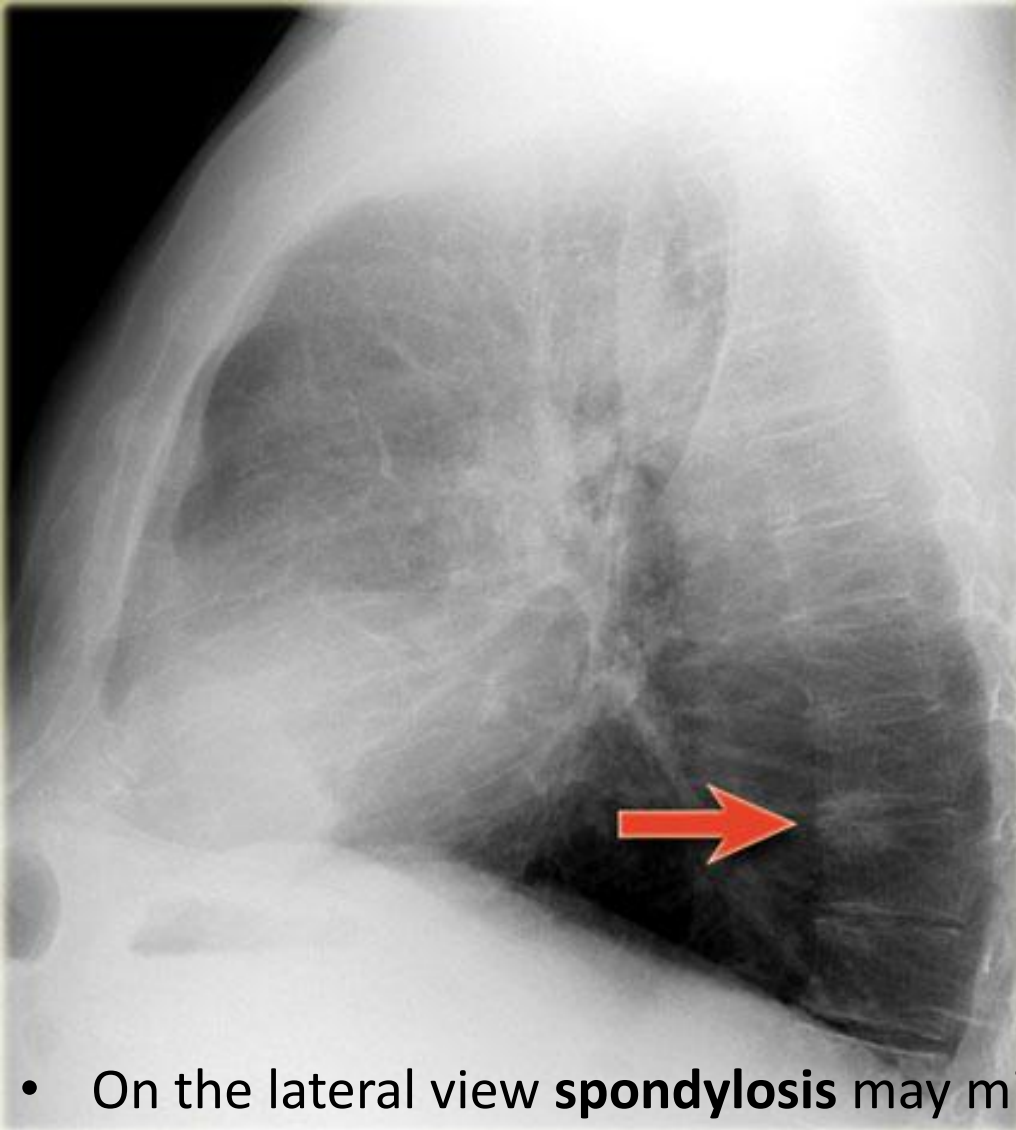
**Once you know how the normal hilar structures look like on a lateral view,
→ it is easier to detect abnormalities.**



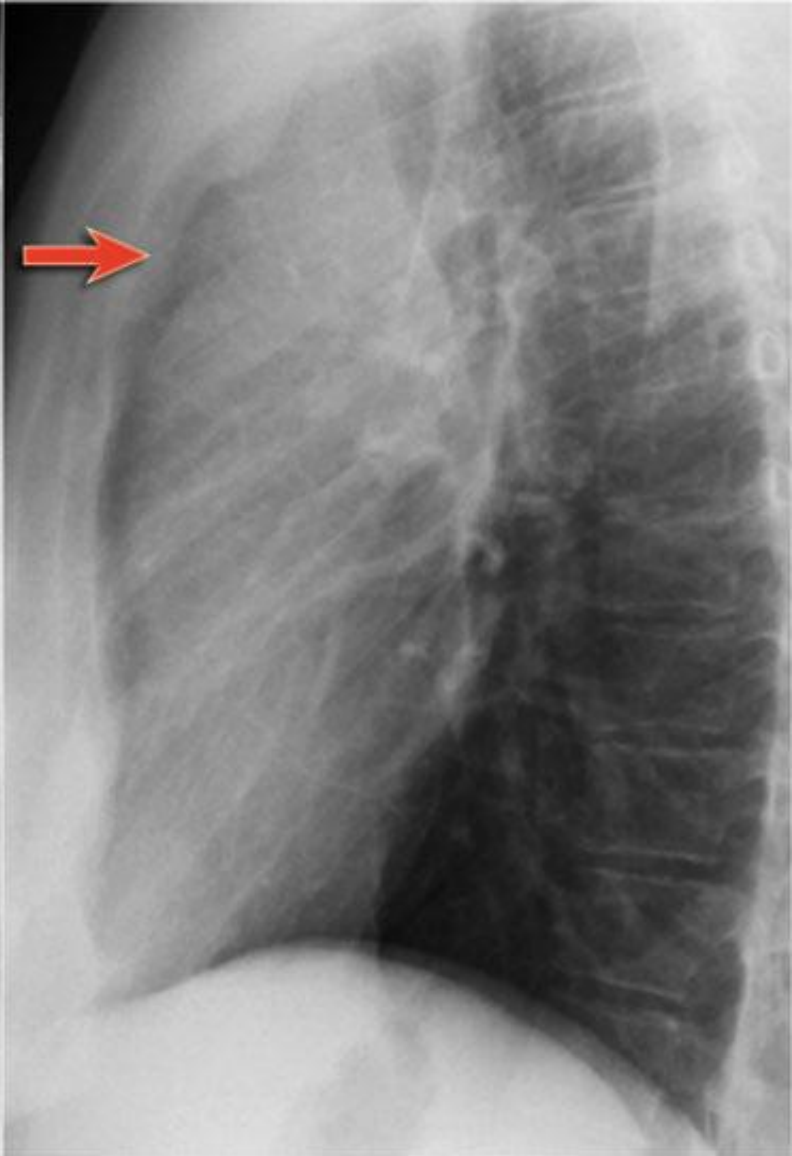
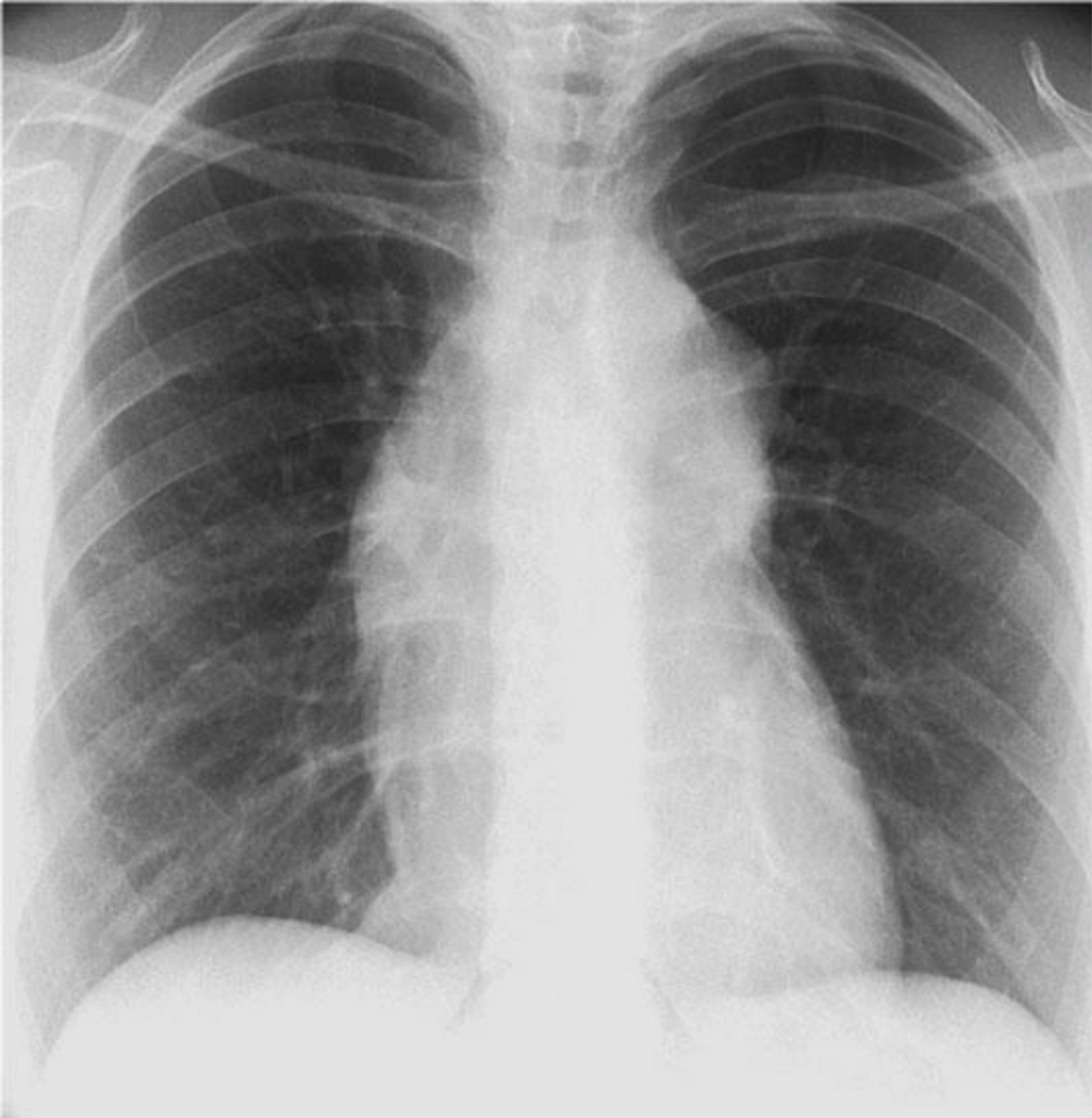
- In this case
PA-view :
hilar
enlargement.
* On the PA-
view it is not
clear
whether this
is due : to
dilated
vessels / or
enlarged
lymph nodes.



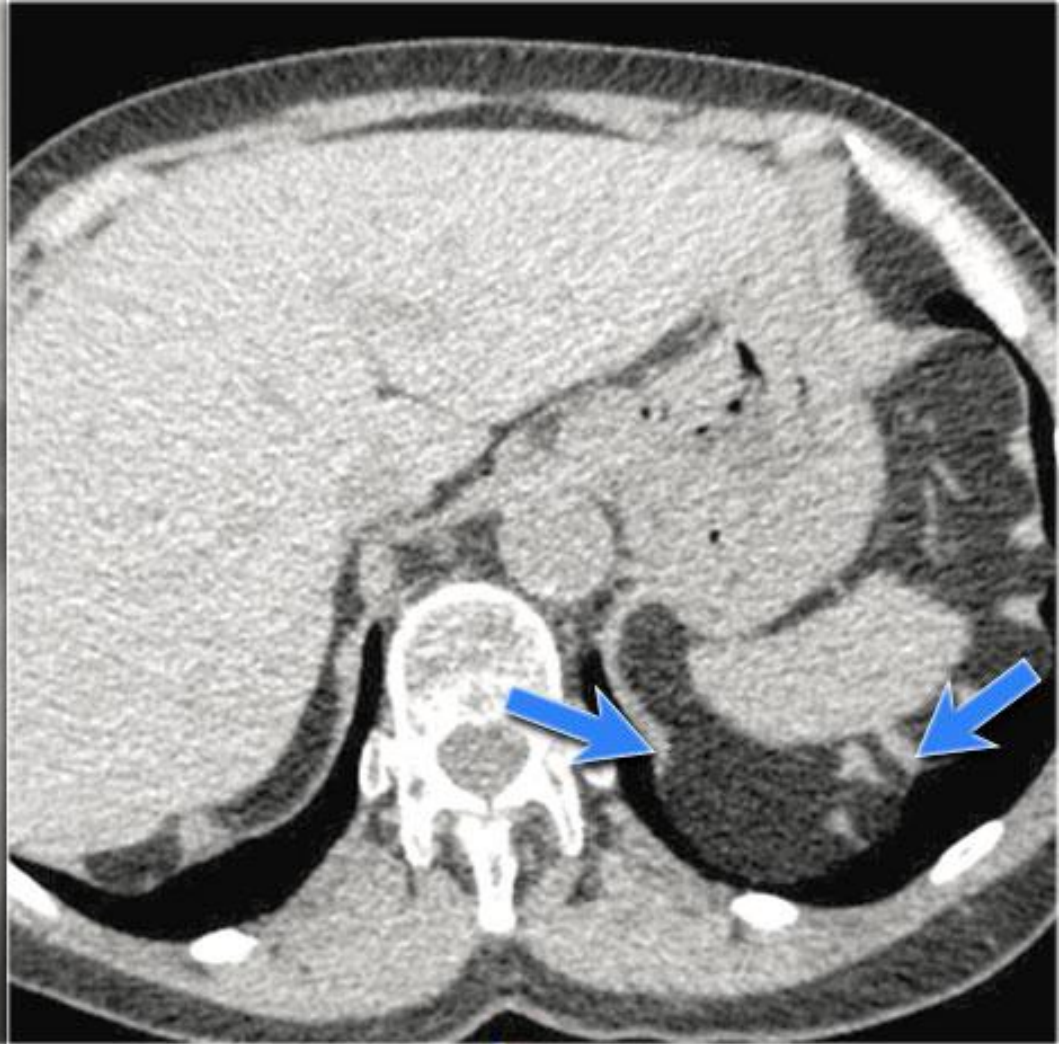
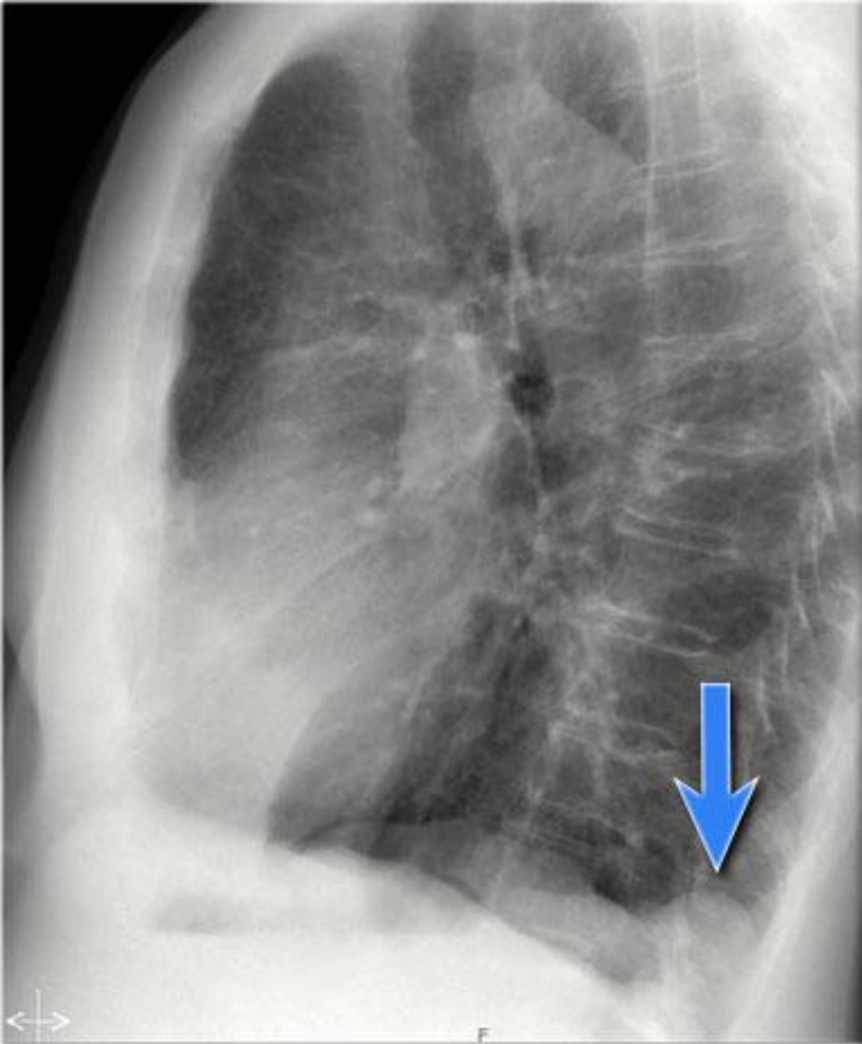
- On lateral view:
- there are round structures in areas where you don't expect any vessels.
- So we can conclude that we are dealing with enlarged lymph nodes.
- This patient has sarcoidosis.
- Notice also the widening of the paratracheal line (or stripe) as a result of enlarged lymph nodes.



- On the lateral view **spondylosis** may mimick a lung mass.
 - Any density in the area of the vertebral bodies should lead you to the PA-film to look for spondylosis, which is usually located on the right side (arrows).
- On the left side the formation of osteophytes is hampered by the pulsations of the aorta.



- On the PA-view the superior mediastinum is widened. The lateral view is helpful in this case because it demonstrates a density in the retrosternal space. Now the differential diagnosis is limited to a mass in the anterior mediastinum (4 T's).
- This was a **Hodgkins lymphoma**.



- A common incidental finding in adults is a **Bochdalek hernia**, which is due to a congenital defect in the posterior diaphragm (arrows). In most cases it only contains retroperitoneal fat and is asymptomatic, but occasionally it may contain abdominal organs.
- Large hernias are sometimes seen in neonates and can be complicated by pulmonary hypoplasia.
- A hernia of **Morgagni** is also a congenital diaphragmatic hernia, but is less common. It is located anteriorly.

Systematic Approach

- *Whenever you review a chest x-ray,
..... always use a systematic approach.*

- We use an inside-out approach from central to peripheral.
= First the Hearth figure is evaluated, → followed by mediastinum and Hili → Subsequently the Lungs, lung borders and → finally the chest wall and Abdomen are examined.
- You have to know the **normal anatomy** and **Variants**.
- Find subtle abnormalities by using the **sihouette sign** and **mediastinal lines**.

Once you see an abnormality use a pattern approach to come up with the most likely diagnosis and differential diagnosis.

Chest X-Ray

systematic
approach

Inside-Out approach

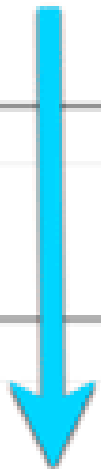
- Heart
- Mediastinum + Hili
- Lungs
- Thoracic wall
- Abdomen



Abnormality

Pattern

Diff diagnosis



Old films

- Always compare with old films,
Actually someone said that the most important radiograph is the old film, since it gives you so much information.
- A lung mass, which hasn't changed in many years
→ is not a lung cancer.
- First study the chest films. Then continue.





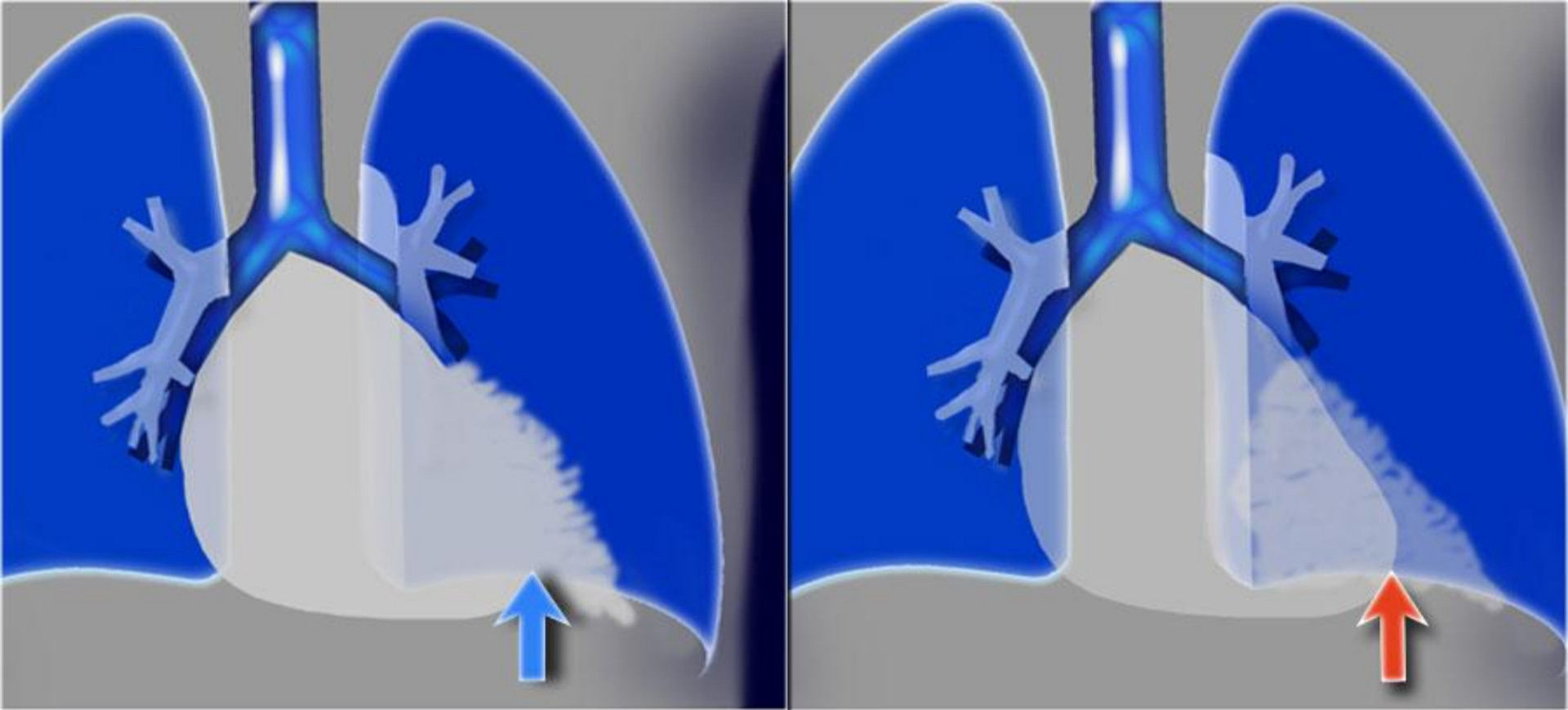
- Based on the CXR that you just saw, you could have made the diagnosis of → **congestive heart failure**, but the findings are very subtle.

However once you compare it to the **old film**, things become more obvious and you will be much more confident in your diagnosis:

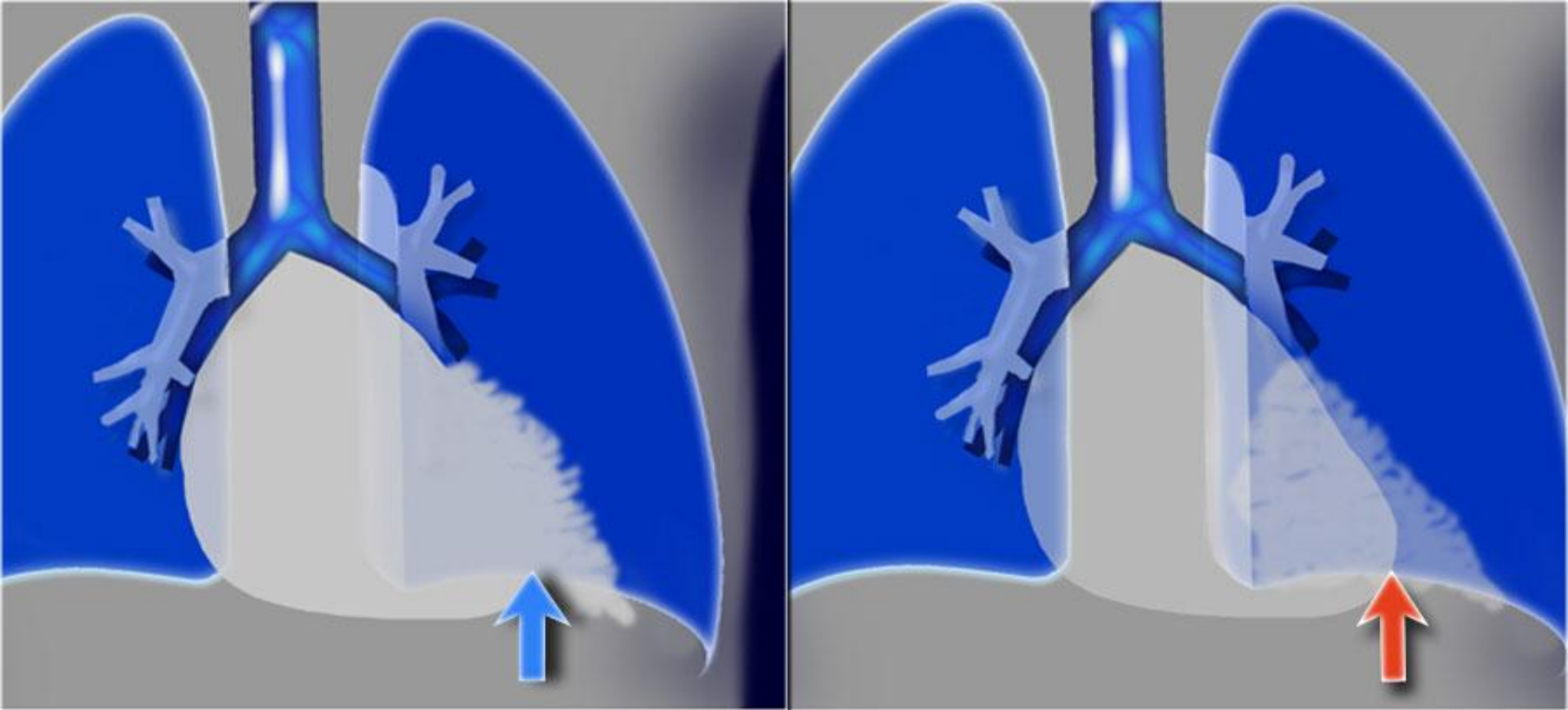
- The **size of the heart** is slightly increased compared to the old film.
- The pulmonary vessels are slightly increased in **diameter** indicating increased pulmonary pressure.
- There are subtle interstitial markings as a result of interstitial edema.
- There is pleural fluid bilaterally.
- Notice that the inferior border of the lower lobes has changed in position.
- ***All these findings indicate the presence of heart failure.***

Silhouette sign

- A very important sign.
- It enables us to : find subtle pathology and to locate it within the chest.
- The loss of the normal silhouette of a structure is called *the silhouette sign*.

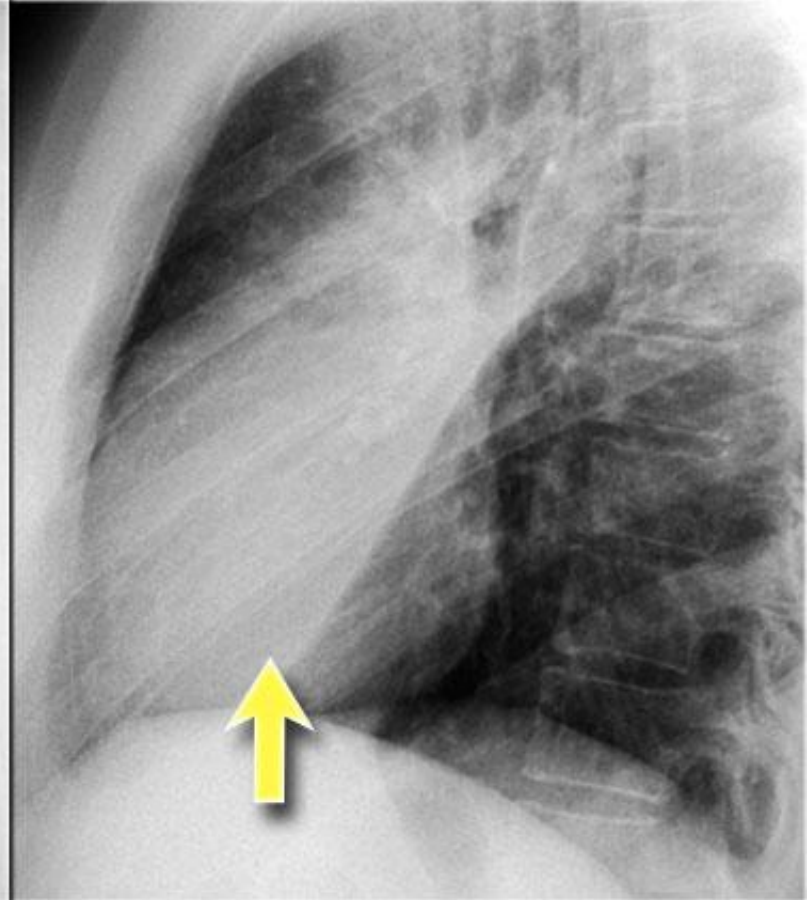
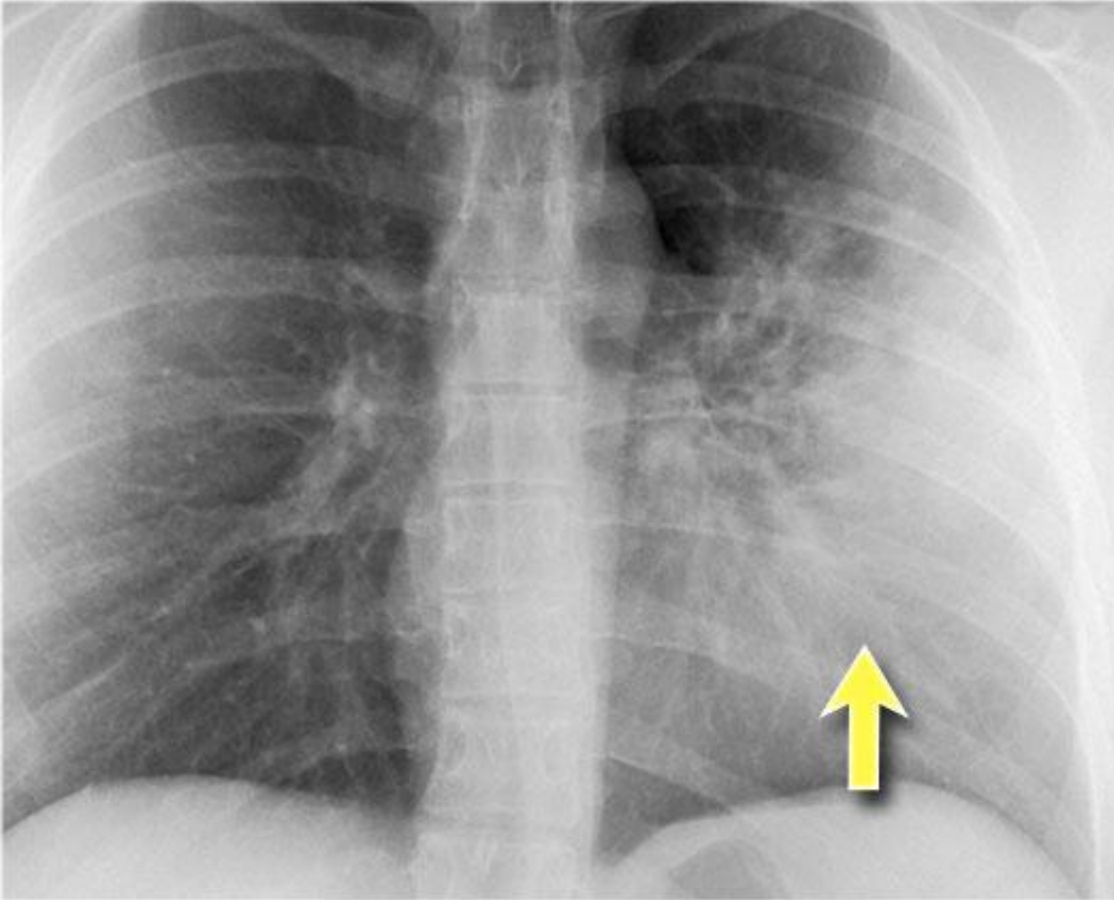


- The heart is :
 - located anteriorly in the chest and it is bordered by the lingula of the left lung.
- The difference in density between the heart and the air in the lung enables us to see the silhouette of the left ventricle.
 When there is **something in the lingula** with the **same 'water density'** as the heart, → the normal silhouette will be lost (blue arrow).
- When there is a **pneumonia in the left lower lobe**, which is **located more posteriorly** in the chest, the left ventricle will still be bordered by air in the lingula and we will still see the silhouette of the heart (red arrow).

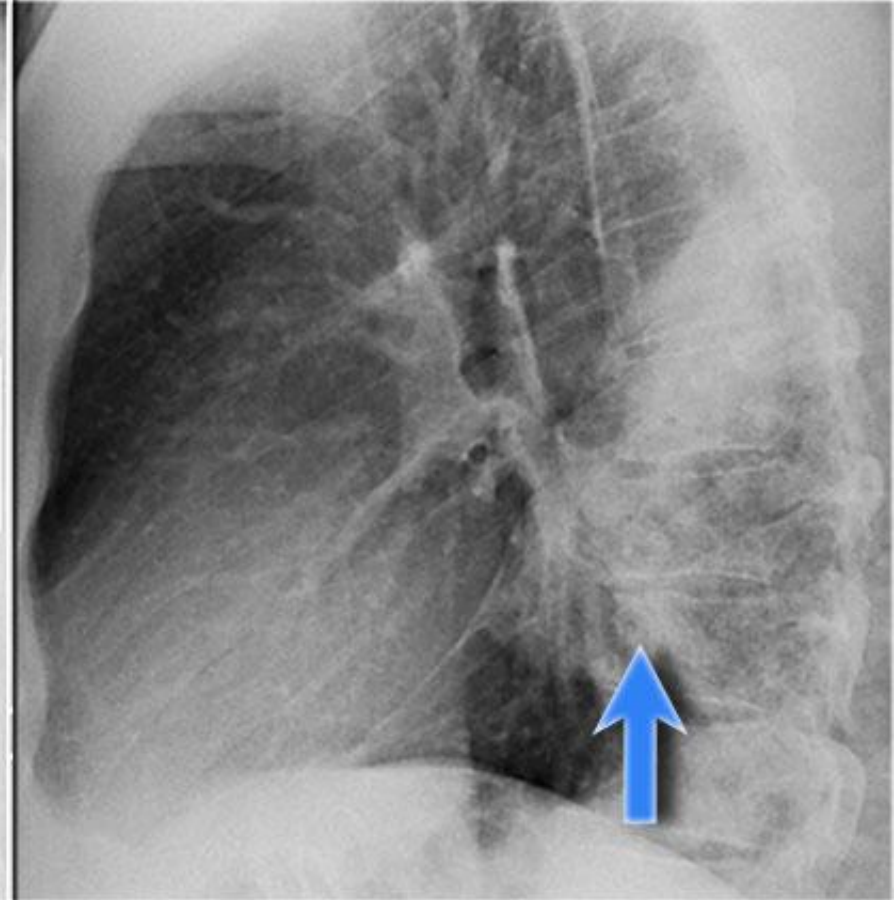
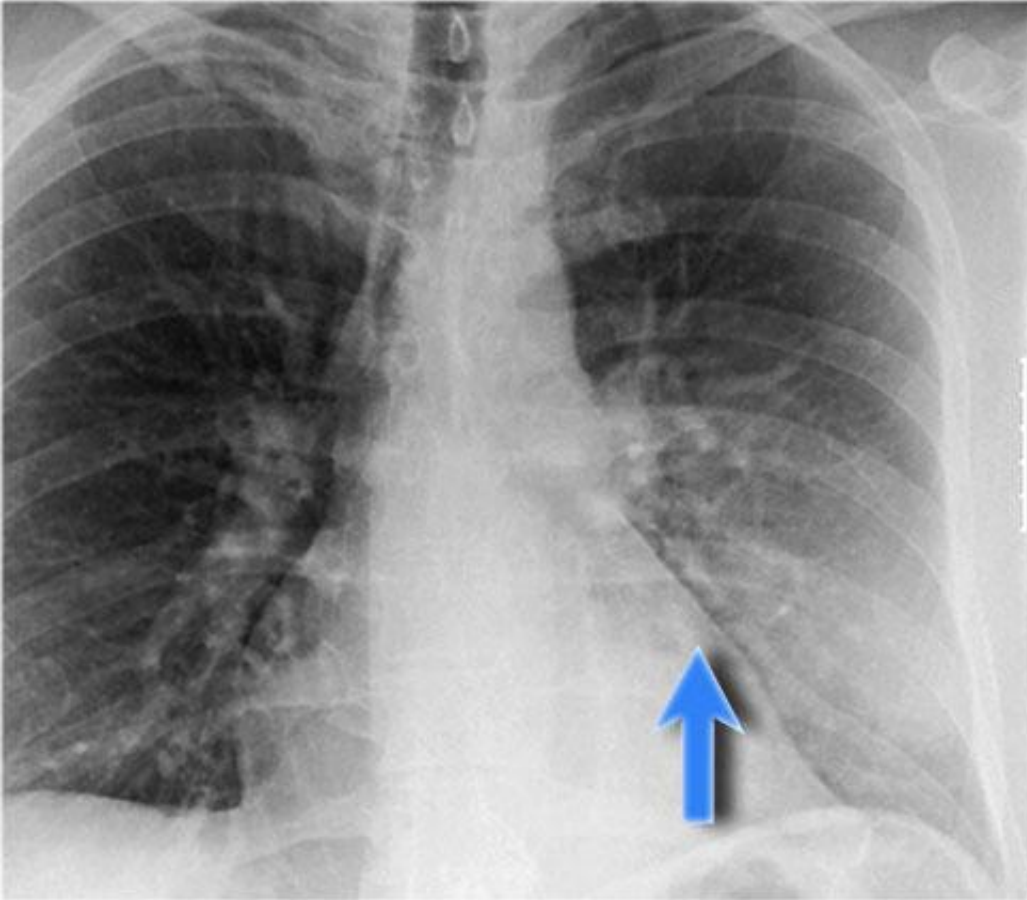


i.e.

- Pathology + masked Rt cardiac border = Lingula Pathology “Ant”
- Pathology + seen Rt cardiac border = Lt Lower Lobe Pathology “post”



- The PA-film shows a silhouette sign of the left heart border. Even without looking at the lateral film, → the pathology must be located anteriorly in the left lung.
- This was a consolidation due to a pneumonia caused by *Sterptococcus pneumoniae*.

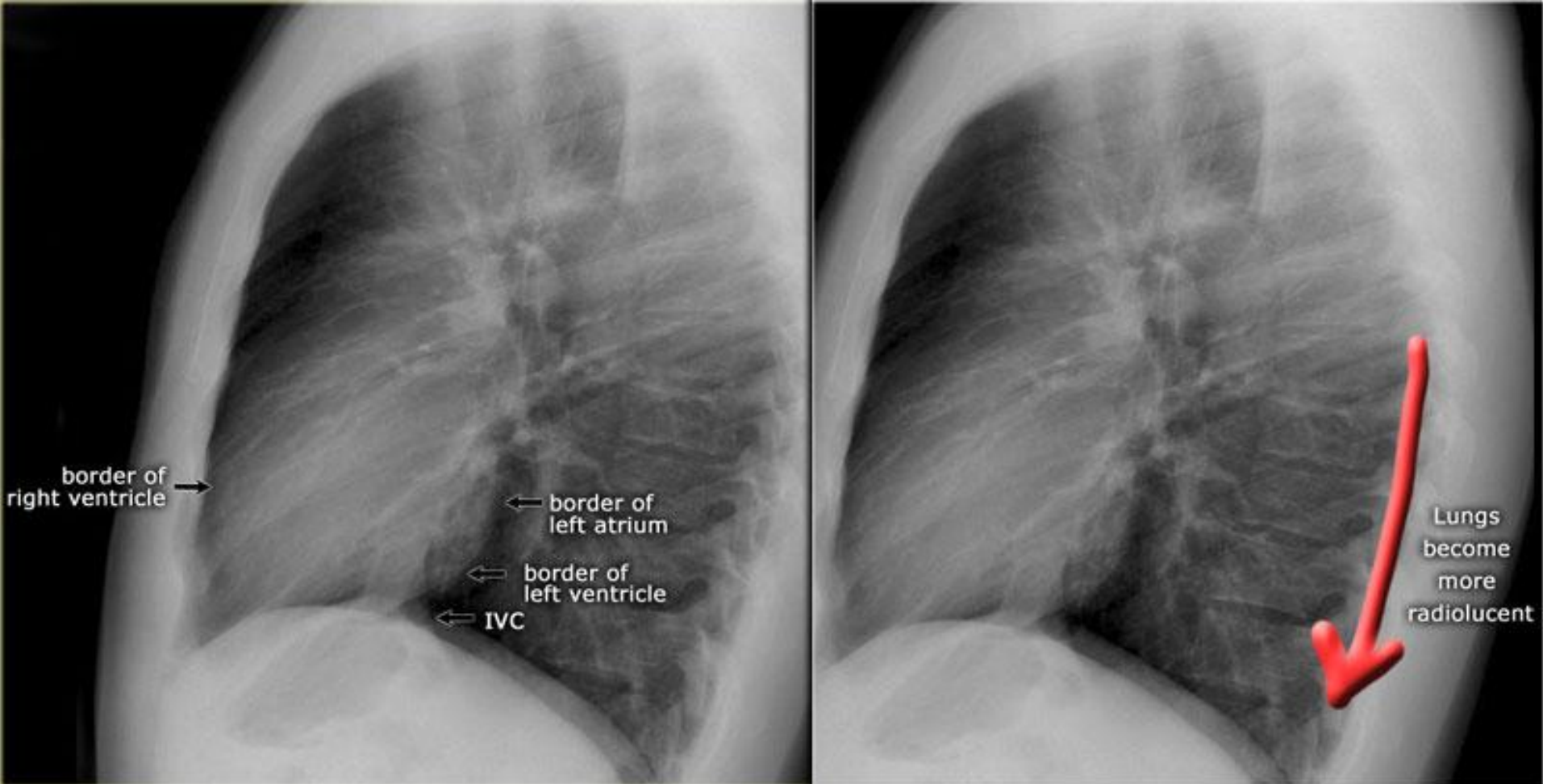


- a consolidation which is located in the left lower lobe.

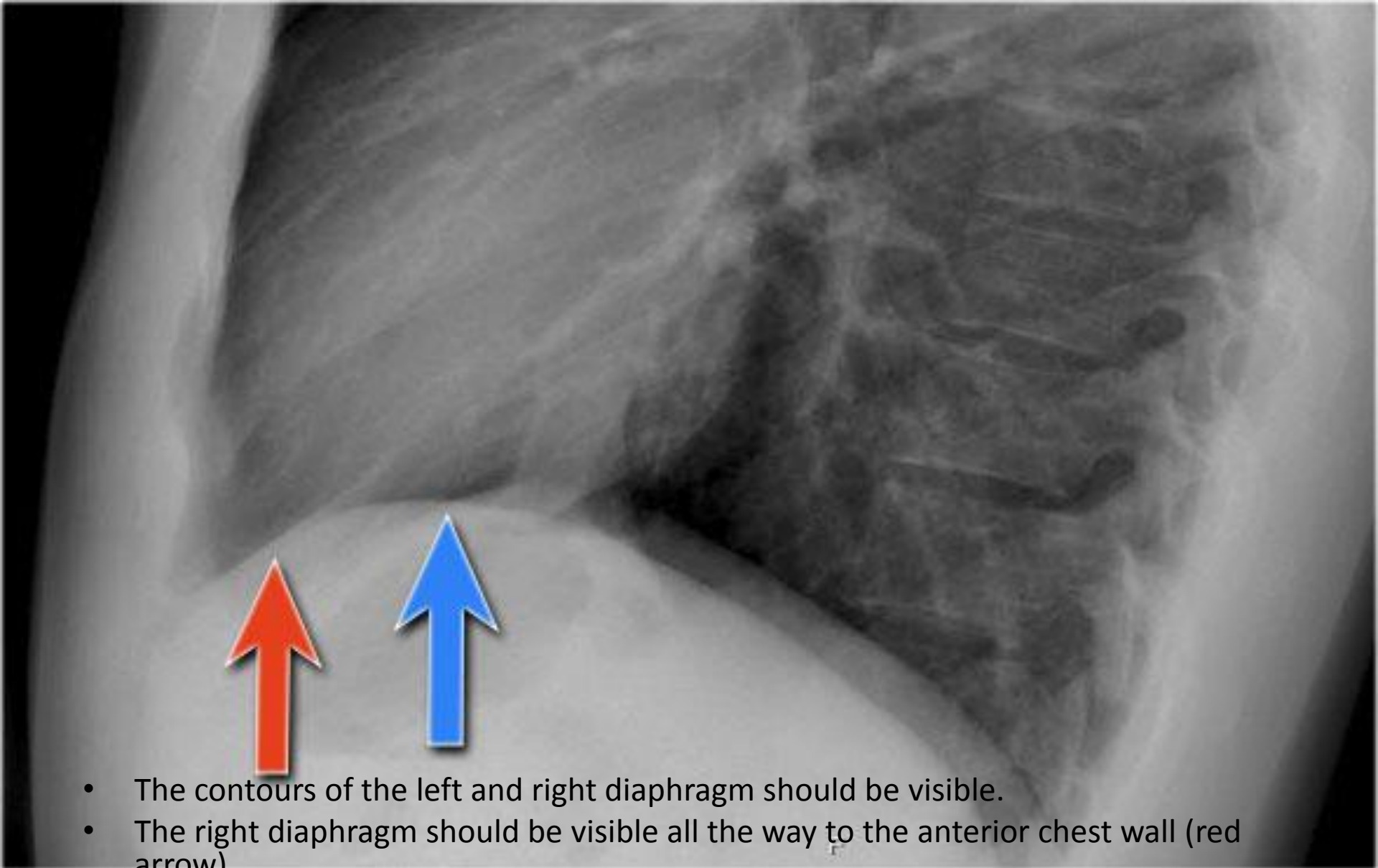
There is a normal silhouette of the left heart border.

LATERAL FILM





- The retrosternal space should be radiolucent, ← since it only contains air.
- Any radiopacity in this area is suggestive of a process in the anterior mediastinum or upper lobes of the lung.

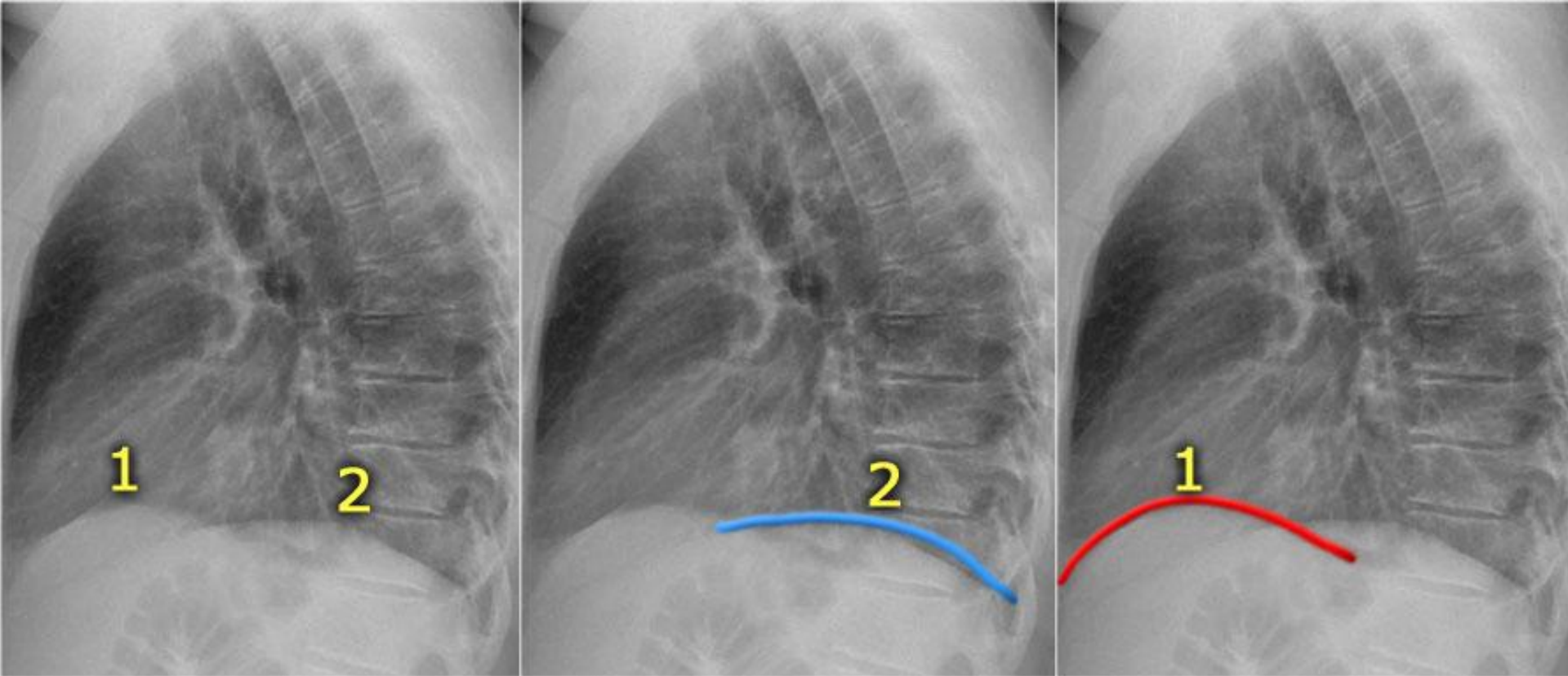


- The contours of the left and right diaphragm should be visible.
- The right diaphragm should be visible all the way to the anterior chest wall (red arrow).

Actually we see the interface between the air in the lungs and the soft tissue structures in the abdomen.

- The left diaphragm can only be seen to a point where it borders the heart (blue arrow).

- On this lateral film there is too much density over the lower part of the spine.
- By only looking at the interfaces of the left and right diaphragm on the lateral film, it is possible to tell on which side the pathology is located.
- First study the lateral film.
Then continue.



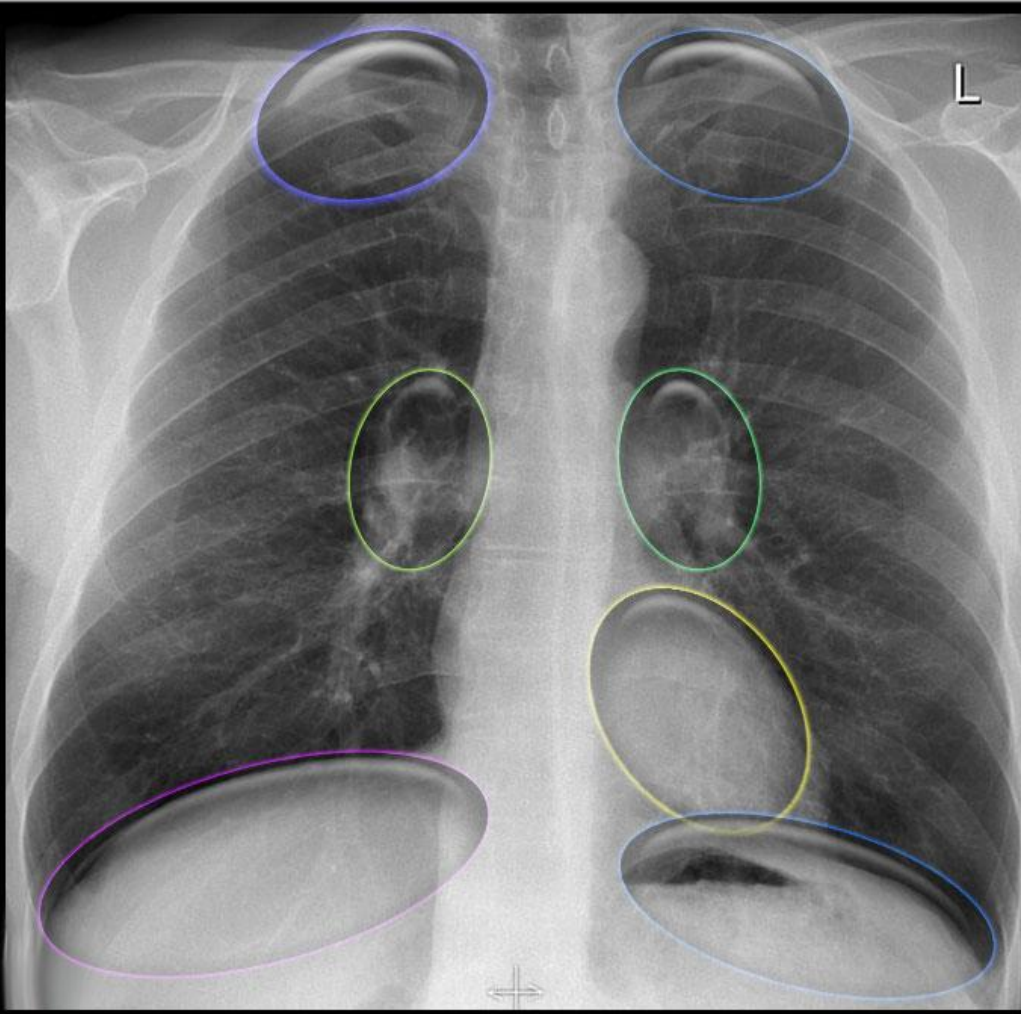
- On a normal lateral chest film the silhouette of the left diaphragm 2- can be seen from posterior up to where it is bordered by the heart, which has the same density (blue arrow).
- One should be able to follow the contour of the right diaphragm -1- from posterior all the way to anterior, because it is only bordered by the lung.
- Here we ***cannot follow the contour of the right diaphragm*** all the way to posterior, which indicates that there is something of ***water-density in the right lower lobe (red arrow)***.



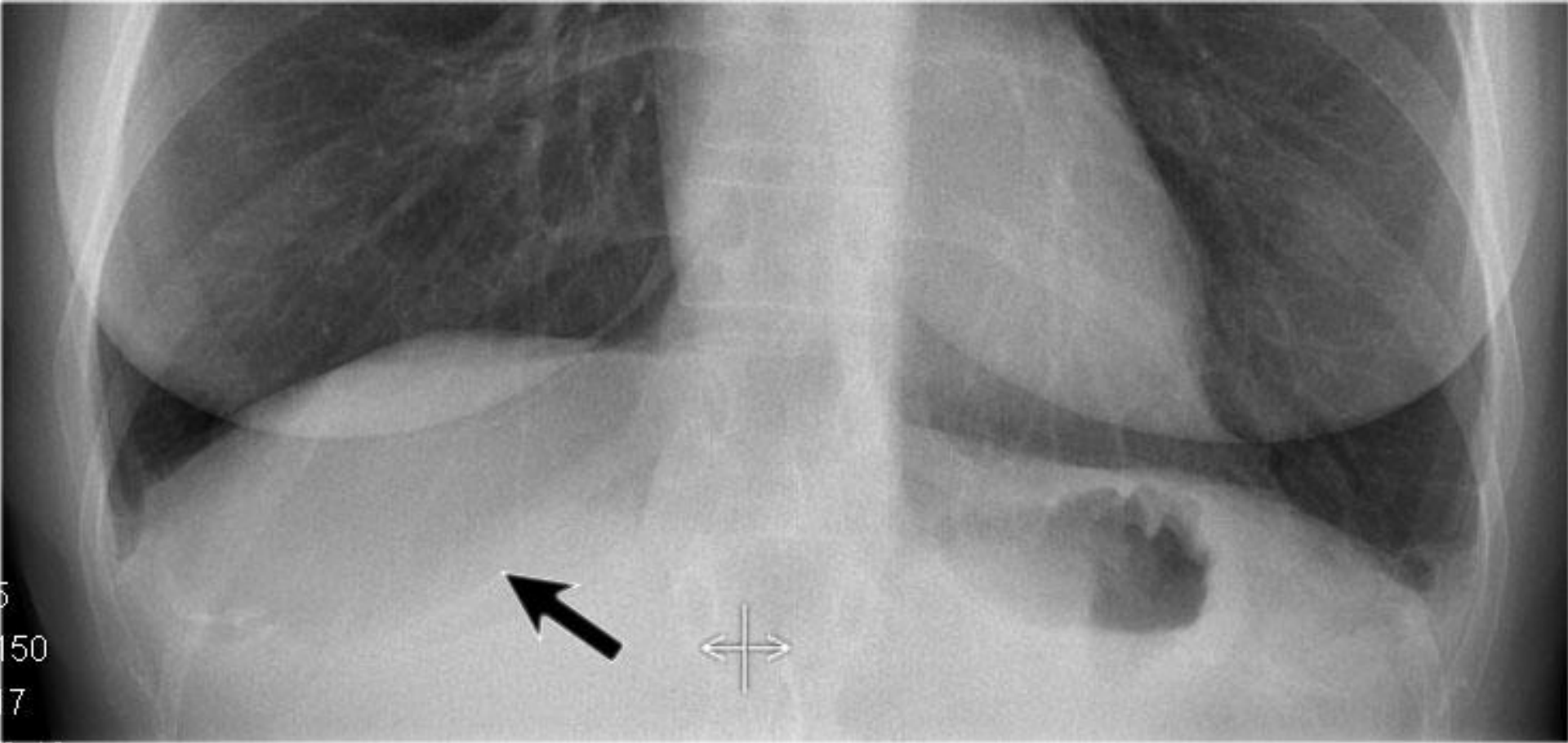
- **On the PA-film:** normal silhouette of the heart border, → so the pathology is not in the anterior part of the chest, which we already suspected by studying the lateral view.
- ***Why do we still see the silhouette of the right diaphragm on the PA-film?***
- What we see is actually the highest point of the right diaphragm, which is anterior to the pneumonia in the right lower lobe.
The pneumonia does not border the highest point of the diaphragm.

Hidden areas

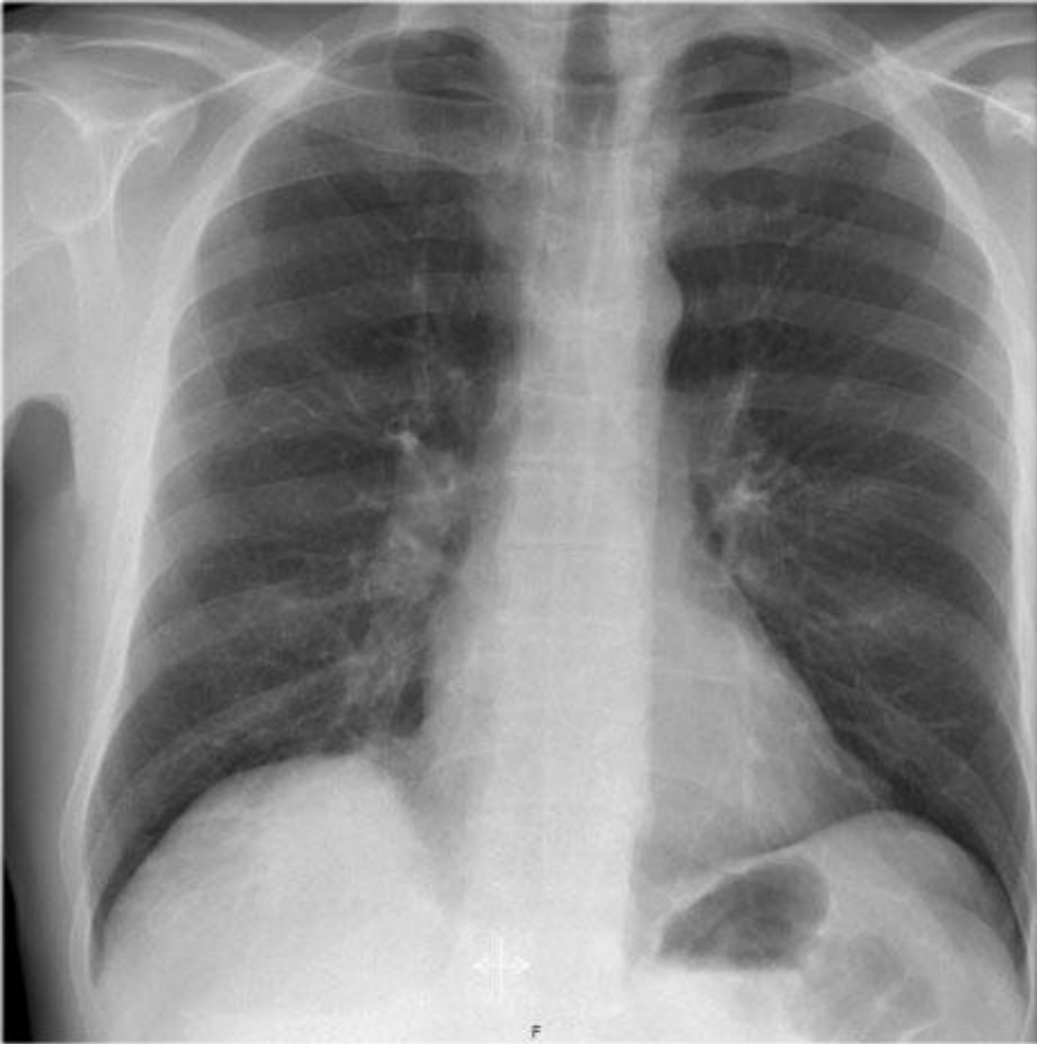
- Apical zones
- Hilar zones
- Retrocardial zone
- zone below the dome of diaphragm



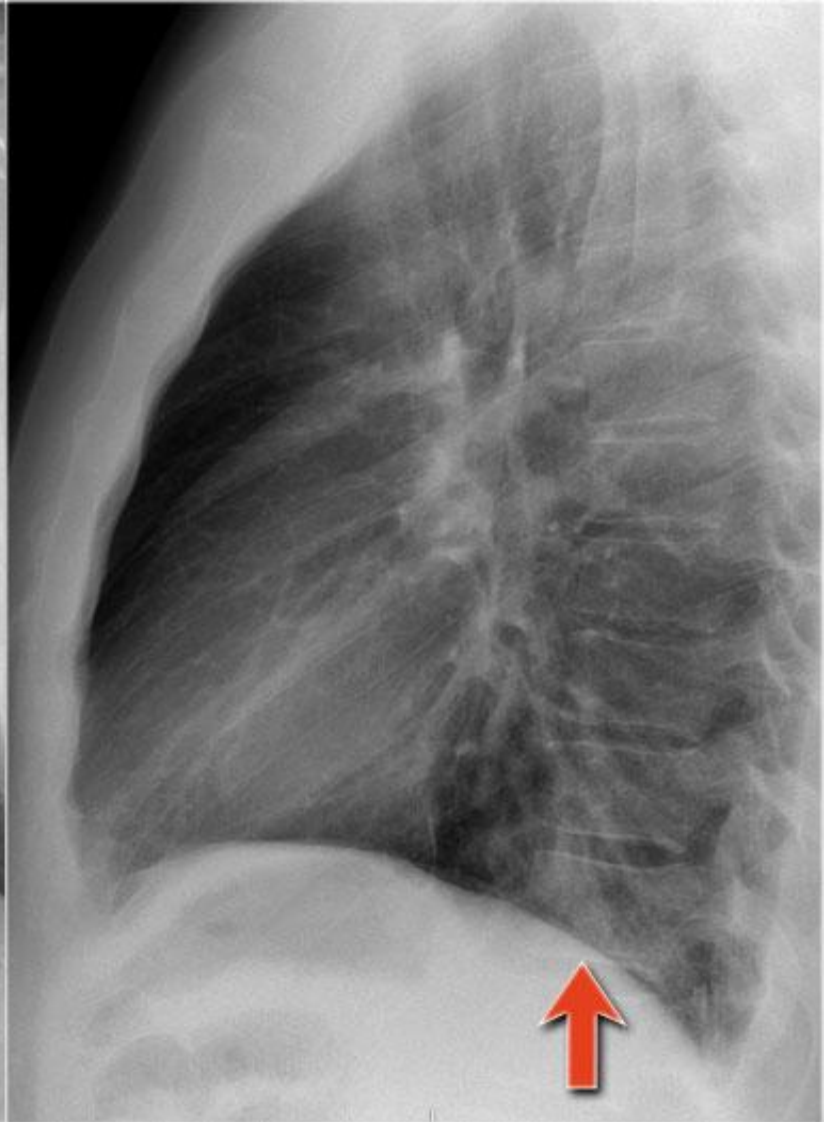
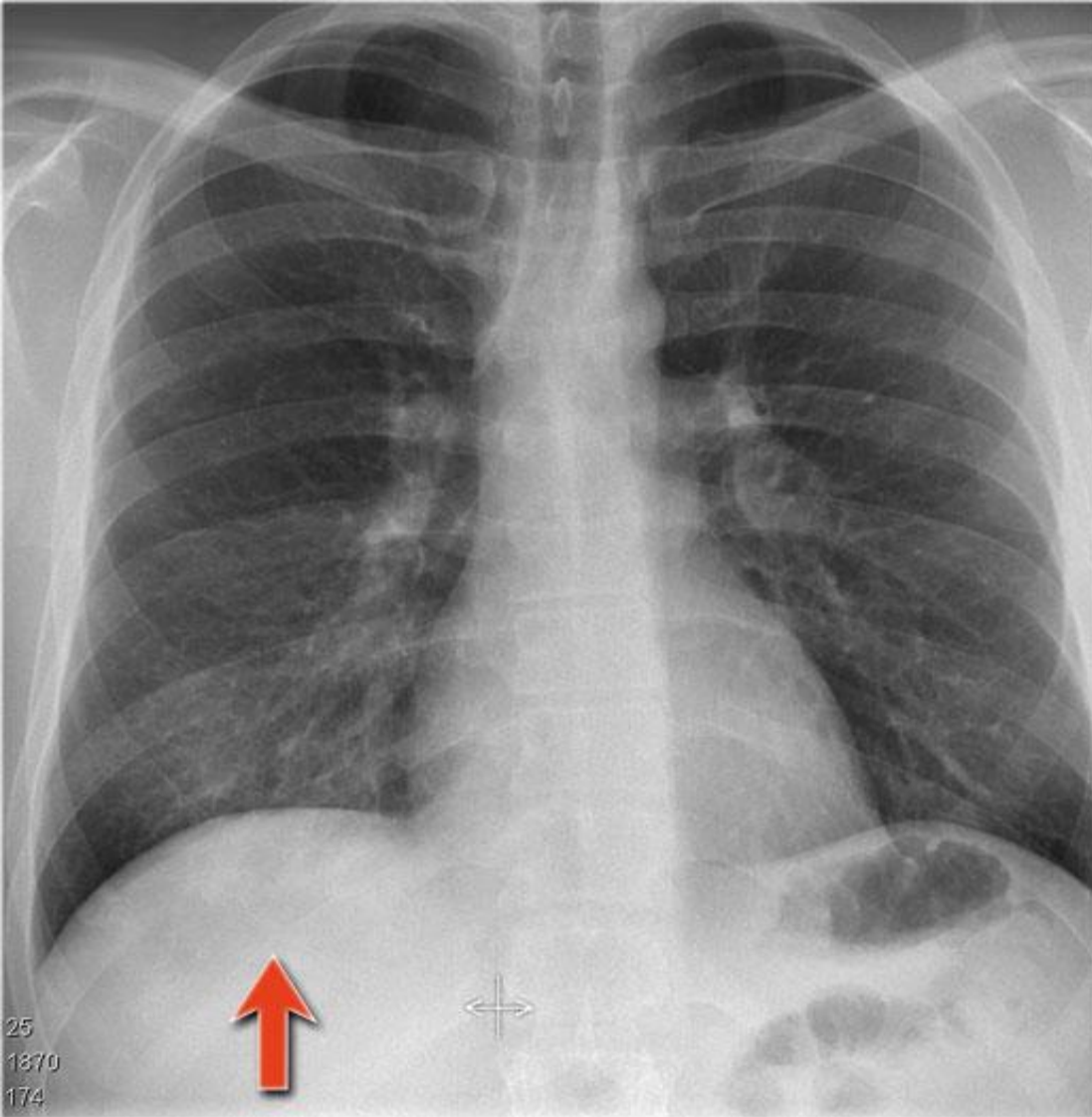
Some areas that need special attention,
pathology in these areas can easily be overlooked:
These areas are also known as *the hidden areas*.



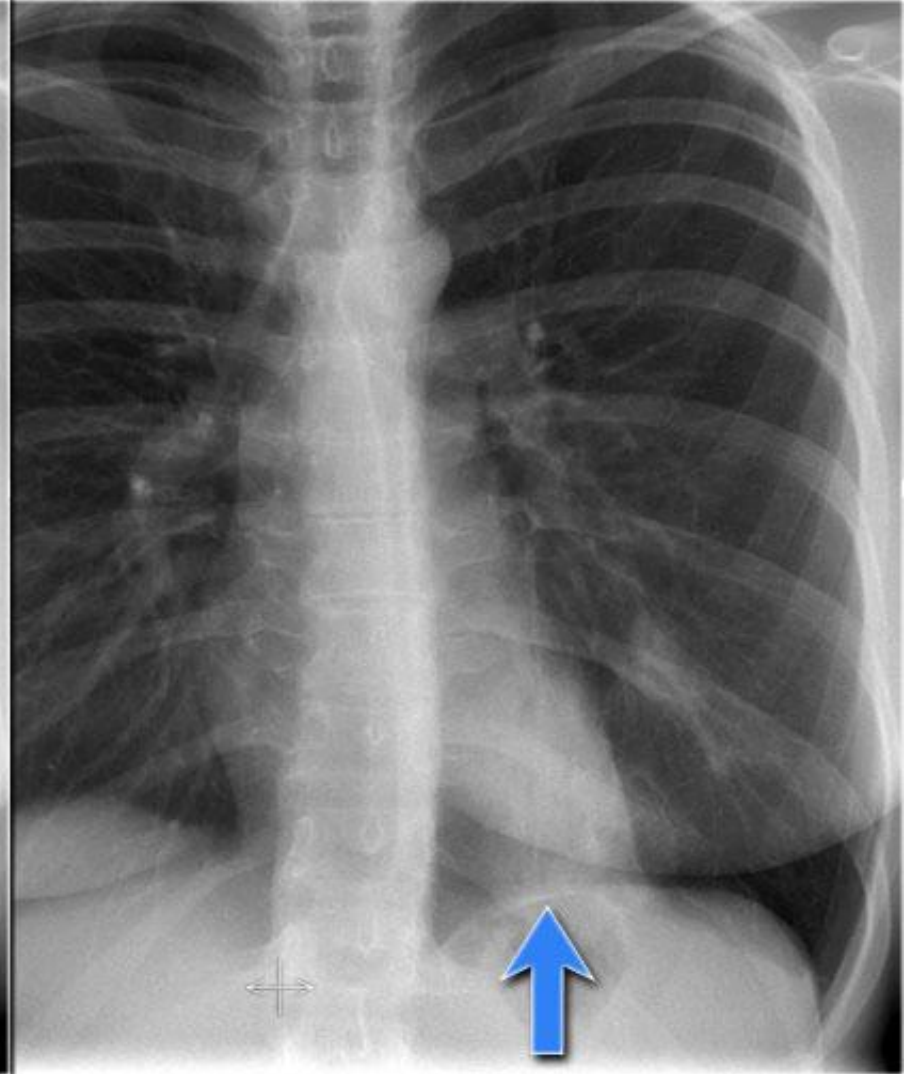
- Notice : quite some lung volume below the dome of the diaphragm, which will need your attention (arrow).



- **An example** of a **large lesion in the right lower lobe**, which is difficult to detect on the PA-film, unless when you give special attention to the hidden areas.
- Click on the image for an enlarged view.



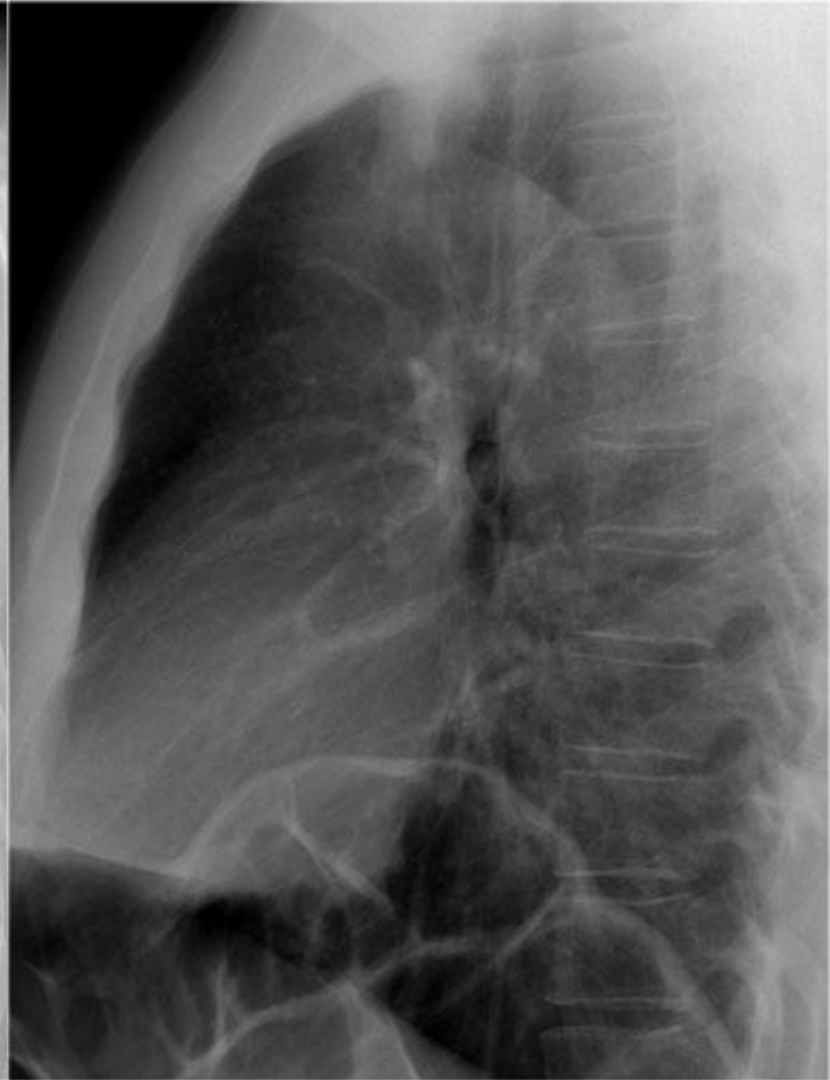
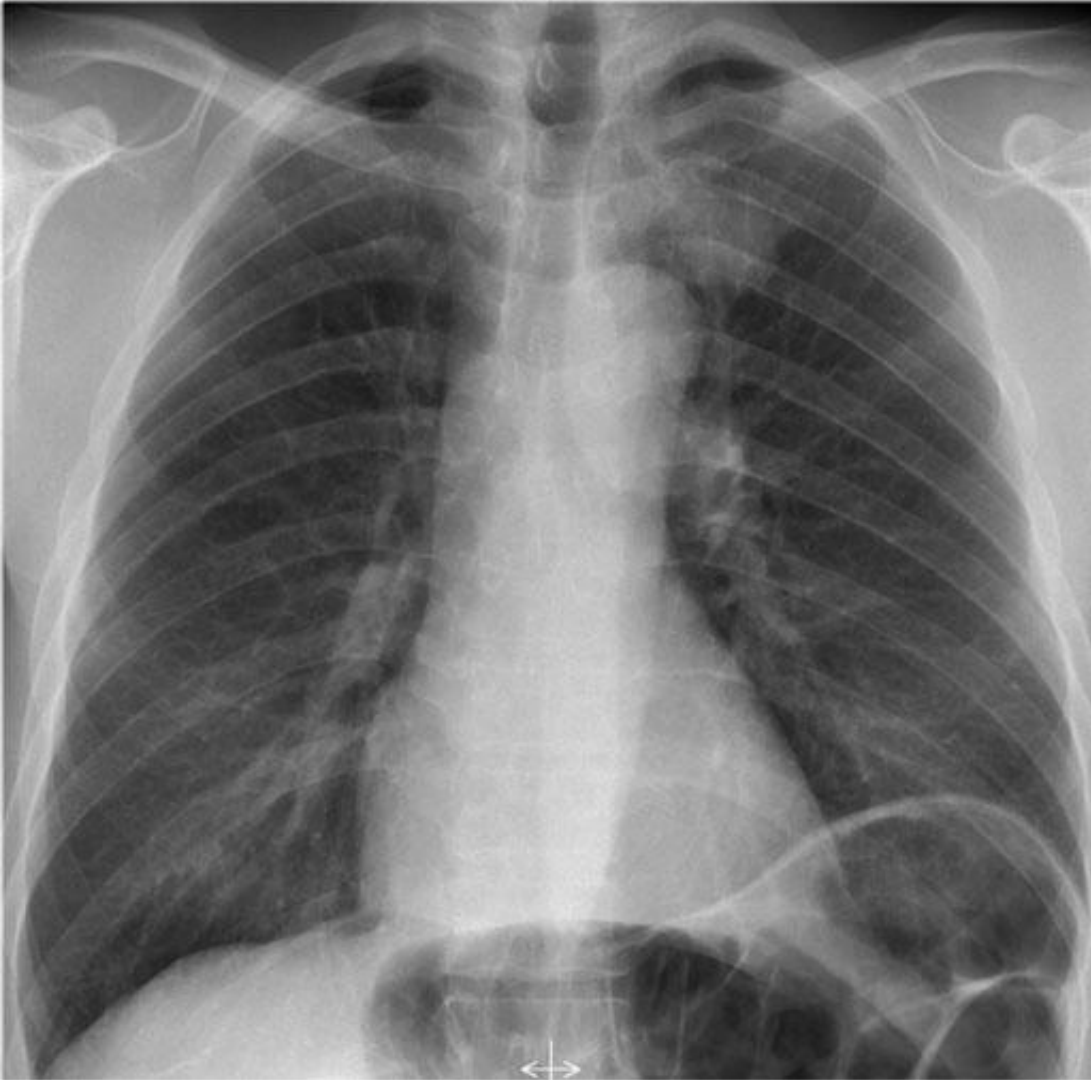
- **Pneumonia** in the right lower lobe mainly below the level of the dome of the diaphragm (red arrow).
- Notice the increase in density on the lateral film in the lower vertebral region.



- Notice the subtle increased density in the area behind the heart that needs special attention (blue arrow).
→ a lower lobe pneumonia.

- We know that in some cases there is an extra joint in the anterior part of the first rib which may simulate a mass.

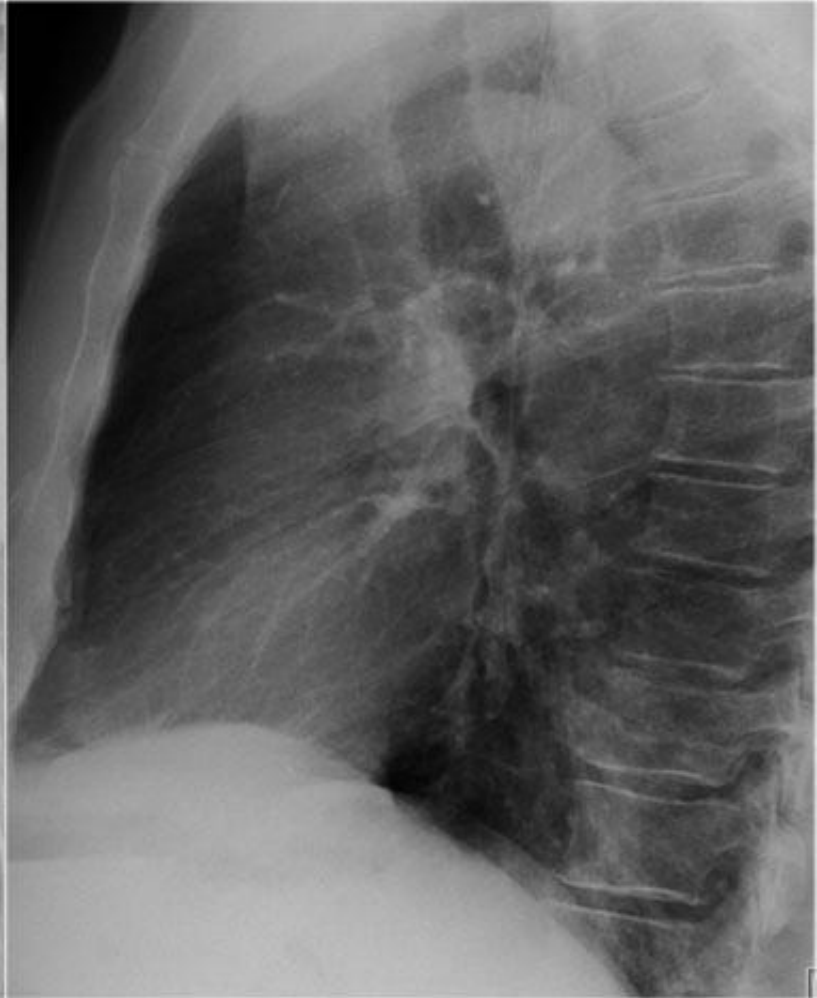
However this is also a hidden area where it can be difficult to detect a mass.



- In this case a **small lung cancer** is seen behind the left first rib. Notice that it is also seen on the lateral view in the retro-sternal area.
-Continue with the PET-CT.



- The PET-CT demonstrates the tumor (arrow) which has already spread to the bone and liver. The diagnosis was made by a biopsy of an osteolytic metastasis in the iliac bone.



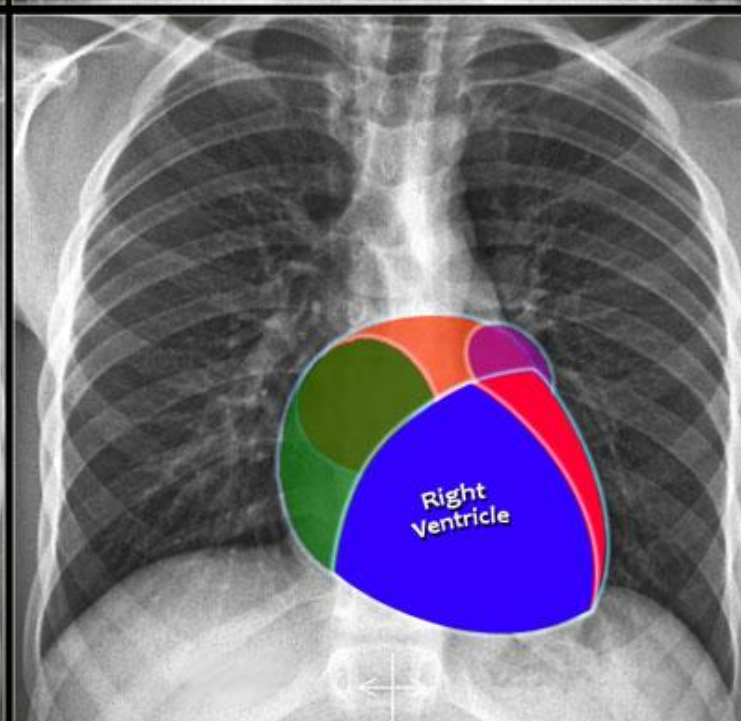
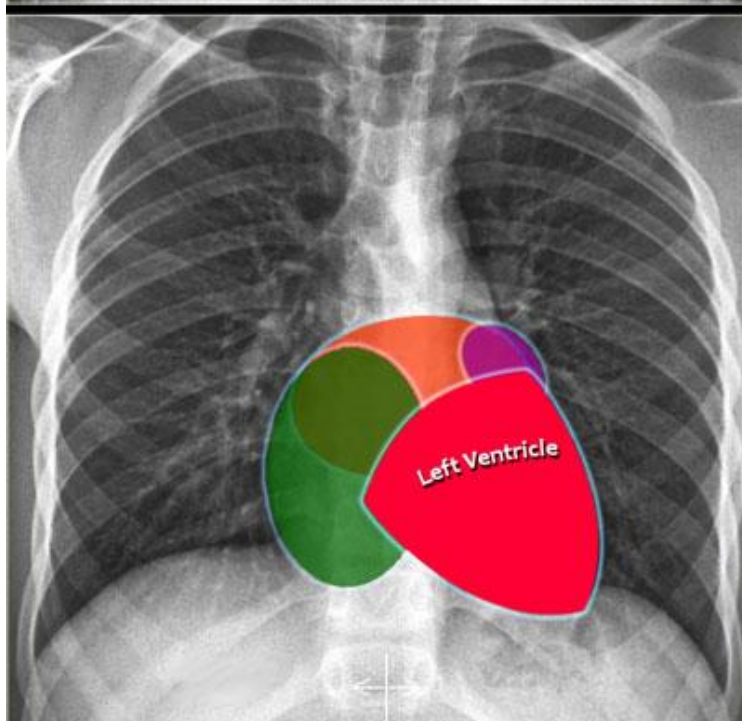
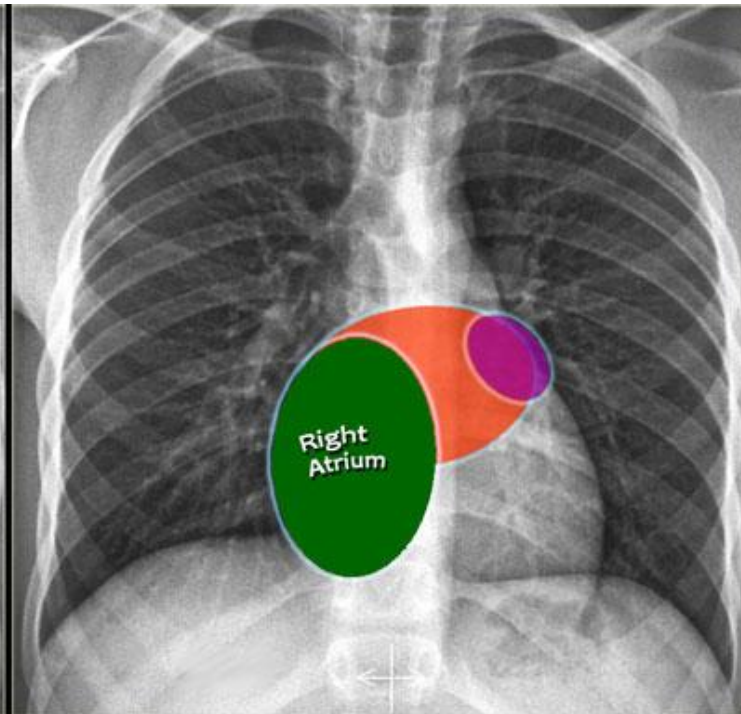
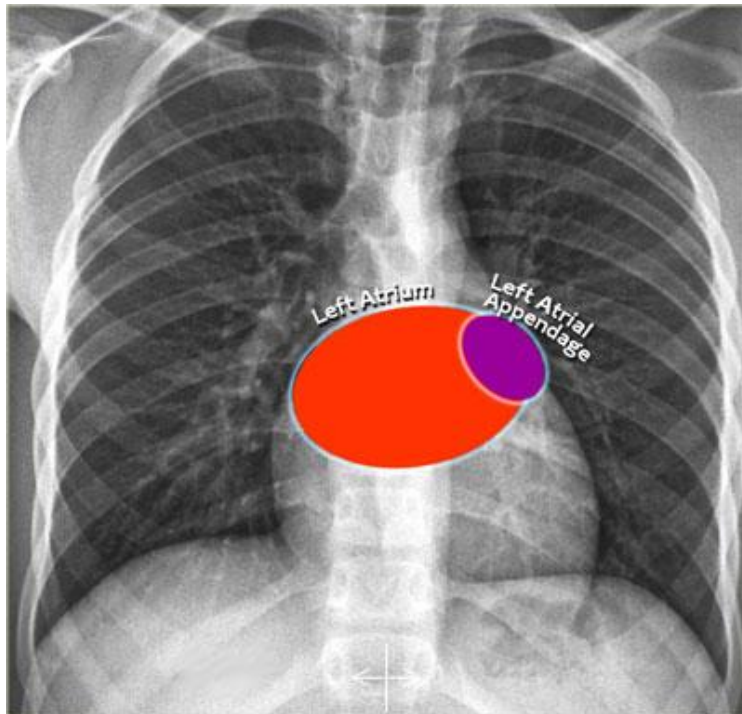
- There is a subtle consolidation in the left lower lobe in the hidden area behind the heart.
Again there is **increased density over the lower vertebral region.**

Heart and Pericardium

- On a chest film only the outer contours of the heart are seen.

In many cases we can only tell whether the heart figure is normal or enlarged and it will be difficult to say anything about the different heart compartments.

- However it can be helpful to know where the different compartments are situated.



Left Atrium

- Most posterior structure.
- Receives blood from the pulmonary veins that run almost horizontally towards the left atrium.
- Left atrial appendage (in purple) can sometimes be seen as a small outpouching just below the pulmonary trunk.
- Enlargement of the left atrium results on the PA-view in outpouching of the upper heart contour on the right and an obtuse angle between the right and left main bronchus. On the lateral view bulging of the upper posterior contour will be seen.

Right Atrium

- Receives blood from the inferior and superior vena cava.
- Enlargement will cause an outpouching of the right heart contour.

Left Ventricle

- Situated to the left and posteriorly to the right ventricle.
- Enlargement will result on the PA-view in an increase of the heart size to the left and on the lateral view in bulging of the lower posterior contour.

Right Ventricle

- Most anterior structure and is situated behind the sternum.
- Enlargement will result on the PA-view in an increase of the heart size to the left and can finally result in the left heart border being formed by the right ventricle.

Left Atrium

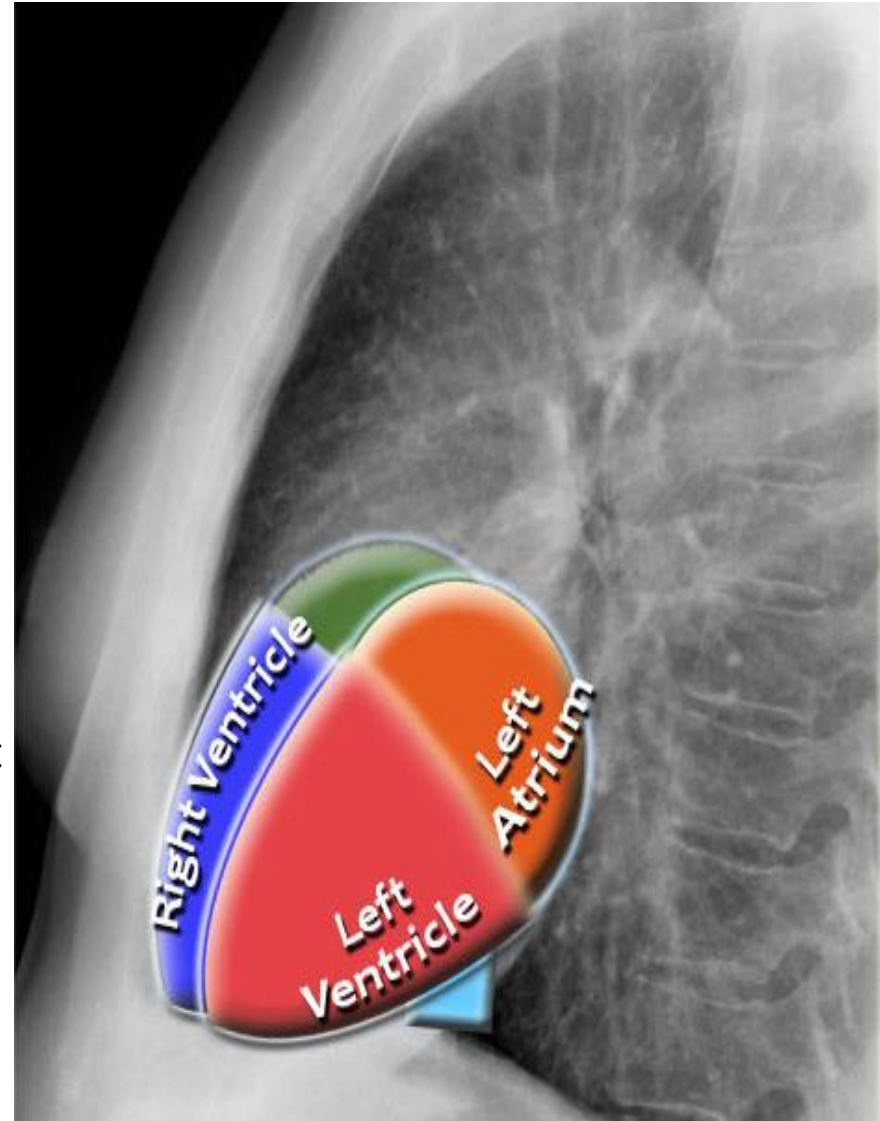
- The upper posterior border of the heart is formed by the left atrium.
- Enlargement will result in bulging of the upper posterior contour

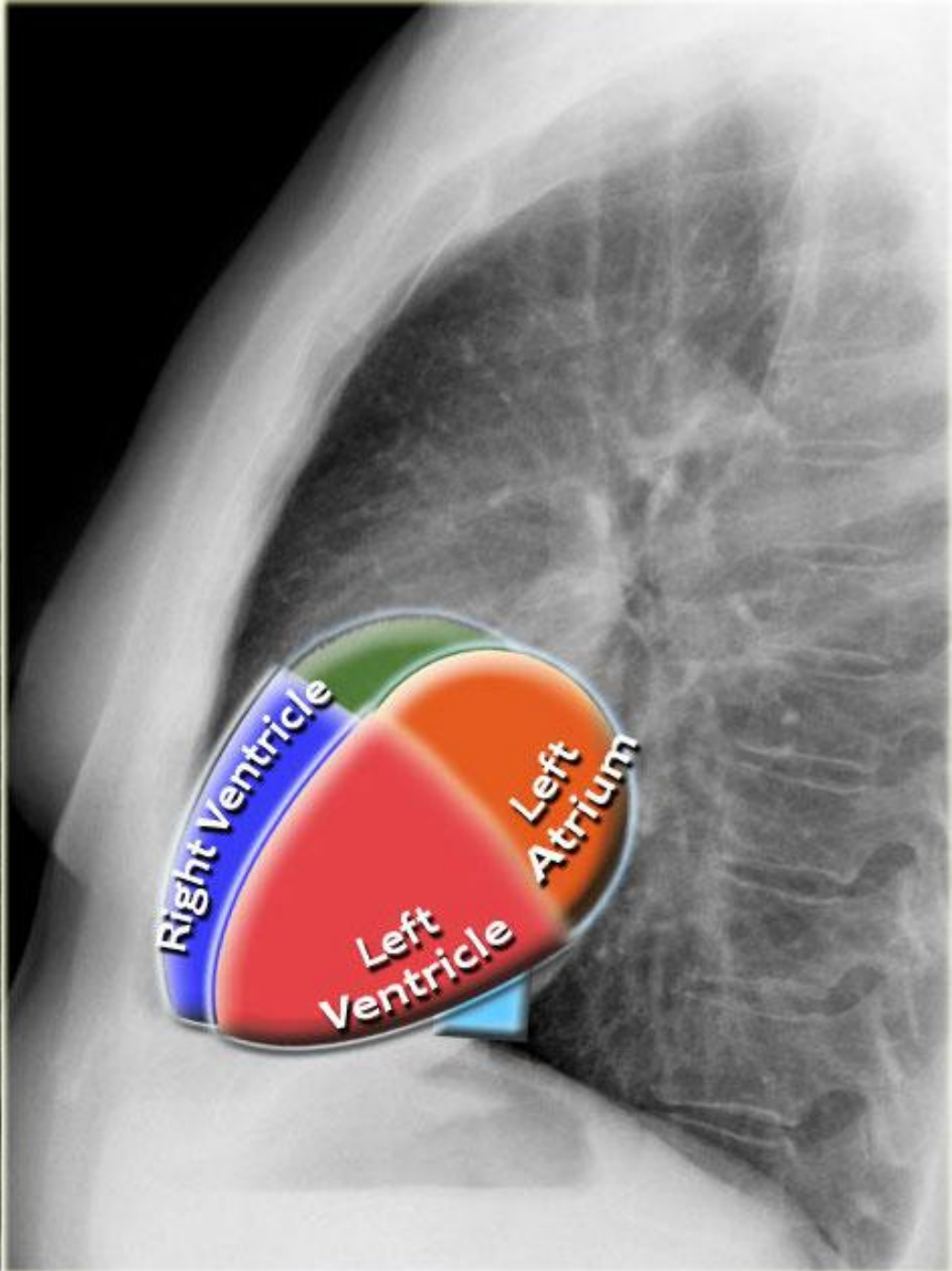
Left Ventricle

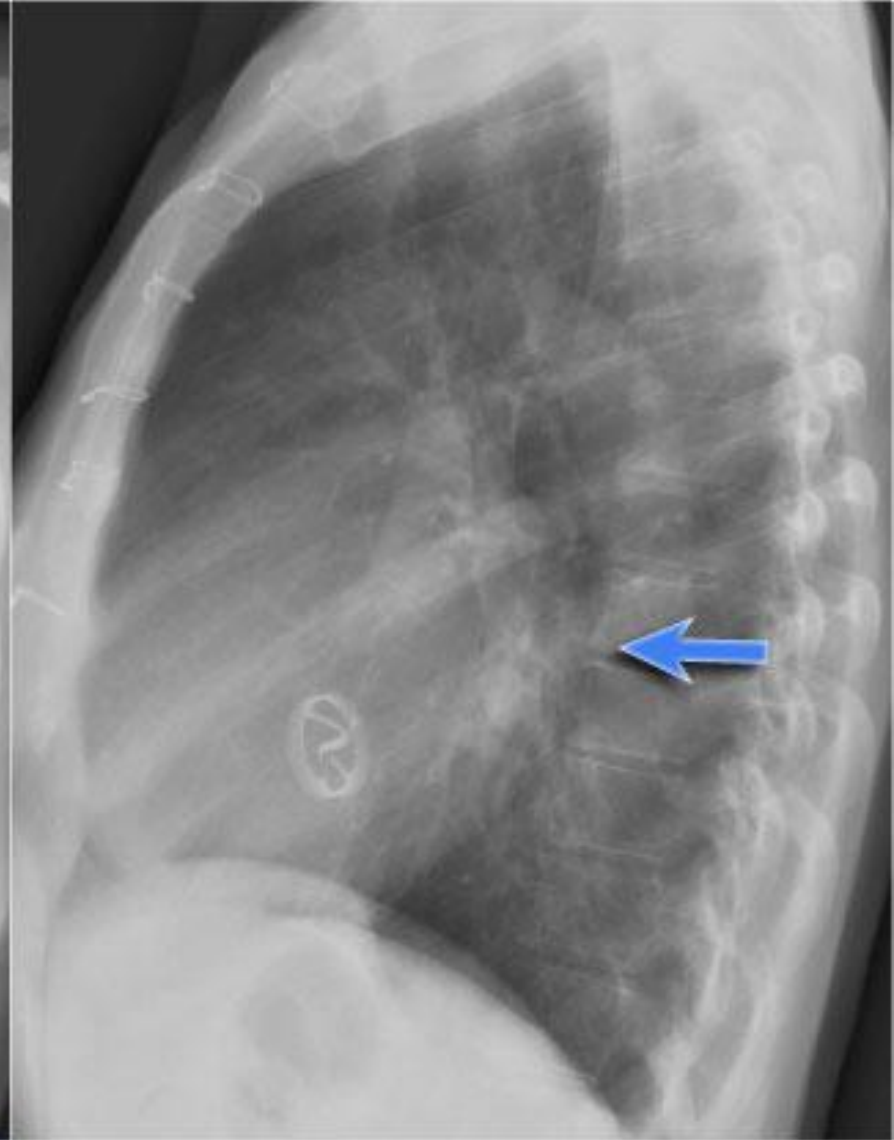
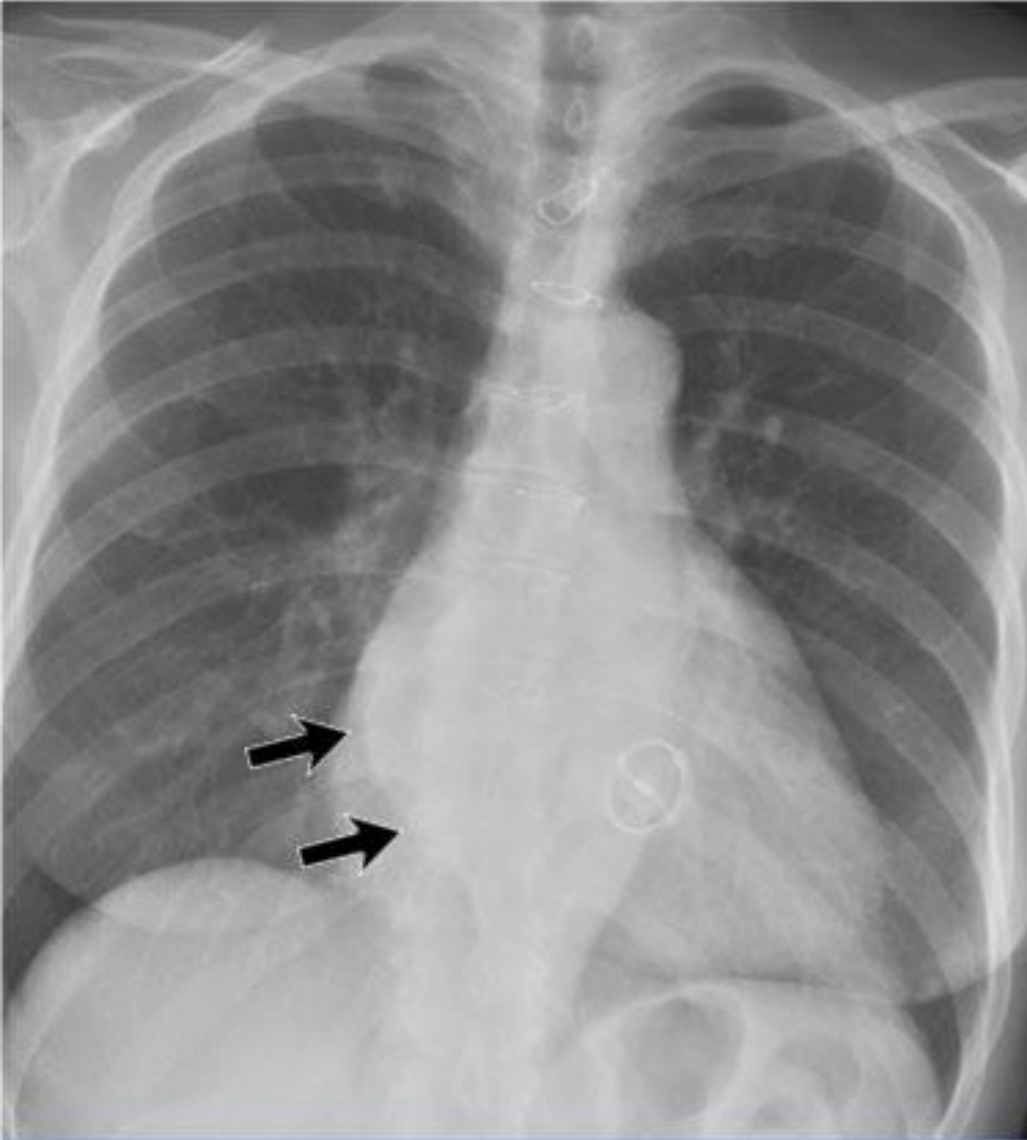
- Forms the lower posterior border.
- Enlargement will displace the contour more posteriorly.

Right Ventricle

- The lower retrosternal space is filled by the right ventricle.
- Enlargement of the right ventricle will result in more superior filling of this retrosternal space.



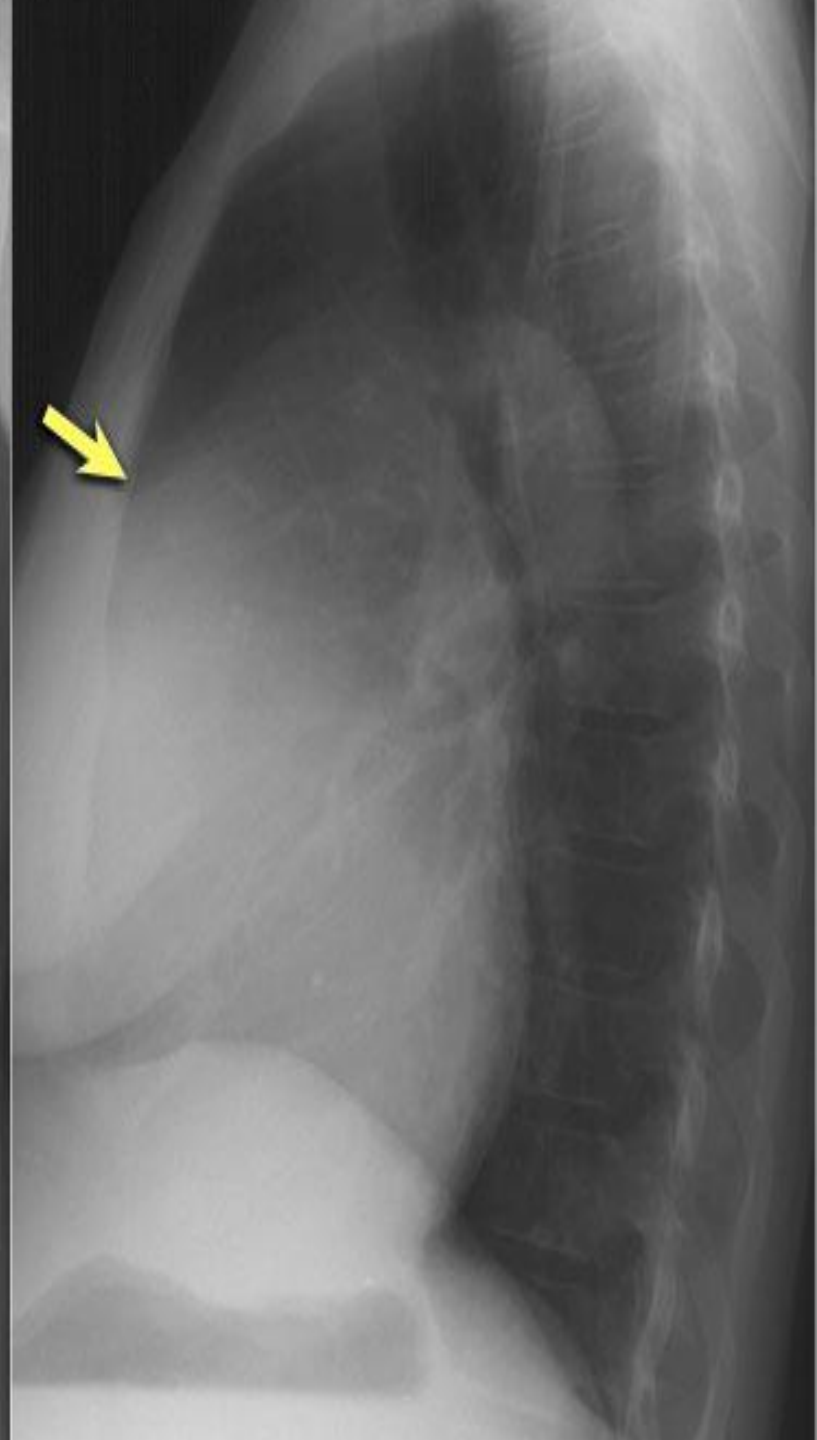
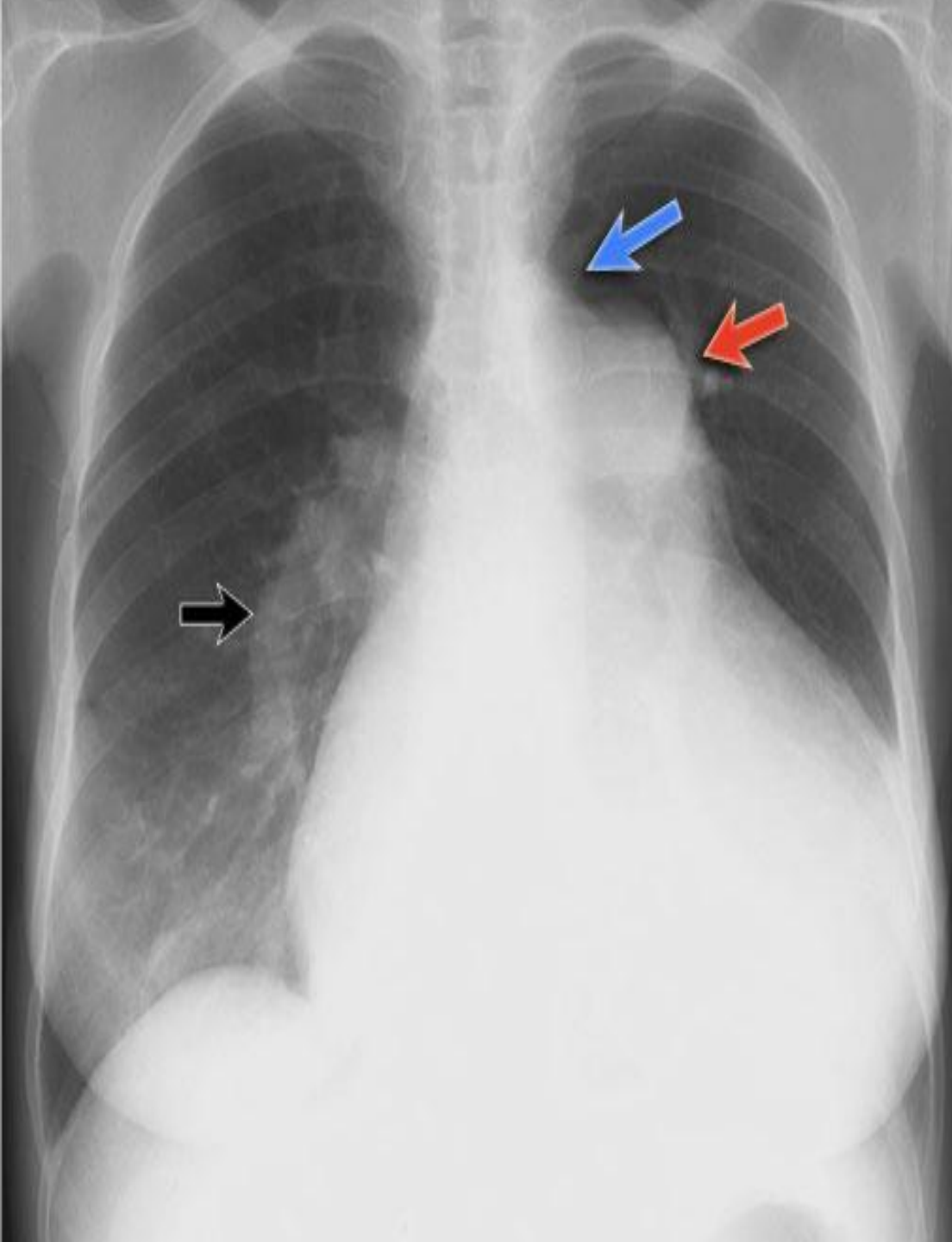




Left Atrium enlargement

This is a patient with longstanding mitral valve disease and mitral valve replacement.

- Extreme dilatation of the left atrium has resulted in bulging of the contours (blue and black arrows).



On these chest films :

- the heart is extremely dilated.

Notice that it is especially the right ventricle that is dilated. This is well seen on the lateral film (yellow arrow).

- There is a small aortic knob (blue arrow),
- while the pulmonary trunk and the right lower pulmonary artery are dilated.

All these findings are probably the result of a left-to-right shunt with subsequent development of pulmonary hypertension.



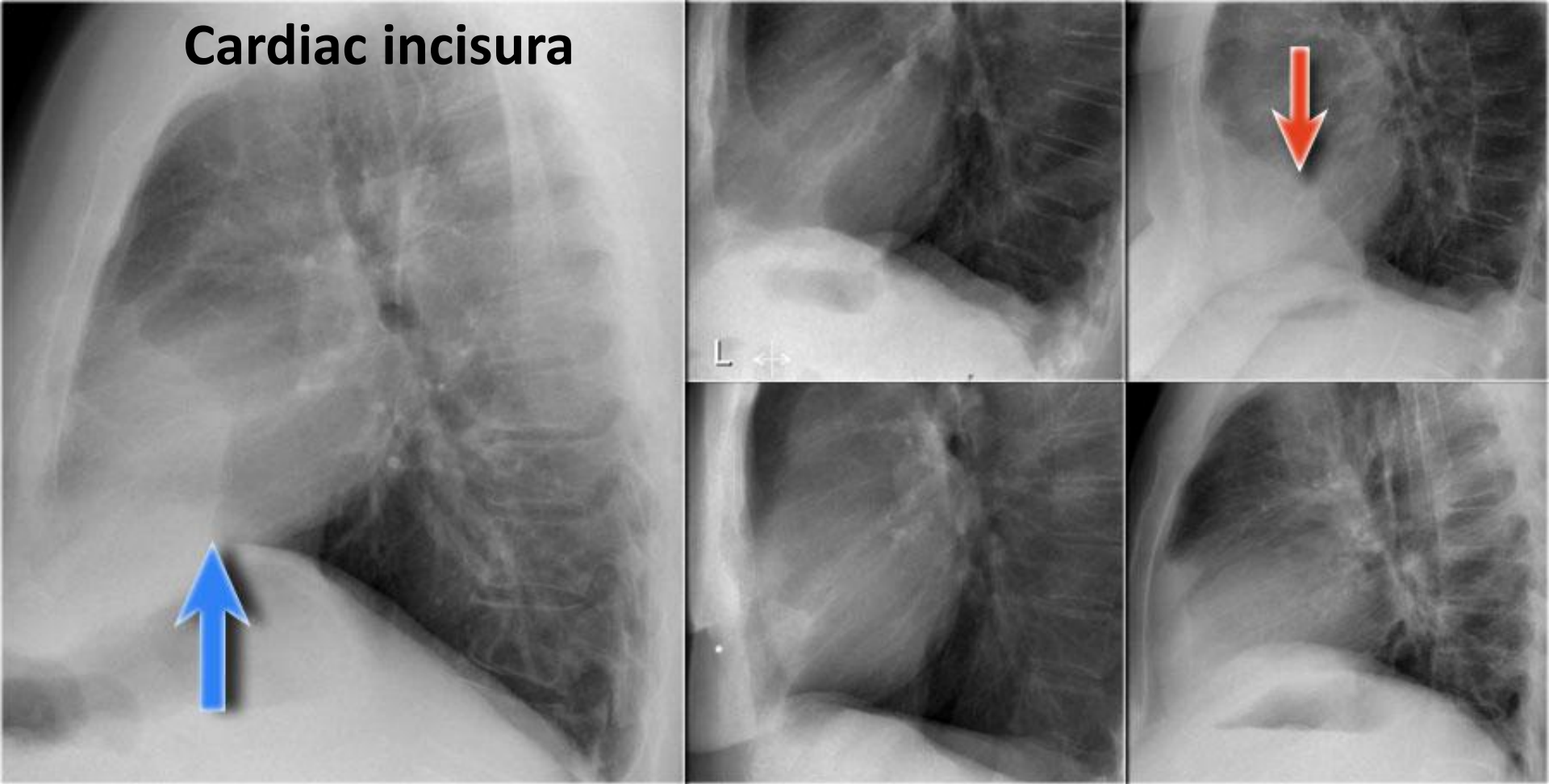
The location of the cardiac valves is best determined on the lateral radiograph.

A line is drawn on the lateral radiograph from the carina to the cardiac apex.

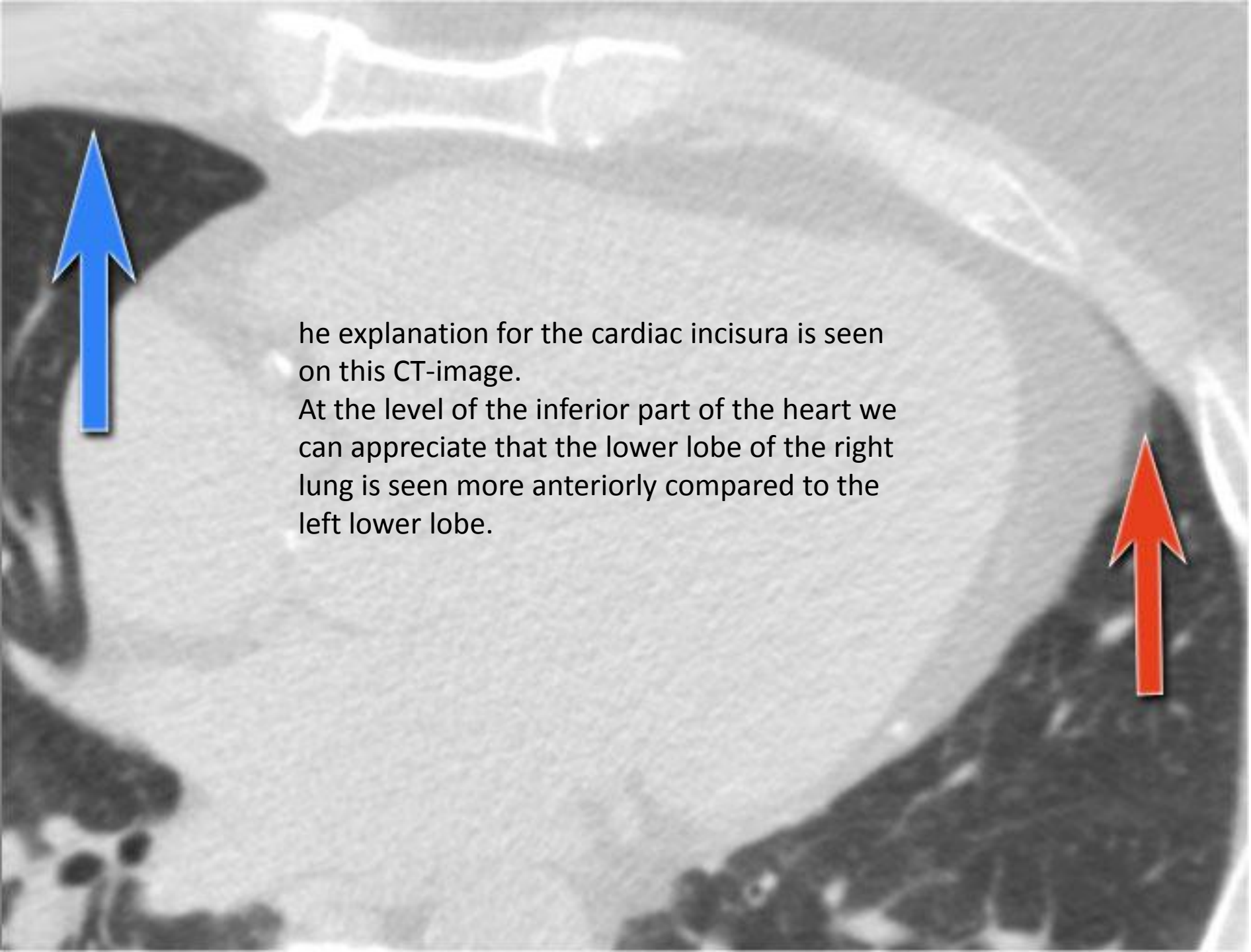
The pulmonic and aortic valves generally sit above this line and the tricuspid and mitral valves sit below this line (4).

On this lateral view you can get a good impression of the enlargement of the left atrium.

Cardiac incisura



- On the right side of the chest the lung will lie against the anterior chest wall. On the left however the inferior part of the lung may not reach the anterior chest wall, since the heart or pericardial fat or effusion is situated there.
- This causes a density on the anteroinferior side on the lateral view which can have many forms.
It is a normal finding, which can be seen on many chest x-rays and should not be mistaken for pathology in the lingula or middle lobe.

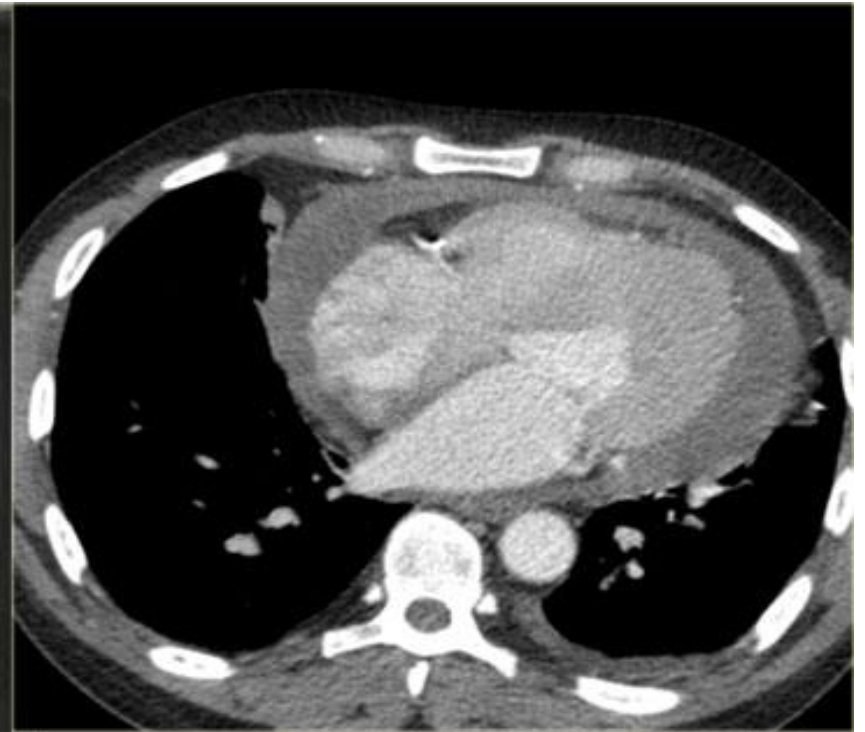
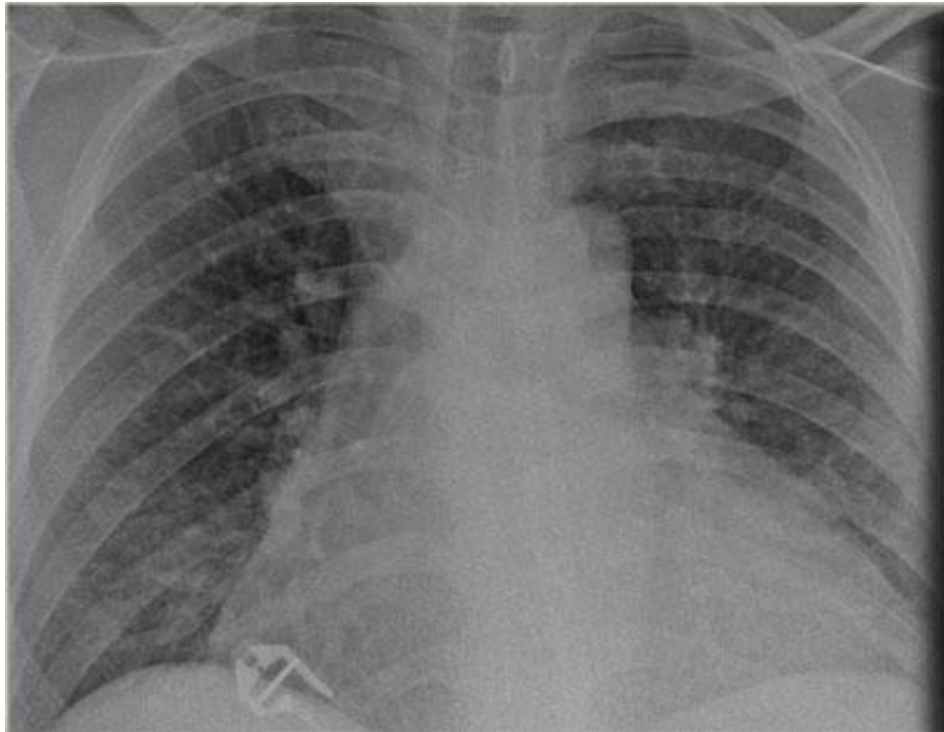


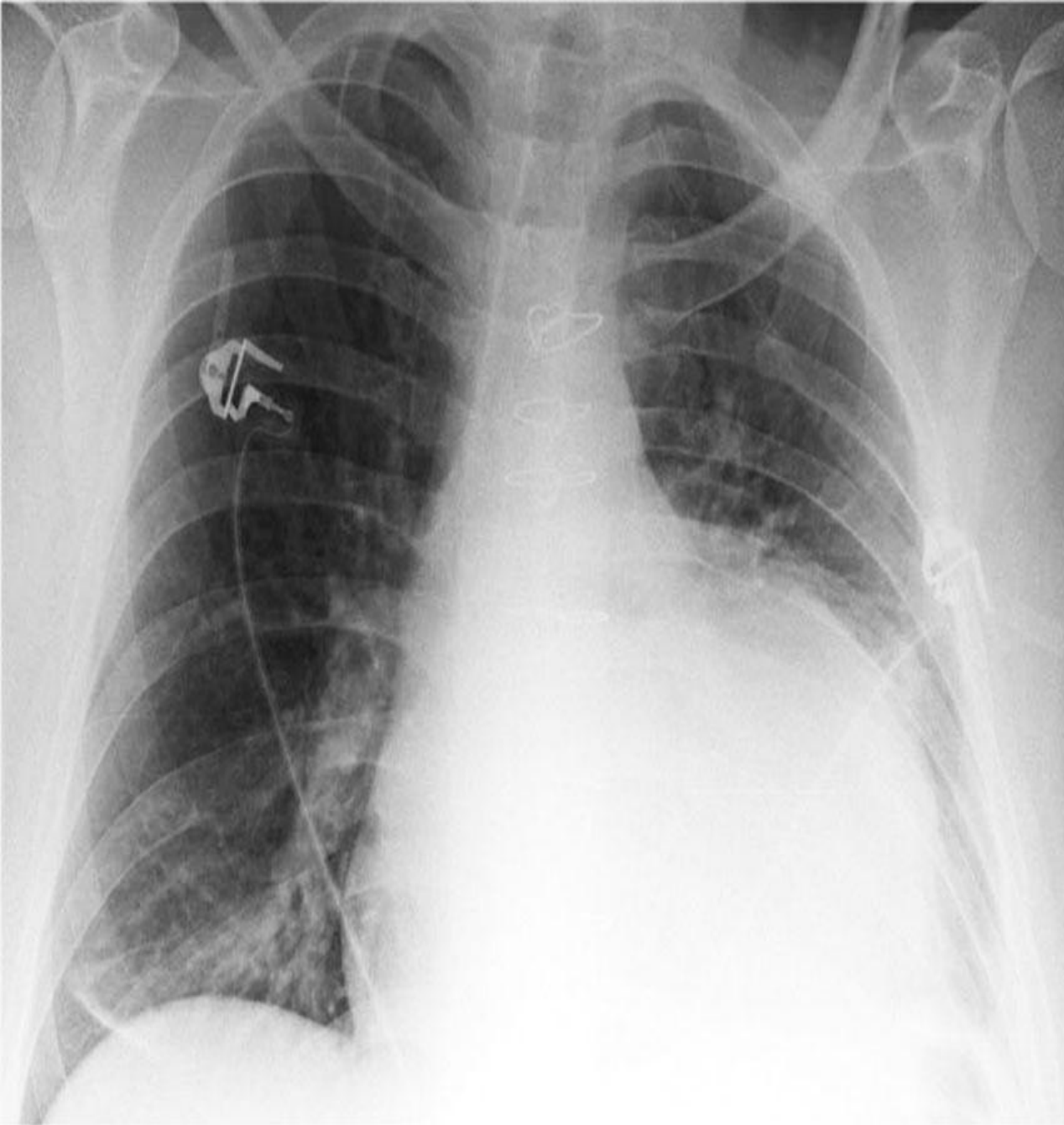
he explanation for the cardiac incisura is seen on this CT-image.

At the level of the inferior part of the heart we can appreciate that the lower lobe of the right lung is seen more anteriorly compared to the left lower lobe.

Pericardial effusion

- Whenever we encounter a large heart figure, we should always be aware of the possibility of pericardial effusion simulating a large heart.
- On the chest x-ray it looks as if this patient has a dilated heart while on the CT it is clear, that it is the pericardial effusion that is responsible for the enlarged heart figure.





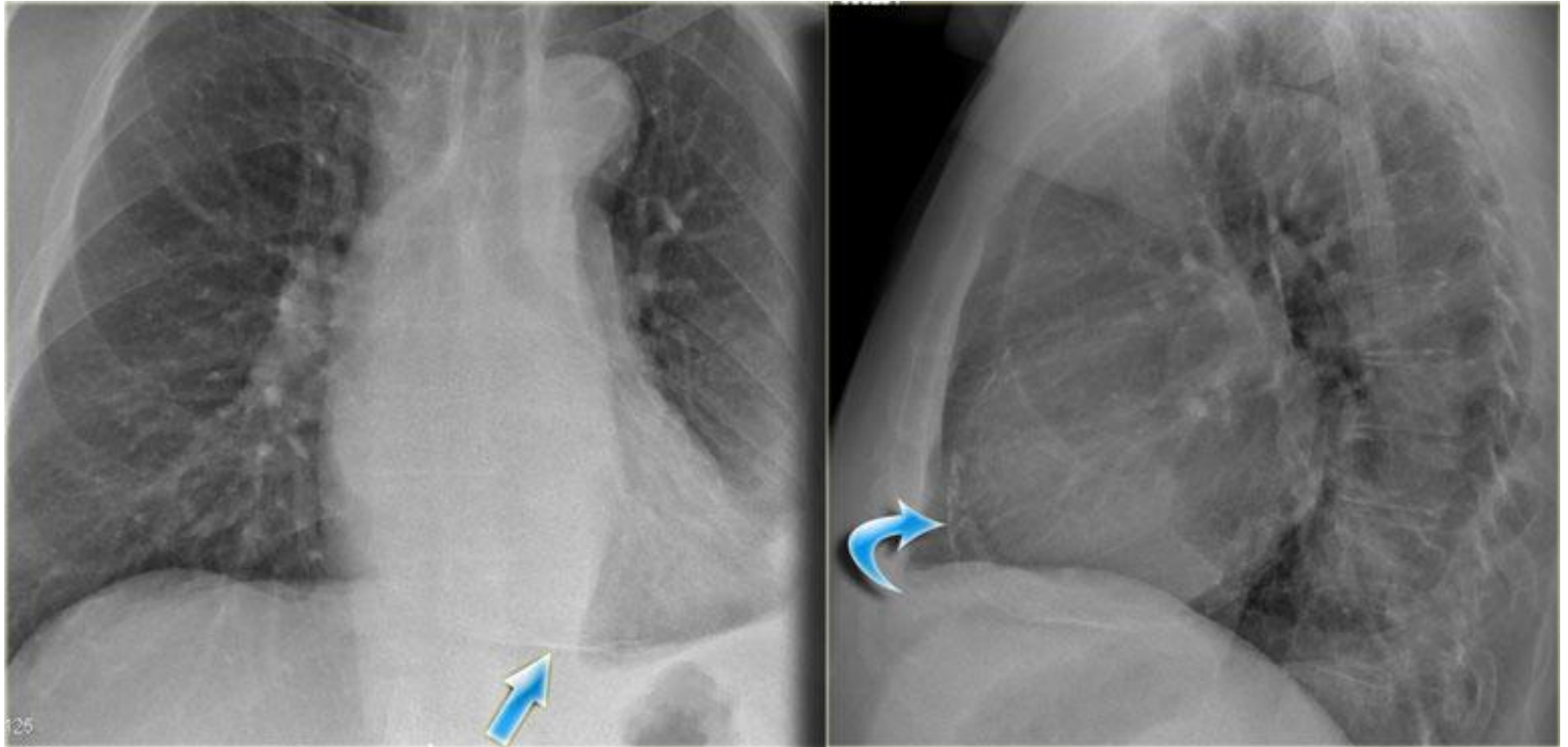
- Especially in patients who had recent cardiac surgery an enlargement of the heart figure can indicate pericardial bleeding.
- This patient had a change in the heart configuration and pericardial bleeding was suspected. Ultrasound demonstrated only a minimal pericardial effusion. Continue with the CT.



Right ventricle

- There is a large pericardial effusion, which is **located posteriorly** to the left ventricle (blue arrow). The left ventricle is filled with contrast and is compressed (red arrow). At surgery a large hematoma in the posterior part of the pericardium was found.
- Notice that on the anterior side there is only a minimal collection of pericardial fluid, which explains why the ultrasound examination underestimated the amount of pericardial fluid.

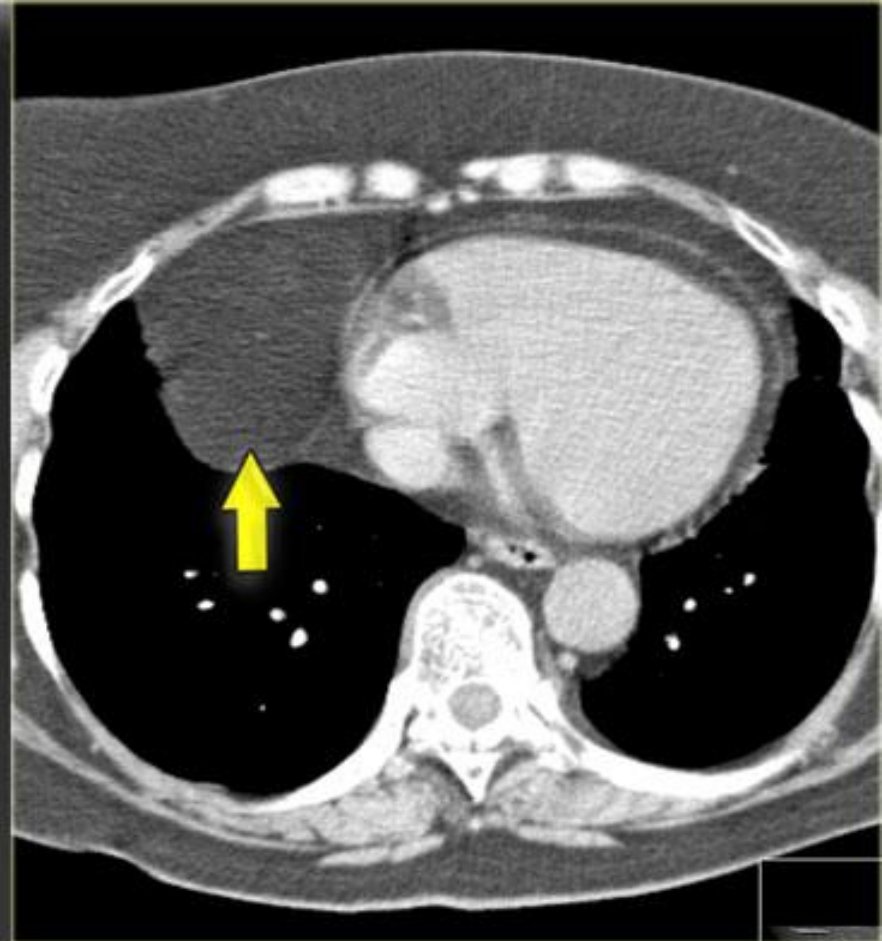
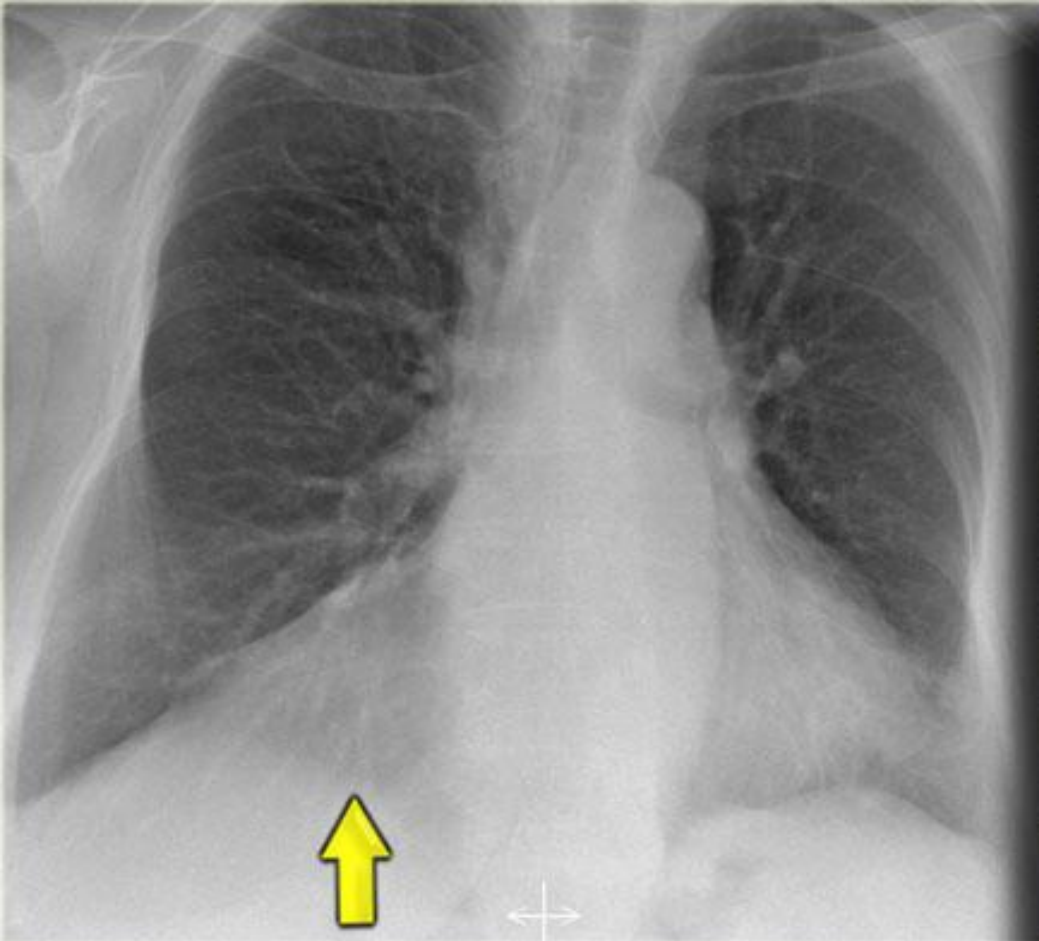
Calcifications



- Detection of calcifications within the heart is quite common.
- The most common are **coronary artery** calcifications and **valve** calcifications.
- Here we see pericardial calcifications which can be associated with constrictive pericarditis.



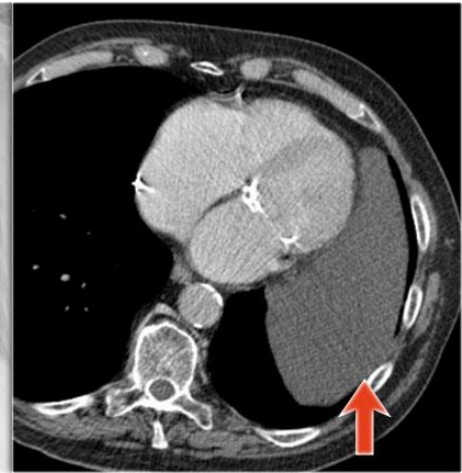
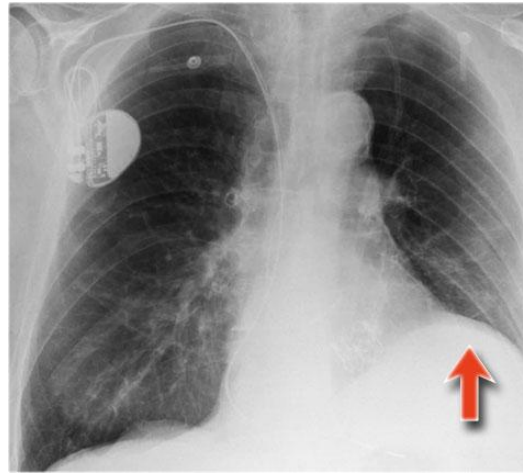
- In this case there are calcifications that look like pericardial calcifications, but these are **myocardial calcifications** in an infarcted area of the left ventricle.
- Notice that they follow the contour of the left ventricle.

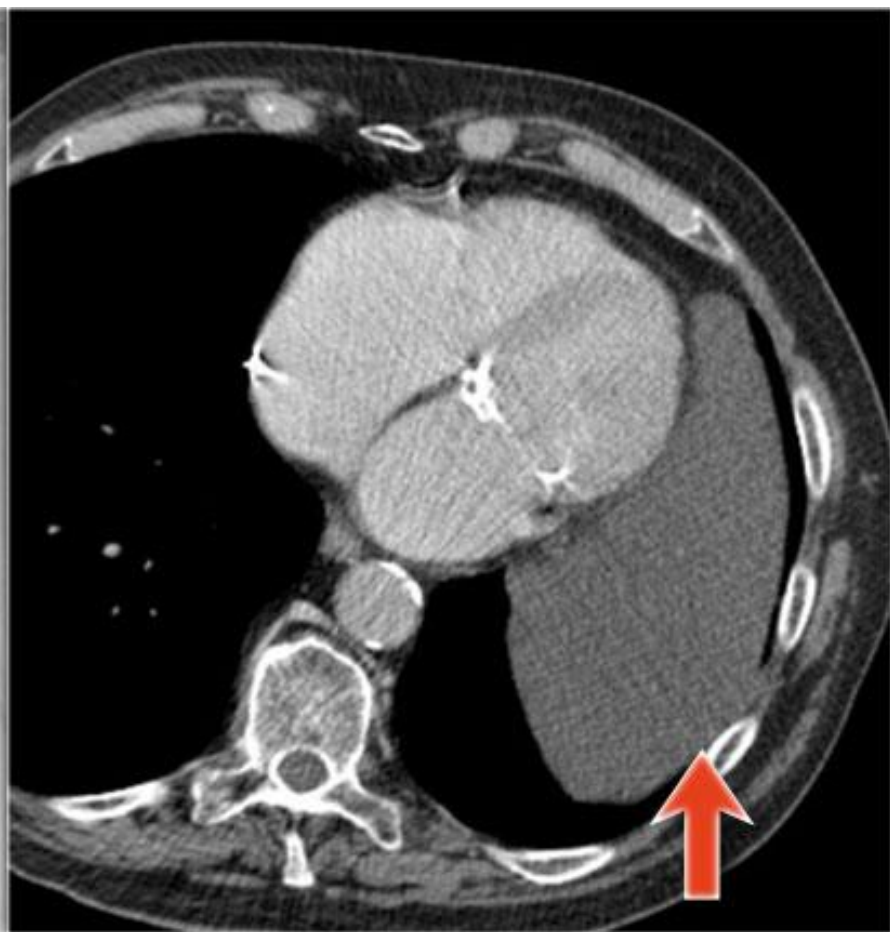


- **Pericardial fatpad**
- Pericardial fat depositions are common. Sometimes a large fat pad can be seen (figure).
- Necrosis of the fat pad has pathologic features similar to fat necrosis in [epiploic appendagitis](#).
It is an uncommon benign condition, that manifests as acute pleuritic chest pain in previously healthy persons

Pericardial cyst

- **Pericardial cysts** are connected to the pericardium and usually contain clear fluid.
- The majority arise in the anterior cardiophrenic angle,
 - more frequently on the rightside,
 - but they can be seen as high as the pericardial recesses at the level of the proximal aorta and pulmonary arteries .
- Most patients are asymptomatic.
- On the chest x-ray it seems as if there is a elevated left hemidiaphragm.
- On CT however there is a cyst connected to the pericardium.





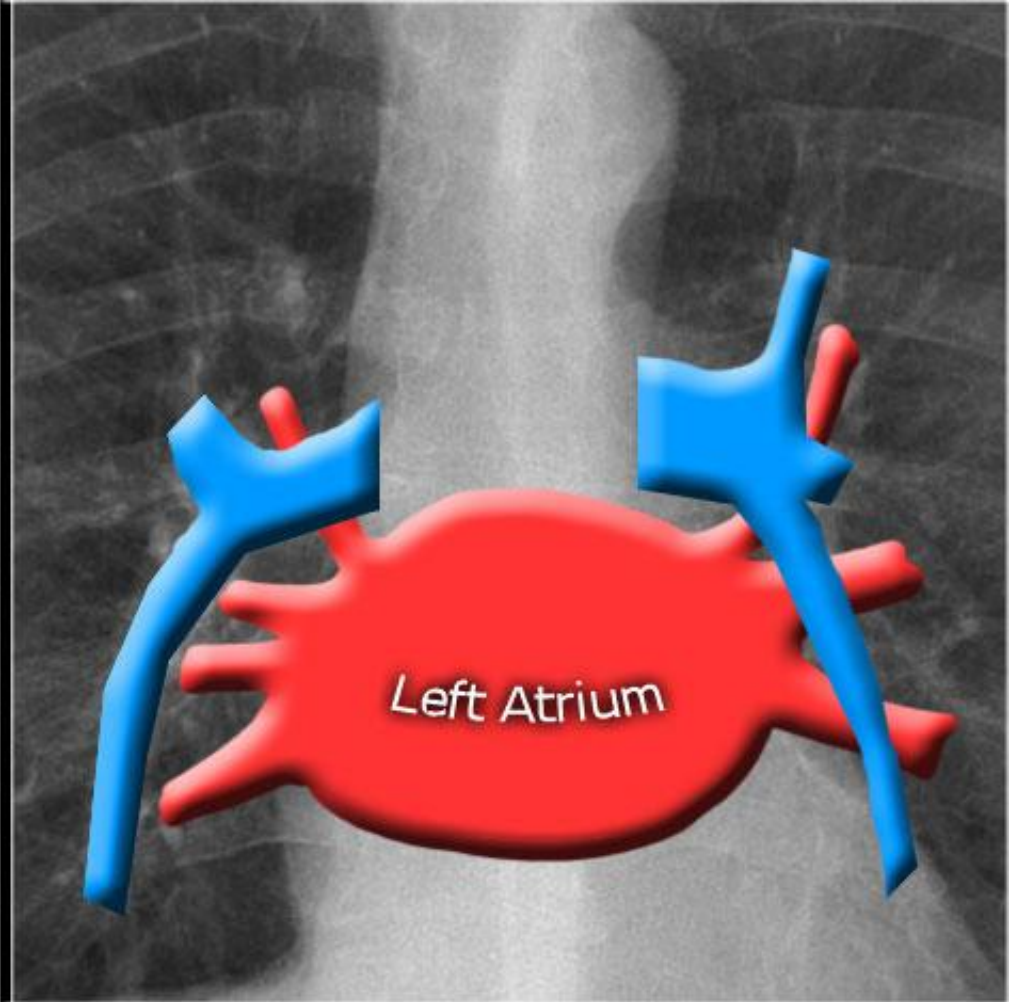
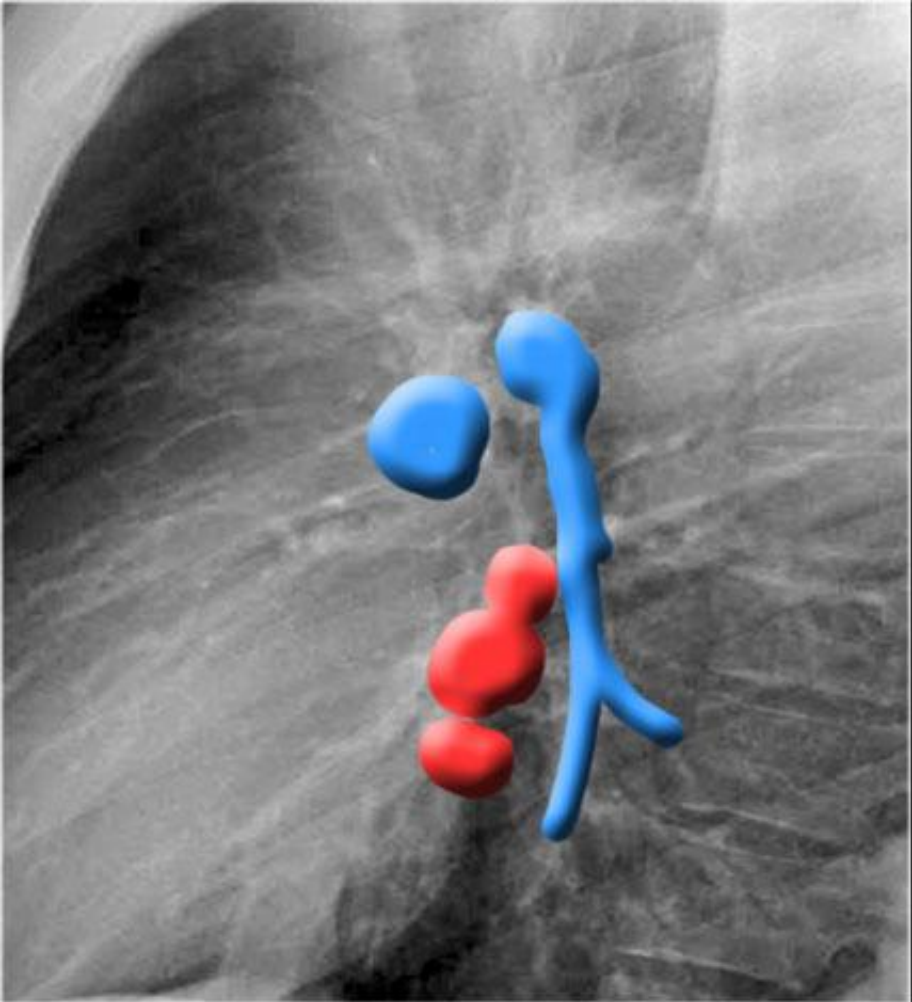
Hili



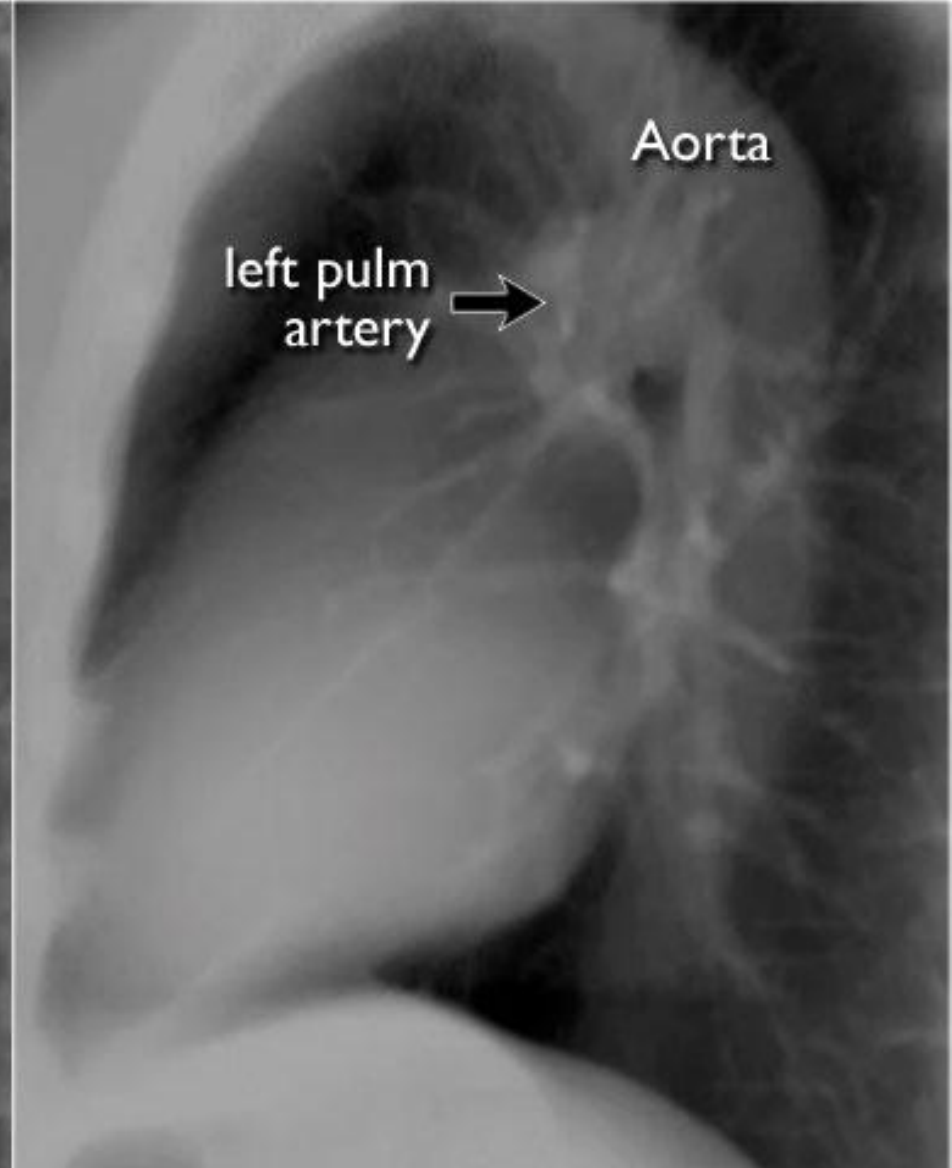
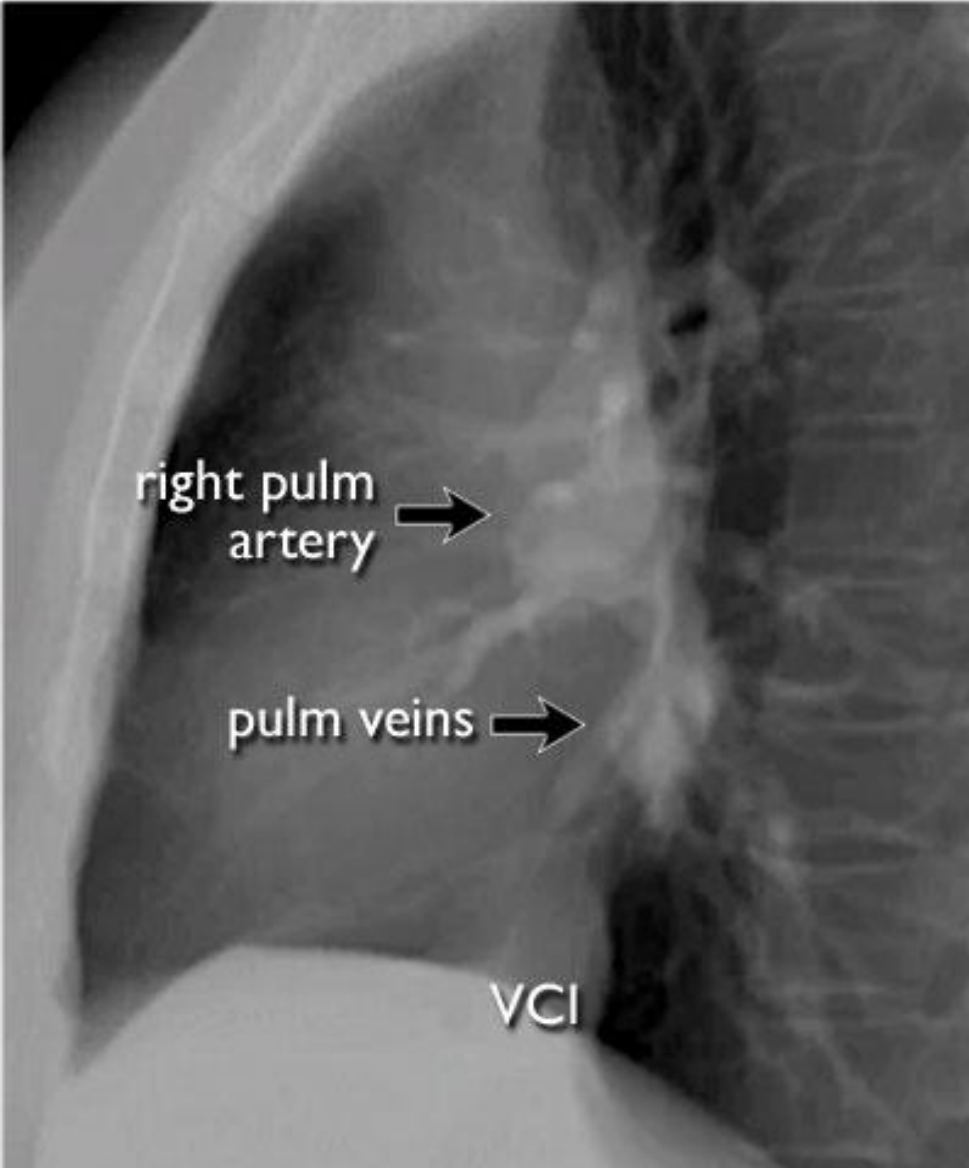
- The normal hilar shadow is for 99% composed of vessels - pulmonary arteries and to a lesser extent veins.
- The vessel margins are smooth and the vessels have branches.

Lt hilum never be
lower than Rt hilum.

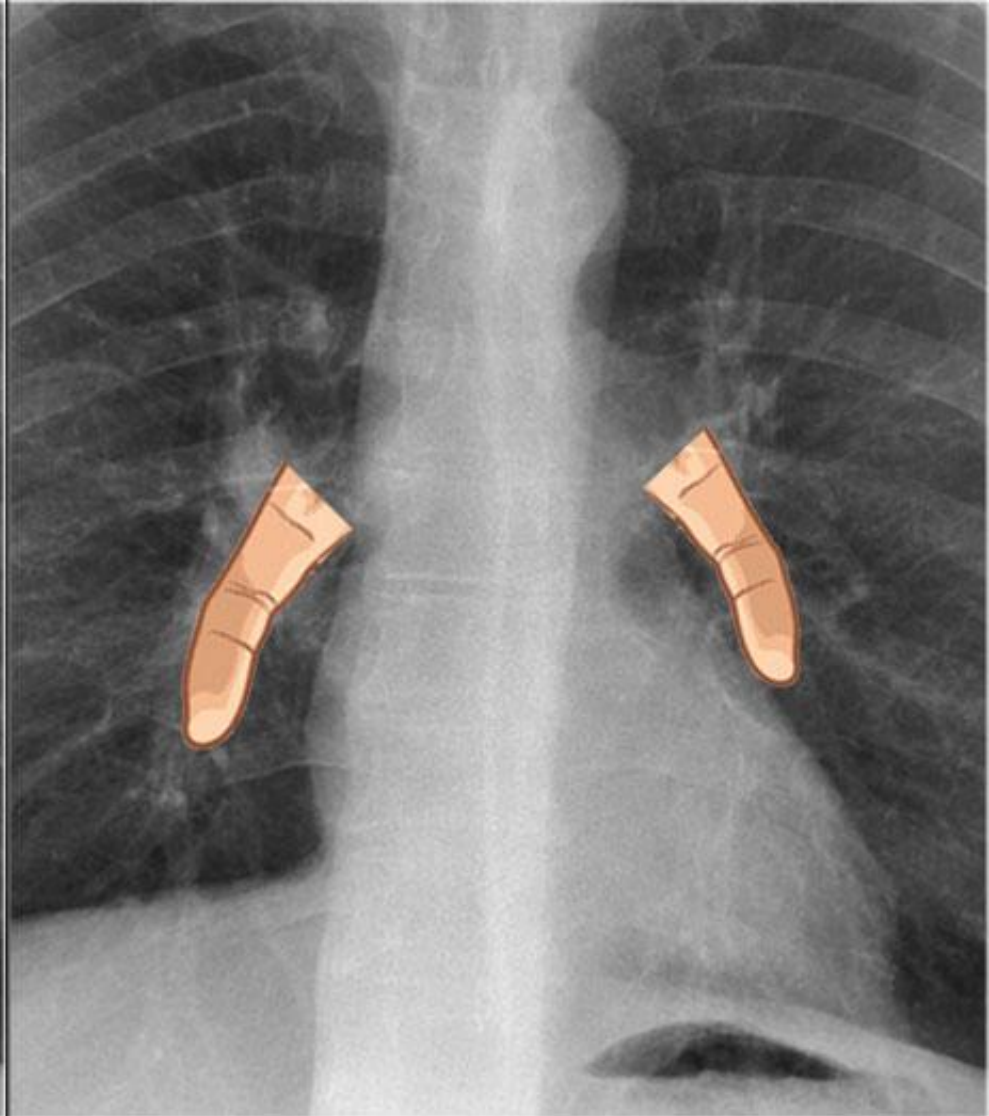
- The left pulmonary artery runs over the left main bronchus,
- while the right pulmonary artery runs in front of the right main bronchus, which is usually lower in position than the left main bronchus.
- Hence the **left hilum** is **higher** than the right.
- Only in a minority of cases the right hilus is at the same level as the left, but never higher.



- In this illustration :
- lower lobe arteries are coloured blue “because they contain oxygen-poor blood”.
- They have a **more vertical** orientation,
- while the **pulmonary veins** run **more horizontally** towards the left atrium, which is located below the level of the main pulmonary arteries.



- Both pulmonary arteries and veins can be identified on a lateral view and should not be mistaken for lymphadenopathy.
- Sometimes the pulmonary veins can be very prominent.



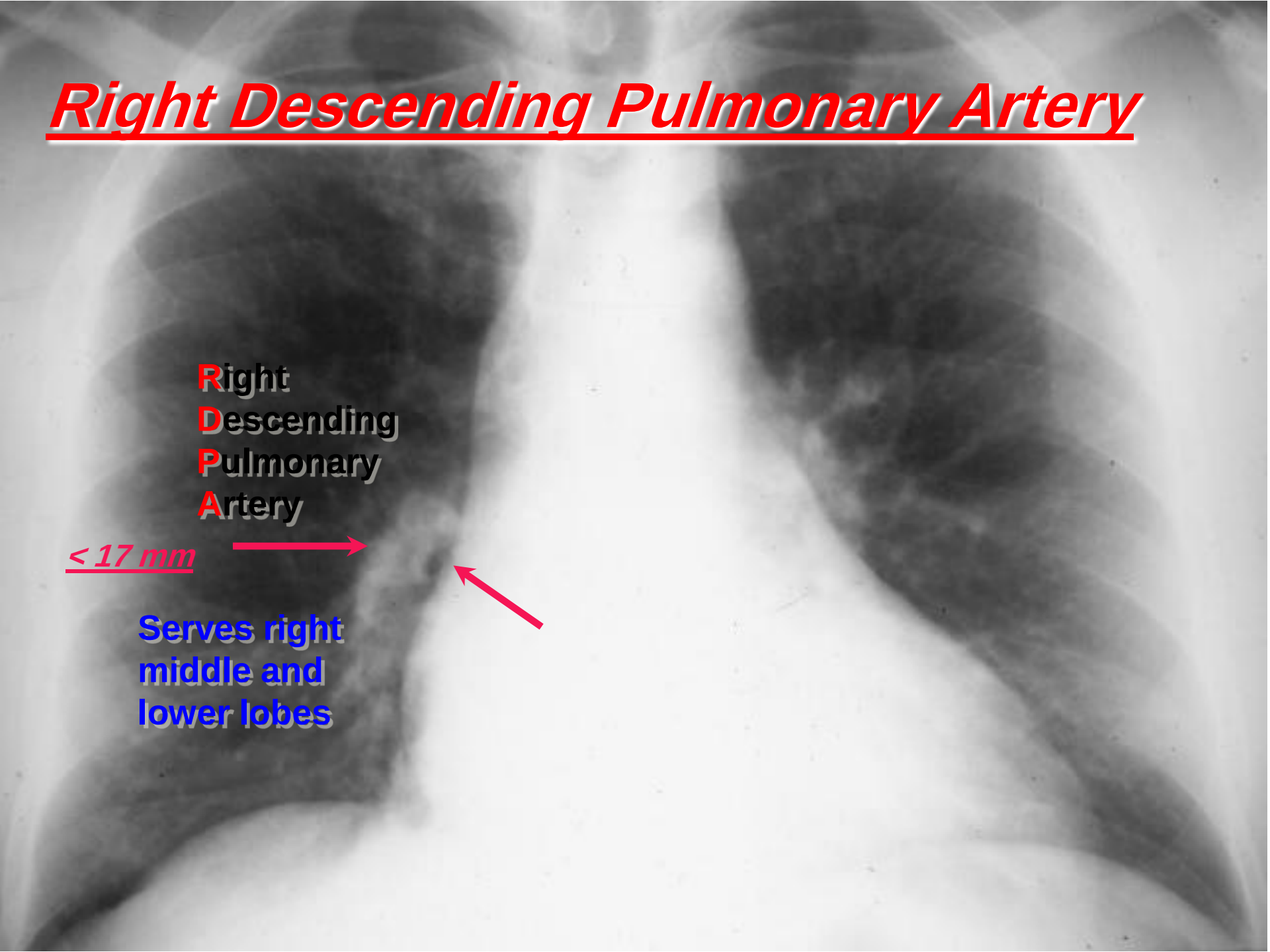
- The lower lobe pulmonary arteries extend inferiorly from the hilum. They are described as little fingers, because each has the size of a little finger .
- On the right side the little finger will be visible in 94% of normal CXRs and on the left side in 62% of normals.

Right Descending Pulmonary Artery

Right
Descending
Pulmonary
Artery

< 17 mm

**Serves right
middle and
lower lobes**





- Study the CXR of a 70-year old male who fell from the stairs and has severe pain on the right flank..
- Notice on the PA-film the absence of the little finger on the right and on the lateral view the increased density over the lower vertebral column.
- *What is your diagnosis?*

Hilar enlargement

The table summarizes the causes of hilar enlargement.

= Normal hili are:

- Normal in position - left higher than right
- Equal density
- Normal branching vessels

Hilar Enlargement

Unilateral

Infection

- TB - viral - bacterial

Tumor

- Lungca - Lymphoma
- Metastases
 - Lung - head/neck
 - thyroid - testis

Vascular

- pulmonary arterial aneurysm / stenosis

Bilateral

Sarcoidosis / Silicosis

Infection

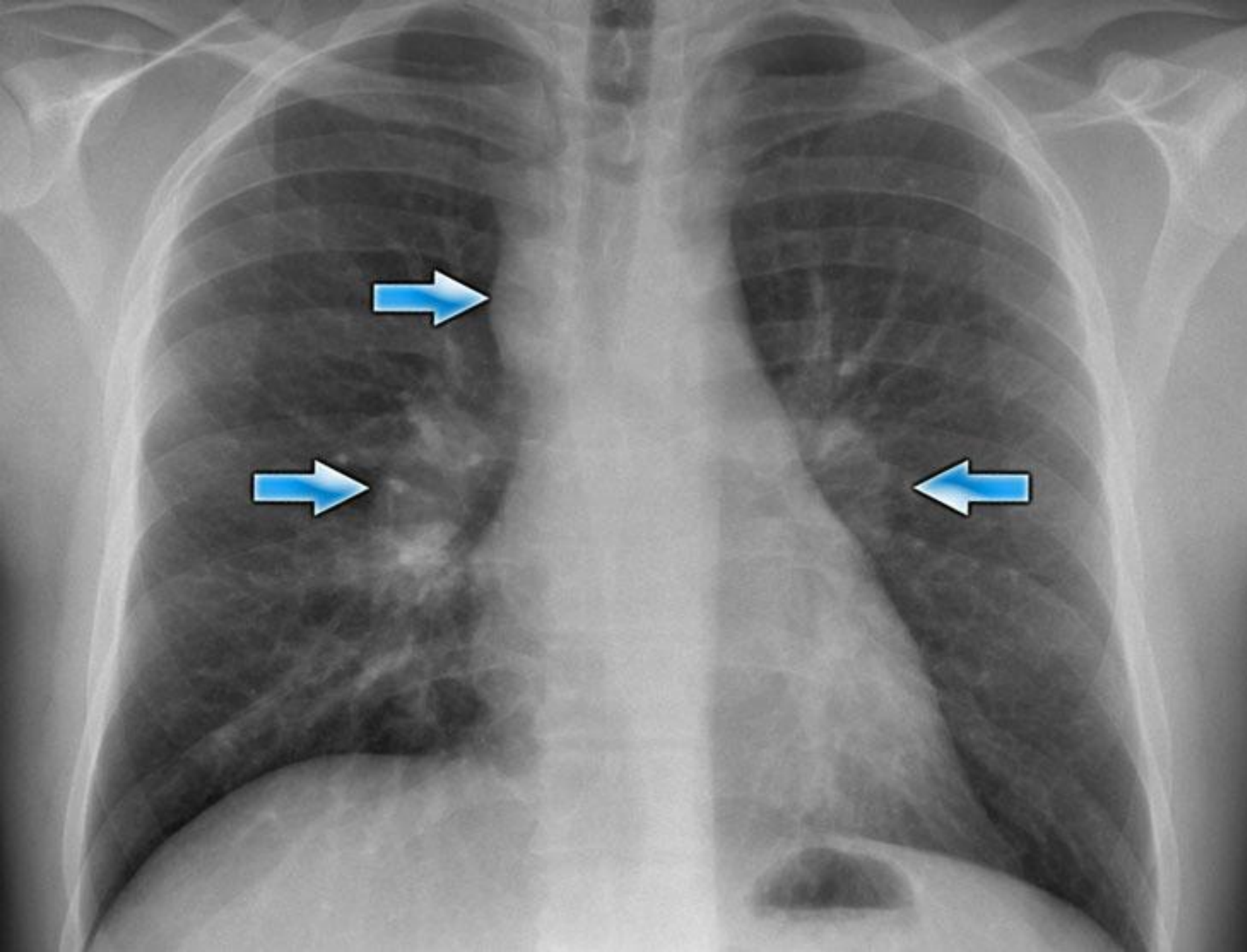
- TB - viral - bacterial

Tumor

- Lymphoma
- Metastases

Vascular

- pulmonary arterial hypertension
 - COPD - Mitral valve disease
 - left-to-right shunt
 - recurrent embolism



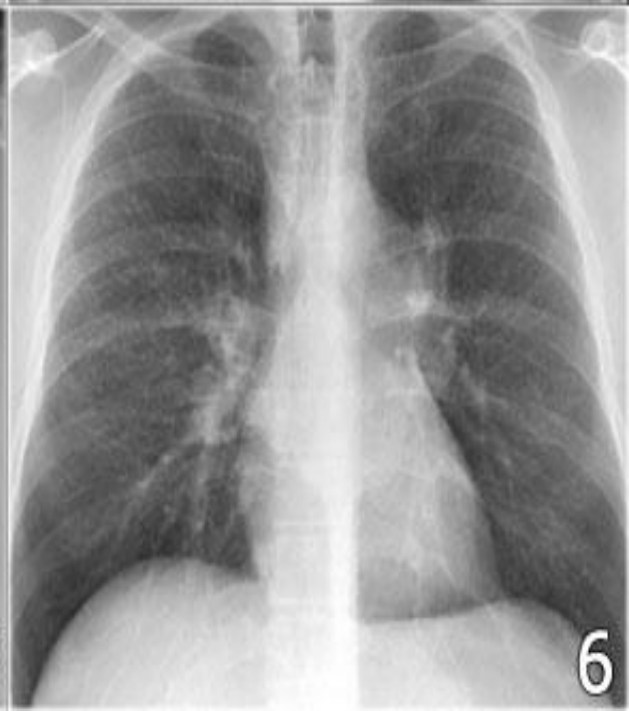
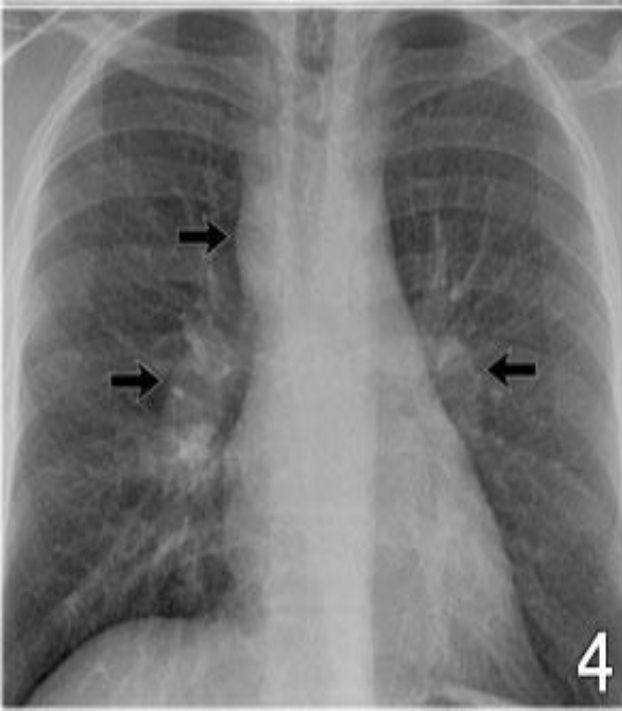
- Enlargement of the hili is usually due to: **lymphadenopathy** or **enlarged vessels**.
- In this case there is an enlarged hilar shadow on both sides.

This could be the result of enlarged vessels or enlarged lymph nodes.

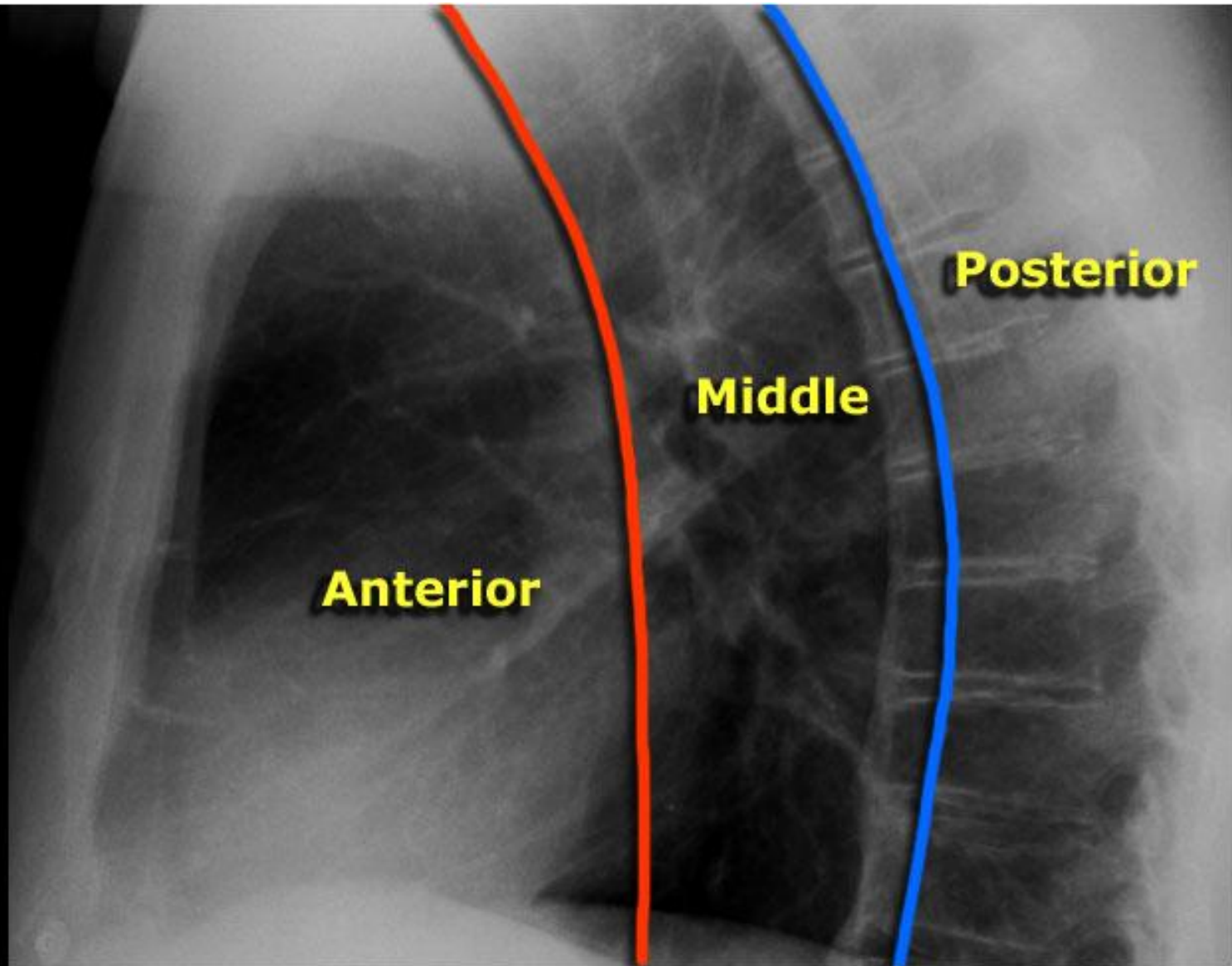
A very helpful finding in this case is the mass on the right of the trachea.

- This is known as the 1-2-3 sign in sarcoidosis, i.e. enlargement of left hilum, right hilum and paratracheal.

- Here some more examples of sarcoidosis.
1. Lymphadenopathy and groundglass appearance of the lungs
 2. Lymphadenopathy, 1-2-3 sign
 3. Bulky lymphadenopathy
 4. 1-2-3 sign
 5. Nodular lung pattern, no lymphadenopathy
 6. Hilar and paratracheal lymphadenopathy



Mediastinum



divided into an anterior, middle and posterior compartment, each with it's own pathology.

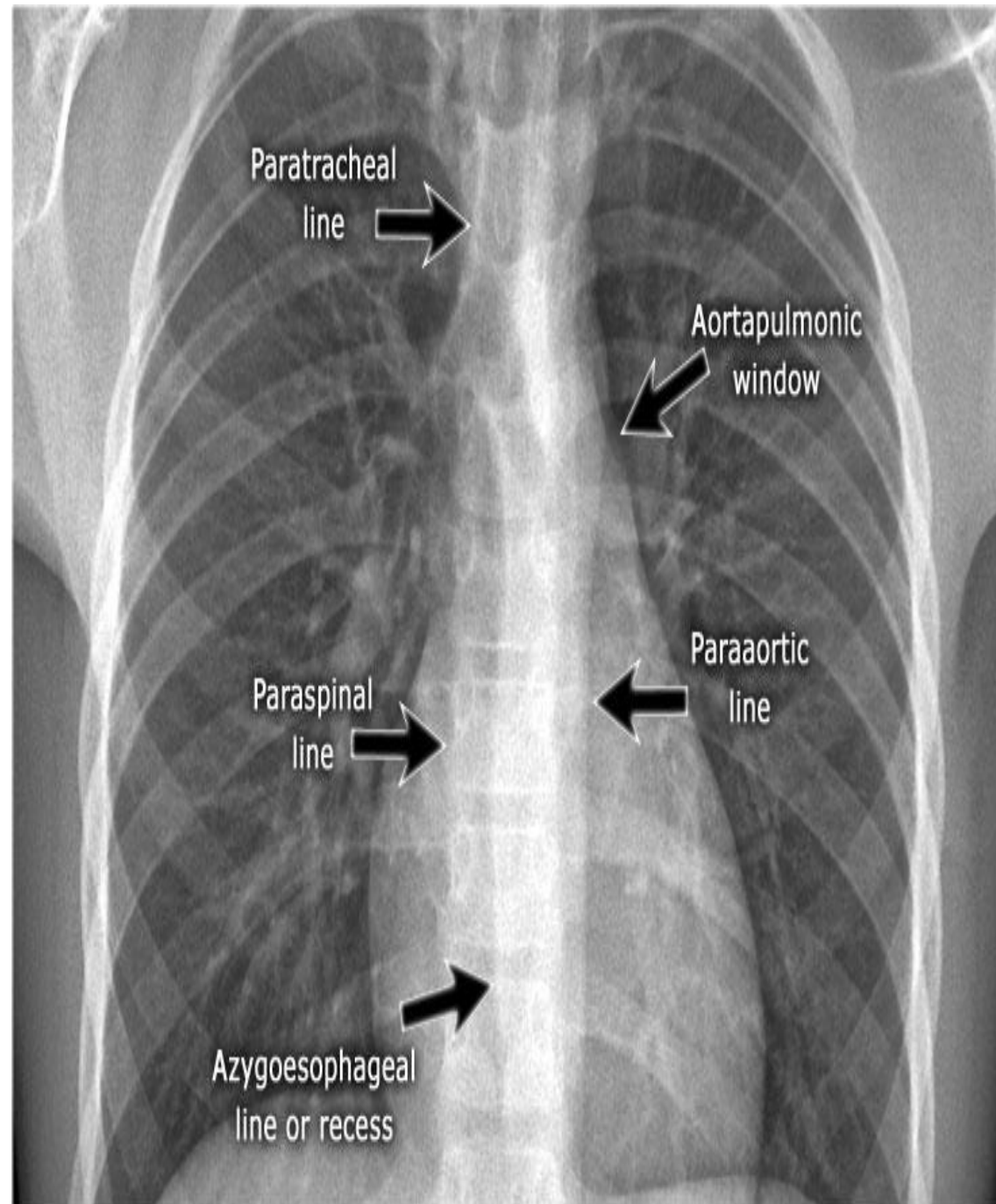
Mediastinum

	Lesions	Fluid	Fat	Vascular
Anterior	Thymic Lymphoma Germ Cell Goiter	Thymic Cyst Thymoma Pericardial Cyst Germ Cell Lymphoma	Germ cell-b Thymolipoma Fat Pad	Thyroid Cardiac Coronary
Middle	Lymph nodes Duplication Cyst Arch anomaly	Duplication Cyst Necrotic nodes Pericard recess Retroperitoneal	Lipoma Esophageal FV polyp	Arch anomaly Azygous Vein Vascular nodes
Posterior	Neurogenic Bone and marrow	Neuroenteric Cyst Schwannoma Meningocele	Extramedullary Hematopoiesis	Desc Aorta
> 1 compartment	Infection Hemorrhage Lung Cancer	Lymphangioma Mediastinitis	Liposarcoma	Hemangioma

- Mediastinal masses are discussed in more detail in [Mediastinal masses](#).
- Here is just a brief overview.

Mediastinal lines

Mediastinal lines or stripes are interfaces between the soft tissue of mediastinal structures and the lung. Displacement of these lines is helpful in finding mediastinal pathology,



Azygoesophageal recess

displacement

Esophagus

- Hiatal hernia
- Dilatation - Achalasia / Sclerodermia
- Mass - carcinoma

Lymph nodes

- Subcarinal - station 7

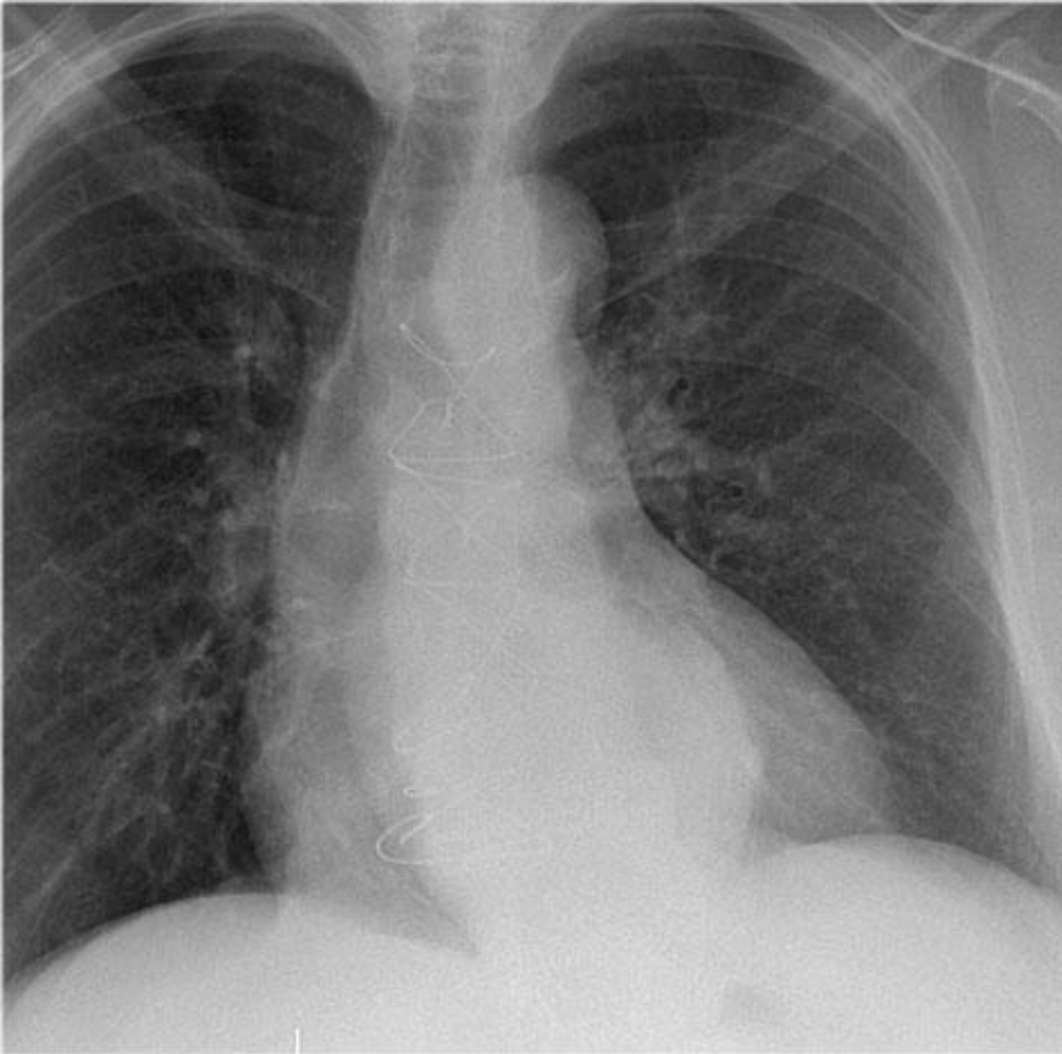
Left atrium

- Dilatation - mitral stenosis

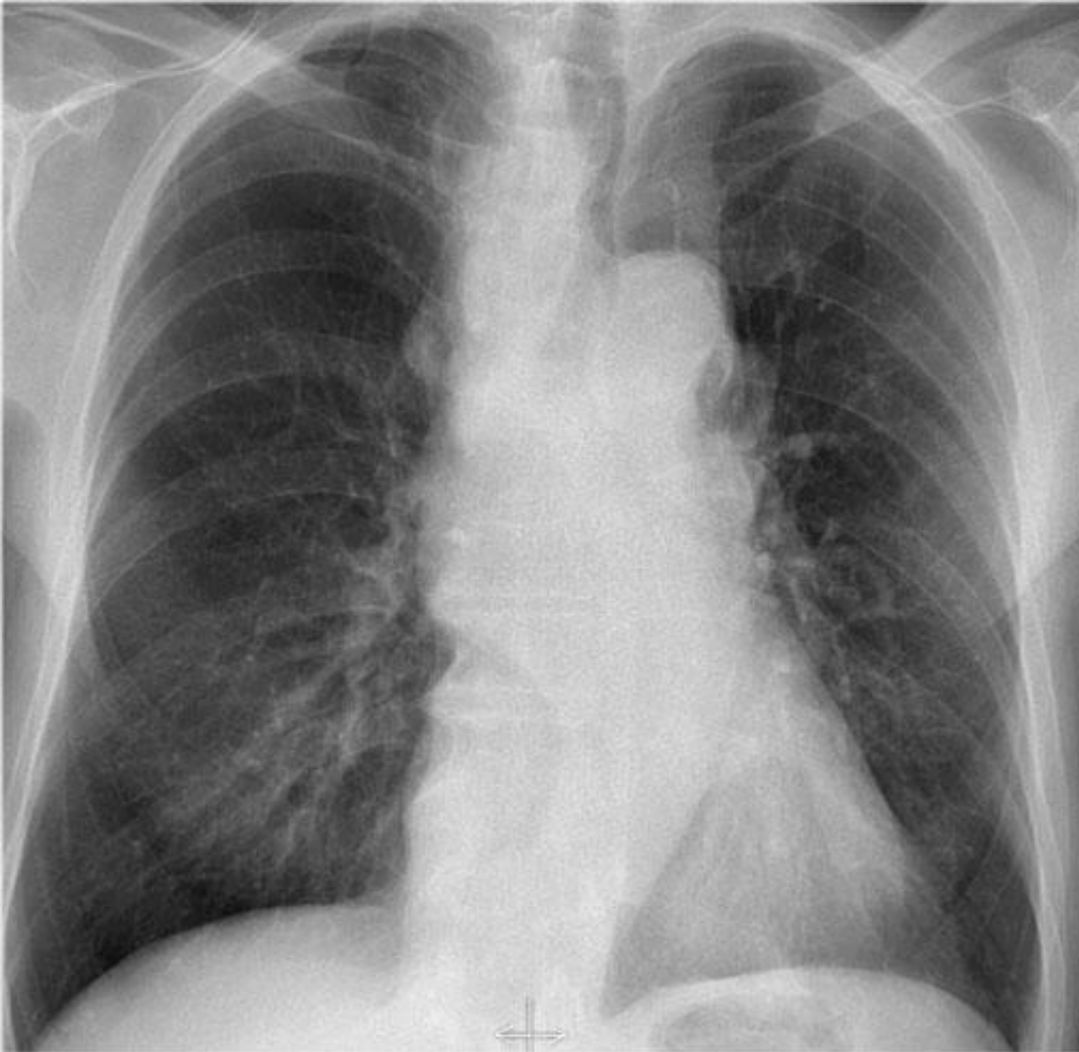
Cyst

- Bronchogenic cyst

- **Azygoesophageal recess**
- The most important mediastinal line to look for is the azygoesophageal line, which borders the azygoesophageal recess.
- This line is visible on most frontal CXRs.
- The causes of displacement of this line are summarized in the table.

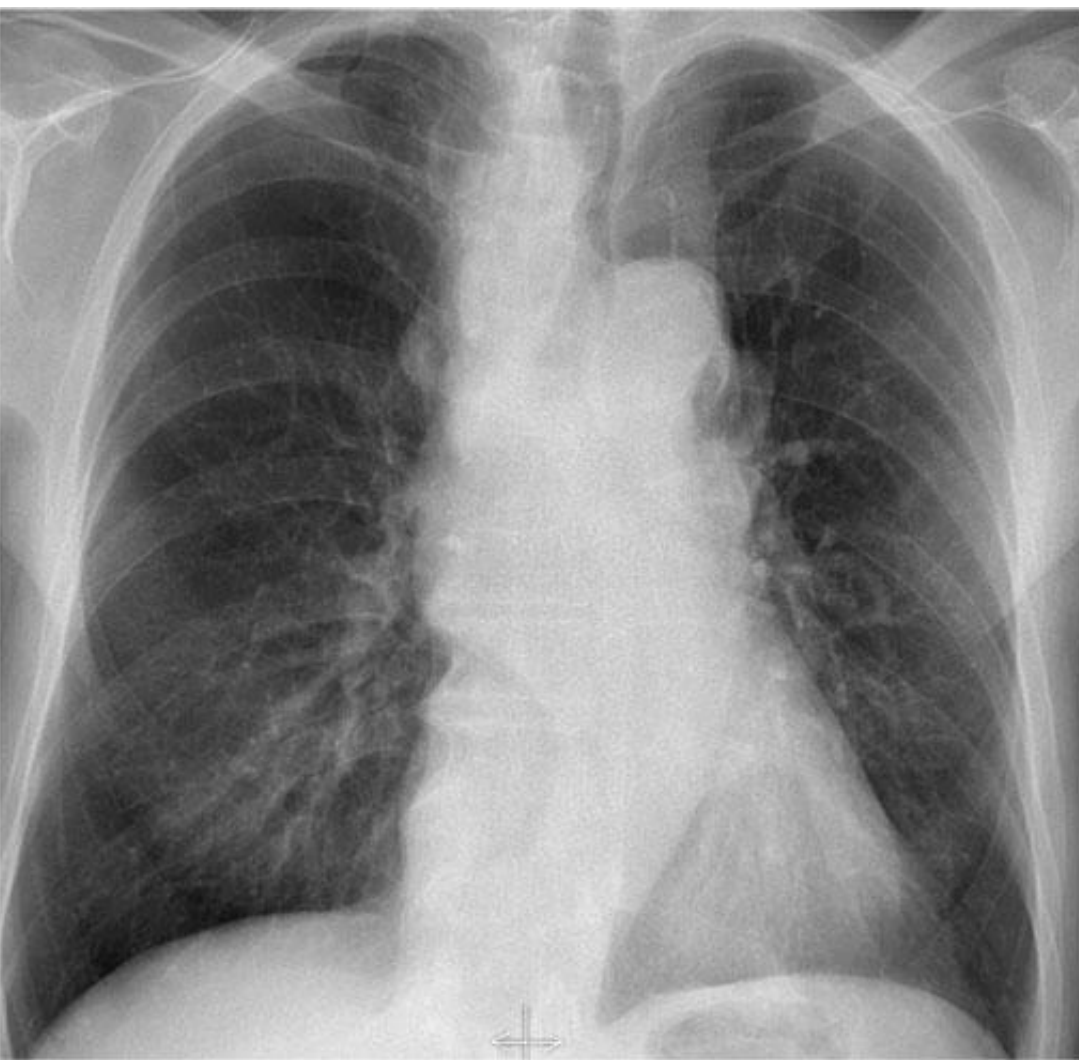


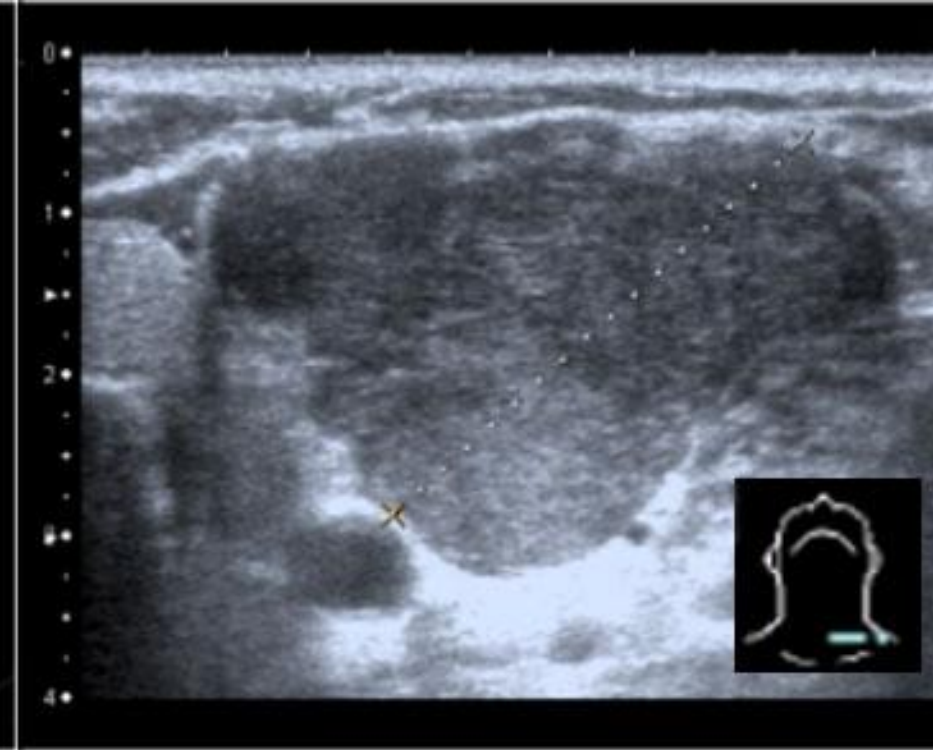
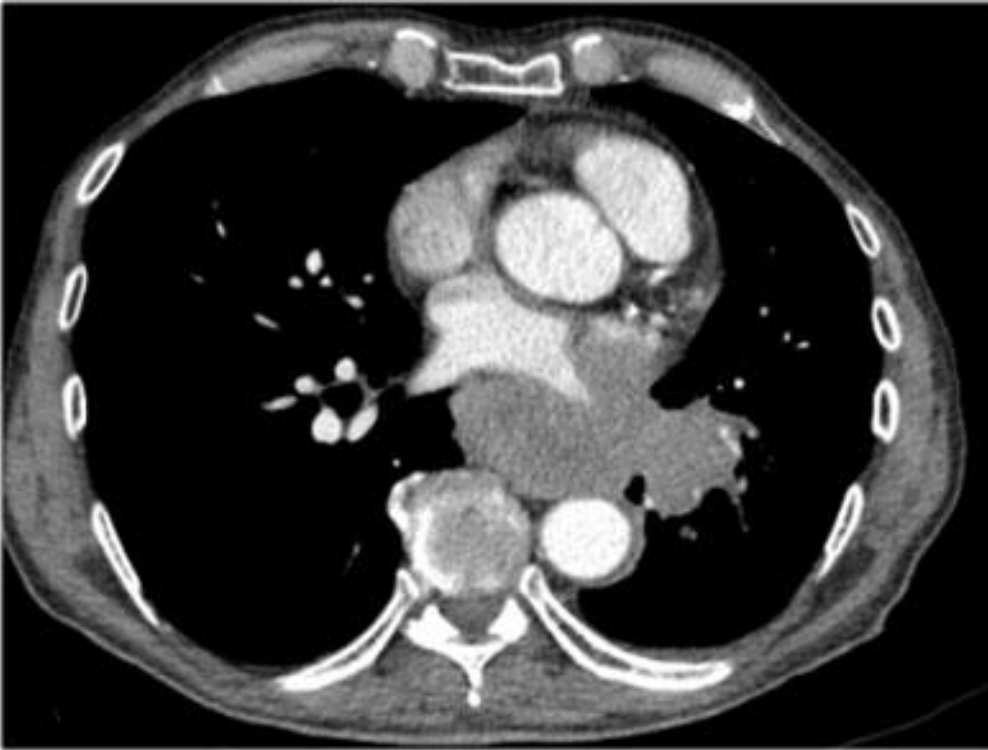
- A hiatal hernia is the most common cause of displacement of the azygoesophageal line.
- Notice the air within the hernia on the lateral view.



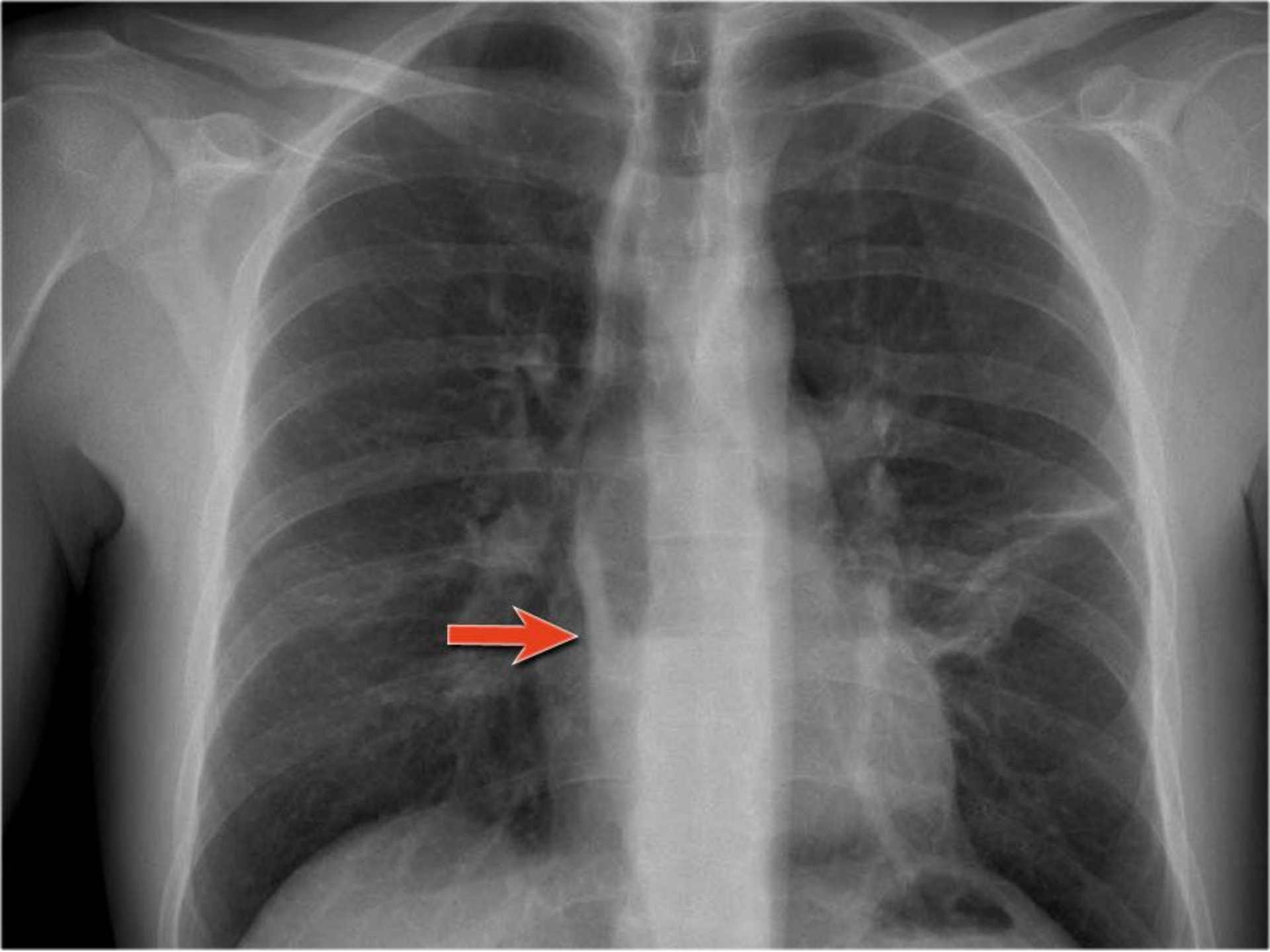
- Another common cause of displacement of the azygoesophageal line is **subcarinal lymphadenopathy**.
- Notice the displacement of the upper part of the azygoesophageal line on the chest x-ray in the area below the carina. This is the result of massive lymphadenopathy in the subcarinal region (station 7).
- There are also nodes on the right of the trachea displacing the right paratracheal line.

- On the PET we can appreciate the massive lymphadenopathy far better than on the CXR.
- There are also lymphomas in the neck.
this is an important finding, since these nodes are accessible for biopsy.
- Continue with images of CT and ultrasound.

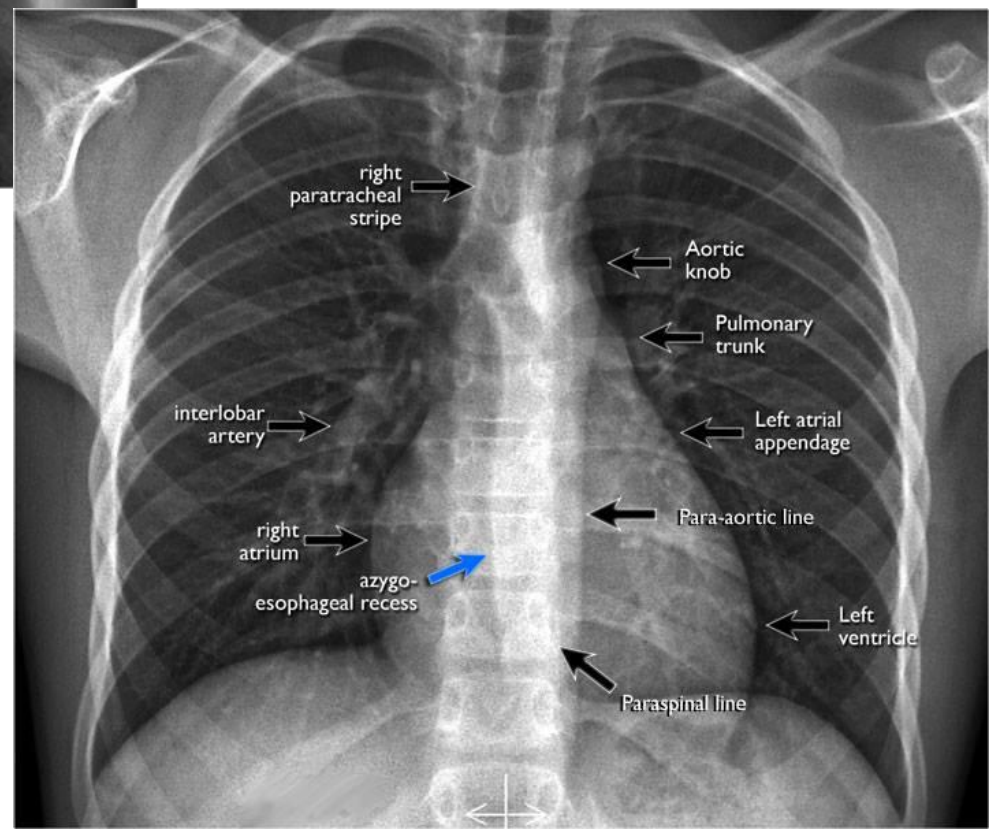
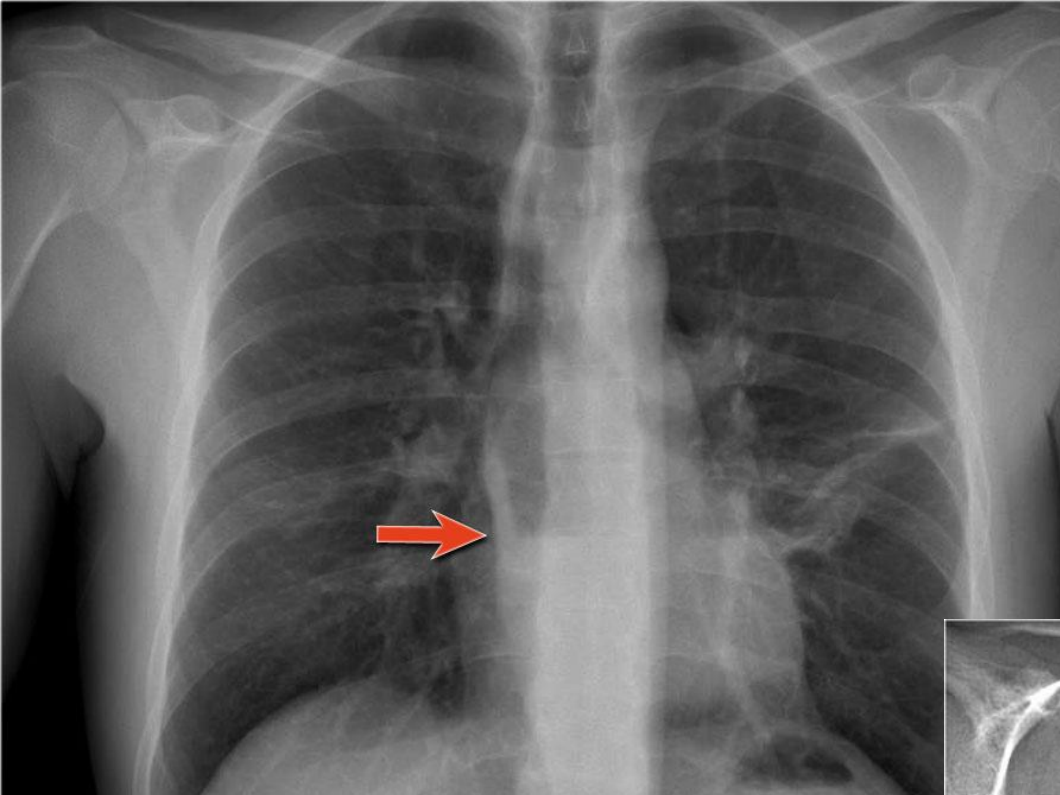


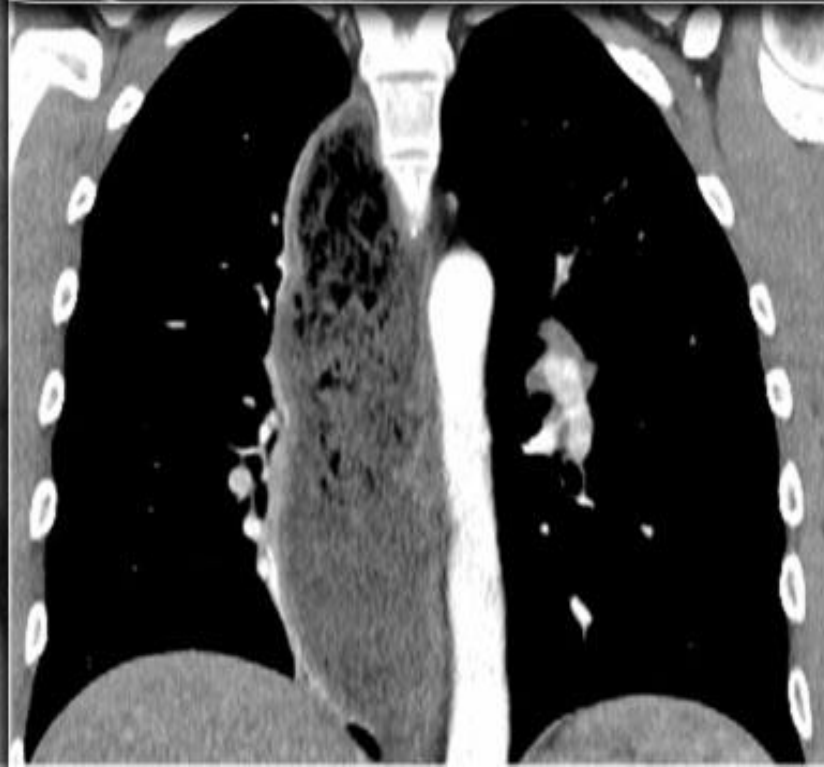
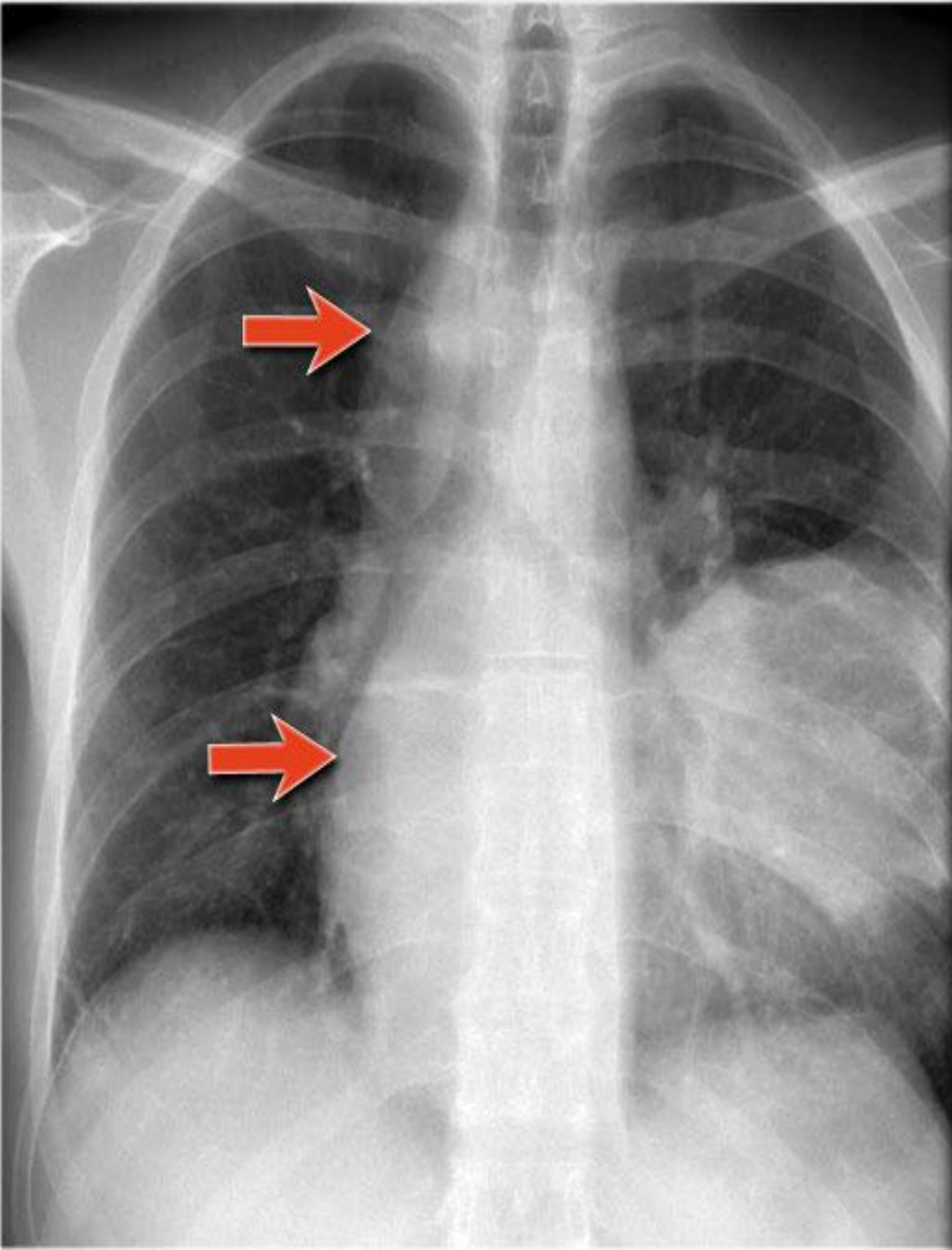


- Here we see a CT-image.
The azygo esophageal recess is displaced by lymph nodes that compress the left atrium.
- The final diagnosis of **small cell lung cancer** was made through a biopsy of a lymph node in the neck.



- Notice the following:
- There is displacement of the azygoesophageal line both superiorly and inferiorly.
- There is an air-fluid level (arrow).
Combined with the above this must be a dilated esophagus with residual fluid. The final diagnosis was achalasia.
- The density on the left in the region of the lingula is the result from prior aspiration pneumonia.





- Here we have a prior CXR of this patient.
- The AP-film shows a right paratracheal mass.

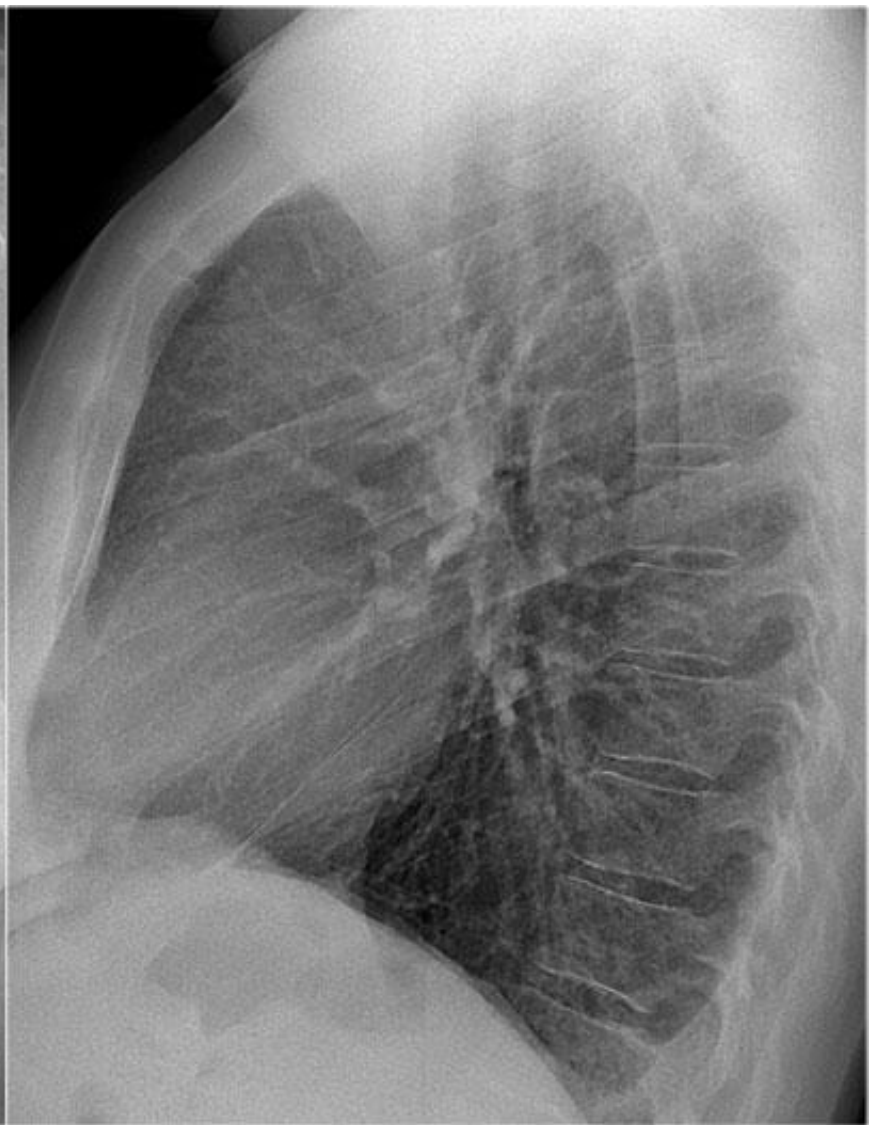
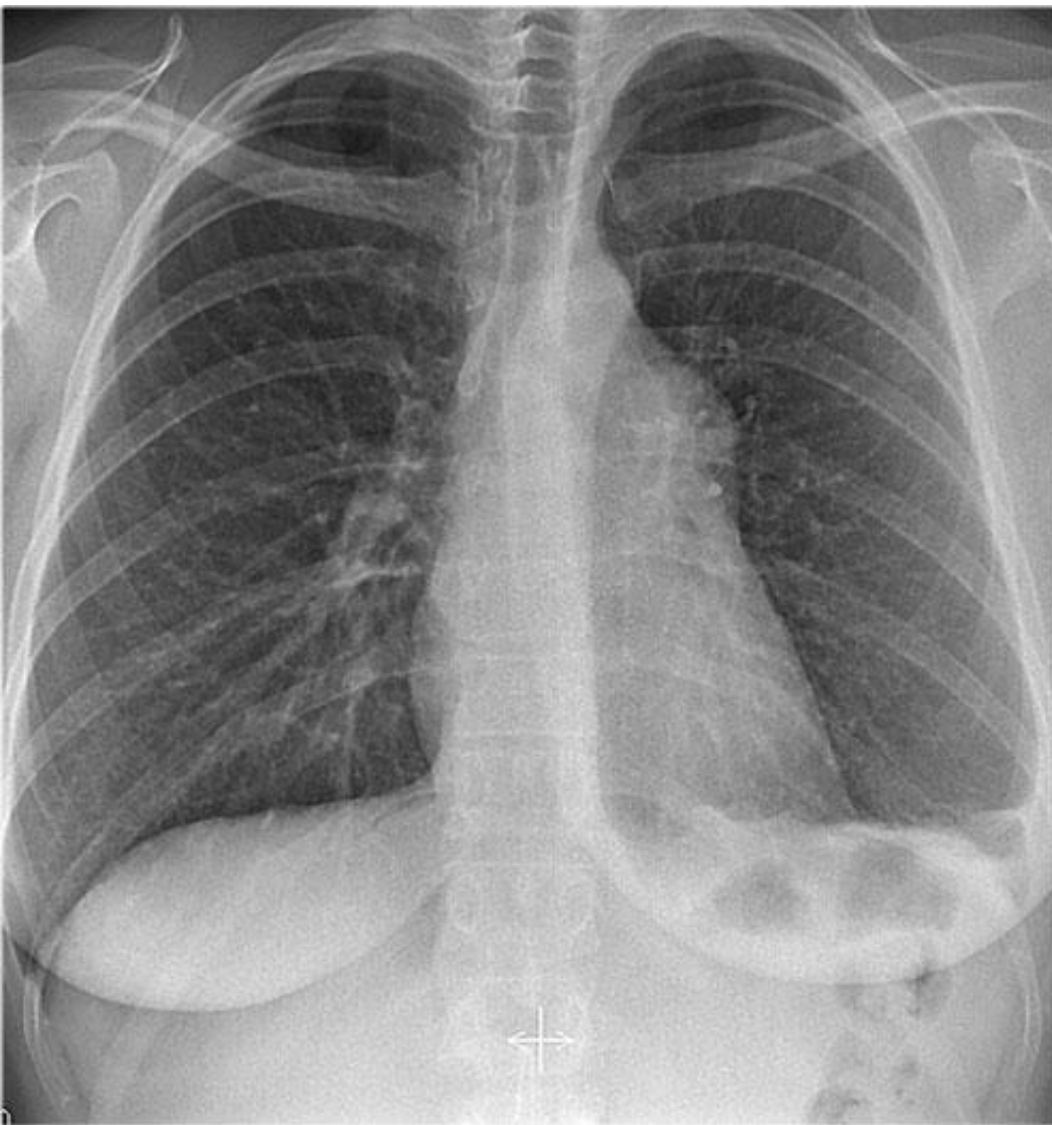
The azygoesophageal recess is not identified, because it is displaced and parallels the border of the right atrium.

The large round density in the left lung is the result of aspiration.

- Notice the massive dilatation of the esophagus on the CT.

Aortopulmonary window

- The aortopulmonary window is the interface below the aorta and above the pulmonary trunk and is concave or straight laterally.
- Here the AP-window is convex laterally due to a mass that fills the retrosternal space on the lateral view.





- On the CT-images a mass in the anterior mediastinum is seen.
- Final diagnosis: Hodgkins lymphoma.



- Here another case.
On the PA-film a mass is seen that fills the aortopulmonary window.



- The PET better demonstrates the extent of the lymphnode metastases in this patient.
- Final diagnosis: small cell lungcarcinoma.

Lungs

- Lung abnormalities mostly present as areas of *increased* density, which can be divided into the following patterns:
 1. Consolidation
 2. Atelectasis
 3. Nodule or mass - solitary or multiple
 4. Interstitial
- Less frequently areas of *decreased density* are seen as in emphysema or lungcysts.
- These lungpatterns will discussed in more detail in an article that will be published soon: Chest X-Ray - Lung disease.

Consolidation

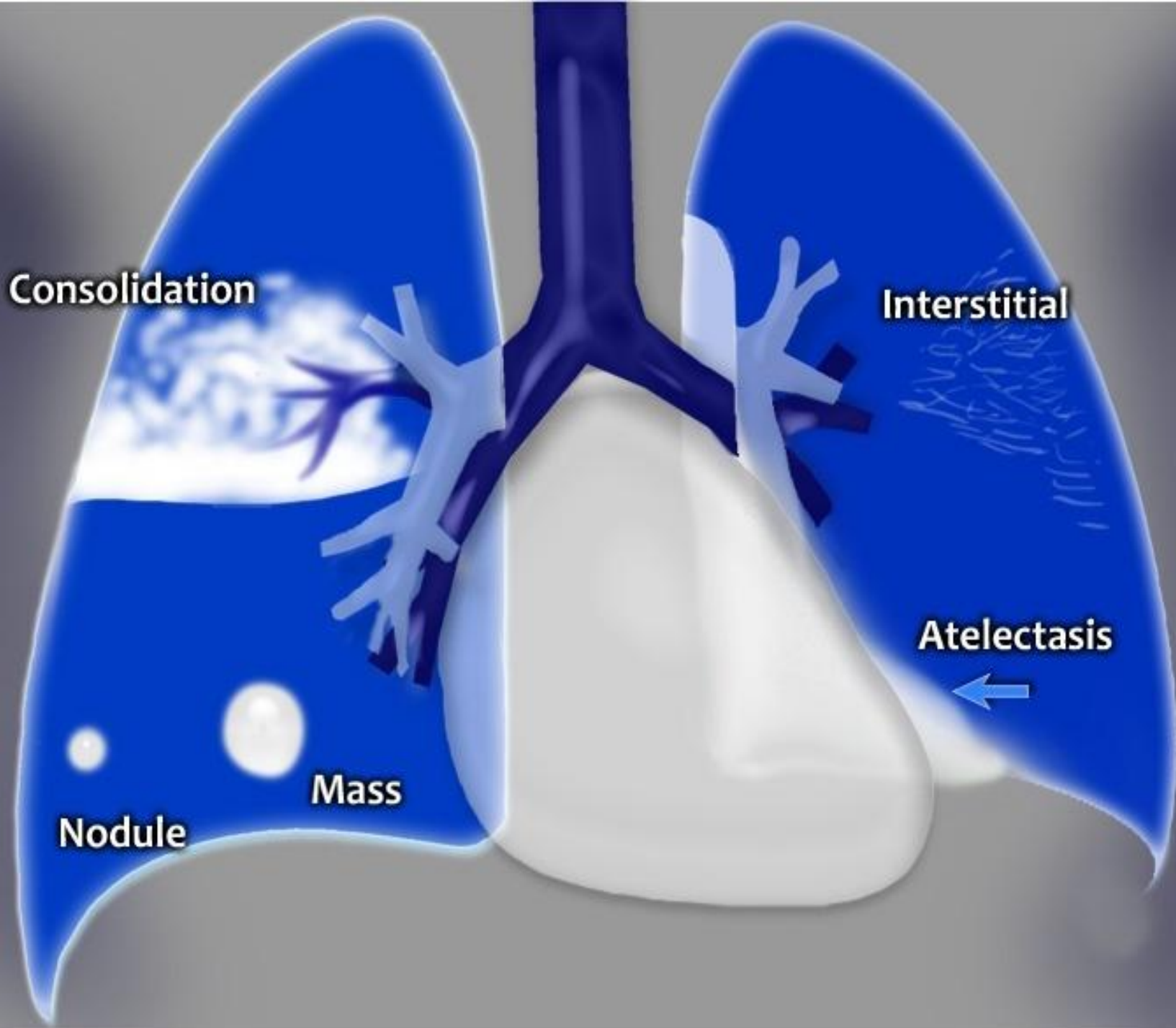
Interstitial

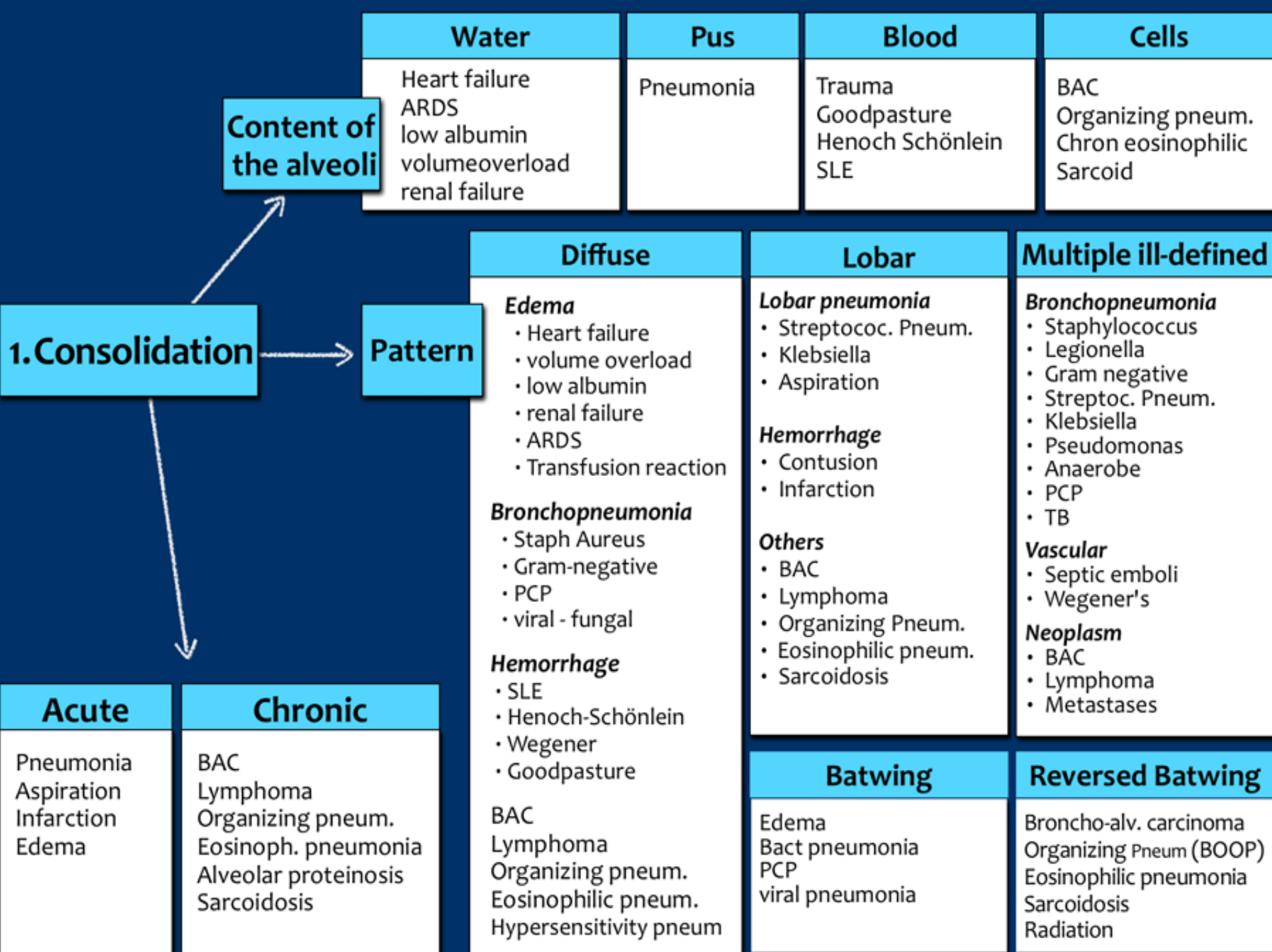
Atelectasis



Mass

Nodule





2. Atelectasis

```
graph LR; A[2. Atelectasis] --> B[Resorption]; A --> C[Relaxation]; B --> B1[Mucus]; B --> B2[Tumor]; B --> B3[Foreign body]; C --> C1[Pleural effusion]; C --> C2[Pneumothorax]; C --> C3[Round atelectasis];
```

Resorption

Mucus
Tumor
Foreign body

Relaxation

Pleural effusion
Pneumothorax
Round atelectasis

3. Nodule - Mass



Nodule < 3cm

Granuloma
- Fungal
- TB

Lungca
Metastasis
Hamartoma

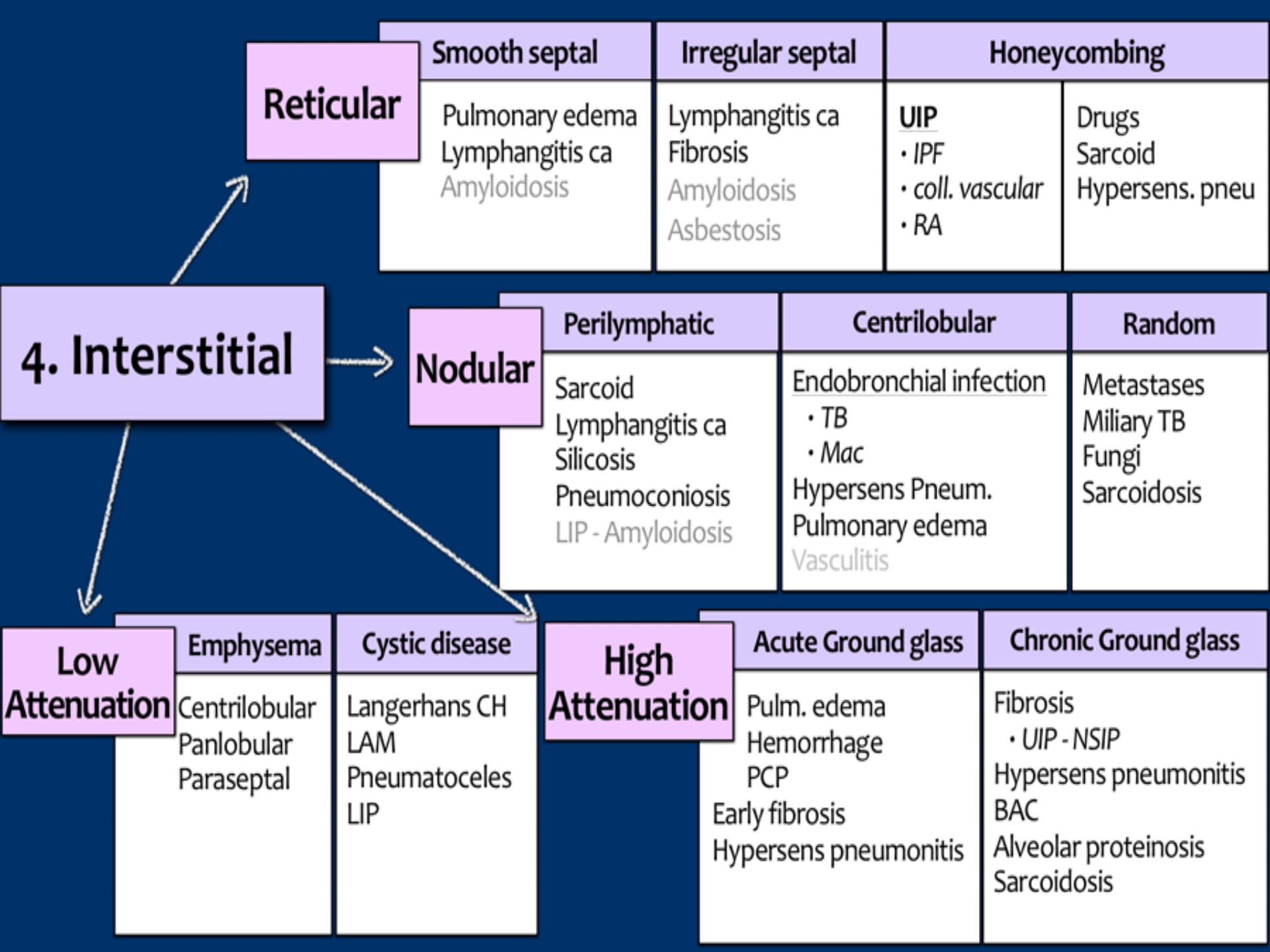
Mass > 3cm

Lungca
Granuloma
Hamartoma

Multiple masses

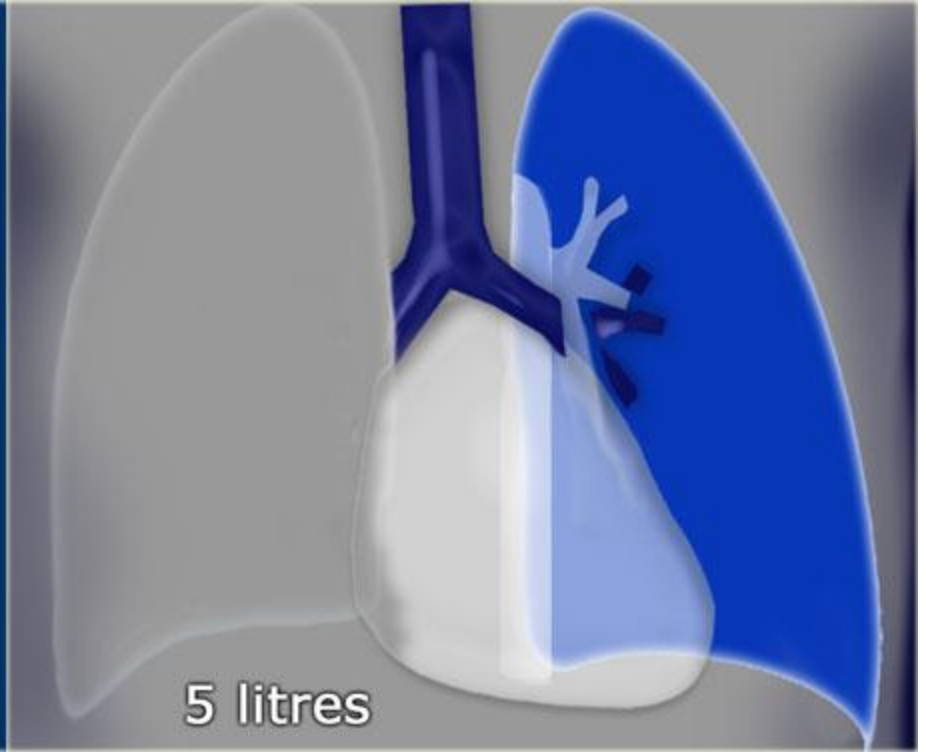
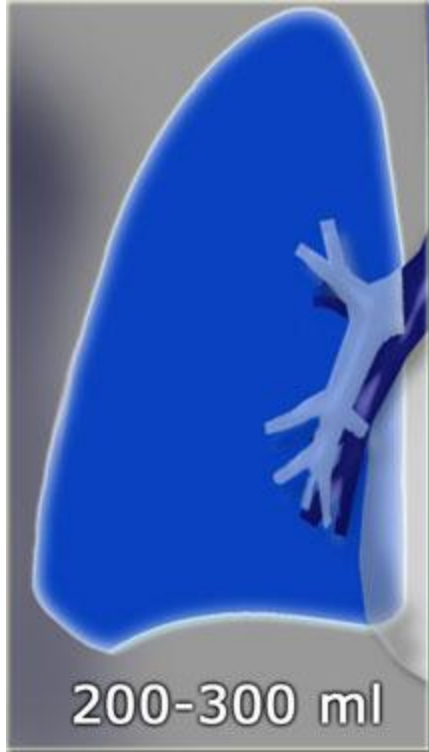
Infection:
- TB
- Histoplasmosis
- Fungi
- Sept emboli

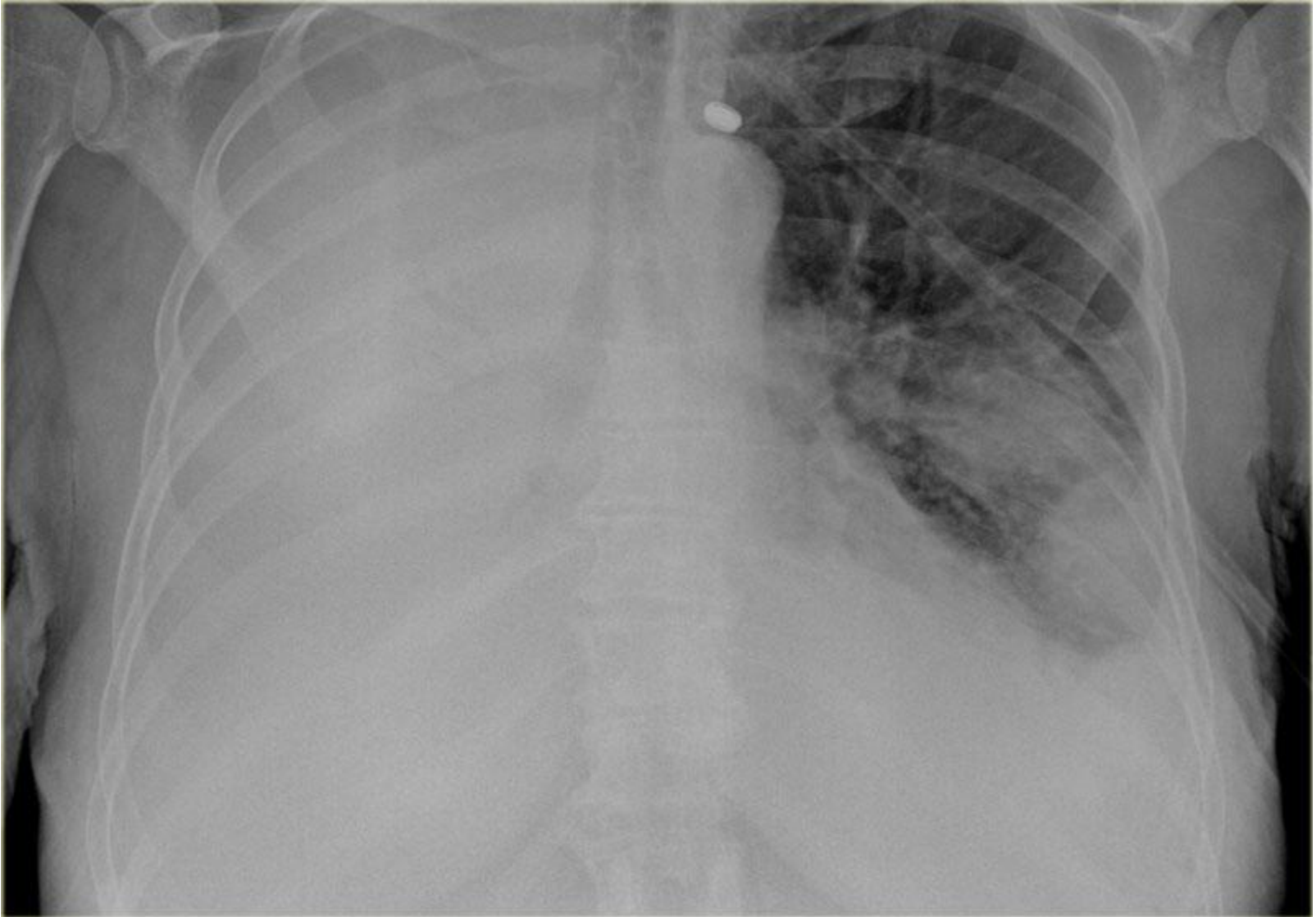
Metastases
BAC
Sarcoidosis
Wegener
RA
Rendu-Osler



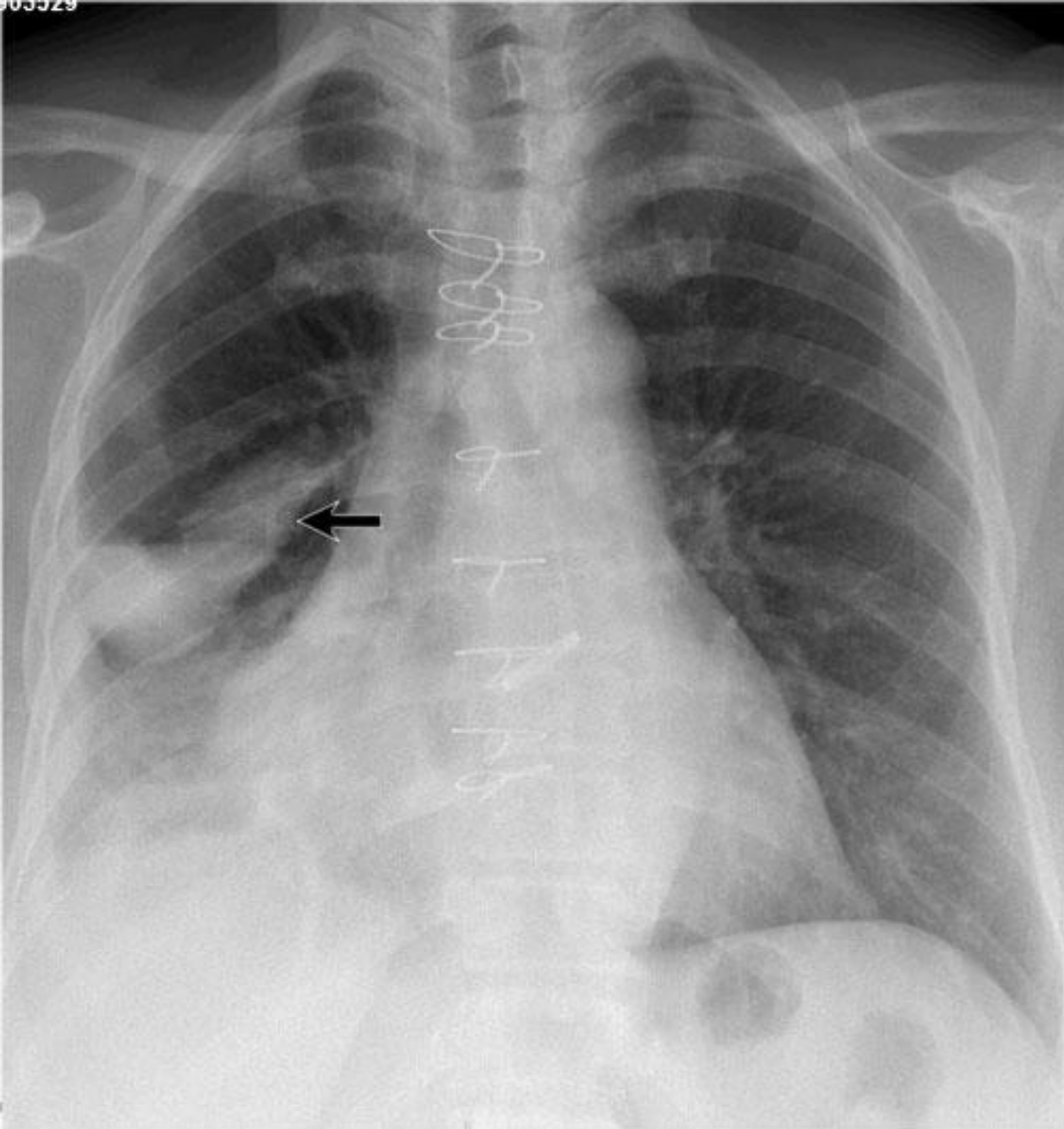
Pleura

- **Pleural fluid**
- It takes about 200-300 ml of fluid before it comes visible on an CXR (figure).
- About 5 liters of pleural fluid are present when there is total opacification of the hemithorax.





- Total opacification of the right hemithorax in a patient with pleuritis carcinomatosa on both sides.
- On the right there is only some air visible in the major bronchi creating an air bronchogram within the compressed lung.



- Pleural fluid may become encysted.
- Here we see fluid entrapped within the fissure. This can sometimes give the impression of a mass and is called 'vanishing tumor'.

Pneumothorax

causes

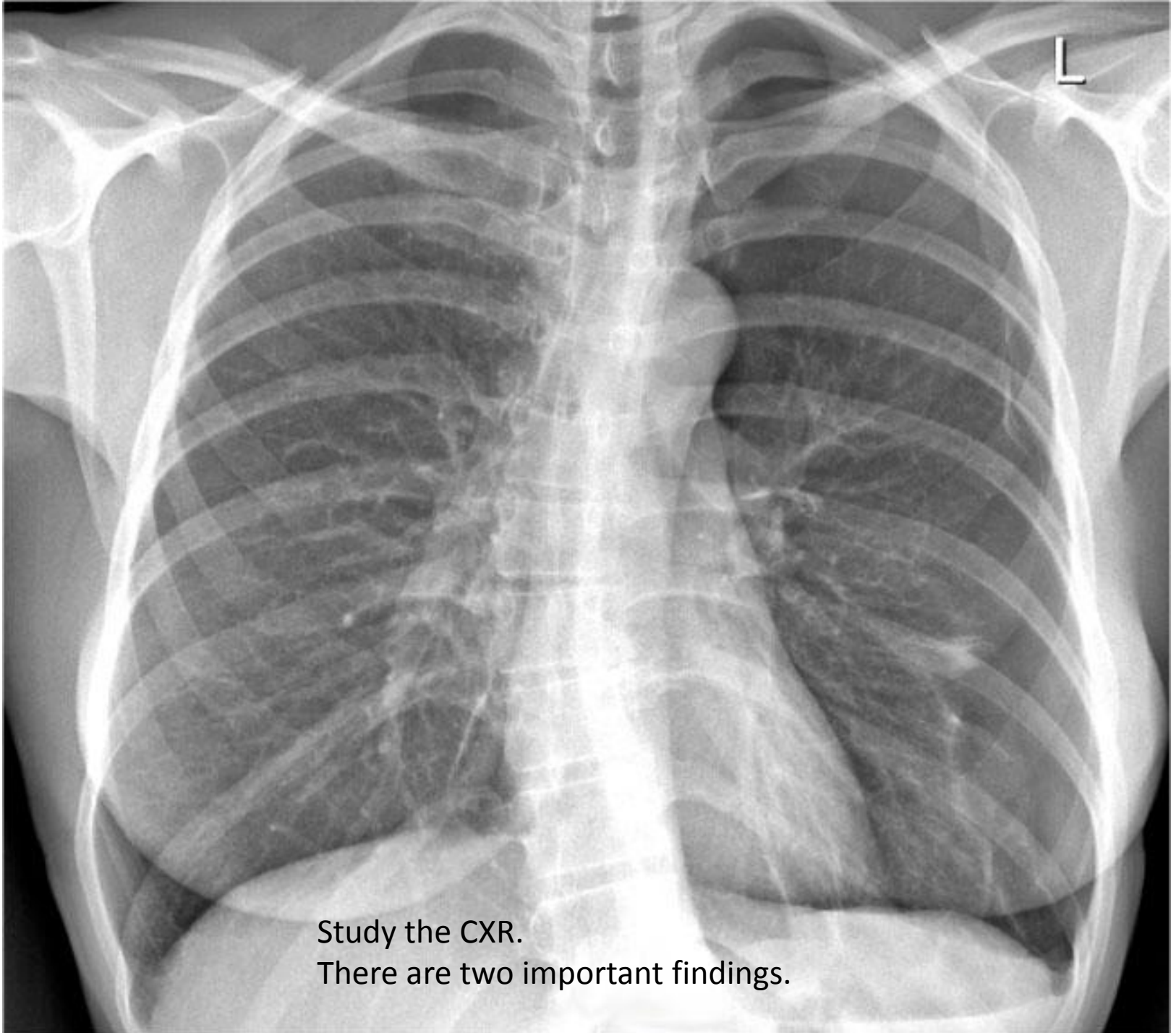
Spontaneous - subpleural bleb

Lung disease

- COPD
- Asthma
- Cystic disease: LAM - Langerhans CH

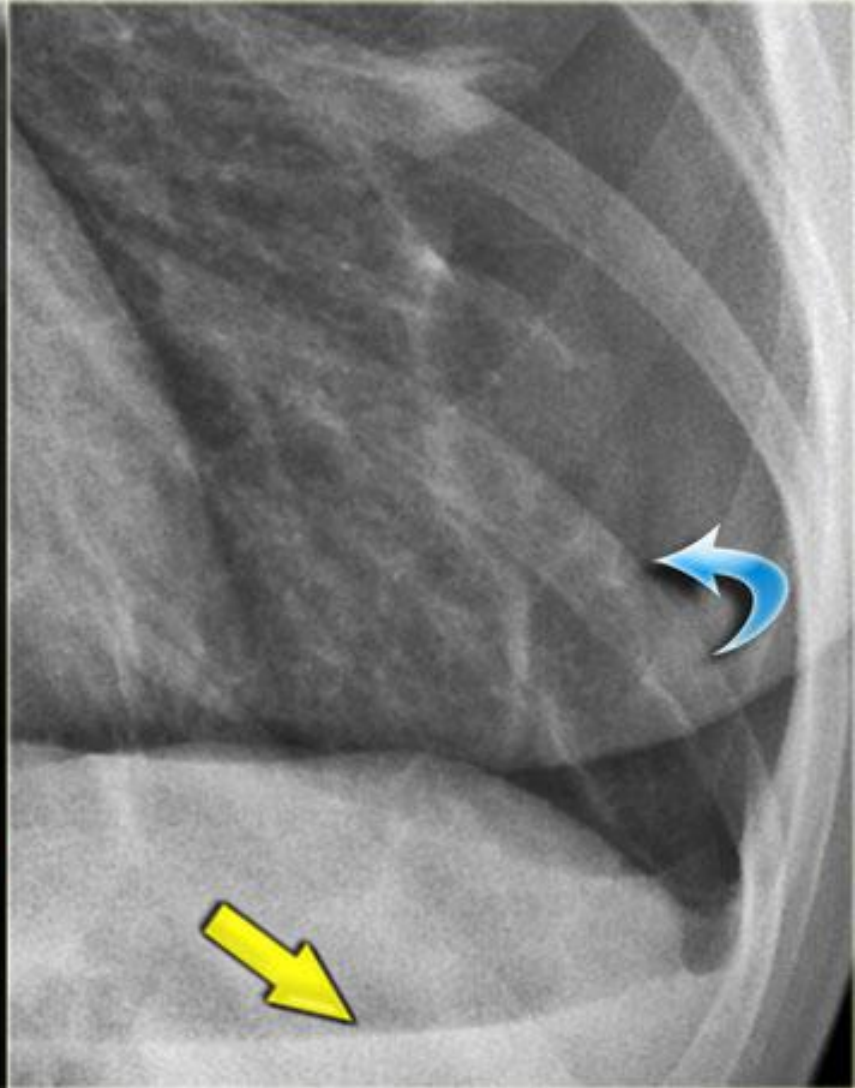
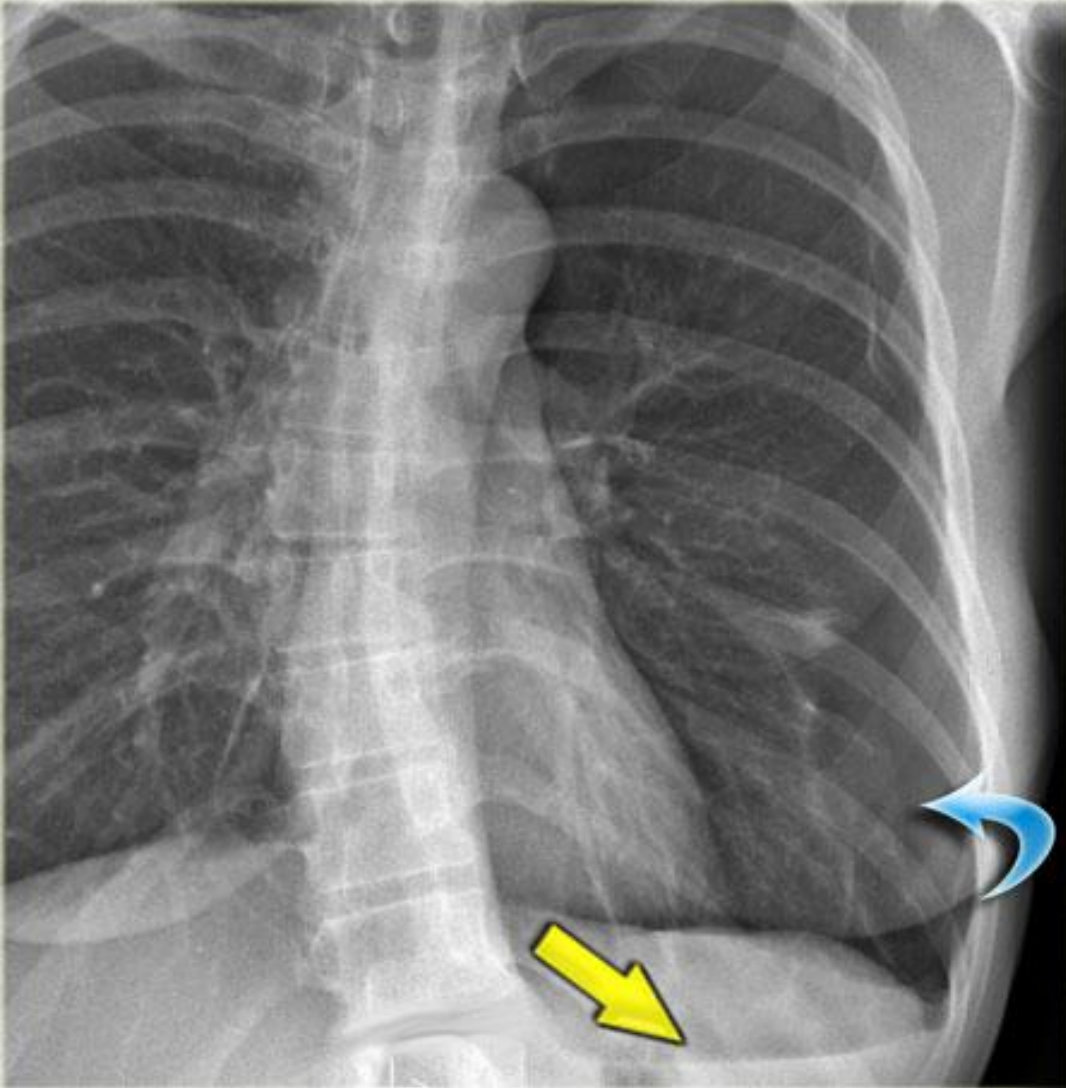
Trauma

- Biopsy - central line - mechan. ventilation
- Cocaine - marijuana



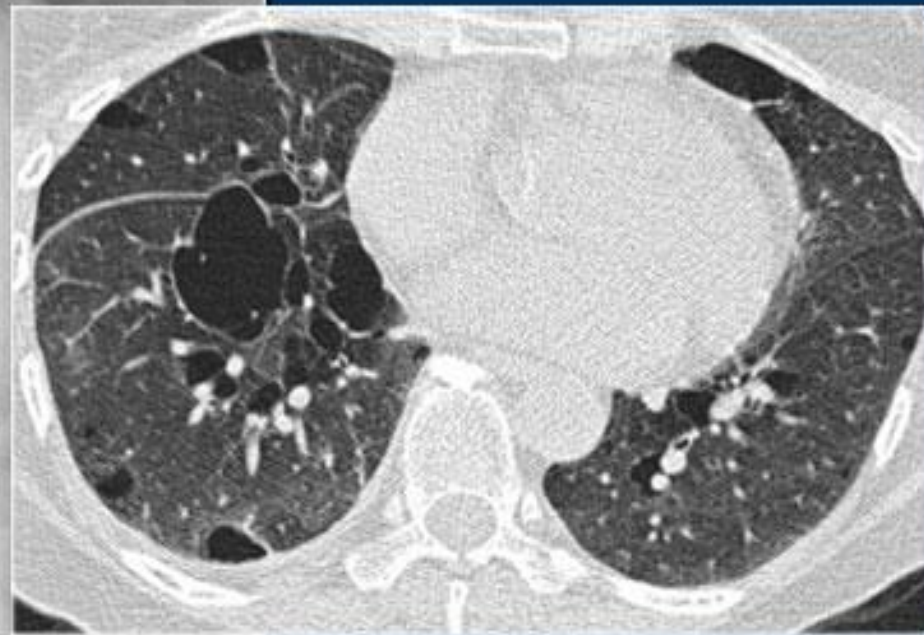
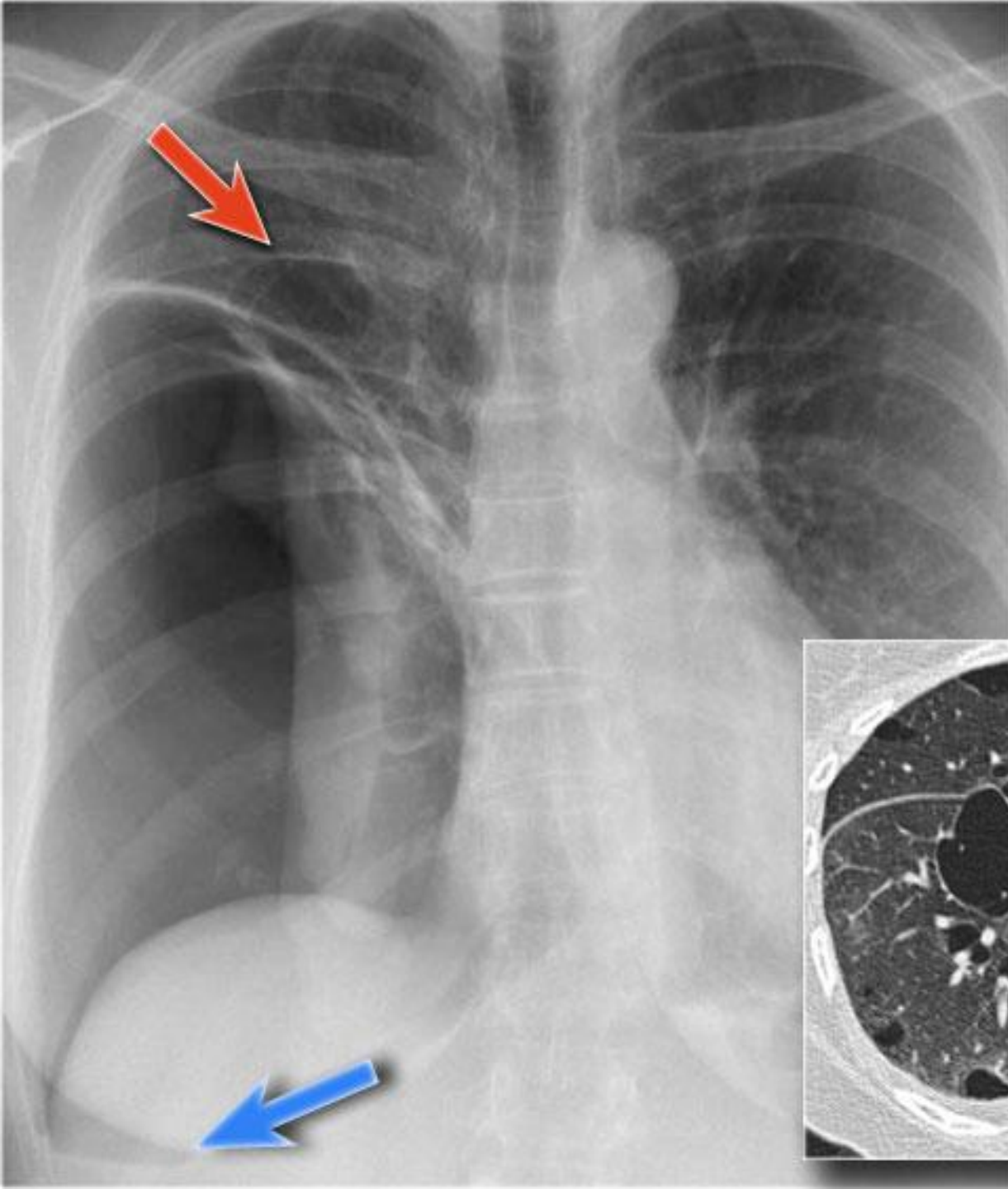
Study the CXR.
There are two important findings.

- The retracted visceral pleura is seen (blue arrow) which indicates that there is a pneumothorax.
- There is a horizontal line visible (yellow arrow).
Normally there are no straight lines in the human body unless when there is an air-fluid level.
This means that there is a hydro-pneumothorax.
- When a pneumothorax is small, this air-fluid level can be the only key to the diagnosis of a pneumothorax.





- Study the CXR.
- There are 3 important findings.
- Notice that the mediastinum is slightly displaced to the left.
Does this mean that there is a tension pneumothorax?
- Do you have an idea about the cause of the pneumothorax?



- There is a hydropneumothorax.
Notice the air-fluid level (blue arrow).
- The upper lobe is still attached to the chest wall by adhesions.
Maybe this patient was treated for a prior pneumothorax.
- There is a lung cyst in the upper lobe (red arrow).
So we can assume that the pneumothorax has something to do with a cystic lung disease.
- Since this patient is a woman, lymphangioleiomyomatosis (LAM) is a possible diagnosis.
- LAM is a rare lung disease that results in a proliferation of smooth muscle throughout the lungs resulting in the obstruction of small airways leading to pulmonary cyst formation and pneumothorax.
LAM also occurs in patients who have tuberous sclerosis.

Pleural Opacities

Solitary

Fluid collection

Loculated pleural effusion
Organized empyema
Hematoma

Tumor

Mesothelioma
Metastasis
Neural tumor
Lipoma
Solitary fibrous tumor of
the pleura

Multiple

Pleural plaques

asbestos-related

Fluid collection

Loculated pleural effusion

Tumor

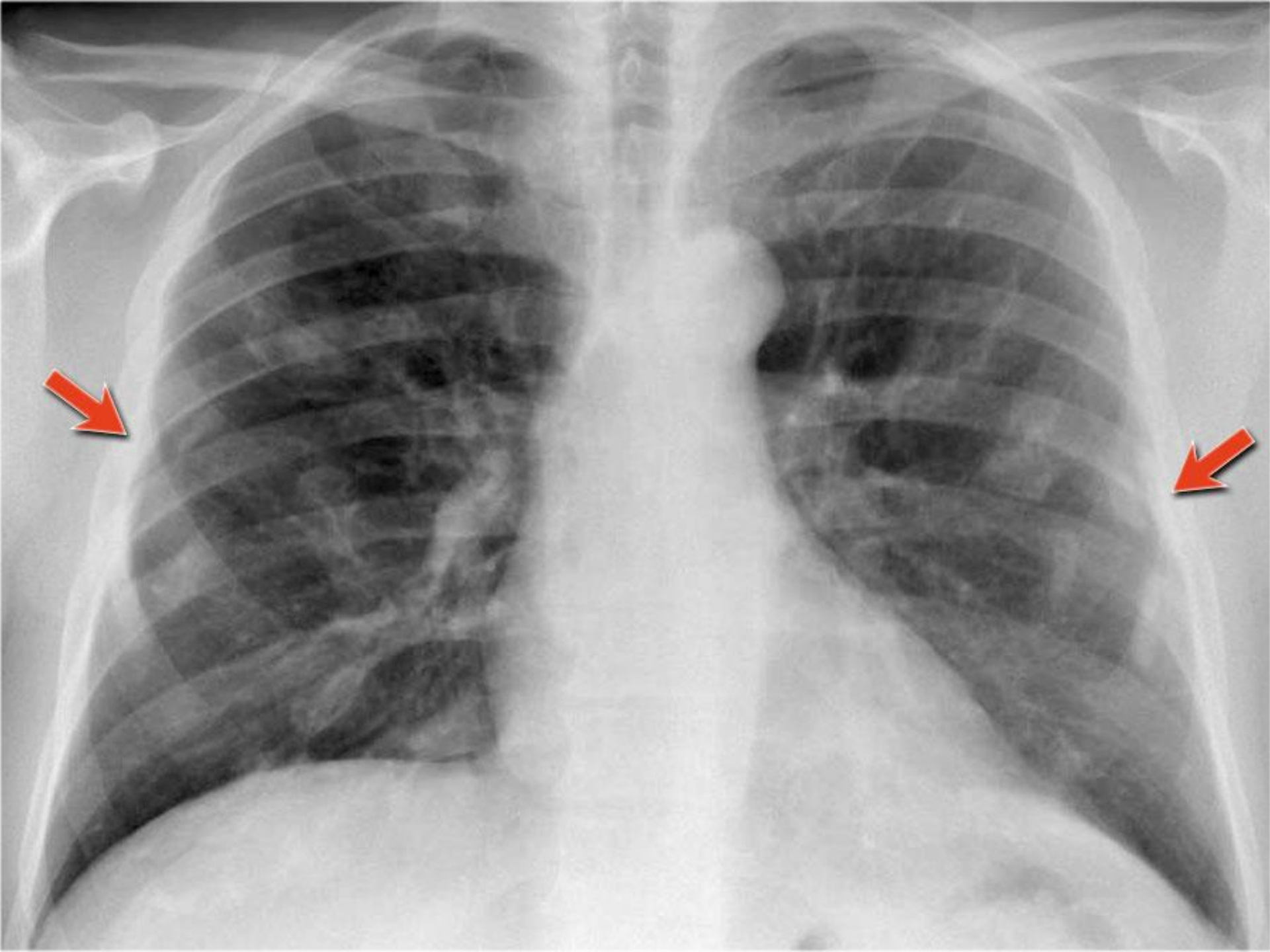
Mesothelioma
Metastases
Neural tumors
Splenosis

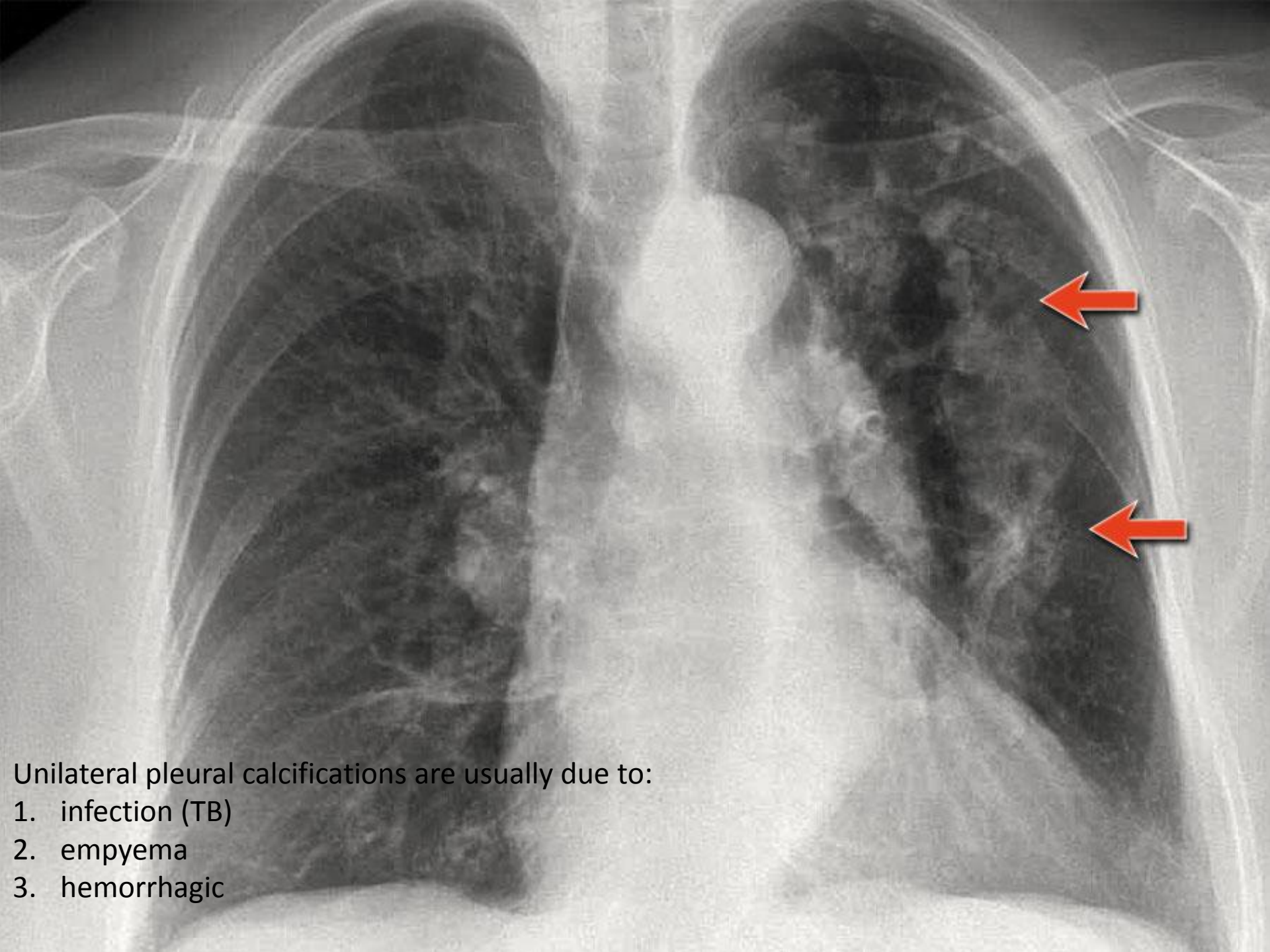
- **Pleural plaques**

The CXR shows multiple opacities.

They have irregular shapes and do not look like a lung masses or consolidations.

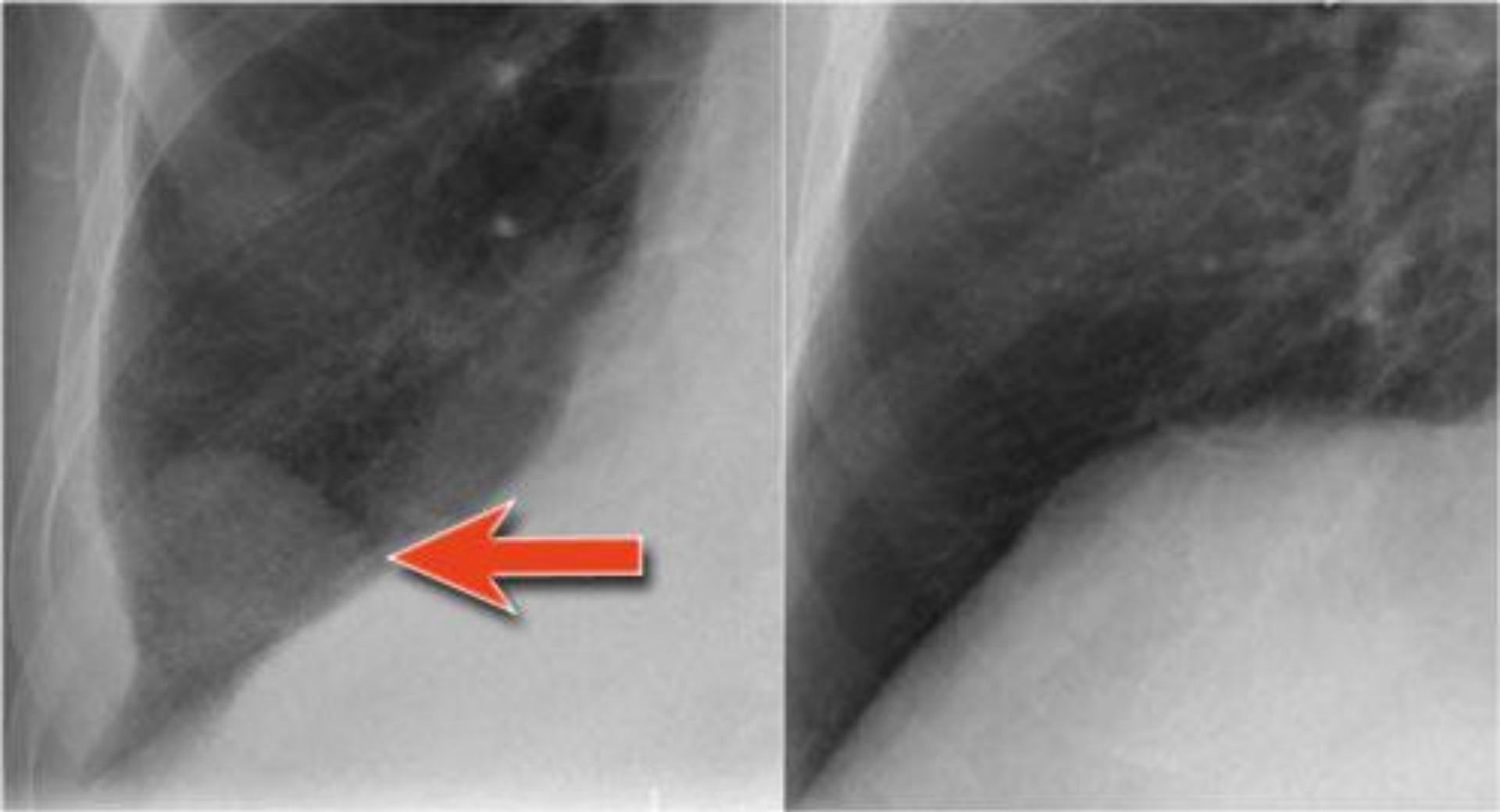
- Some of these opacities are clearly bordering the chest wall (red arrows).
- All these findings indicate that we are dealing asbestos related pleural plaques.
- Asbestos related pleural plaques are usually:
 1. bilateral and extensive.
 2. covering the dome of the diaphragm.





Unilateral pleural calcifications are usually due to:

1. infection (TB)
2. empyema
3. hemorrhagic

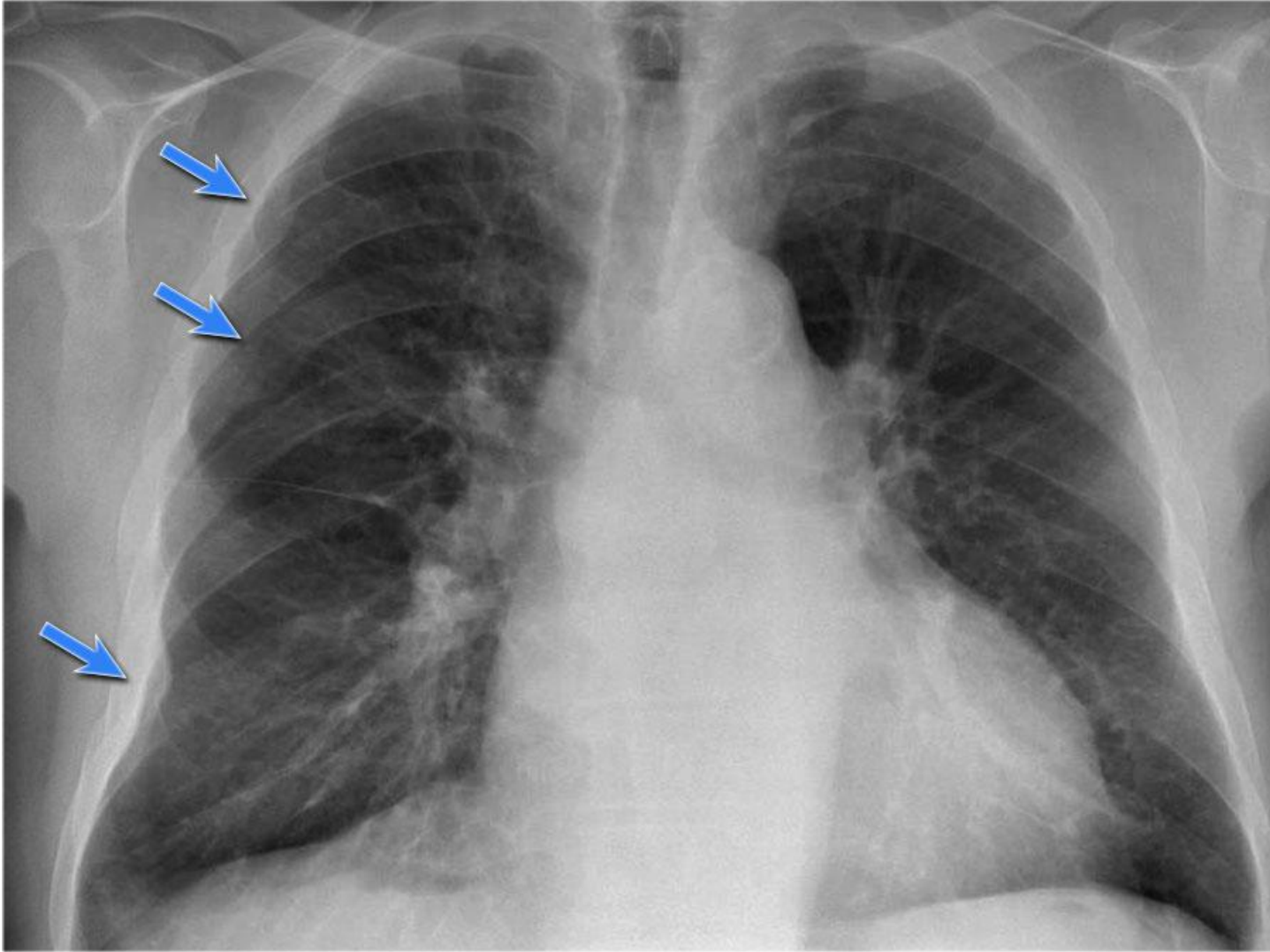


- **Pleural hematoma**

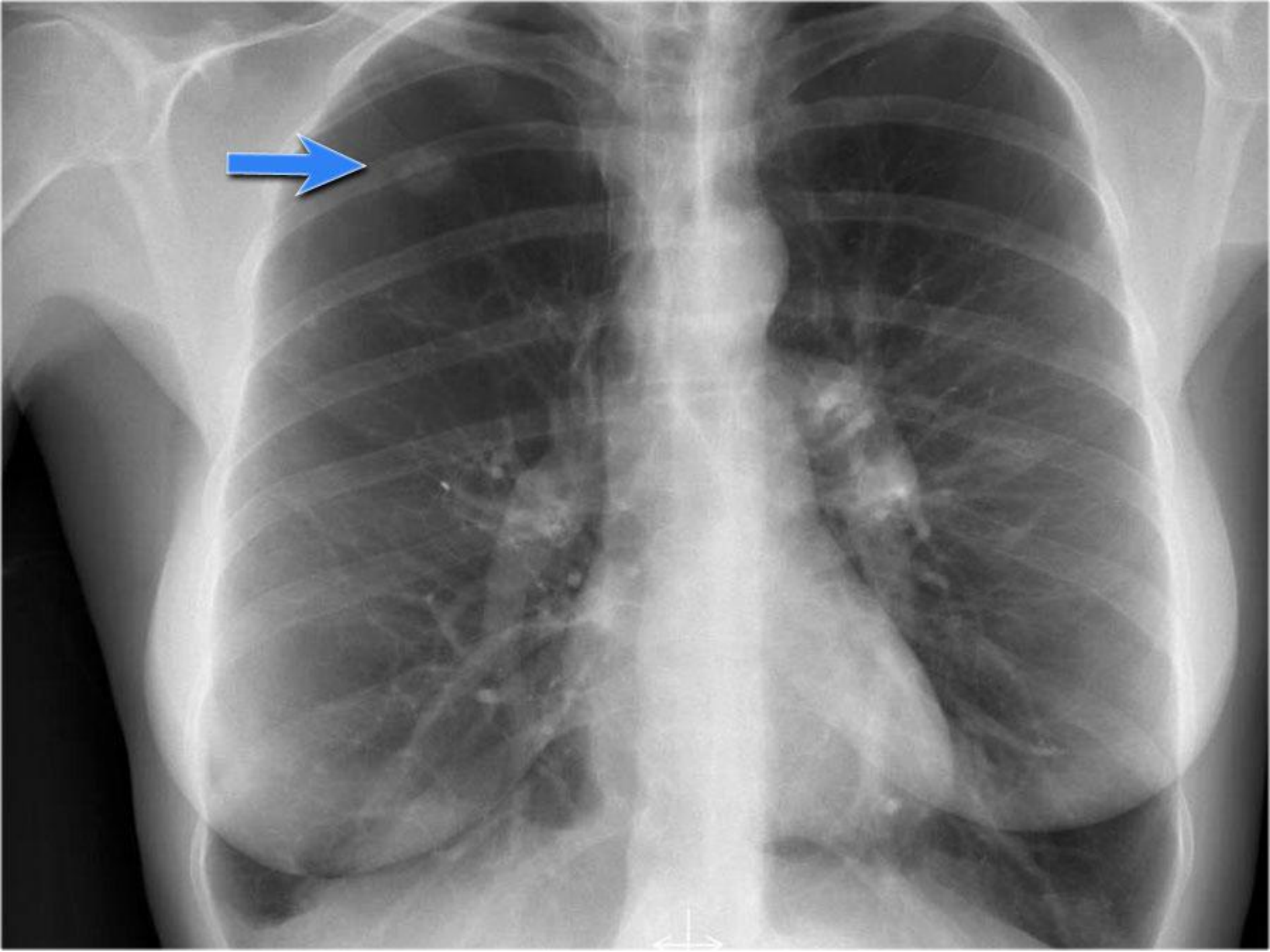
These images are of a patient, who had a pleural opacity after a chest trauma.

- It was believed to be a hematoma and resolved spontaneously.

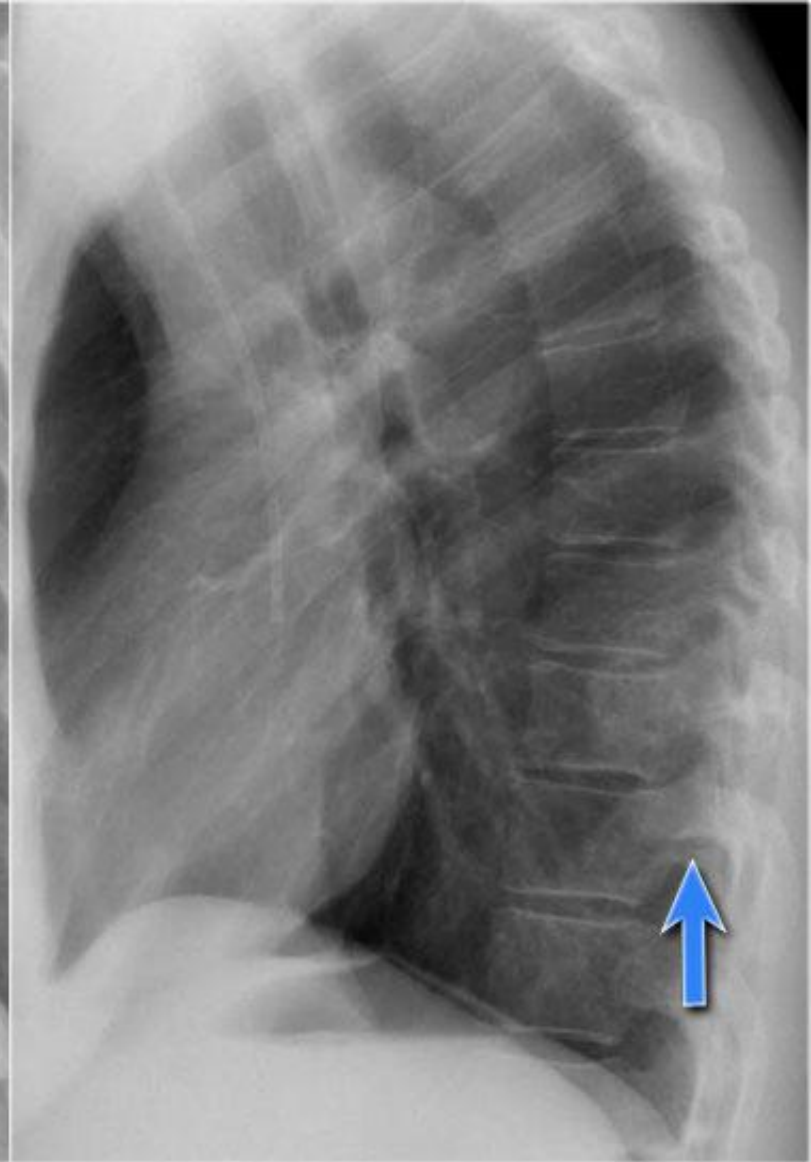
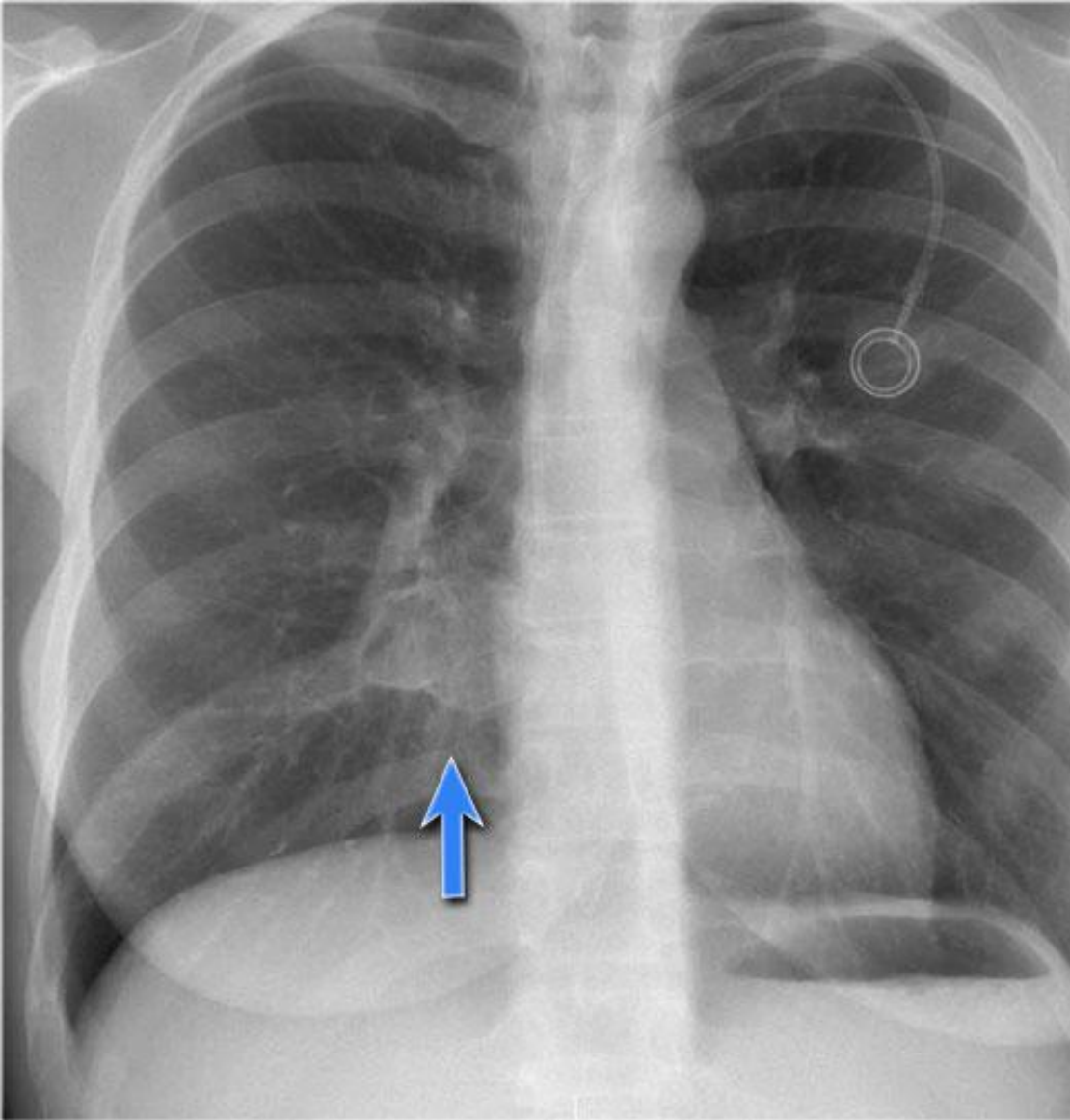
Chest wall



- **Ribfractures**
The most common identified chest wall abnormalities are old ribfractures.
- The CXR shows many rib deformities due to old fractures.



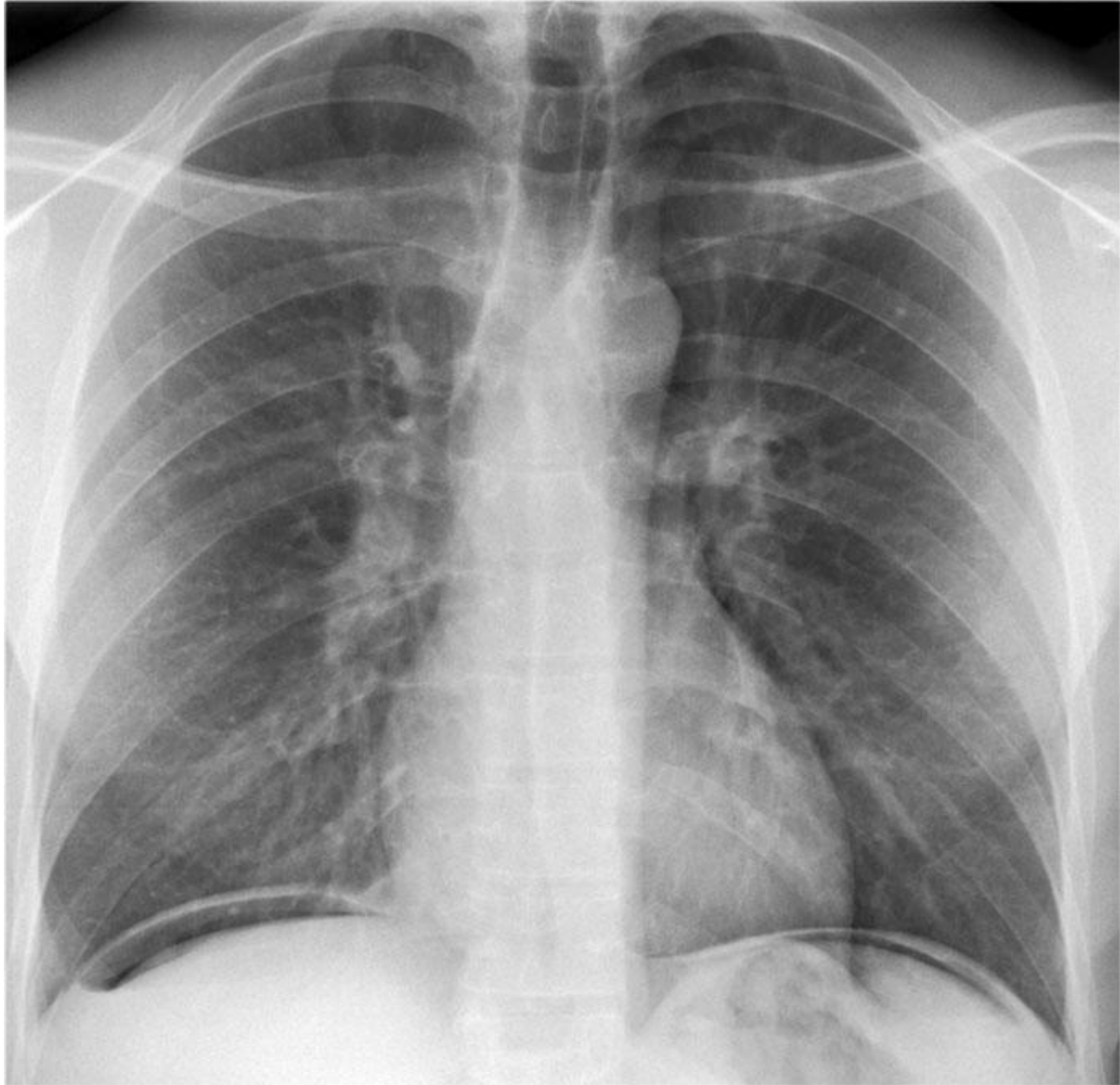
- When a rib fracture heals, the callus formation may create a mass-like appearance (blue arrow).
- Sometimes a CT is necessary to differentiate a healing fracture from a lung mass.
- Notice the large lung volume and the enlarged pulmonary vessels.
Probably we are dealing with pulmonary arterial hypertension in a patient with COPD.

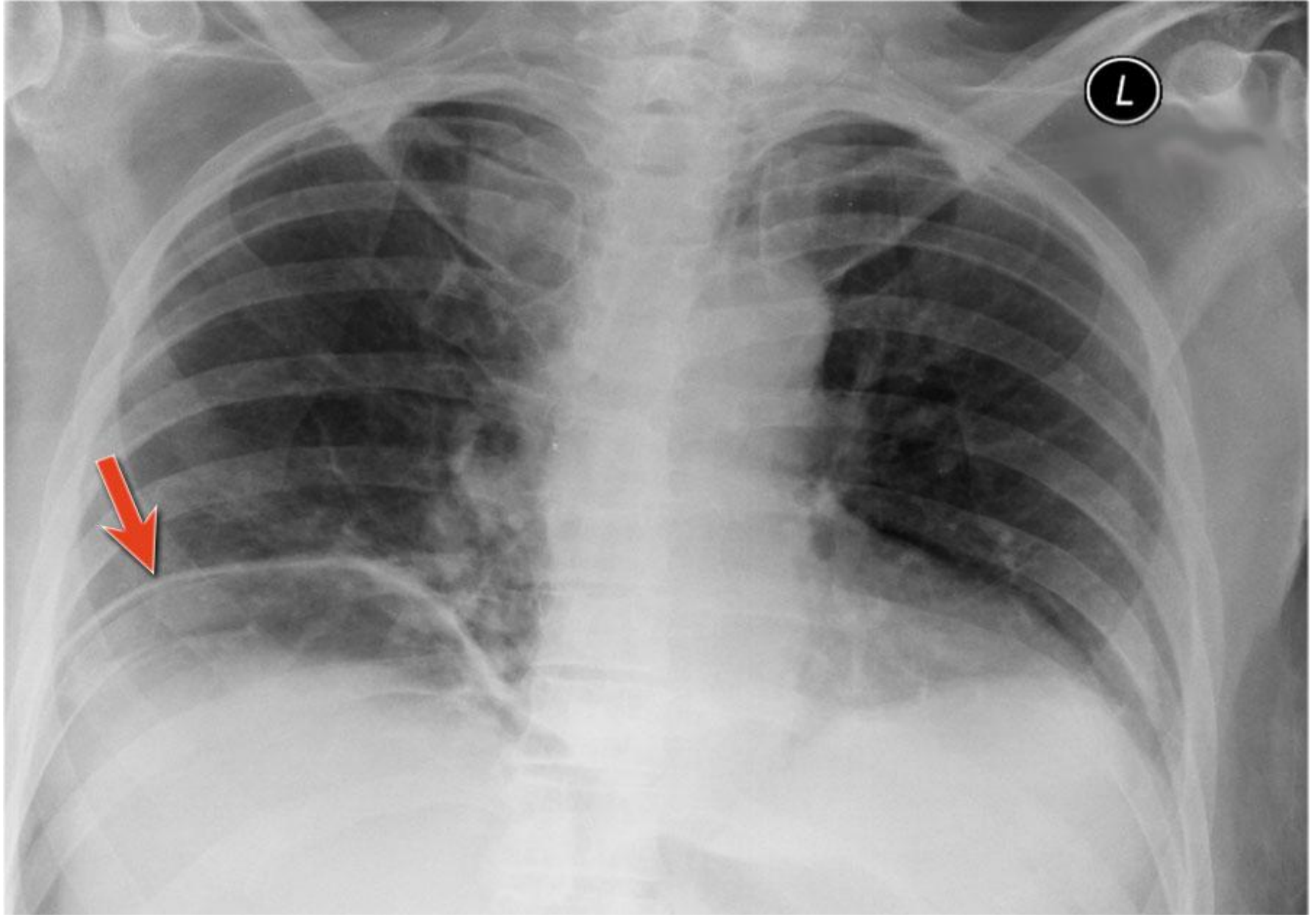


- The second most common chest wall abnormalities that we see on a CXR are metastases in vertebral bodies and ribs.
- Notice the expansile mass in the posterior rib on the right.

Abdomen

- The most obvious finding on this CXR is free air under the diaphragm.
- This finding indicates a bowel perforation, unless when the patient had recent abdominal surgery and there is still some air left in the abdomen, which can stay there for several days.
- There is another subtle finding in the left upper lobe.
A subtle density projecting over the first rib - hidden area - proved to be a lungcarcinoma.





- Here another patient with free abdominal air.
- Notice the very thin regular line which is the diaphragm (arrow).
- At first impression one might think that this is just some plate-like atelectasis due to poor inspiration.

Chest X-Ray

2

Heart Failure

Simone Cremers, Jennifer Bradshaw and Freek Herfkens

Radiology department of the Albert Schweitzer
Hospital in Dordrecht and the Medical Centre
Alkmaar, the Netherlands, the Netherlands

<http://www.radiologyassistant.nl/en/p4c132f36513d4/chest-x-ray-heart-failure.html>

- Congestive heart failure (CHF) is the result of insufficient output because of cardiac failure, high resistance in the circulation or fluid overload.
- **Left ventricle (LV)** failure :
 - is the most common
 - results in decreased cardiac output and increased pulmonary venous pressure.
In the lungs LV failure will lead to :
 - dilatation of pulmonary vessels,
 - leakage of fluid into the interstitium and the pleural space and finally into the alveoli → pulmonary edema.
- **Right ventricle (RV)** failure:
 - is usually the result of long standing LV failure or pulmonary disease
 - causes increased systemic venous pressure → edema in dependent tissues and abdominal viscera.
- In the illustration on the left some of the features, that can be seen on a chest-film in a patient with CHF.

- Increased pulmonary venous pressure is related to the pulmonary capillary wedge pressure (PCWP) and can be graded into stages,
- Each with its own radiographic features on the chest film (Table).
- This grading system provides a logical sequence of ***signs in congestive heart failure.***

* In daily clinical practice however :

- some of these features are not seen in this sequence and sometimes may not be present at all.
- This can be seen in patients with chronic heart failure, mitral valve disease and in chronic obstructive lung disease.

Stage of Congestive Heart Failure

Stage 1
Redistribution
PCWP 13-18 mmHg



Redistribution pulmonary vessels
Cardiomegaly
Broad vascular pedicle
(non acute CHF)

Stage 2
Interstitial edema
PCWP 18-25 mmHg

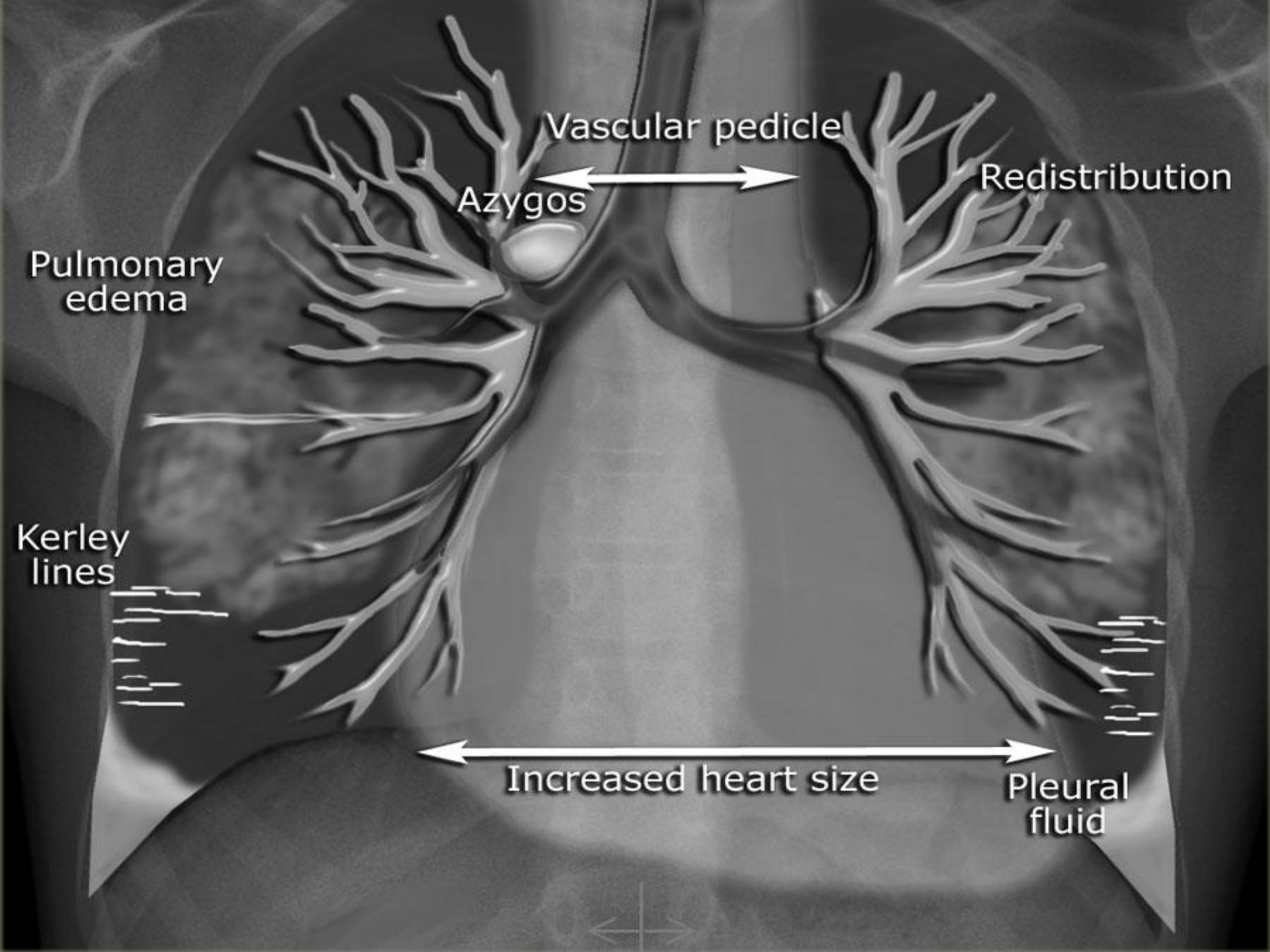


Kerley lines
Peribronchial cuffing
Hazy contour of vessels
Thickened interlobar fissure

Stage 3
Alveolar edema
PCWP > 25 mmHg



Consolidation
Air bronchogram
Cottonwool appearance
Pleural effusion



- In a normal chest film with the patient standing erect,
→ The **pulmonary vessels** supplying the **upper** lung fields are smaller and fewer in number than those supplying the lung bases.

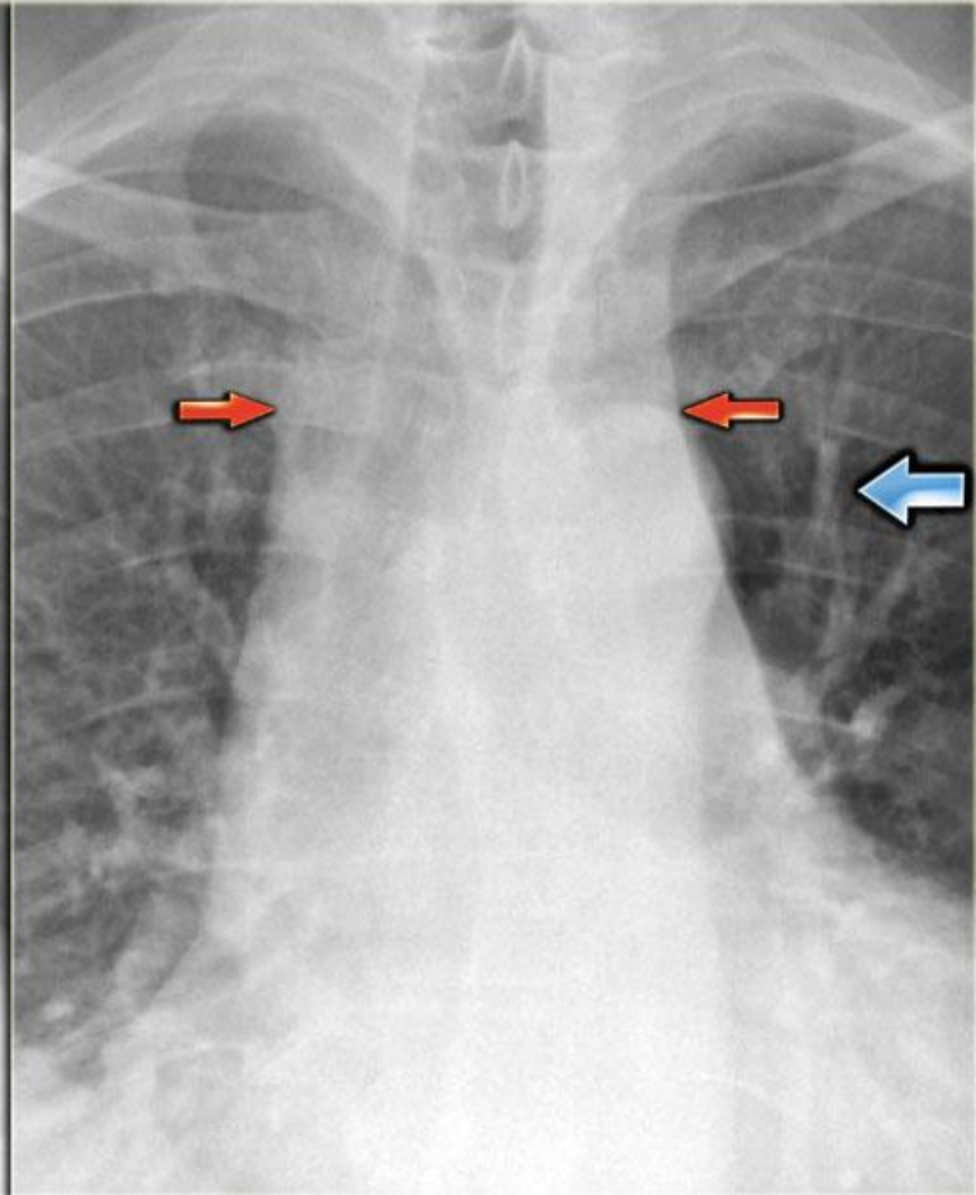
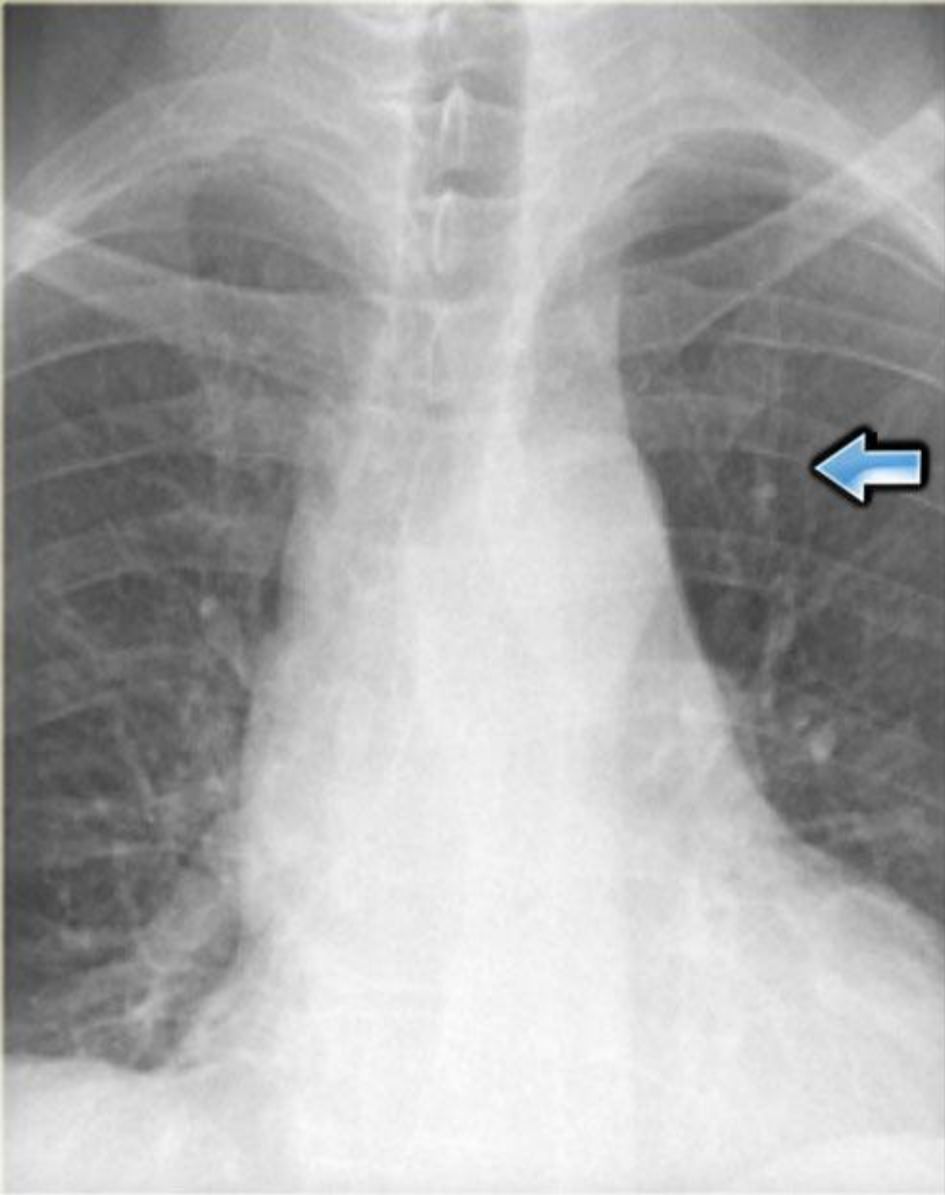
Stage I - Redistribution

→ *redistribution* of pulmonary blood flow.

- First : equalization of blood flow → redistribution of flow from the lower to the upper lobes.
- **The term redistribution** applies to chest x-rays taken in **full inspiration** in the **erect** position.
- **In daily clinical practice** many chest films are taken in a supine or semi-erect position → gravitational difference between the apex and the lung bases will be less.

In the supine position, there will be **equalisation** of blood flow, which may give the **false impression of redistribution**.

In these cases comparison with old films can be helpful.



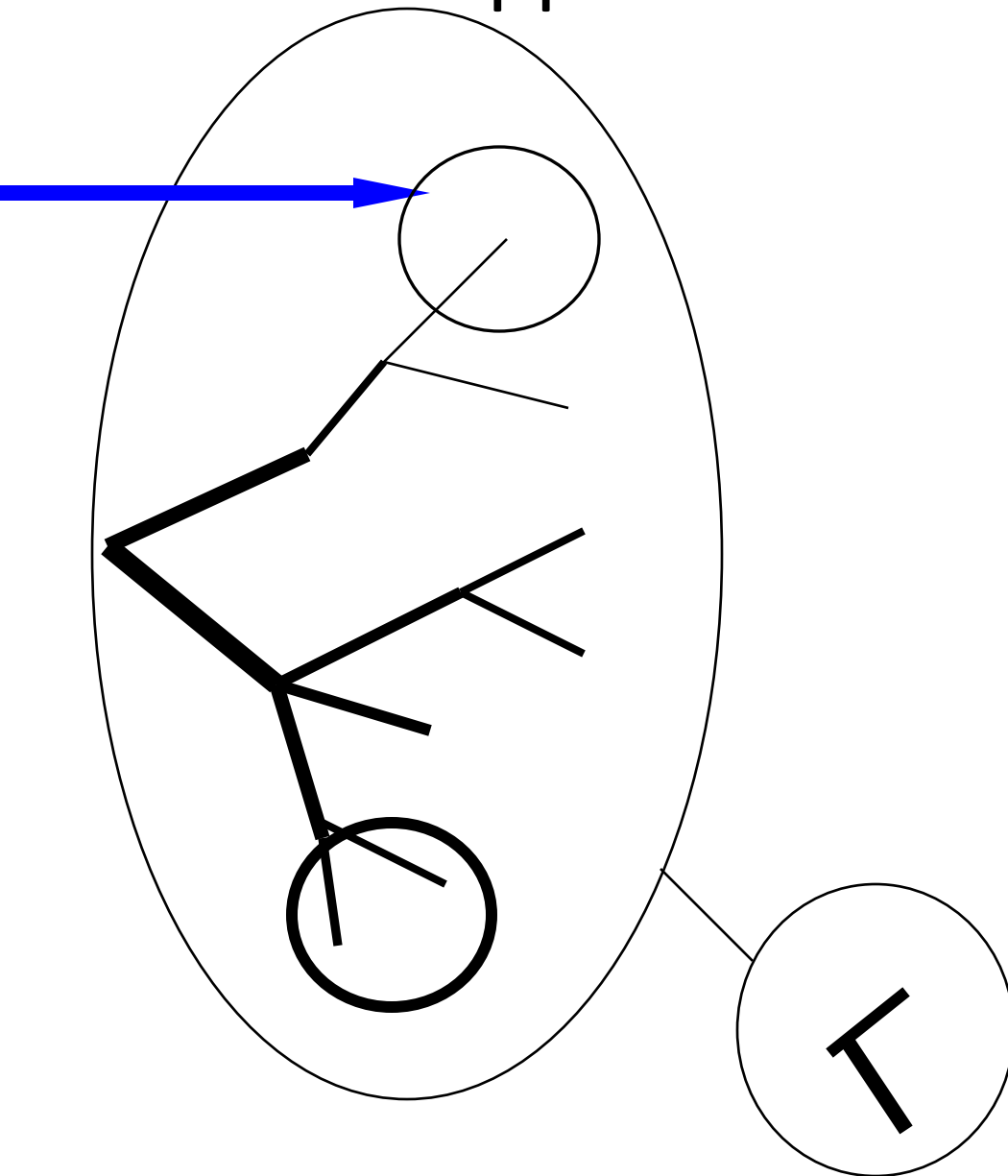
- Views of the upper lobe vessels of a patient in good condition (left) and during a period of CHF (right).
- Notice also the **increased width of the vascular pedicle** (red arrows).

•Artery-to-bronchus ratio

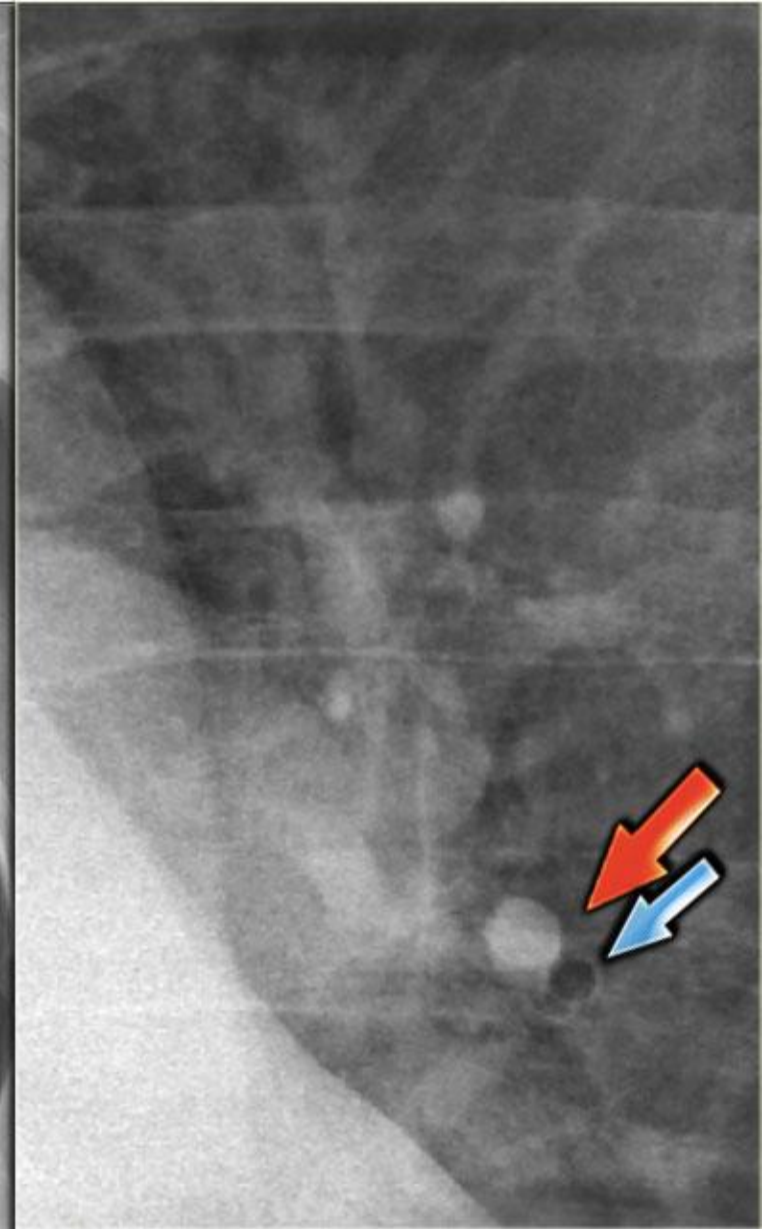
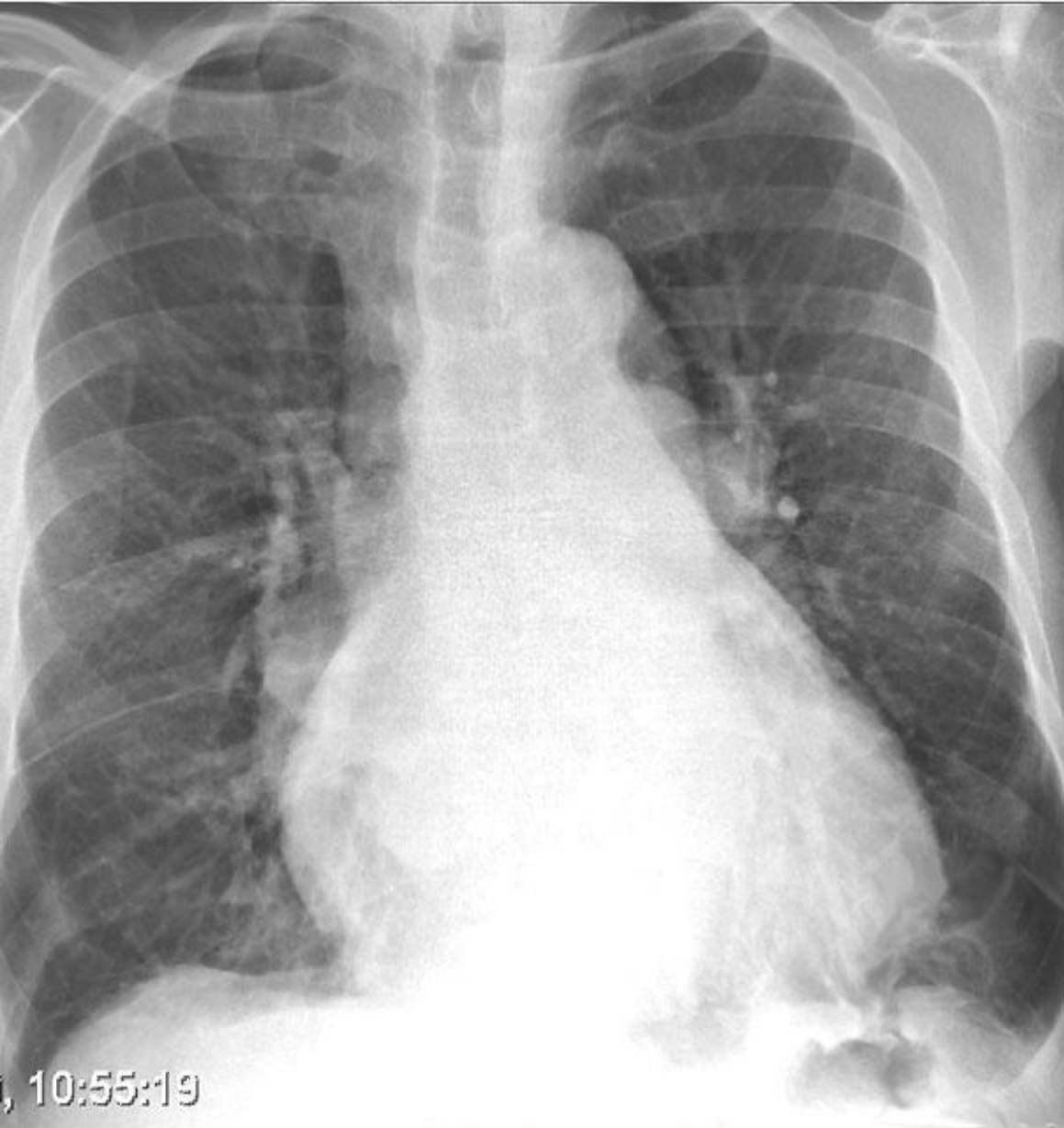
- Normally : vessels in the upper lobes are smaller than the accompanying bronchus with a ratio of 0.85.
- At the level of the hilum they are equal and in the **lower lobes** the arteries are larger with a ratio of **1.35**.
- When there is redistribution of pulmonary blood flow there will be an increased → artery-to-bronchus ratio in the upper and middle lobes.
- This is best visible in the **perihilar region**..... *look at previous film:*
 - On the left a patient with cardiomegaly and redistribution.
 - The upper lobe vessels have a diameter > 3 mm (normal 1-2 mm).
 - Notice the increased artery-to-bronchus ratio at hilar level (arrows).

Normal Distribution of Flow

Upper Versus Lower Lobes



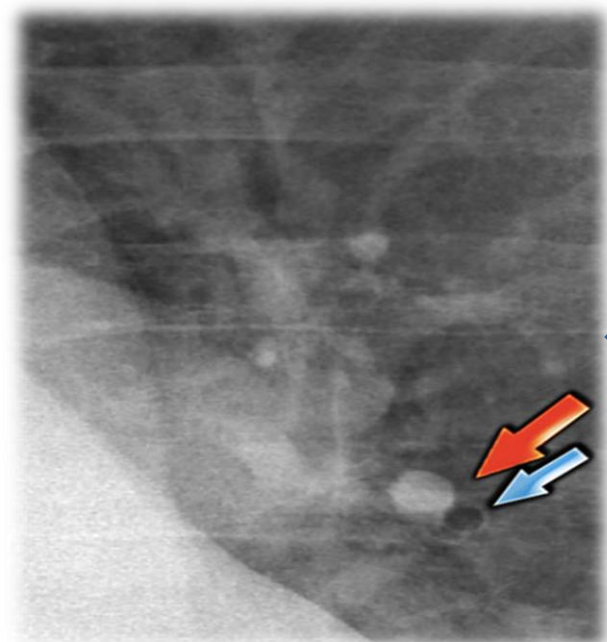
*Size of vessels
at bases is
normally >
size of vessels
at apex*



- Increased artery-to-bronchus ratio in CHF

CXR → Artery-to-Bronchus ratio

	Artery / Bronchus	Normal ratio
UPPER Lobe	Artery < Bronchus	0.85
Hilum	Artery = Bronchus	1
Lower Lobe	Artery > Bronchus	1.35



In Early Heart Failure ,
Redistribution occurs

→ Vessels Diameter in Upper Lobe &
Hilum increased → increase ratio

Summary of Pulmonary Vasculature

- *Normal:*

- RtDPA < 17 mm - Central > Peripheral
- Lower > Upper

- **Pulmonary venous hypertension:**

- Cephalization “Upper > Lower”

- **Pulmonary arterial hypertension:**

- RDPA > 17 mm - Main P.A. out “-15:0”

- **Increased flow:**

- All vessels > Normal +/- RDPA > 17 mm

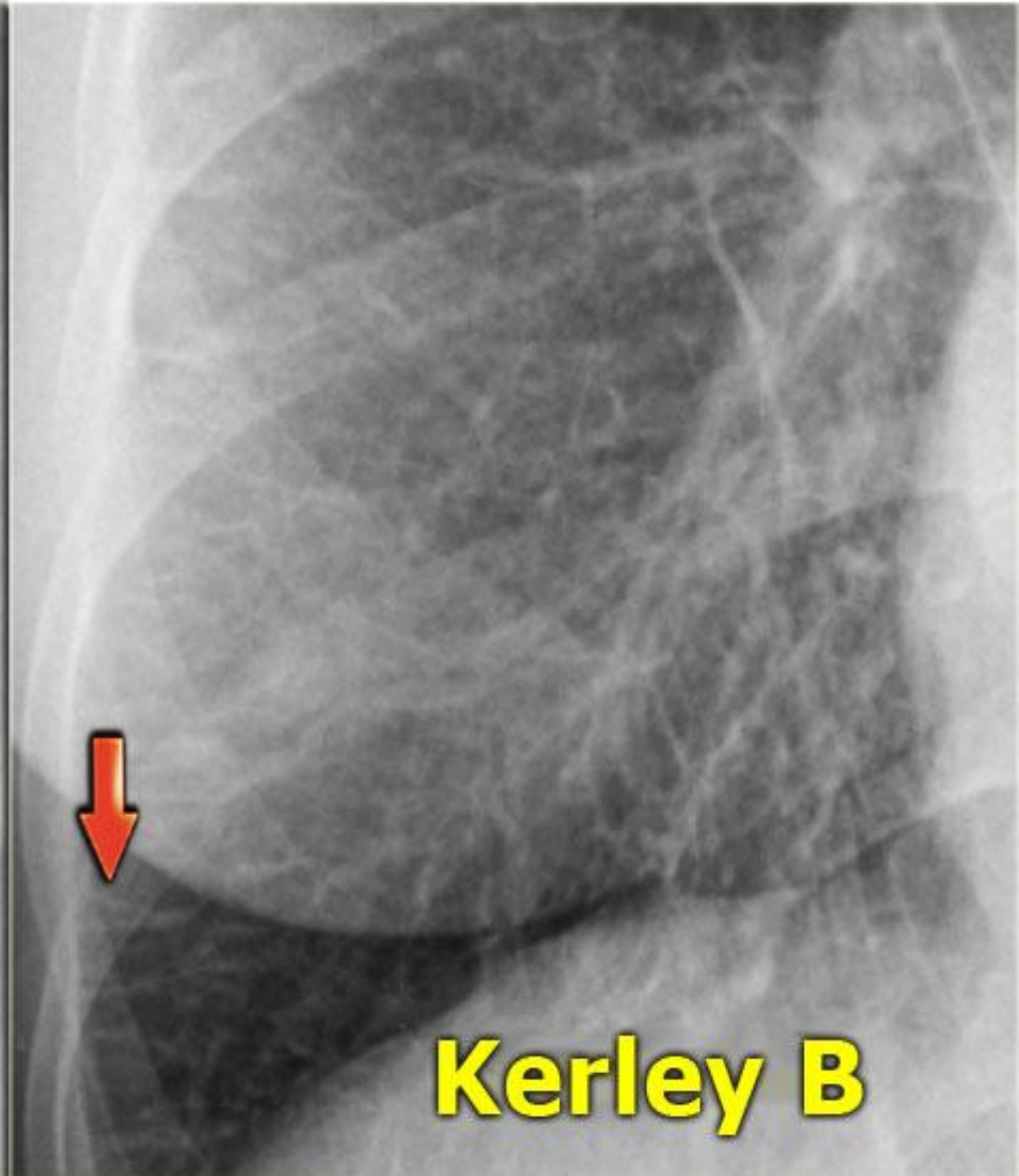
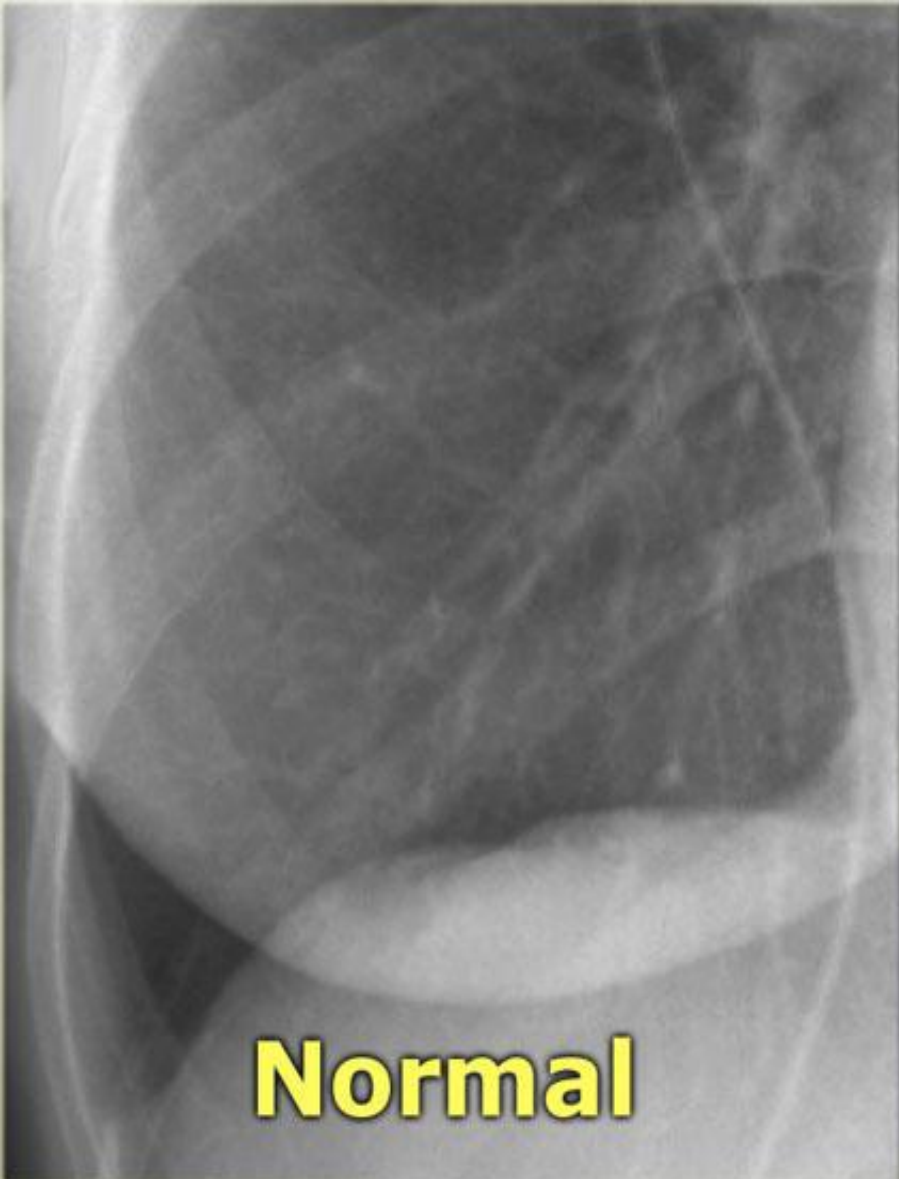
- **Decreased flow:**

- Small Hila - Vessels fewer than normal

Stage II - Interstitial edema

Stage II of CHF is characterized by:

- fluid leakage into the **interlobular** and **peribronchial interstitium**
← increased pressure in the capillaries.
- * When fluid leaks into the peripheral interlobular septa it is seen as **Kerley B** or **septal lines**.
- **Kerley-B lines** are seen as : * horizontal * peripheral * short 1-2 cm lines * near the costophrenic angles. * perpendicular to the pleura

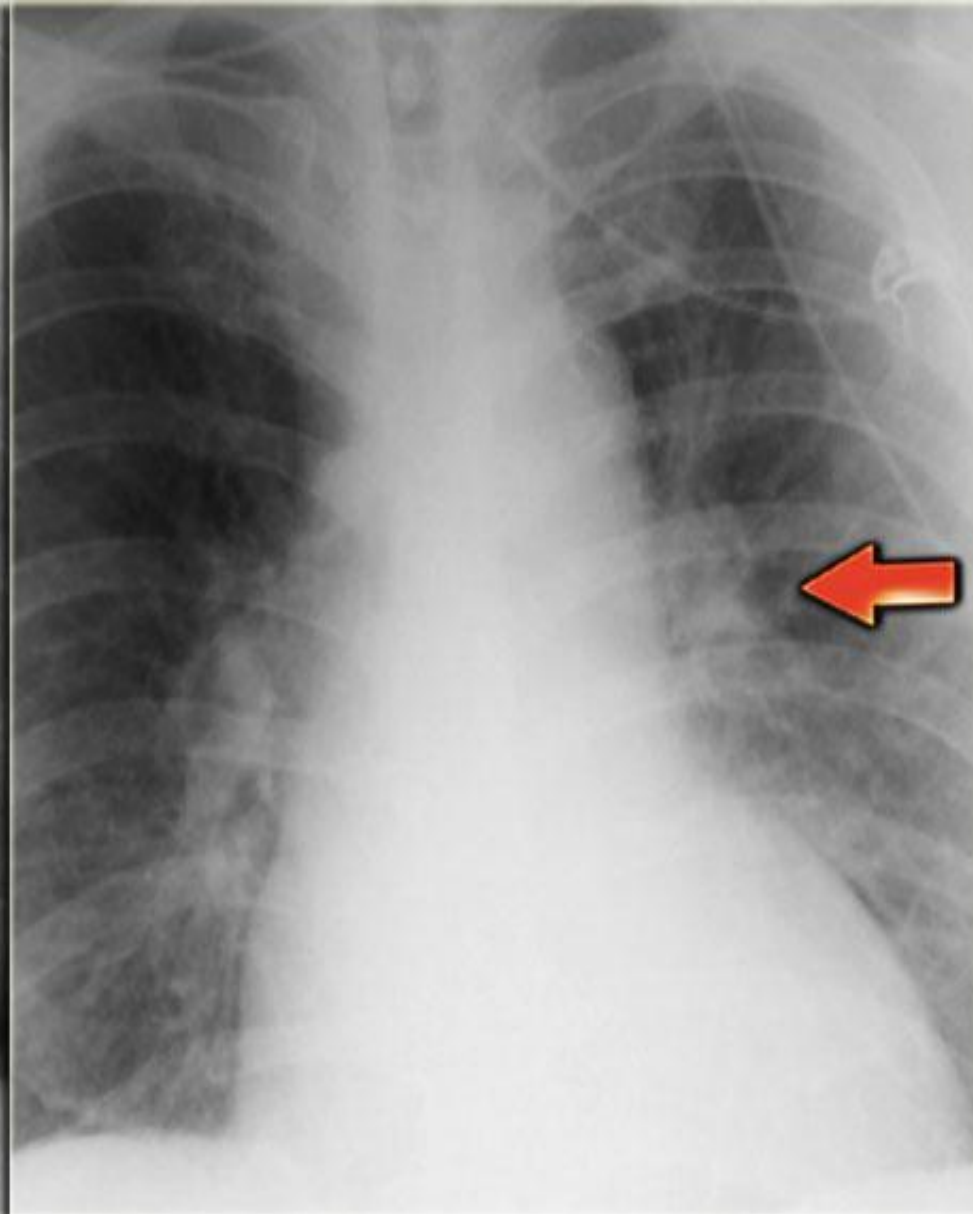
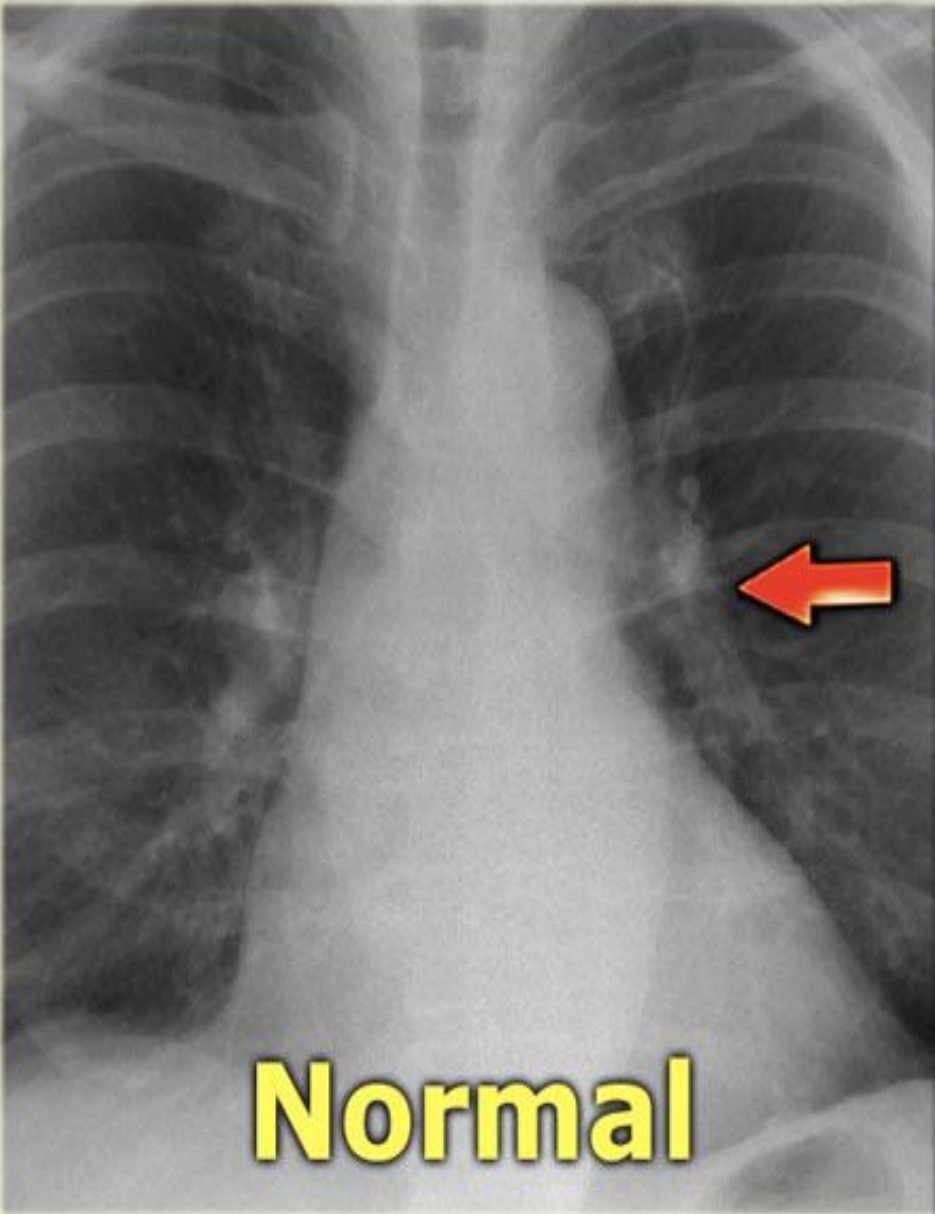


- *LEFT: normal.*
- *RIGHT: CHF stage II, congestion → fluid leak in interlobular septa → Kerley B-lines due to interstitial edema*

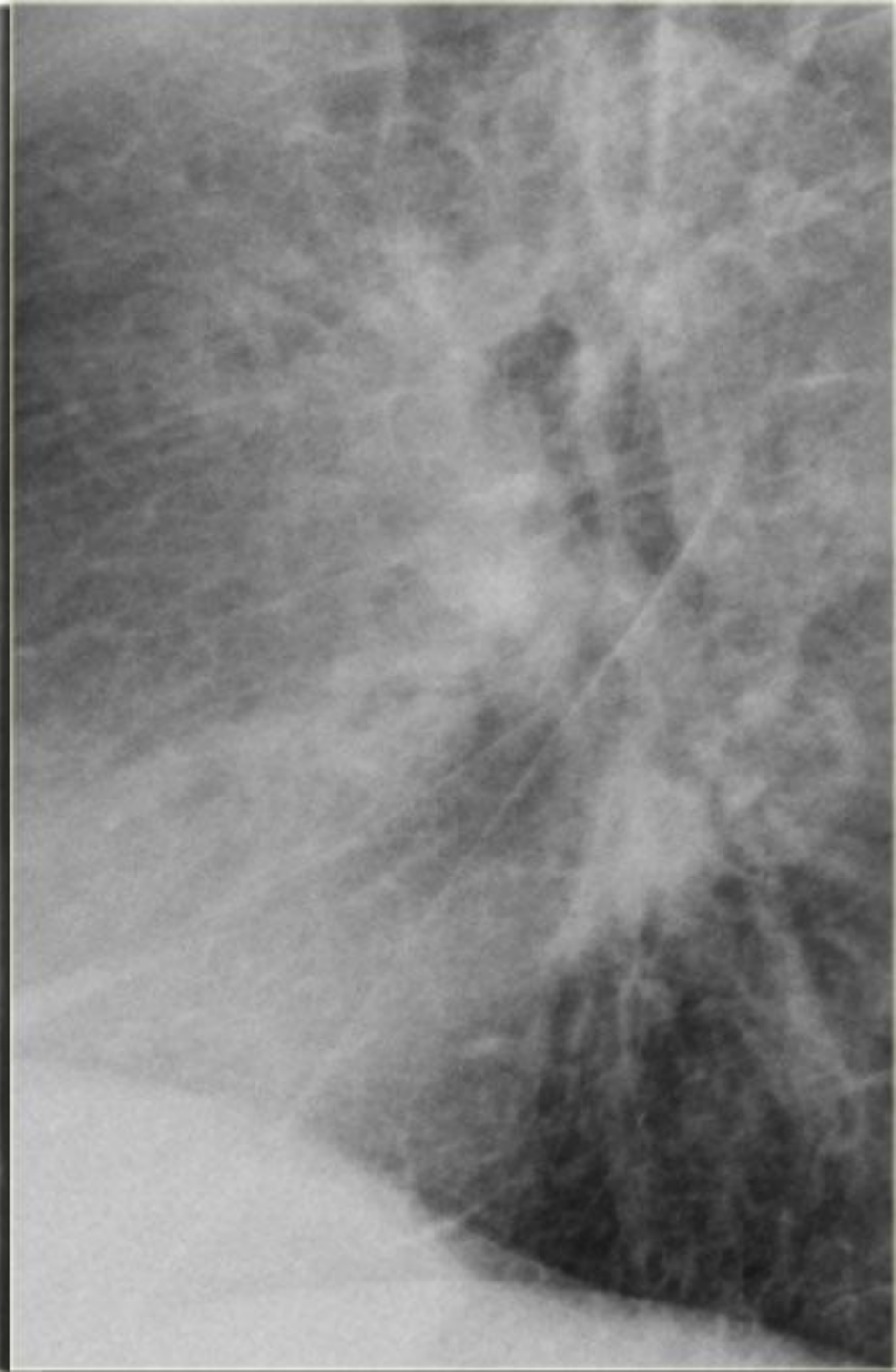
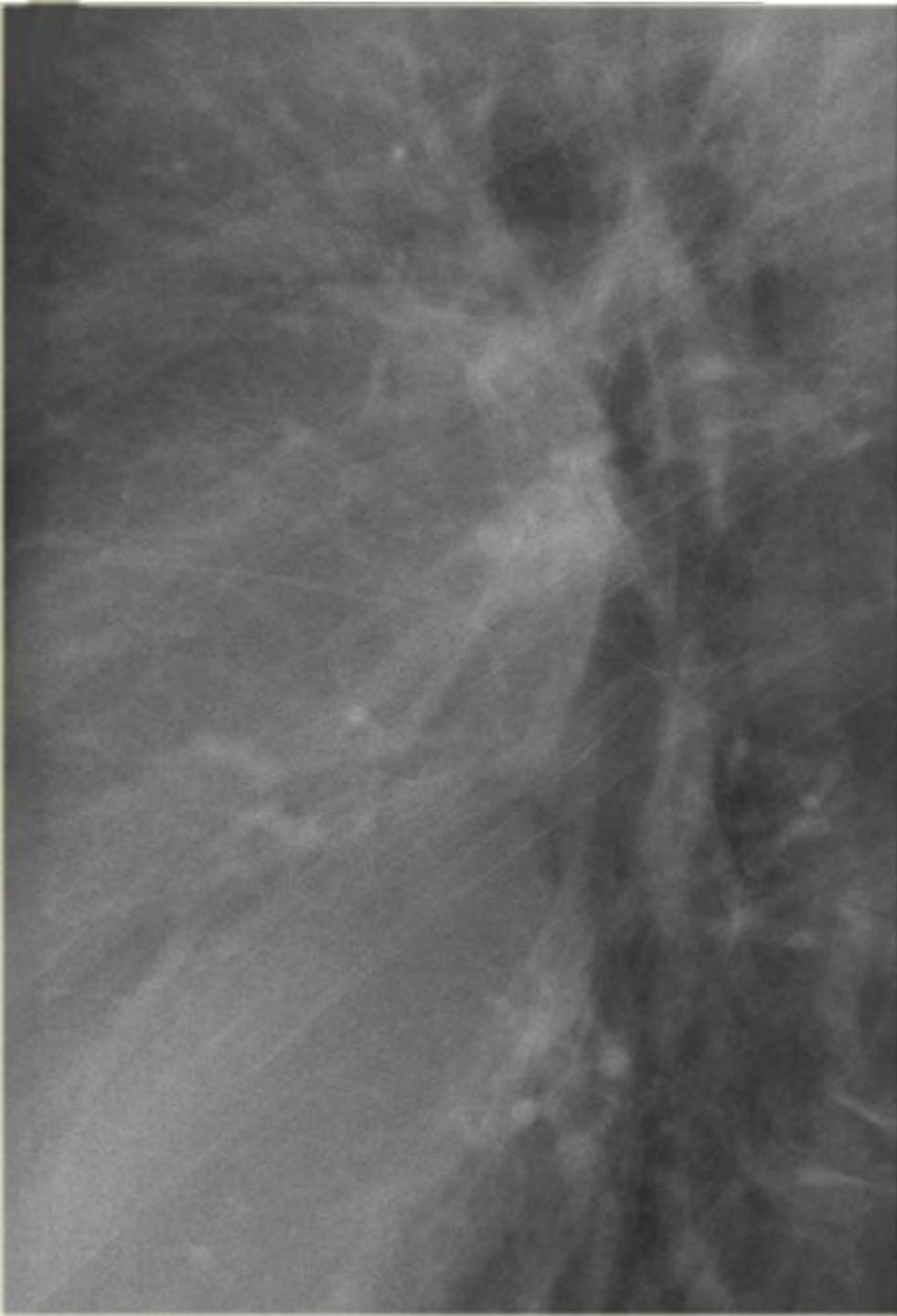
When fluid leaks into the **peri-bronchovascular interstitium** it is seen as **thickening of the bronchial walls** (peribronchial cuffing) and as **loss of definition of these vessels** (perihilar haze).

See next filmOn the left a patient with congestive heart failure.

- increase in the caliber of the pulmonary vessels
- lost their definition because they are surrounded by edema.



- Perihilar haze in interstitial stage of CHF



- On the left another patient with congestive heart failure.

The lateral view nicely demonstrates the increased diameter of the pulmonary vessels and the hazy contours.

Notice also the **septal lines** and the accentuated interstitium.

Furthermore the **fissura major** is **markedly thickened**.

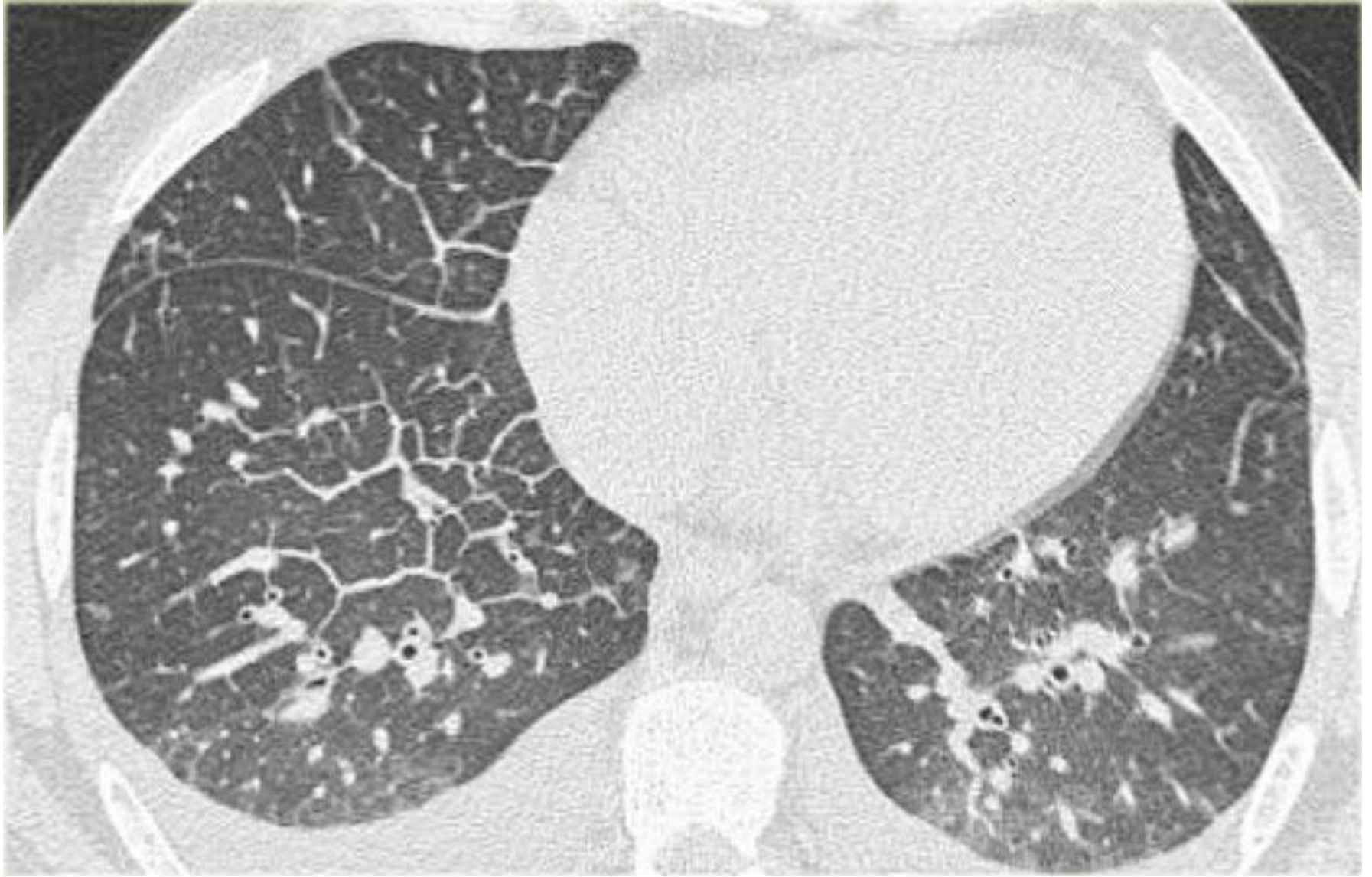
CT will also demonstrate signs of congestive heart failure.

On the image on the left notice the following:

- Thickened septal lines due to interstitial edema
- Subtle ground glass opacity in the dependent part of the lungs (HU difference of 100-150 between the dependent and non-dependent part of the lung).
- Bilateral pleural fluid.

In a patient with a known malignancy lymphangitic carcinomatosis would be high in the differential diagnostic list.

Ground glass opacity is the first presentation of alveolar edema and a precursor of consolidation.



- Thickened septal lines due to interstitial edema in CHF

DIFFUSE LUNG DISEASE

Reticular Pattern

Radiological Finding 7

2 Bronchial

Bronchiectasis

Wall Thickening
"Cuffing"

2 Bee & Spider

Honey Comp

Opacities around
vessel

Cystic Thick wall
Multiple Peripheral
Mm : cm

3 LINES

Septal

Sub-Pleural

Thickened Fissure

Short Lines perpendicular

Curvilinear Parallel to Pleura

Stage III - Alveolar edema

- This stage is characterized by ***continued fluid leakage*** into the interstitium, which cannot be compensated by lymphatic drainage. This eventually leads to **fluid leakage in the alveoli** (alveolar edema) and to leakage into the **pleural space** (pleural effusion).

The distribution of the alveolar edema can be influenced by:

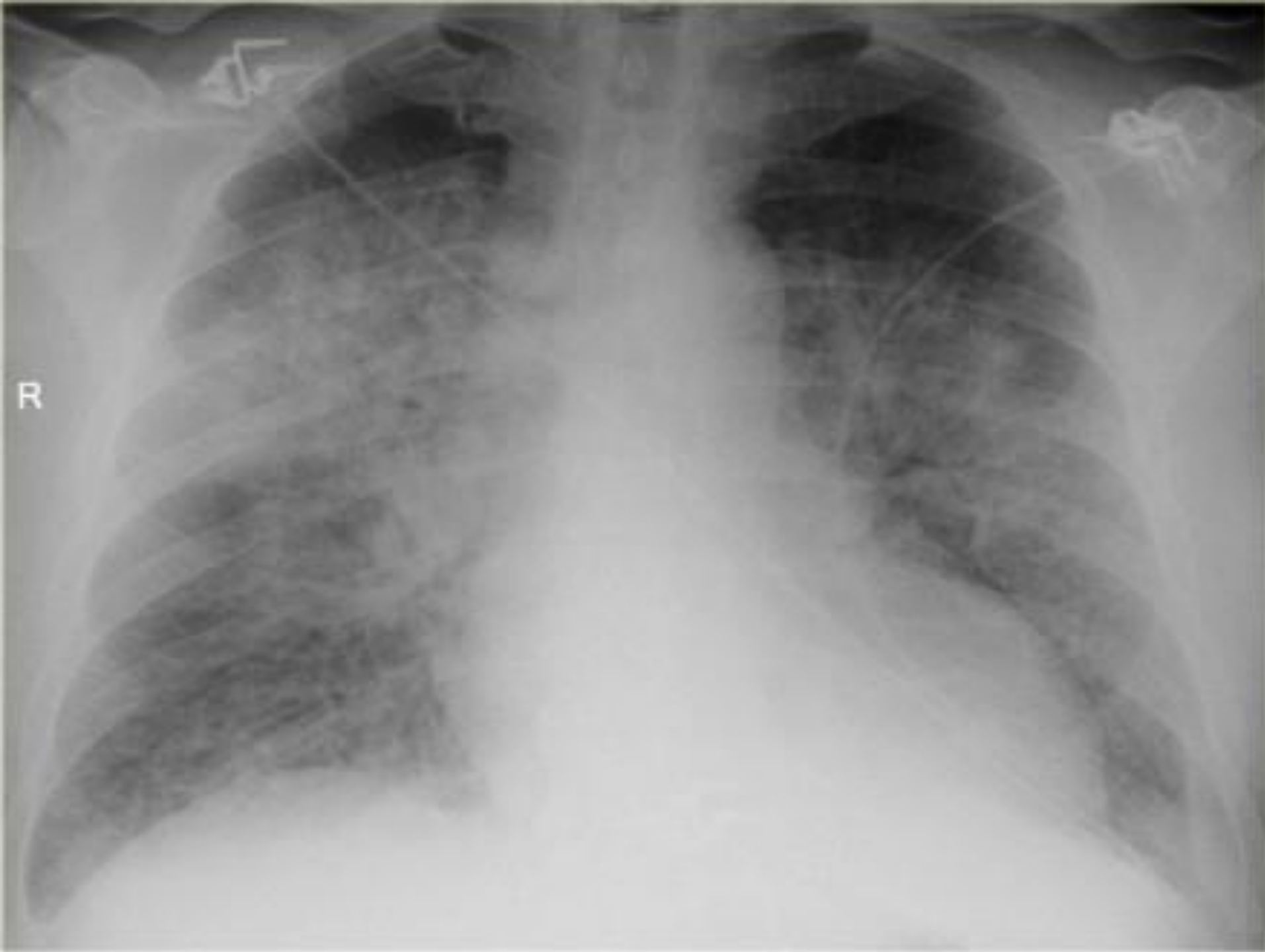
- **Gravity:** supine or erect position and right or left decubitus position
- **Obstructive lung disease**, i.e. fluid leakage into the less severe diseased areas of the lung

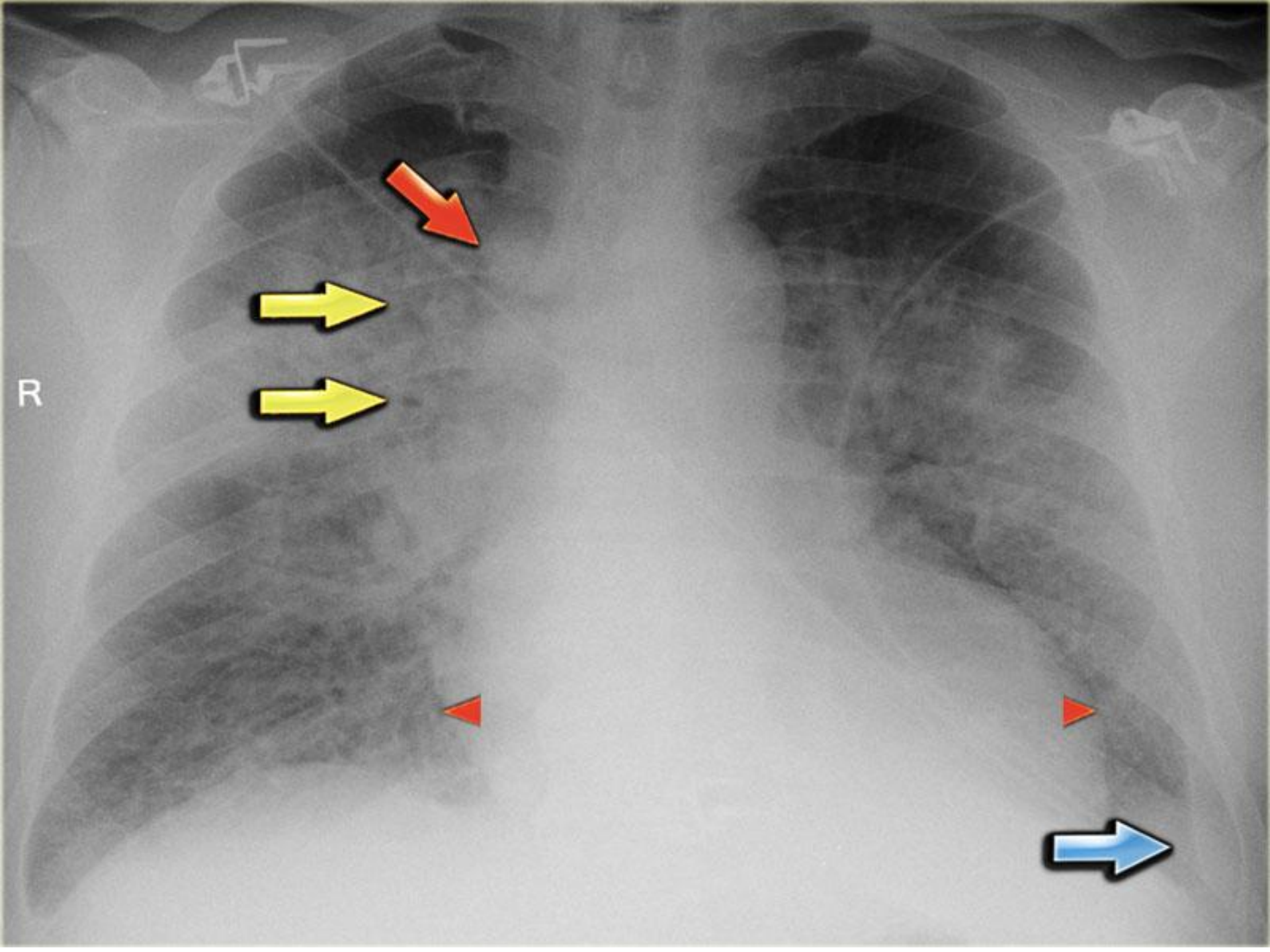
- Next case : a case of severe **dyspnea** due to **acute heart failure**.

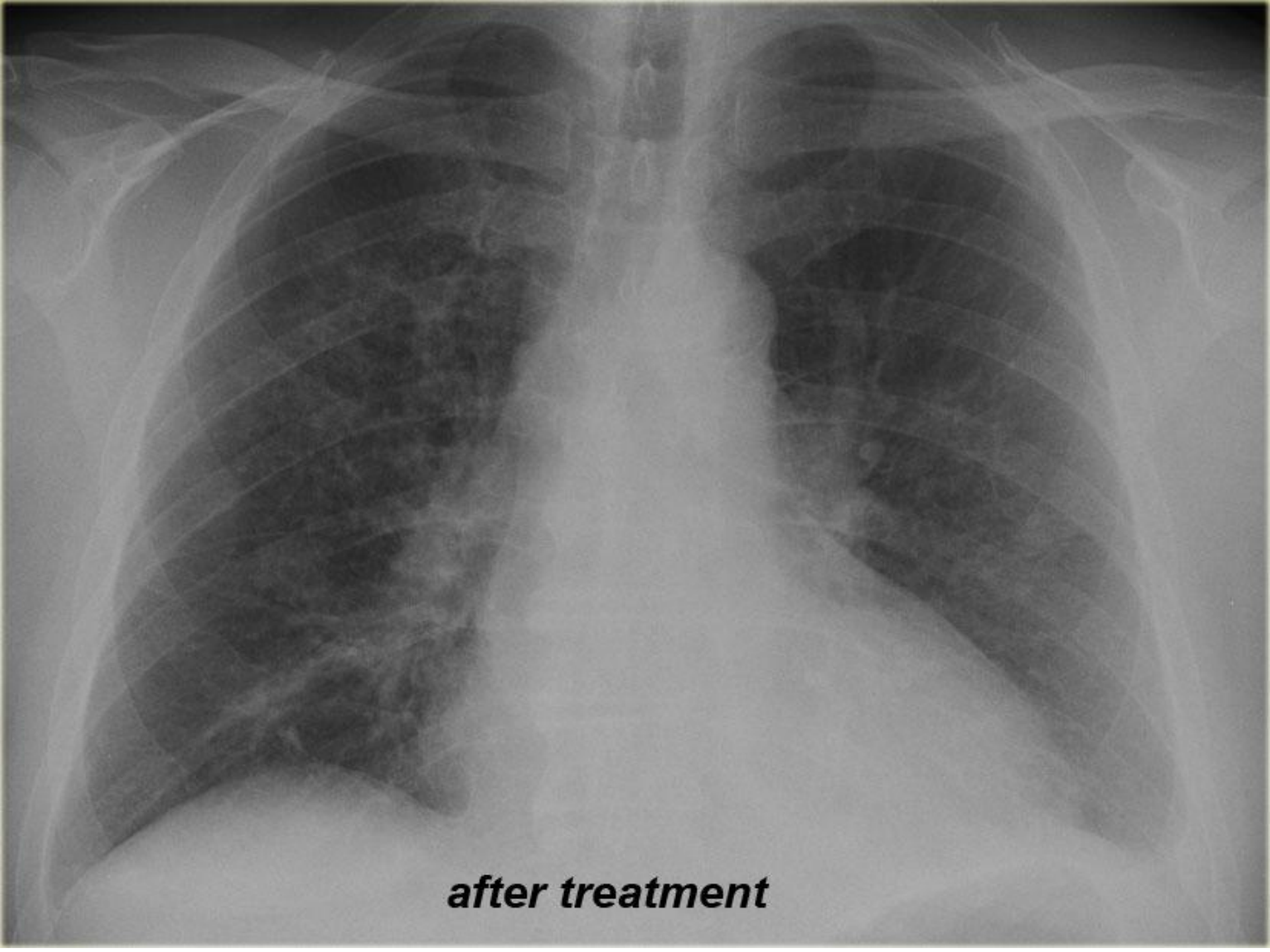
The following signs indicate heart failure:

- alveolar edema with perihilar consolidations and air bronchograms (yellow arrows);
- pleural fluid (blue arrow);
- prominent azygos vein and increased width of the vascular pedicle (red arrow) and
- an enlarged cardiac silhouette (arrow heads).

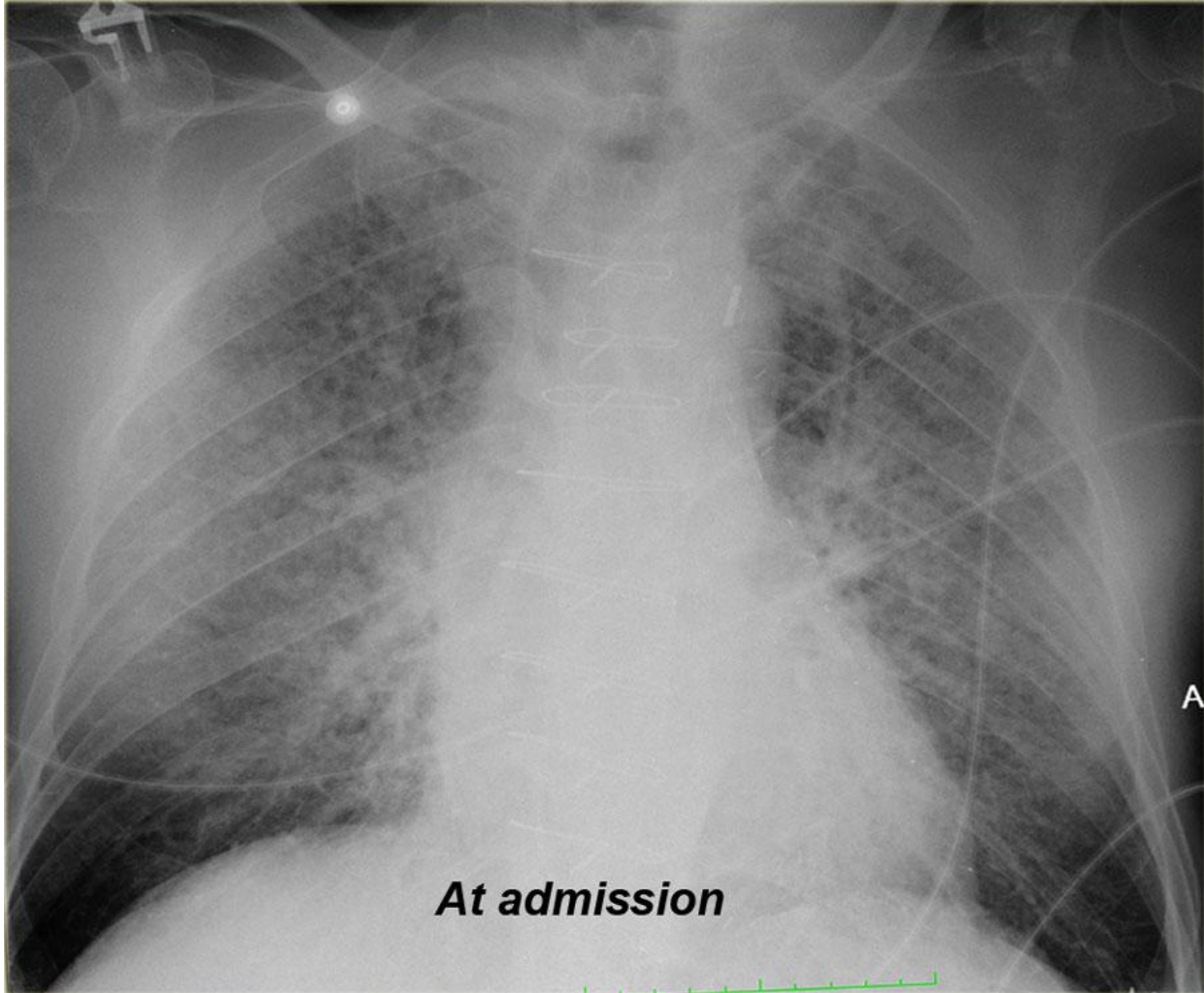
After treatment we can still see an enlarged cardiac silhouette, pleural fluid and redistribution of the pulmonary blood flow, but the edema has resolved.



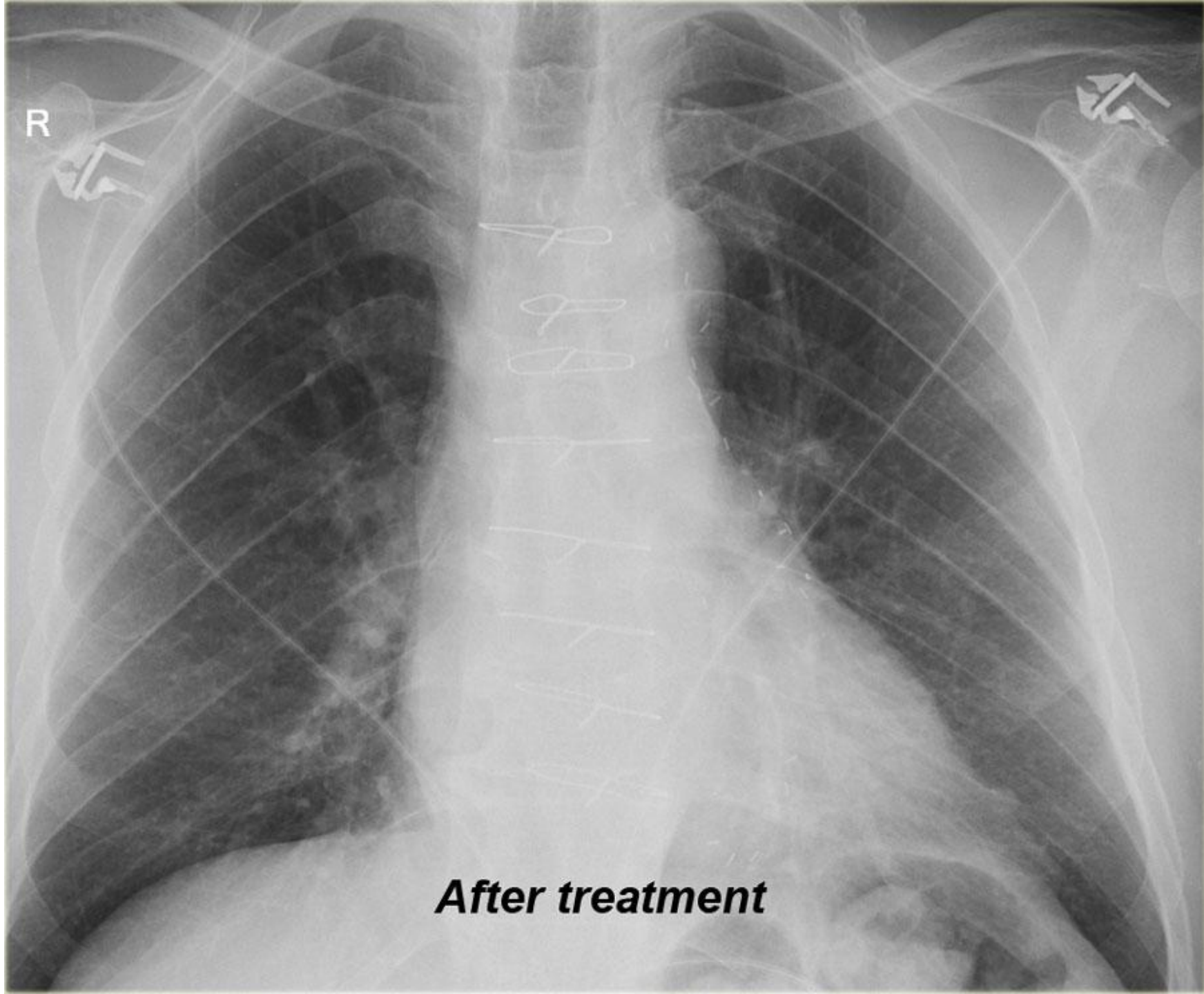




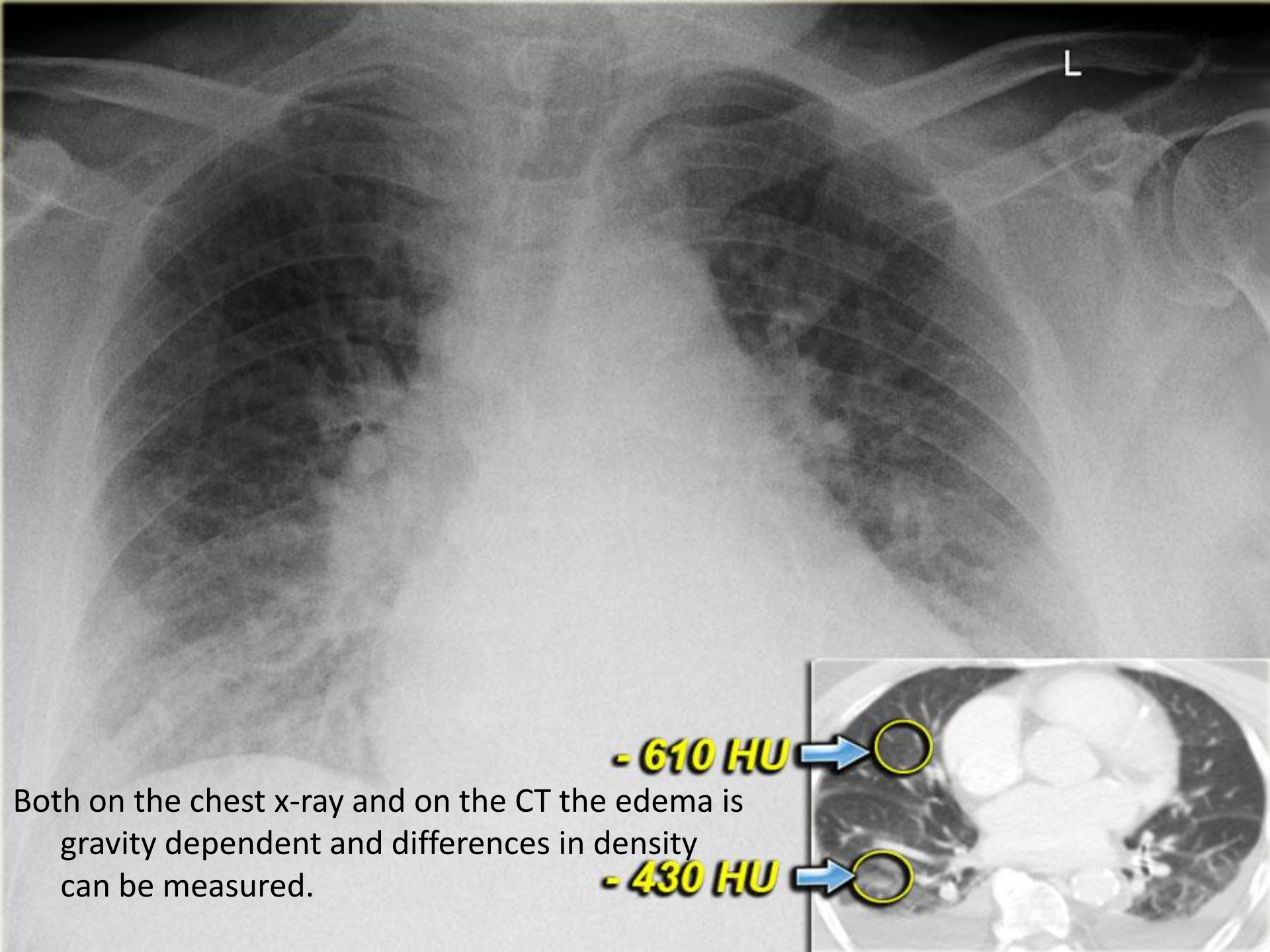
after treatment



- patient with alveolar edema at admission, which resolved after treatment. , you will notice the difference in vascular pedicle width and distribution of pulmonary flow.



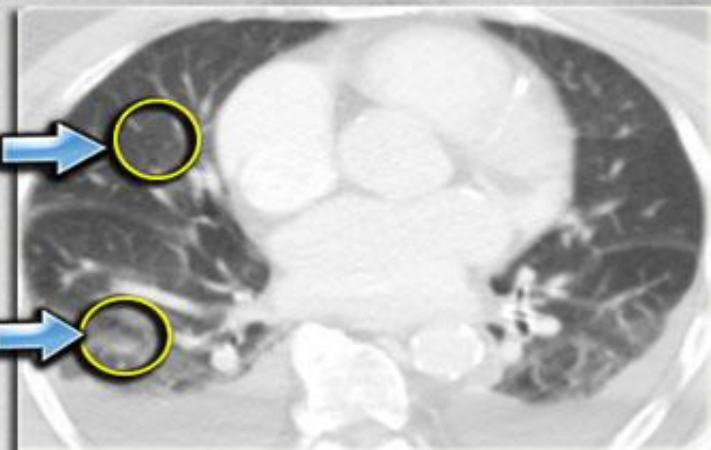
After treatment



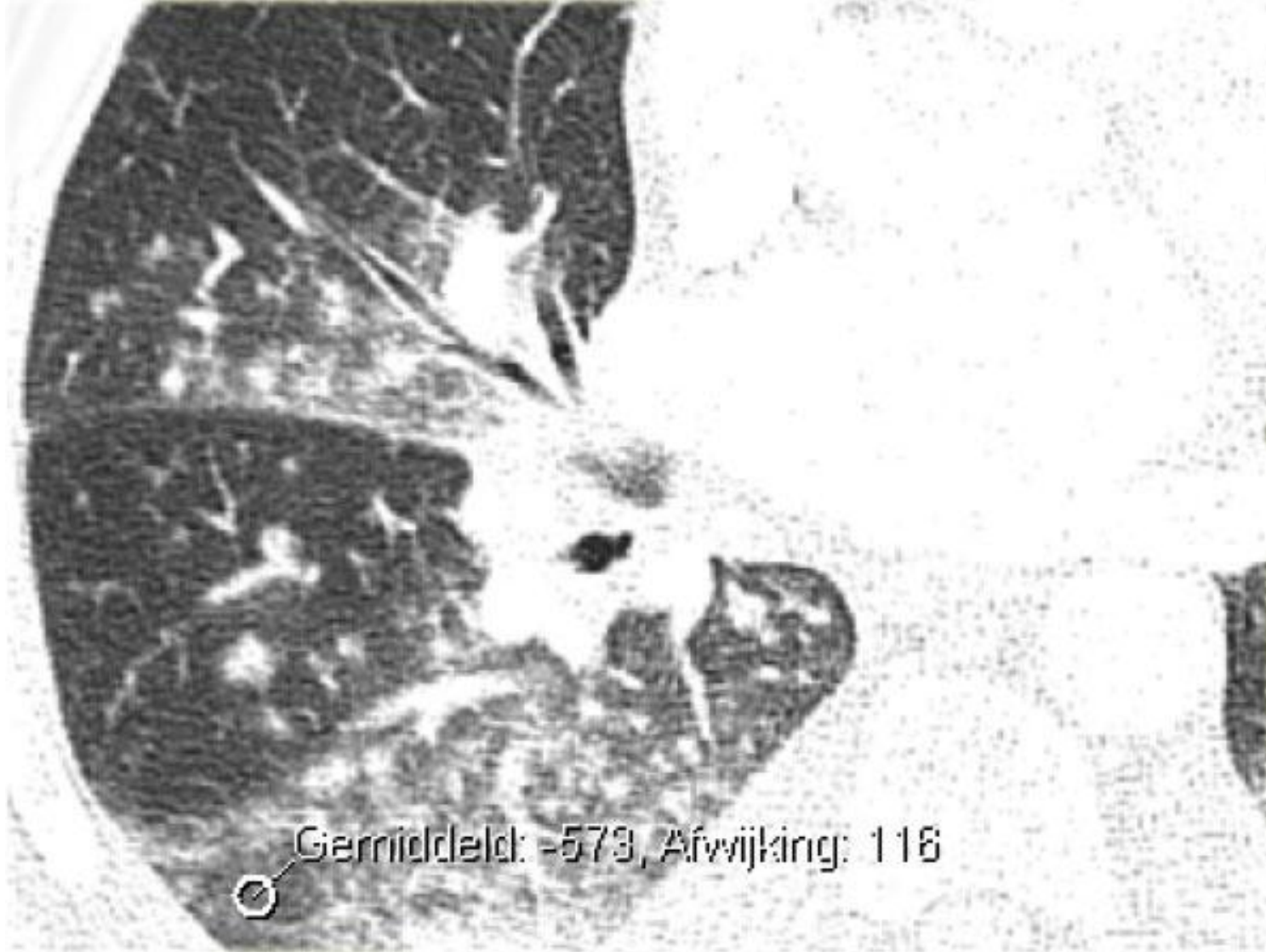
- 610 HU



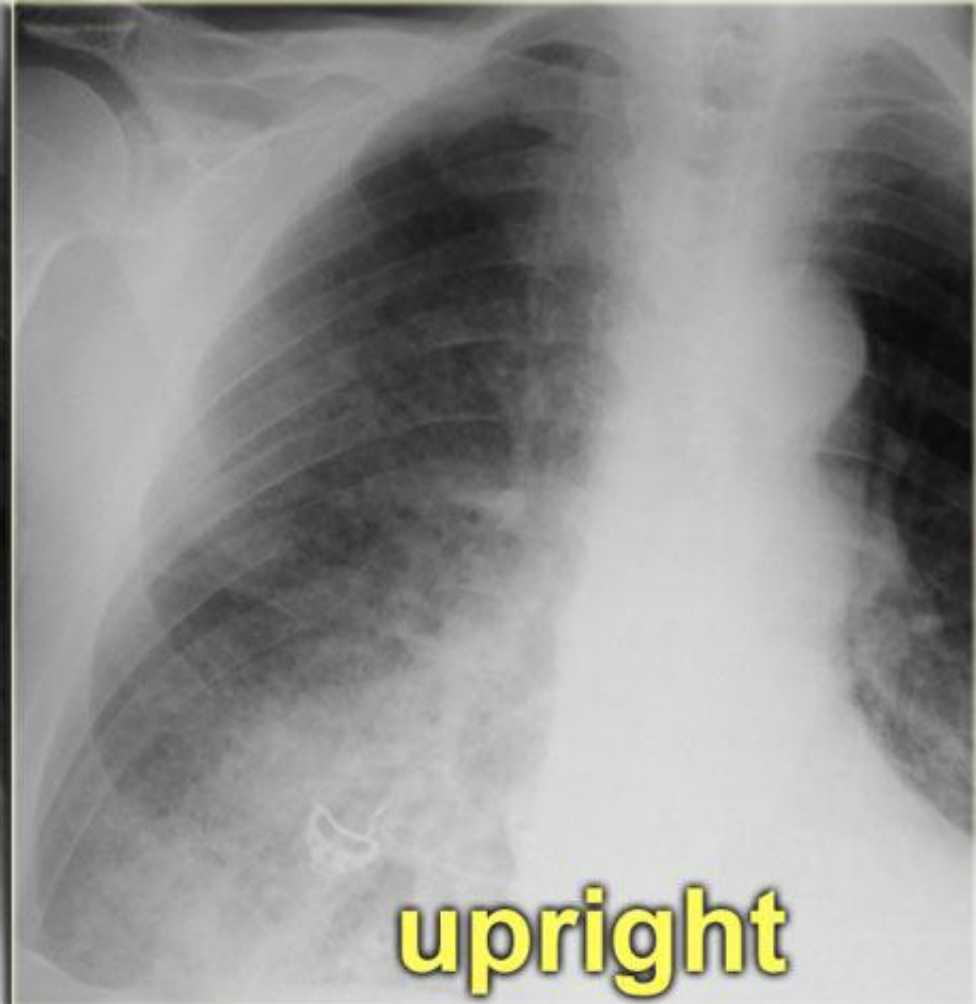
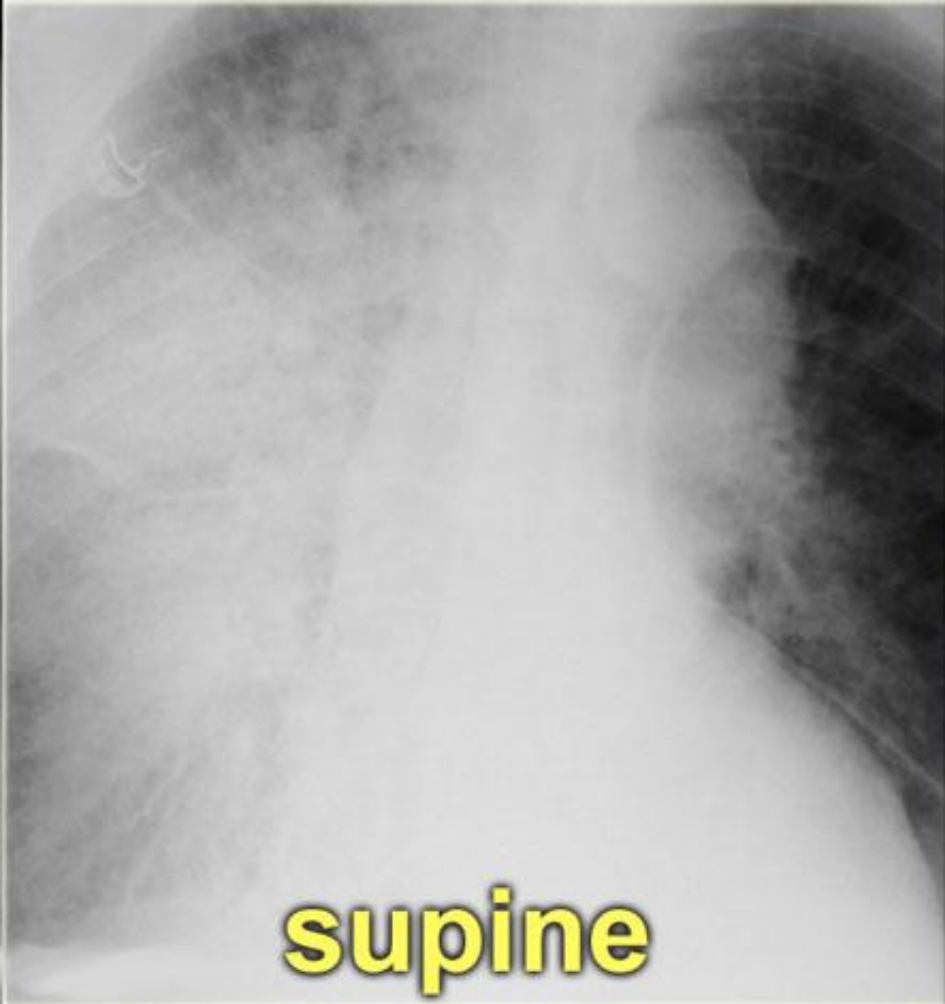
- 430 HU



Both on the chest x-ray and on the CT the edema is gravity dependent and differences in density can be measured.



- Notice that even within each lobe there is a gravity dependent difference in density. This is only seen when the **consolidations are the result of transudate** like in CHF. This is not seen when the **consolidations are the result of exudate** due to infection, blood due to hemorrhage or when there is a capillary leak like in ARDS.



- a patient who first had a chest film in a **supine position**. Notice the **pulmonary edema**, which is almost exclusively seen in the right lung.
A possible explanation for this phenomenon could be, that the patient had been lying on his right side for a while before the x-ray was taken.

Cardiothoracic ratio

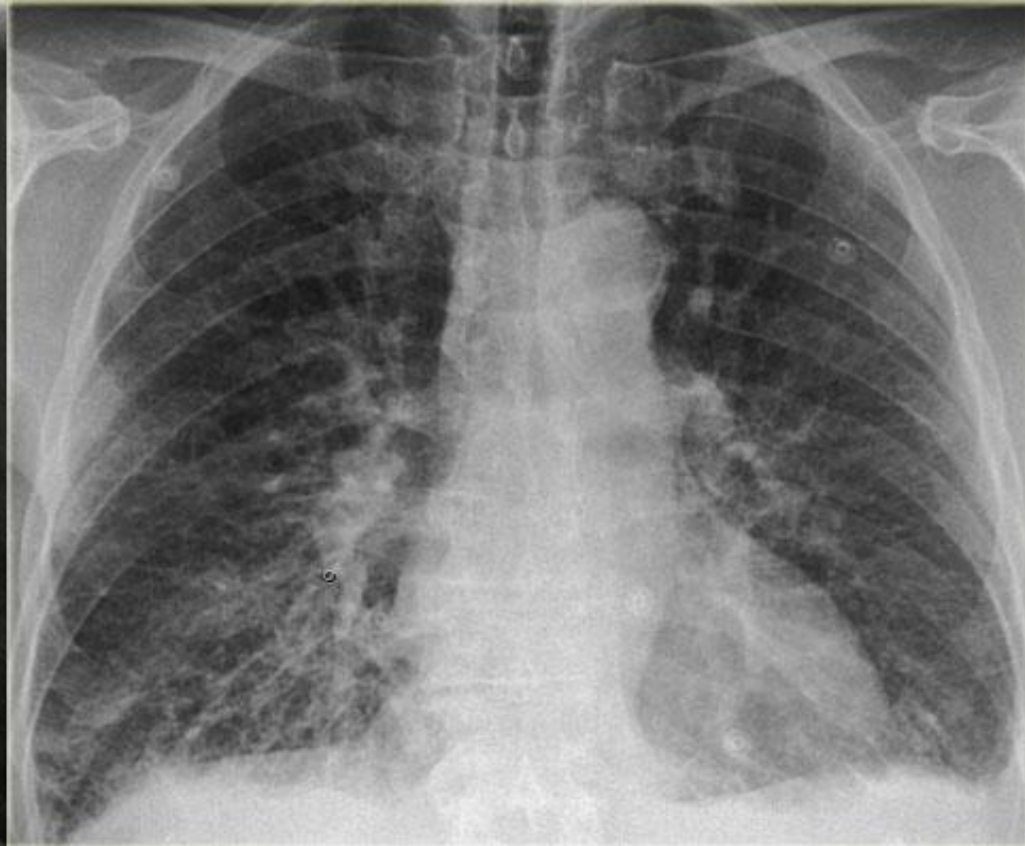
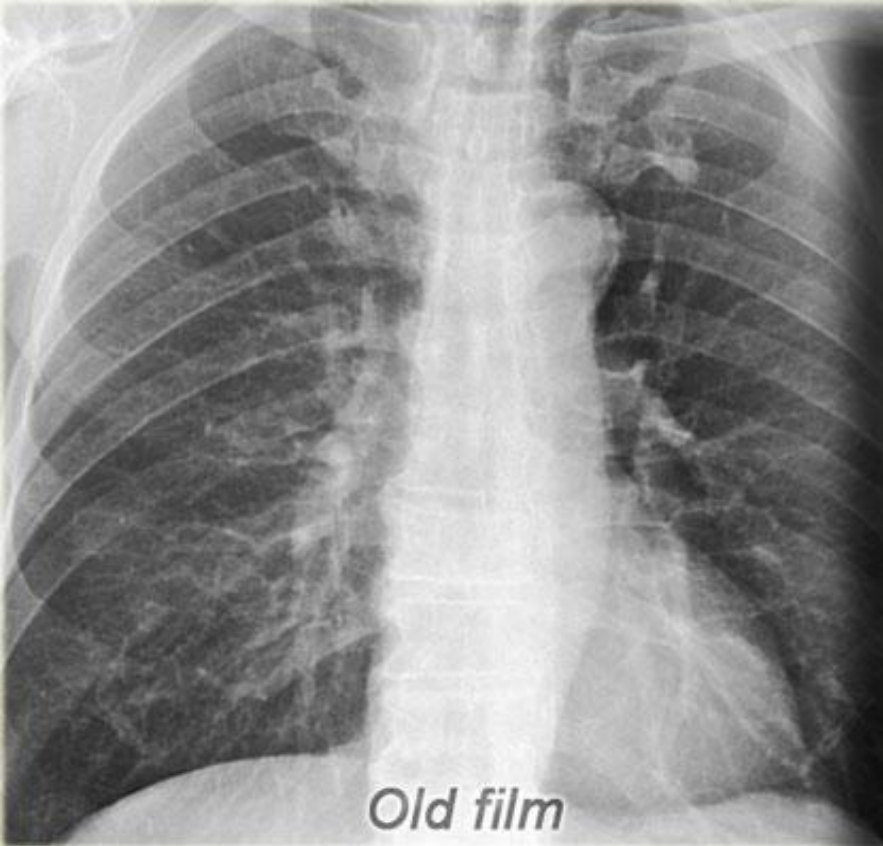
The cardiothoracic ratio (CTR) : is the ratio of the transverse diameter of the heart to the internal diameter of the chest at its widest point just above the dome of the diaphragm as measured on a PA chest film.

- An increased cardiac silhouette is almost always the result of **cardiomegaly**, but occasionally it is due to **pericardial effusion** or even **fat deposition**.
- The heart size is large when the **CTR is > 50%** on a PA chest x-ray.
- A CTR of > 50% has a sensitivity of 50% for CHF and a specificity of 75-80%.
- An **increase in left ventricular volume** of at **least 66%** is necessary before it is noticeable on a chest x-ray.

Other signs of CHF are visible, such as redistribution of pulmonary flow, interstitial edema and some pleural fluid.

On a supine film the cardiac silhouette will be larger due to magnification and high position of the hemidiafragms.

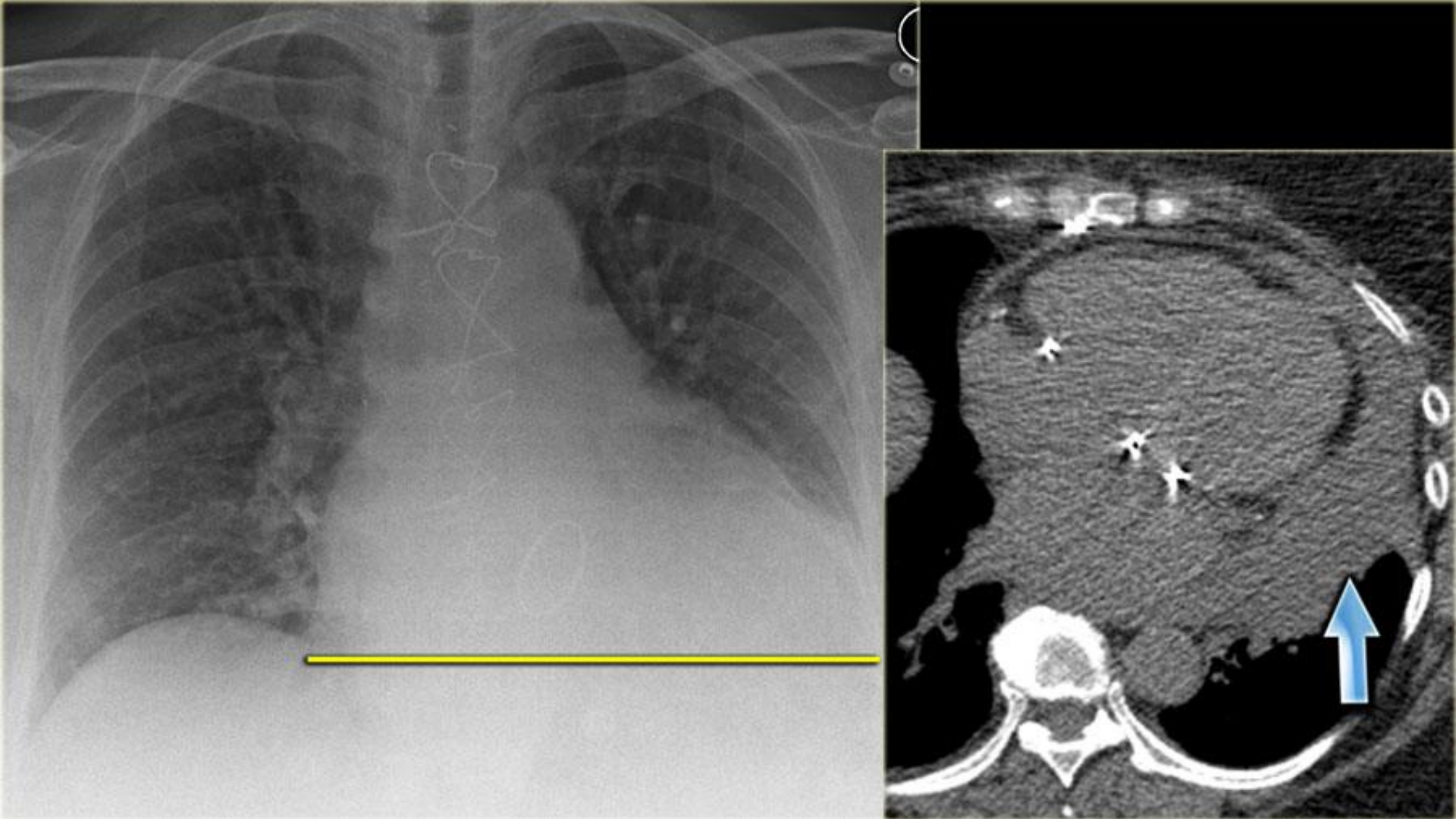
Exact measurements are not that helpful, but comparison to old supine films can be of value.



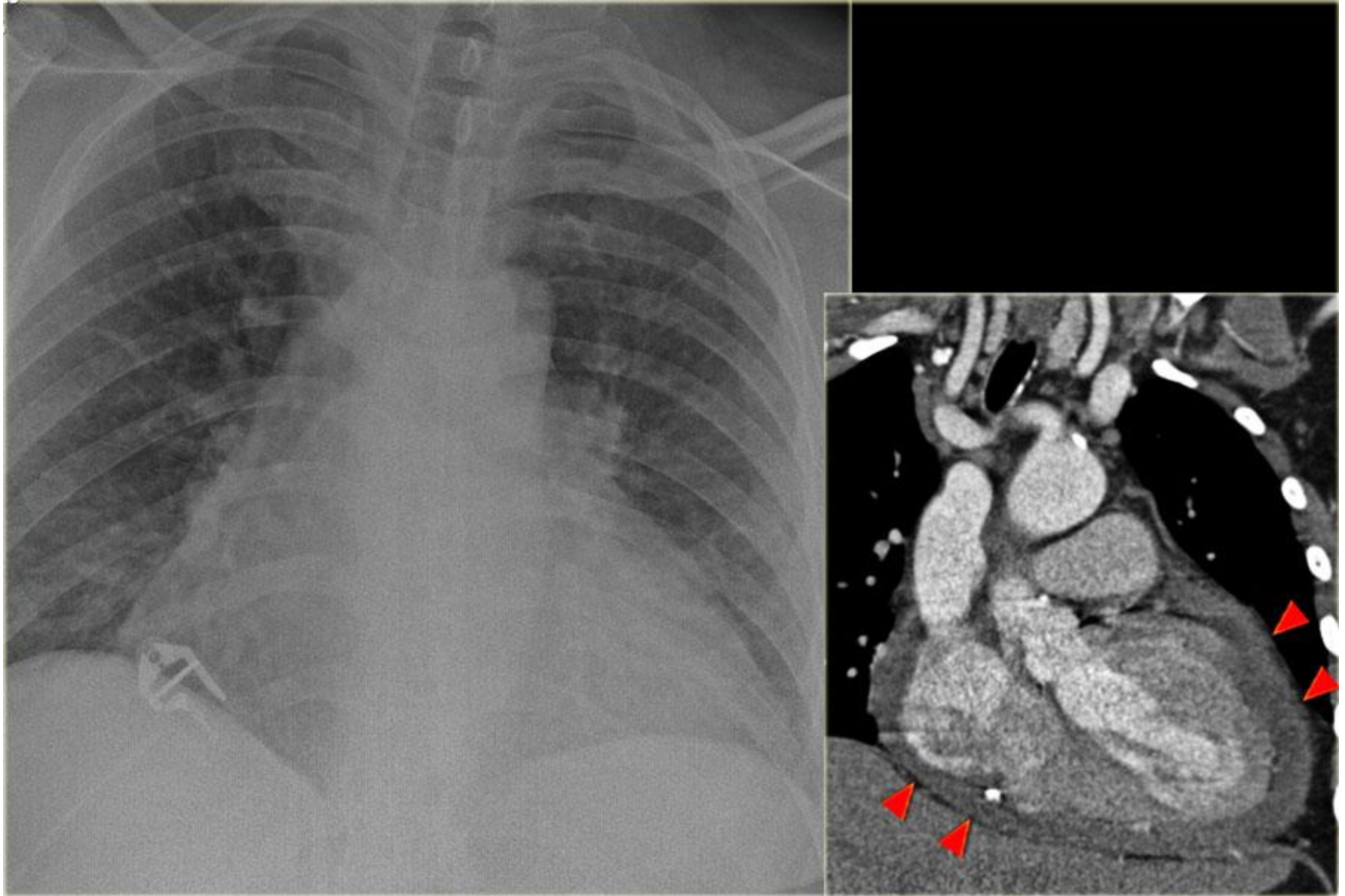
a patient with CHF.

There is an increase in heart size compared to the old film.

- Old film for comparison (left) CHF with redistribution, interstitial edema and some pleural fluid



- Increased CTR due to pericardial effusion



- another patient with a large cardiac silhouette on the chest x-ray due to **pericardial effusion**.
Pericardial effusion is demonstrated on the coronal CT-reconstruction.

Pleural effusion

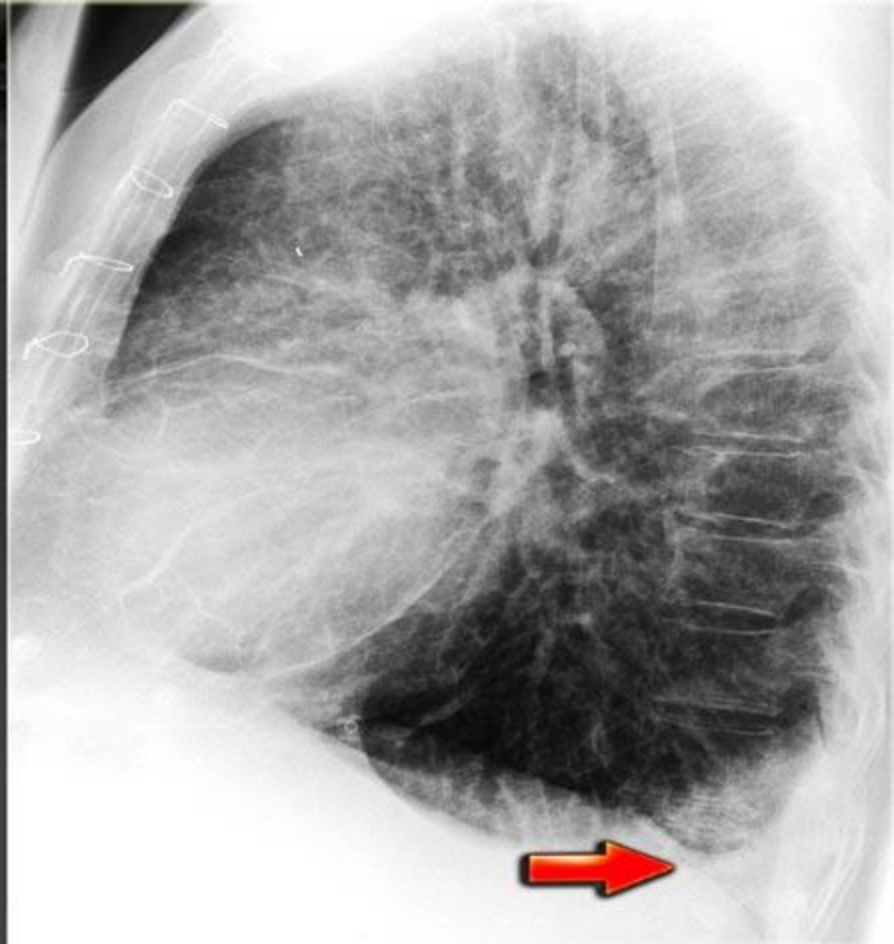
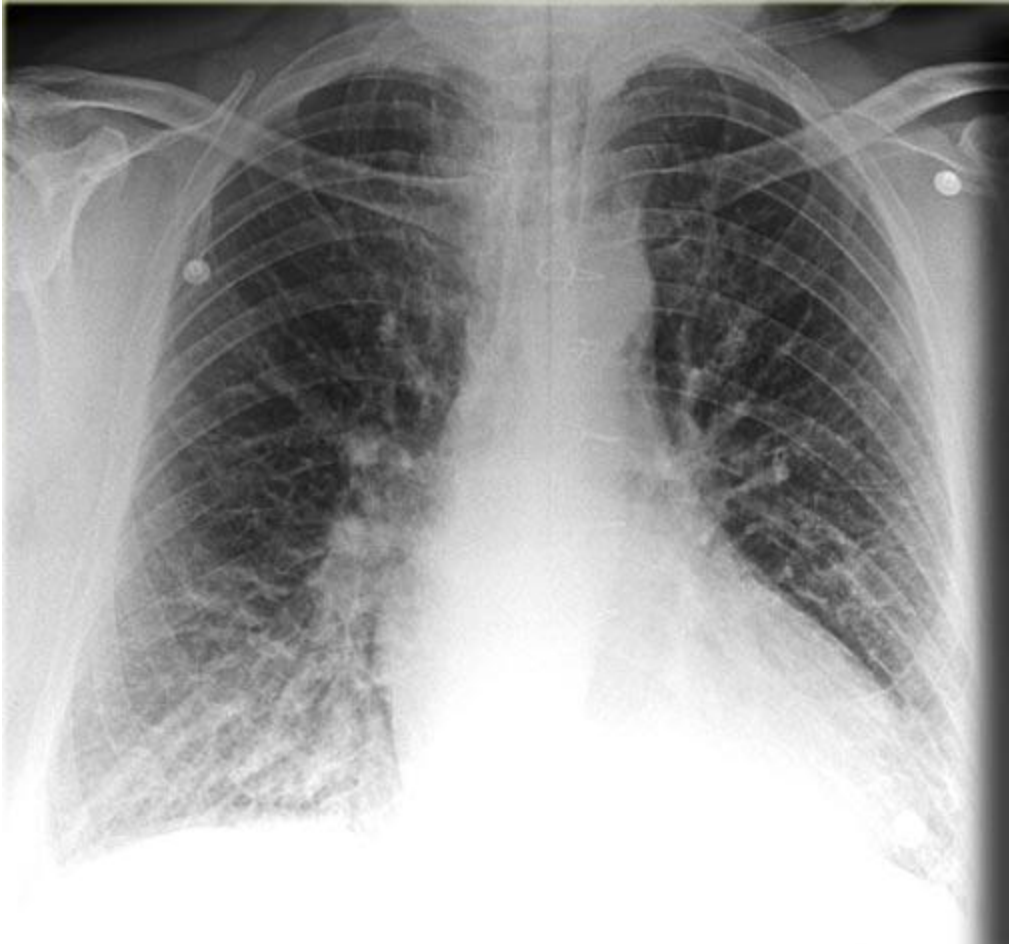
Pleural effusion :

- is bilateral in 70% of cases of CHF.
- When unilateral, it is slightly more often on the right side than on the left side.

= There has to be **at least 175 ml** of pleural fluid, before it will be visible on a PA image as a meniscus in the costophrenic angle.

= On a lateral image effusion of **> 75 ml can be visible**.

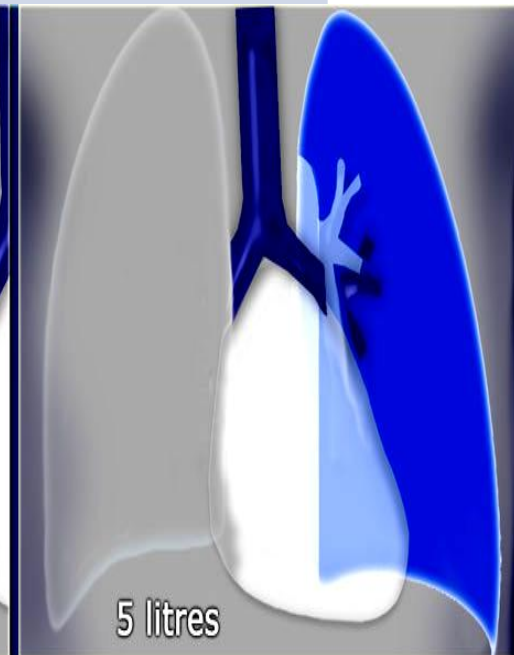
*** If pleural effusion is seen on a supine chest film, it means that there is **at least 500 ml** present.



- a patient who has bilateral pleural effusions. Notice that it is more evident on the lateral view.

Pleural Effusion

CXR View	Effusion Volume	Appearance
<i>P-A View</i>	> 175 ml	meniscus in the costophrenic angle. Just Begin to Visualize
	200 – 300 ml	obliterate CP A
	5 L	Total Opacified Hemithorax
<i>Lateral View</i>	> 75 ml	Begin to visualized

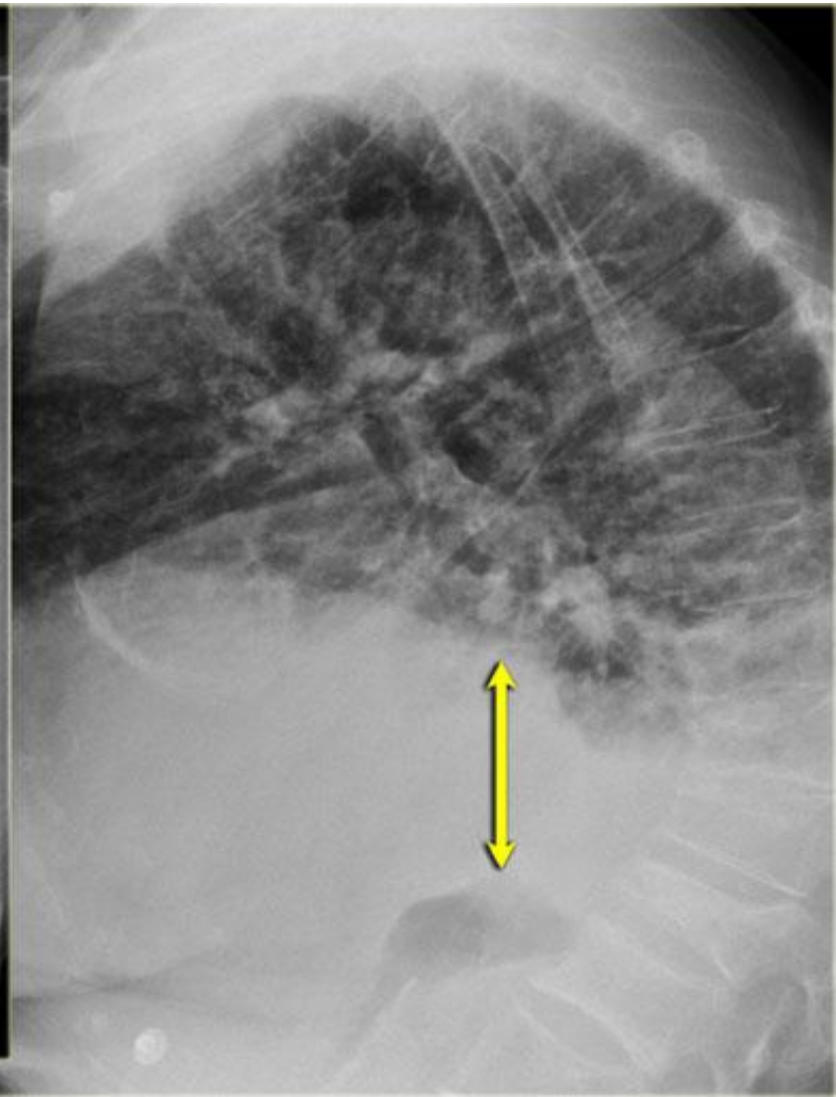
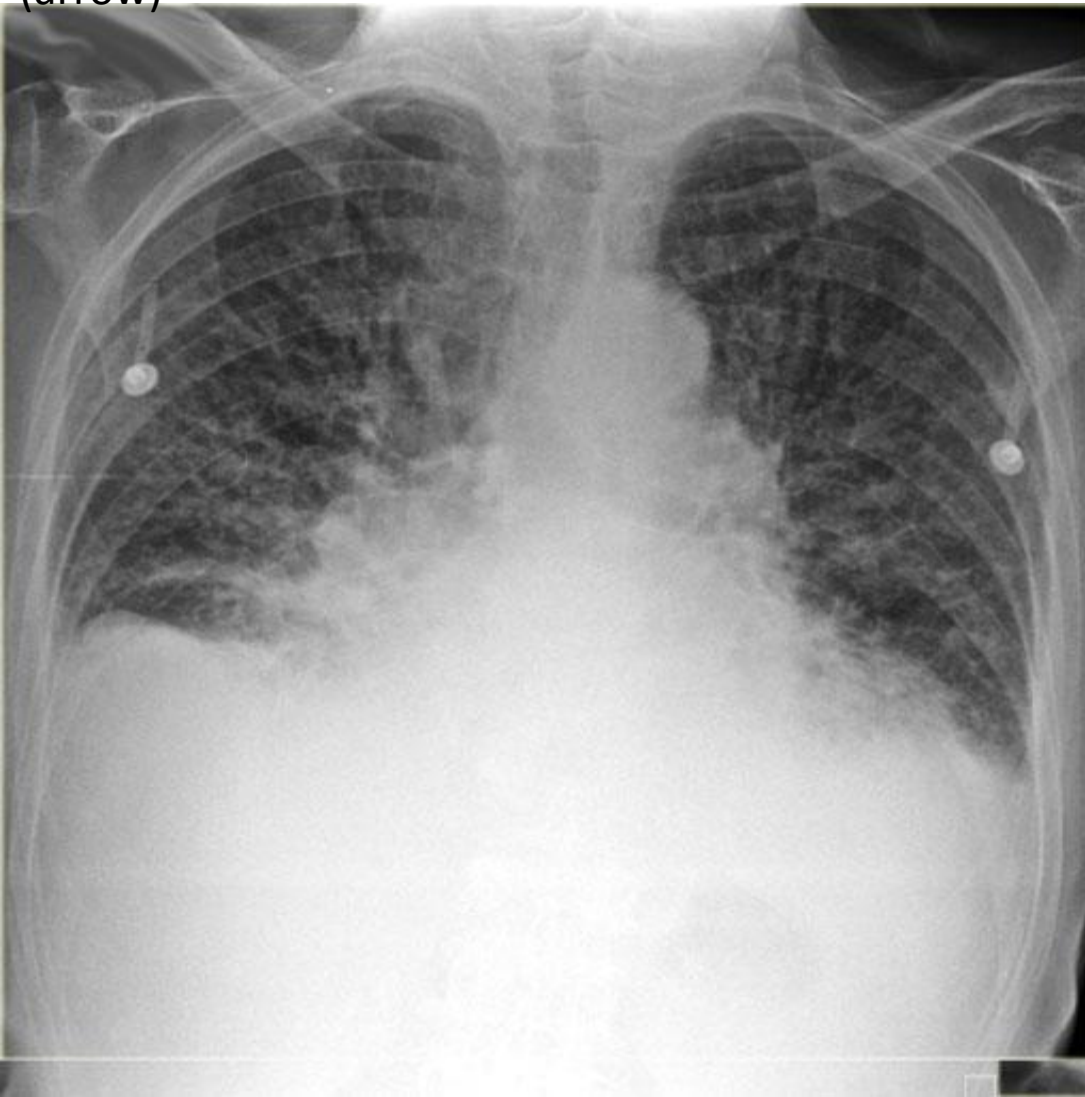


- A subpulmonic effusion may follow the contour of the diaphragm making it tricky to discern.

In these cases, the only way to detect pleural effusion, is when you notice that there is an increased distance between the stomach bubble and the lung.

The stomach is normally located directly under the diaphragm, so, on an erect PA radiograph, the stomach bubble should always appear in close proximity to the diaphragm and the lung.

Subpulmonic pleural effusion with increased distance of the stomach air bubble to the lung base (arrow)



a patient with signs of CHF.

At first glance you might get the impression that there is a high position of the diaphragm.

However when you notice the increased distance of the stomach air bubble to the lung base, you realize that there is a large amount of pleural fluid on both sides (arrow).

Vascular pedicle

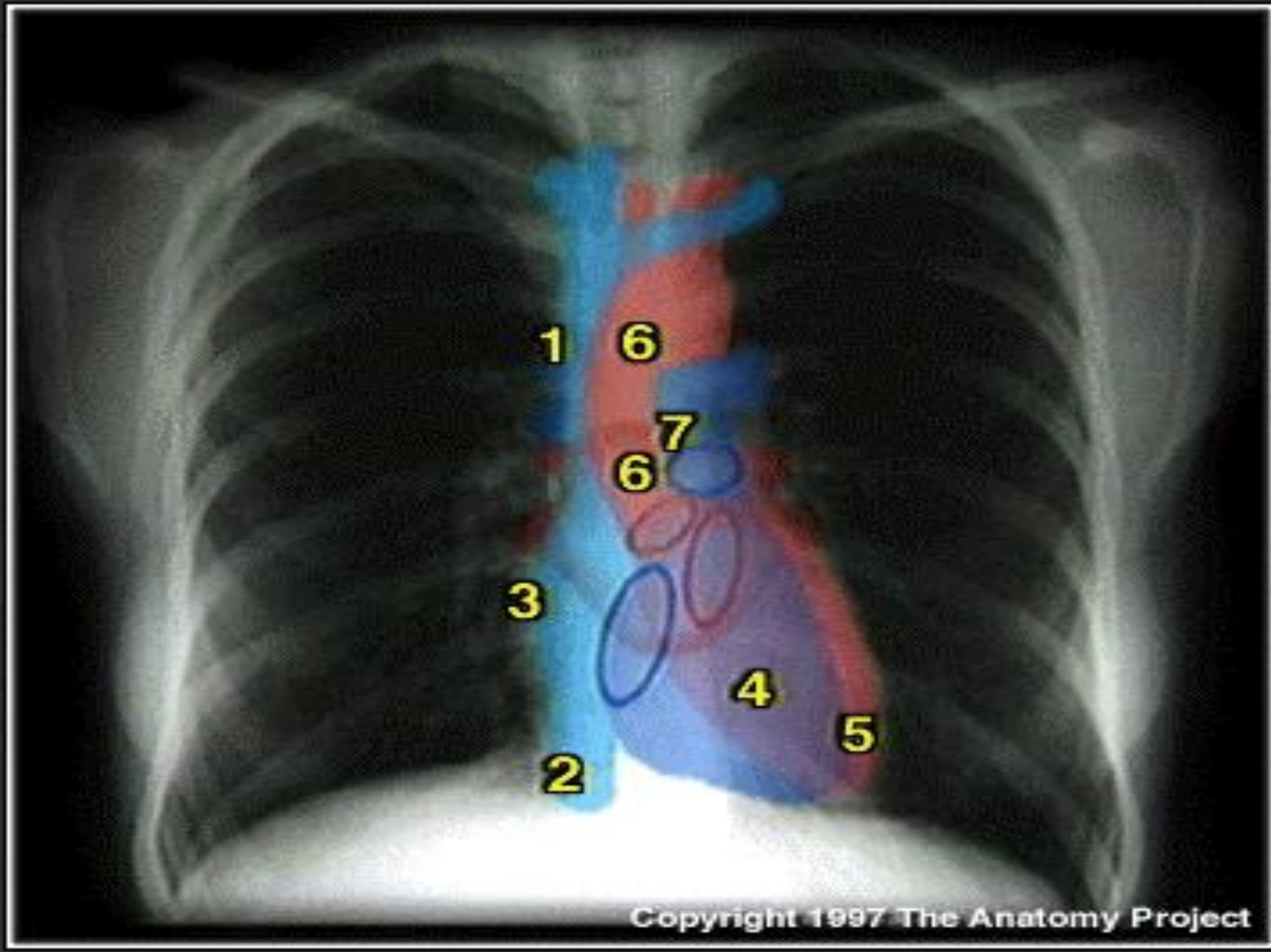
- The vascular pedicle is bordered :
 - on the right → by the superior vena cava and
 - on the left → by the left subclavian artery origin .
- The vascular pedicle is an indicator of the intravascular volume.

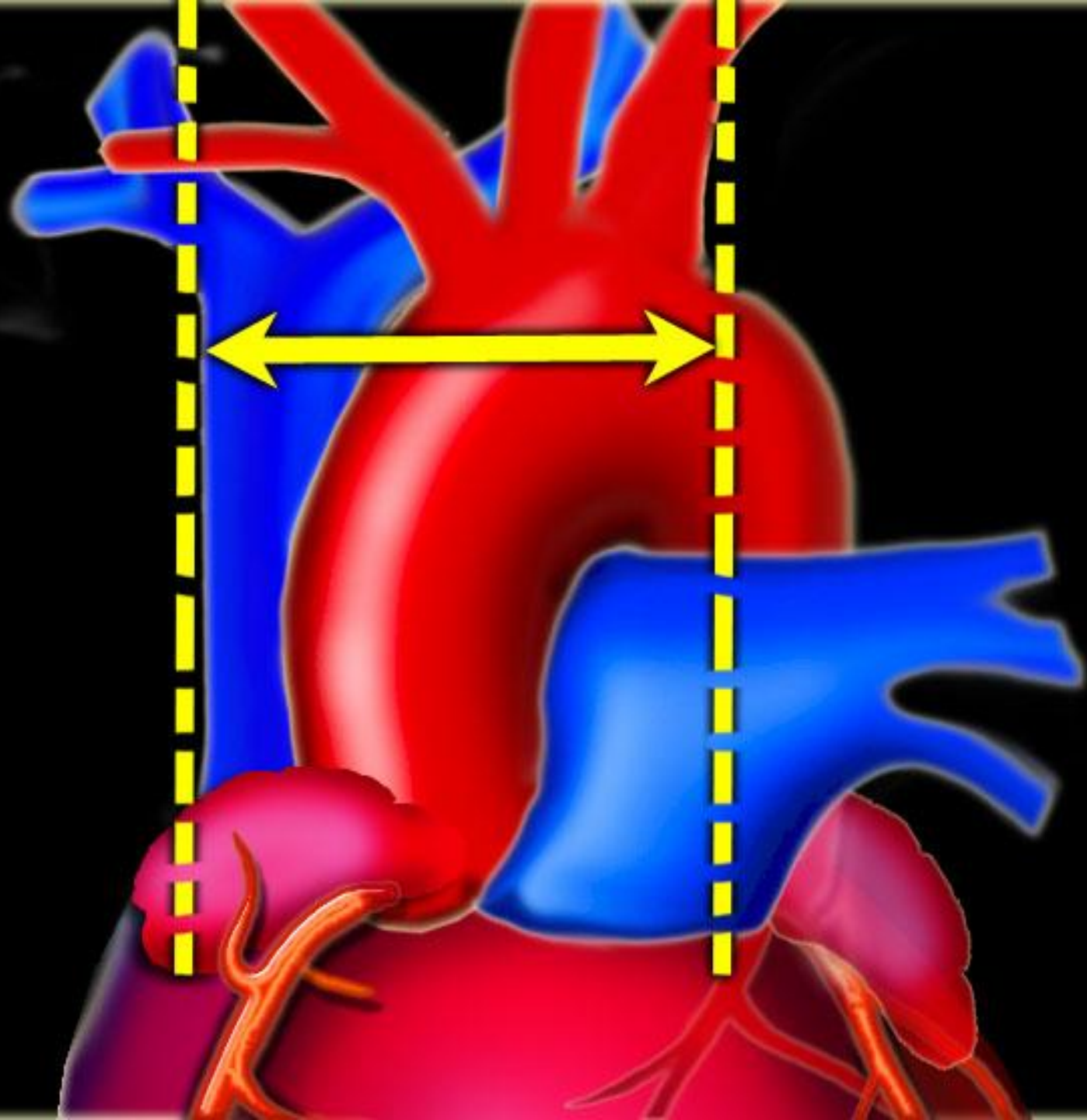
A vascular pedicle width :

- **less than 60 mm** on a PA chest radiograph is seen in 90% of normal chest x-rays.
- **more than 85 mm** is pathologic in 80% of cases.

5 mm increase in diameter corresponds to 1 liter increase of intravascular fluid.

An increase in width of the vascular pedicle is accompanied by an increased width of the azygos vein.

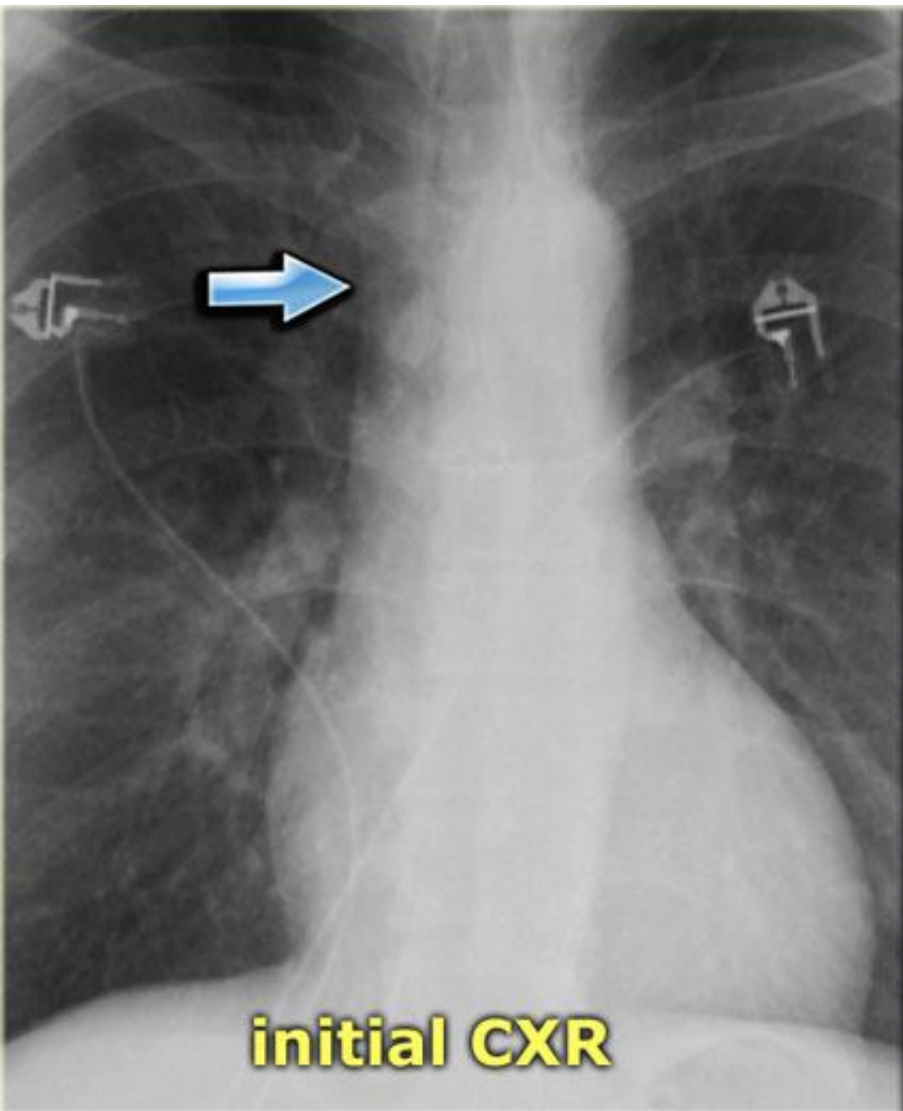




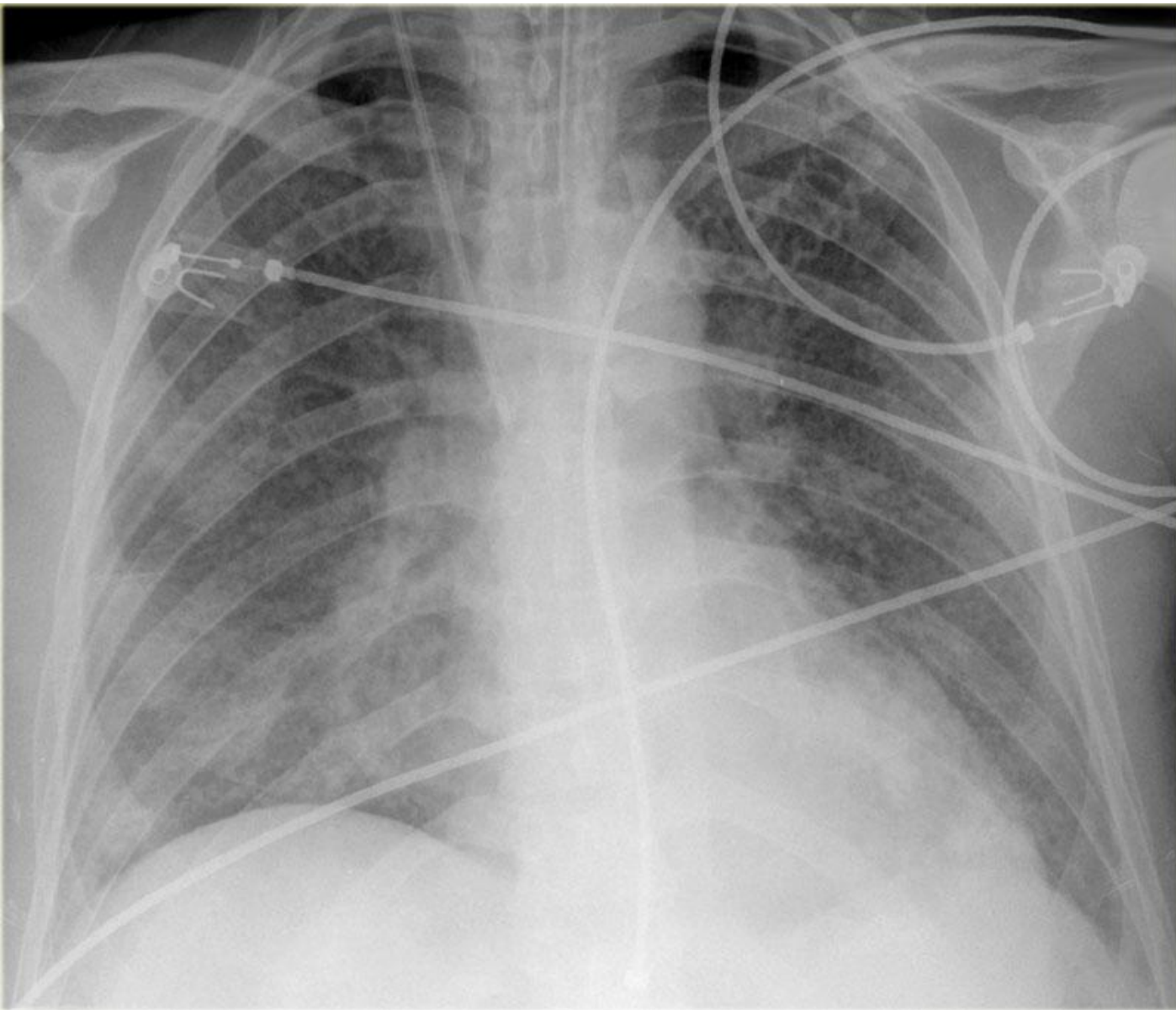
- There are three principal varieties of pulmonary edema: cardiac, overhydration and increased capillary permeability (ARDS).

The vascular pedicle width (VPW) can help in differentiating these different forms of pulmonary edema (6):

- Normal VPW: most common in capillary permeability or acute cardiac failure.
- Widened VPW: most common in overhydration/renal failure and chronic cardiac failure.
- Narrowed VPW: most common in capillary permeability.



Subtle increased width of vascular pedicle (left) and normalisation (right)



• a patient with
ARDS.

There is alveolar
edema in both
lungs.

Notice that the
VPW is normal.

The vessels in the
upper lobes are not
dilated and the
cardiac silhouette is
not enlarged.

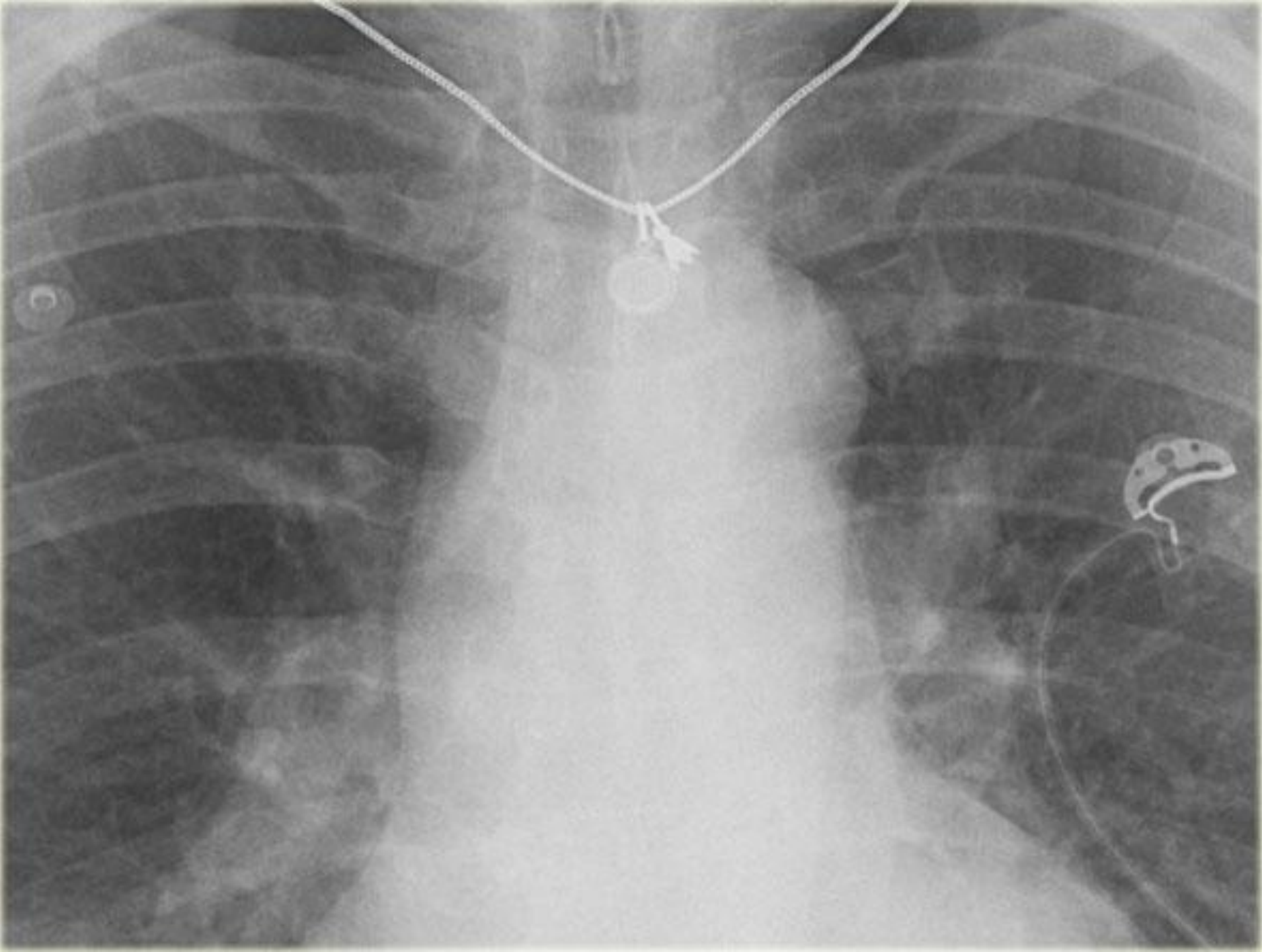
- The VPW is best used as a measure to compare serial chest x-rays of the same patient, as there is a wide range of values for the VPW.

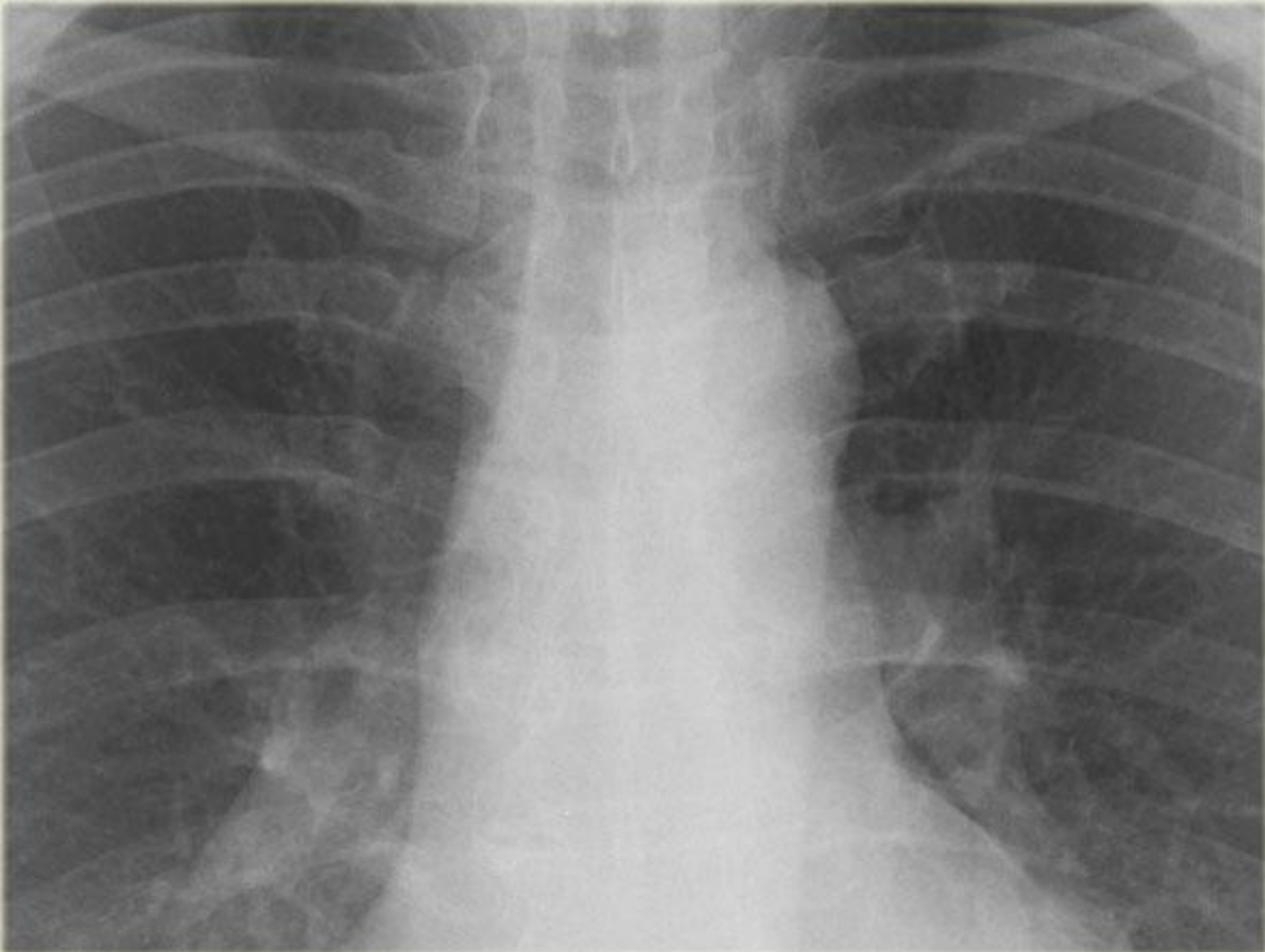
The VPW may increase due to rotation to the right.

On an AP-view the VPW will increase 20% compared to a PA-view.

- Next a patient with subtle signs of congestive heart failure on the initial chest x-ray (image 1/2).

There is a slightly enlarged vascular pedicle, which becomes more obvious when you compare to the chest film after diuretic therapy image 2 .





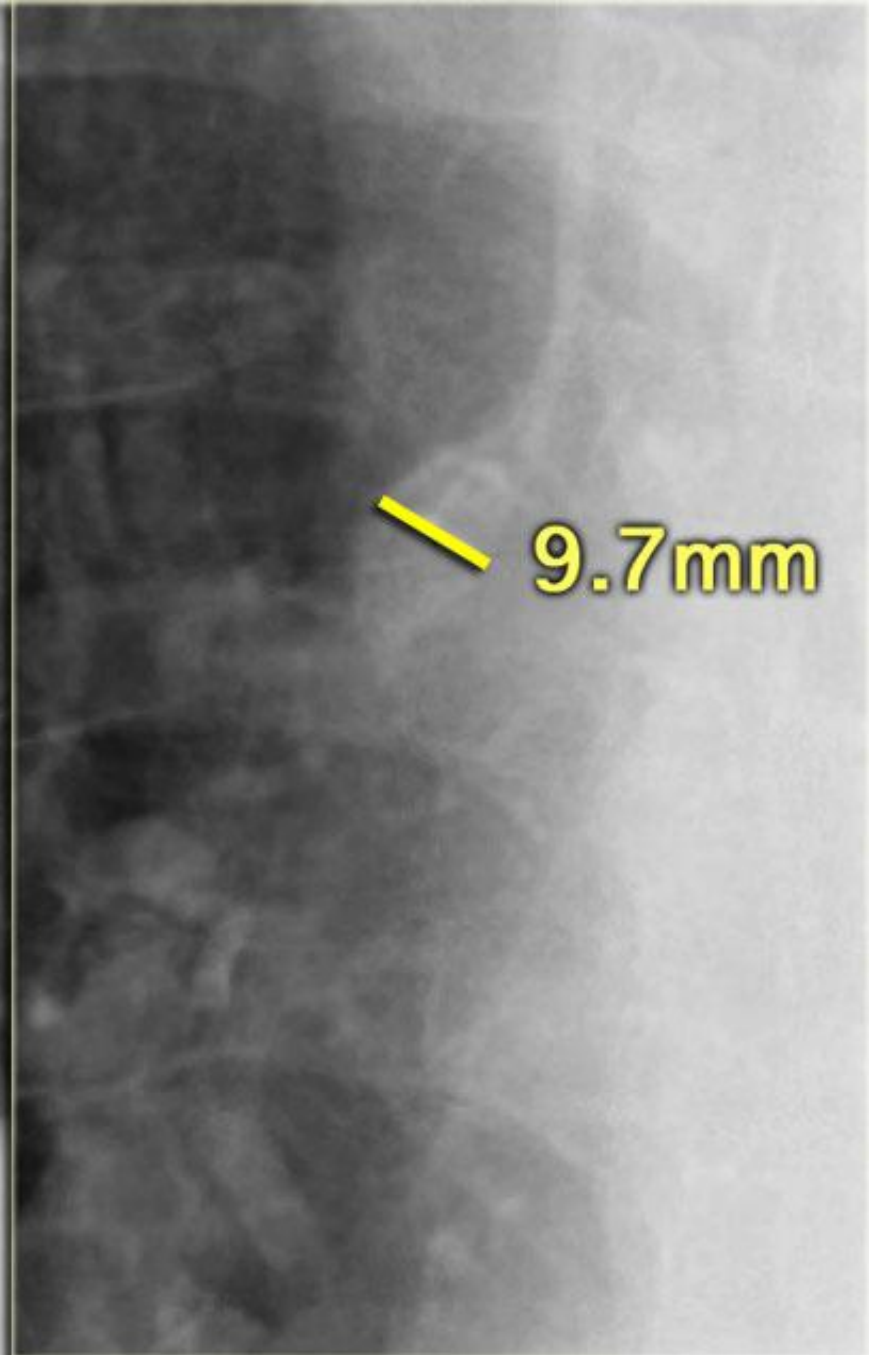
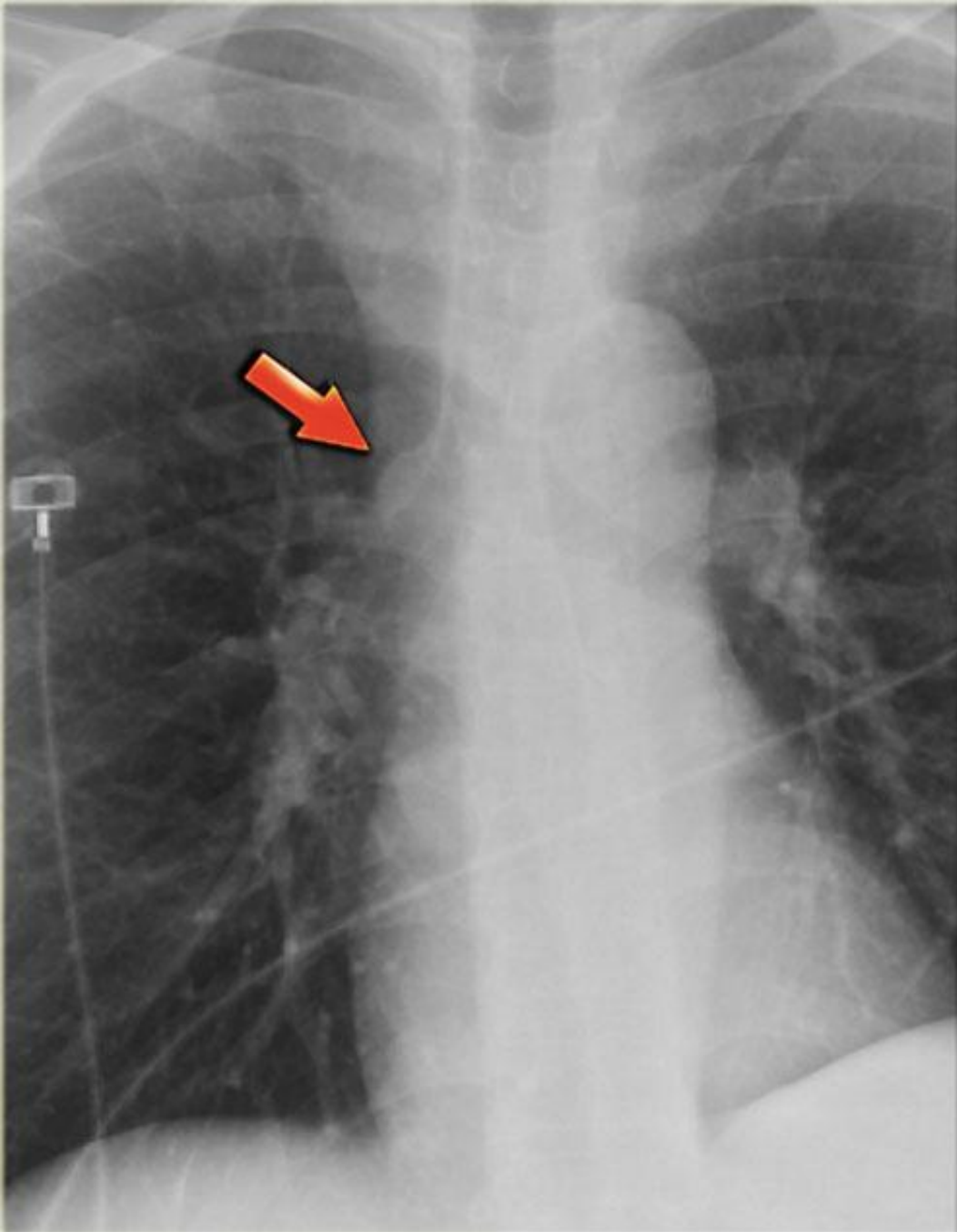
Dilatation of azygos vein

- Dilation of the azygos vein is a sign of increased **right atrial pressure** and is
- usually seen when there is also an increase in the width of the vascular pedicle.
- The diameter of the azygos vein varies according to the positioning.
- **In the standing position** a diameter > 7 mm is most likely abnormal and a
- diameter > 10 mm is definitely abnormal.
- In a **supine** patient > 15 mm is abnormal.

An increase of 3 mm in comparison to previous films is suggestive of fluid overload.

- The difference of the azygos diameter on an inspiration film compared to an expiration film is only 1mm.

This means that the diameter of the azygos is a valuable tool whether or not there is good inspiration.



Right ventricular failure

- RV failure is most commonly caused by longstanding LV failure, which increases the pulmonary venous pressure and leads to pulmonary arterial hypertension, thus overloading the RV.

Other less common causes of RV failure are:

- Severe lung disorder (cor pulmonale)
- Multiple pulmonary emboli
- RV infarction
- Primary pulmonary hypertension
- Tricuspid regurgitation or stenosis, mitral stenosis and pulmonary valve stenosis.
-

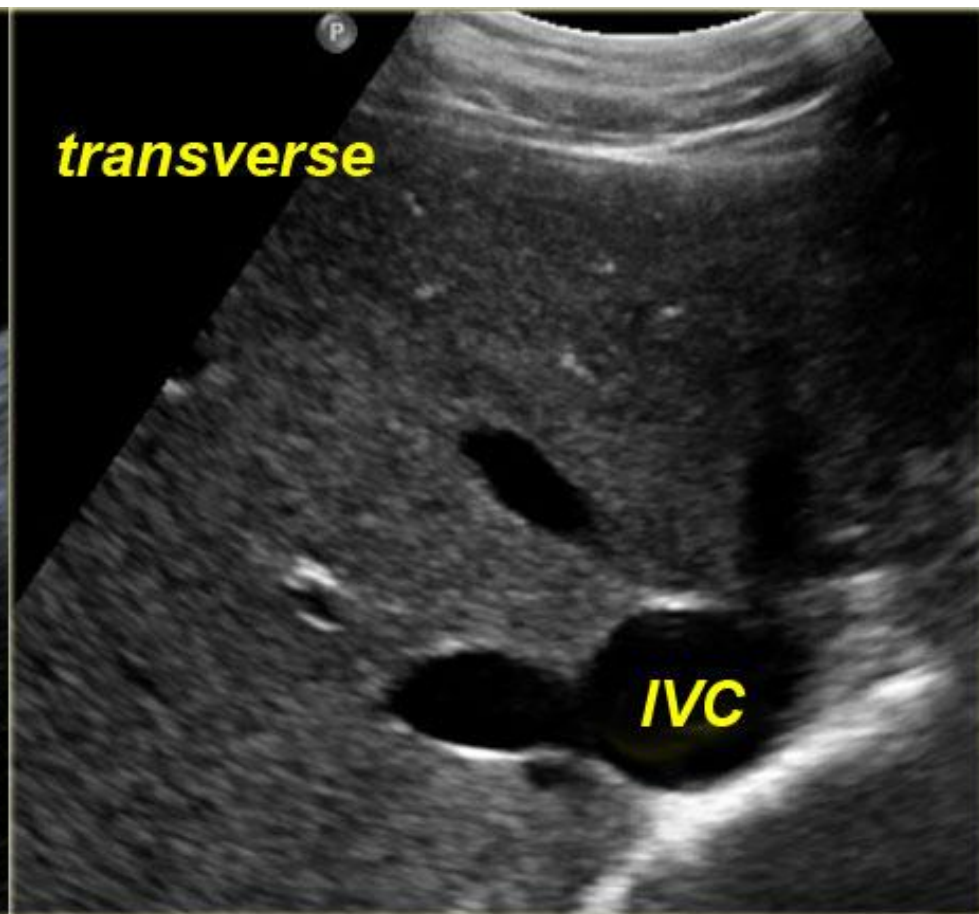
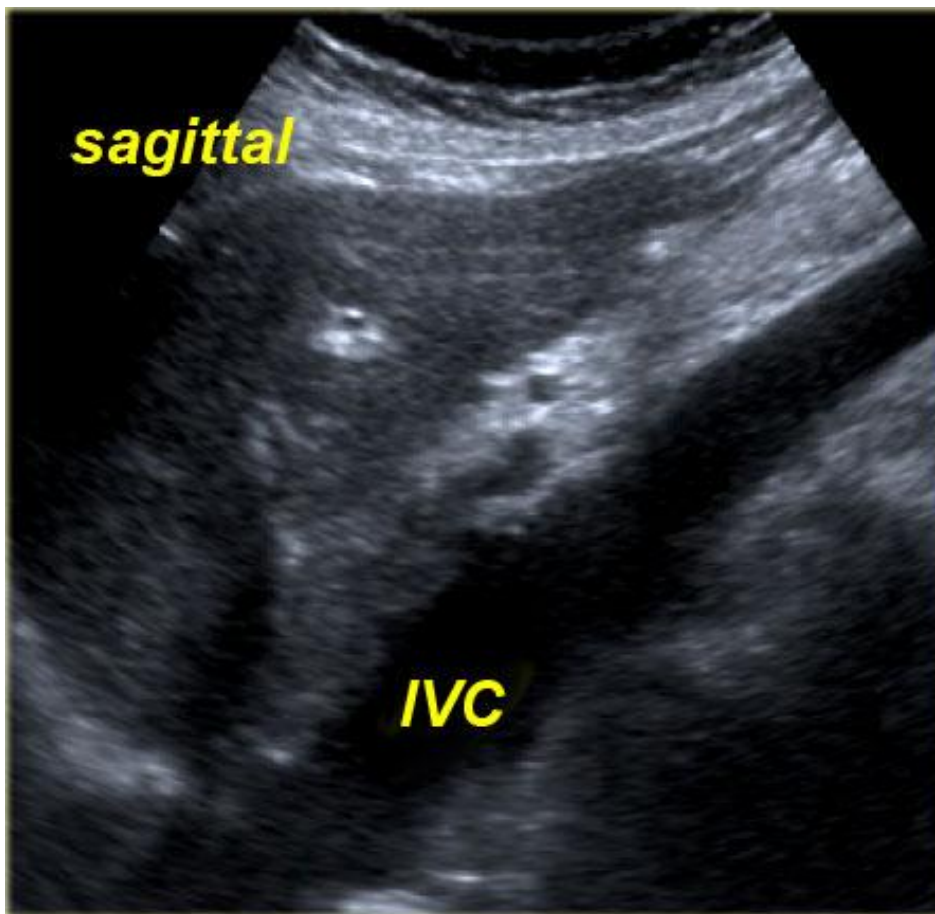
- Radiographic signs of RV failure:

- Increased VPW due to dilatation of the superior vena cava
- Dilatation of azygos vein
- Dilatation of the right atrium
- In many cases there will be both signs of RV and LV failure

-

Sonographic signs of RV failure:

- Dilatation of the inferior vena cava (IVC) and hepatic veins
- Hepatomegaly
- Ascites



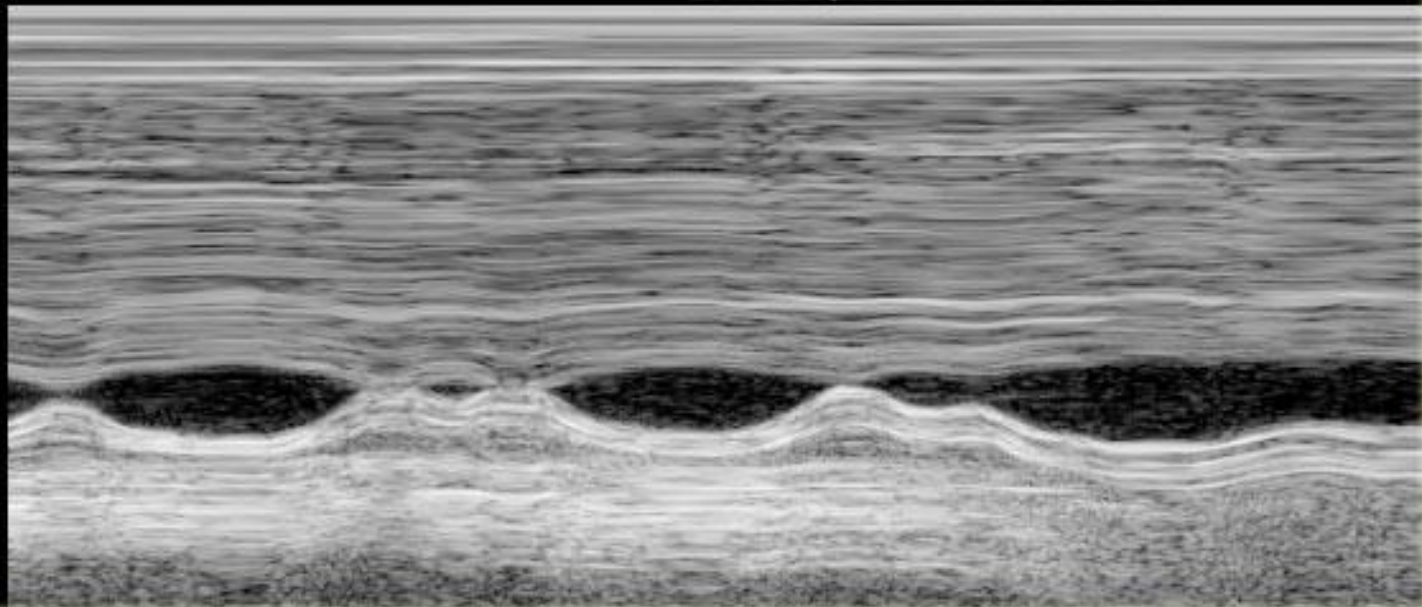
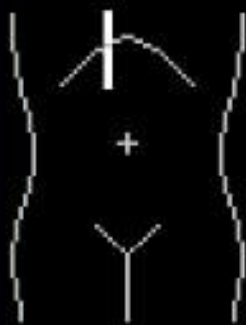
Dilatation of IVC and hepatic veins on US images in a patient with RV failure

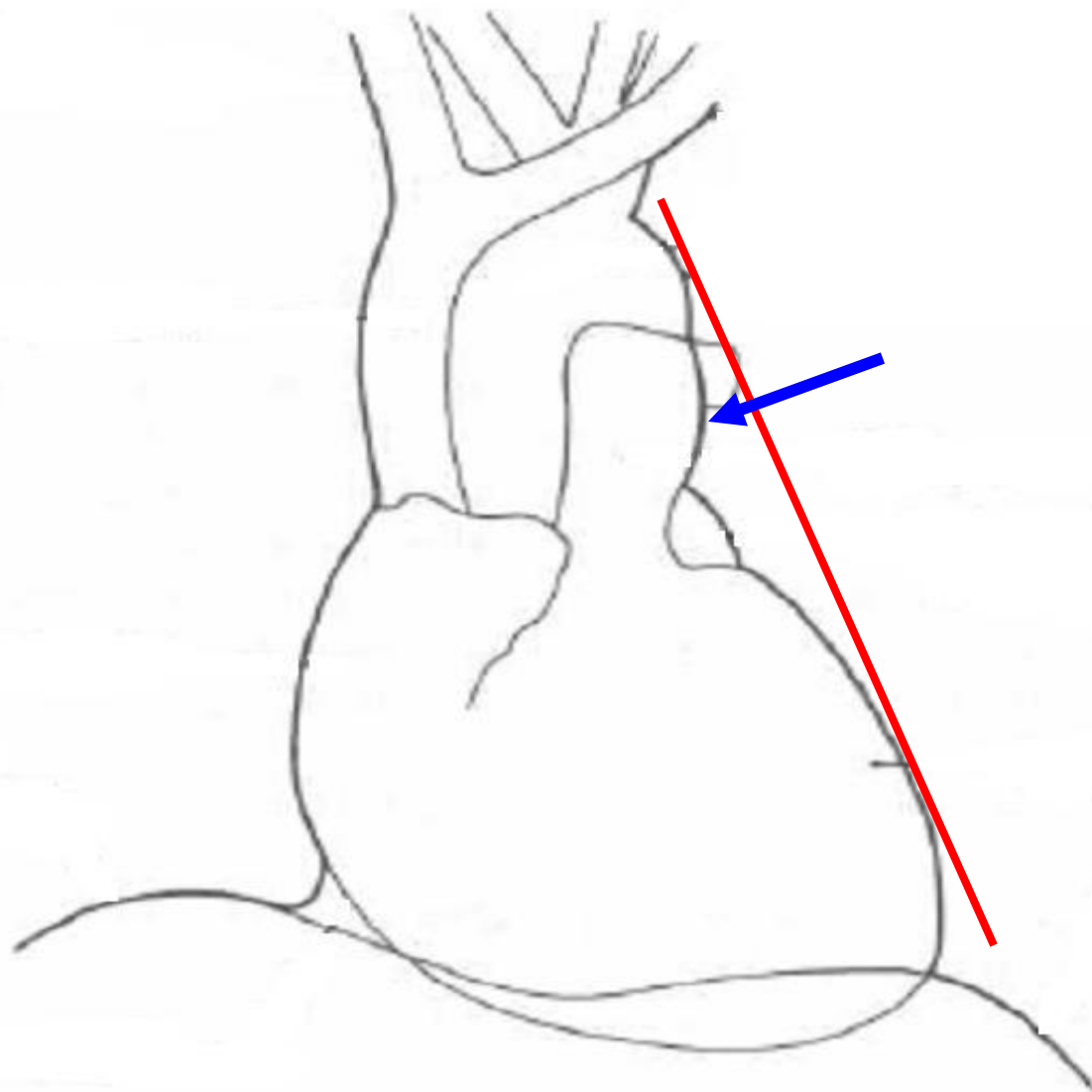
- The indication for ultrasound examination in many of these patients is abnormal liver function tests.

It is therefore important to consider the possibility of RV failure when a patient presents with liver enzyme abnormalities.

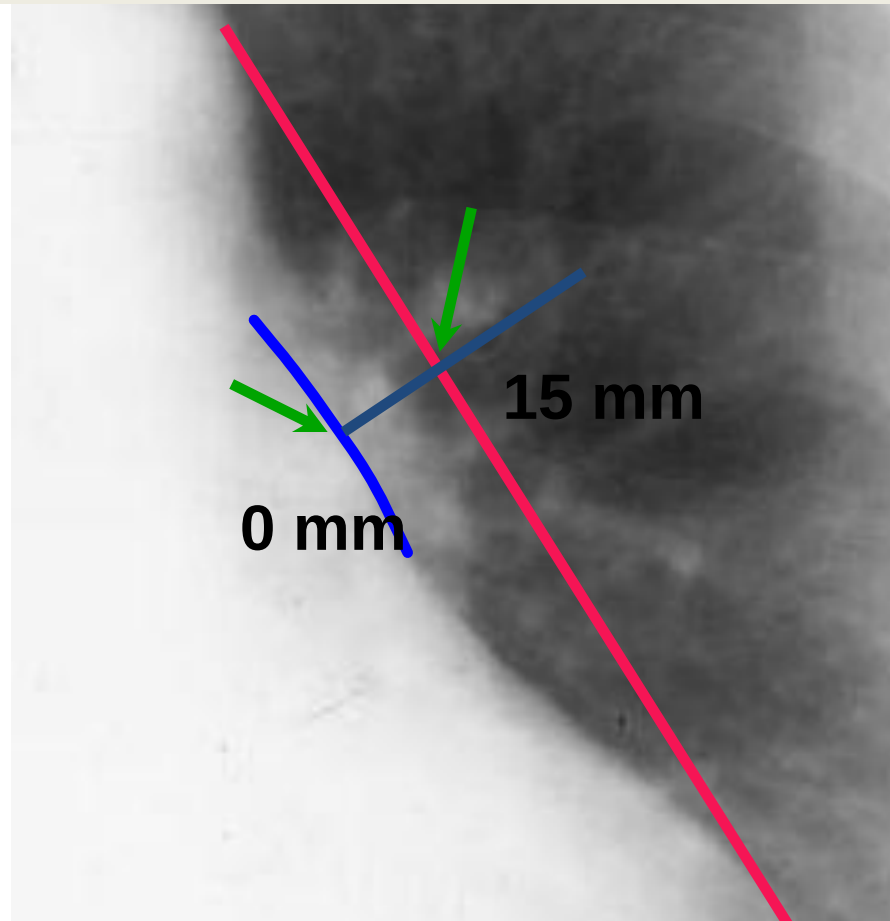
- Under normal conditions dynamic ultrasound will demonstrate changes in caliber of the IVC.

These changes in caliber can be attributed to variations in blood flow in the IVC in accordance with the respiratory and cardiac cycles.

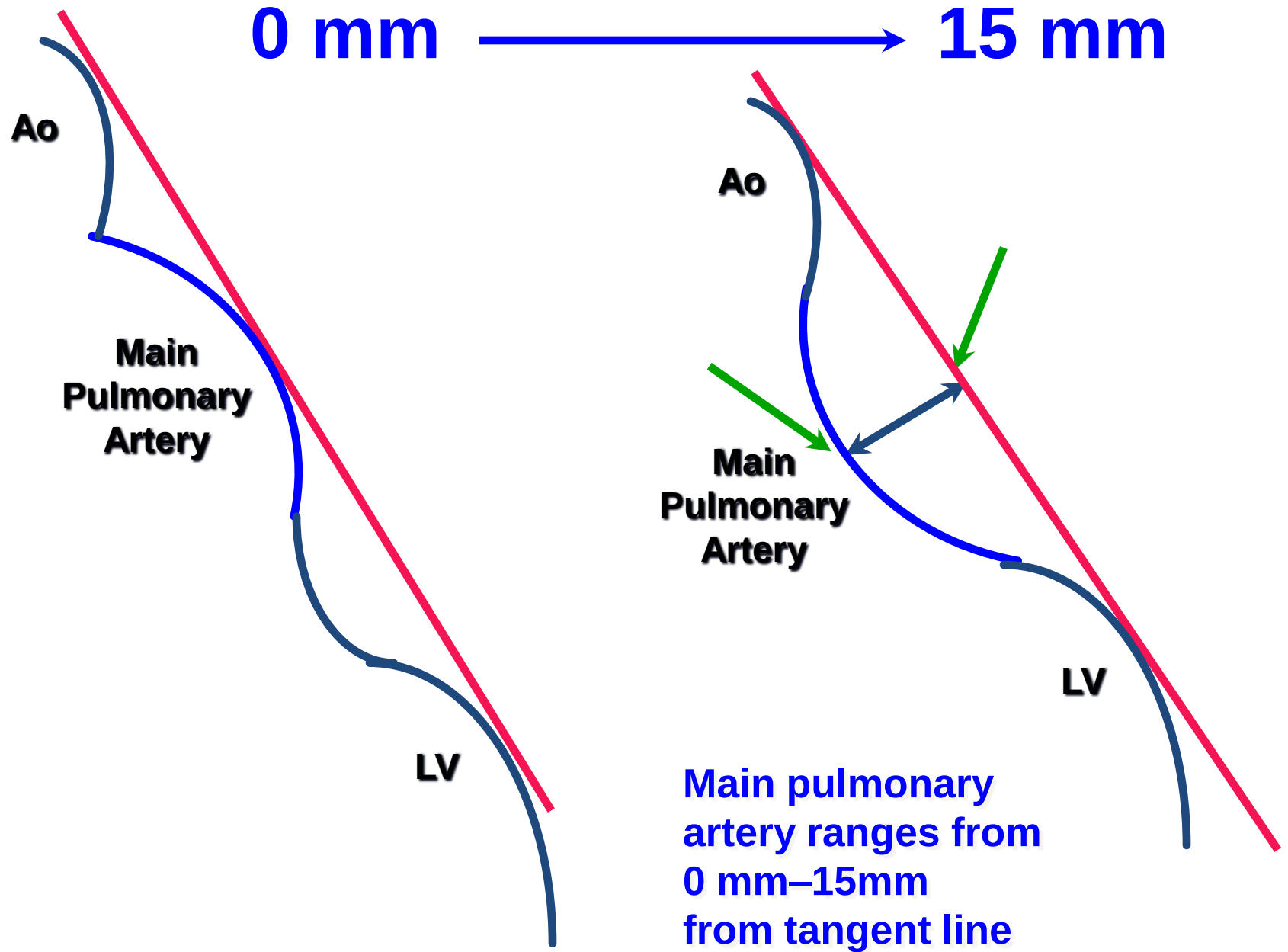




The distance between the tangent and the main pulmonary artery
(between two small green arrows)



Draw a tangent line , from the apex of the left → the aortic knob
(red line) &
measure along a perpendicular to that tangent line (yellow line)



Chest X-Ray

3

Lung disease

Four-Pattern Approach

Robin Smithuis

Radiology Department of the Rijnland
Hospital, Leiderdorp, the Netherlands

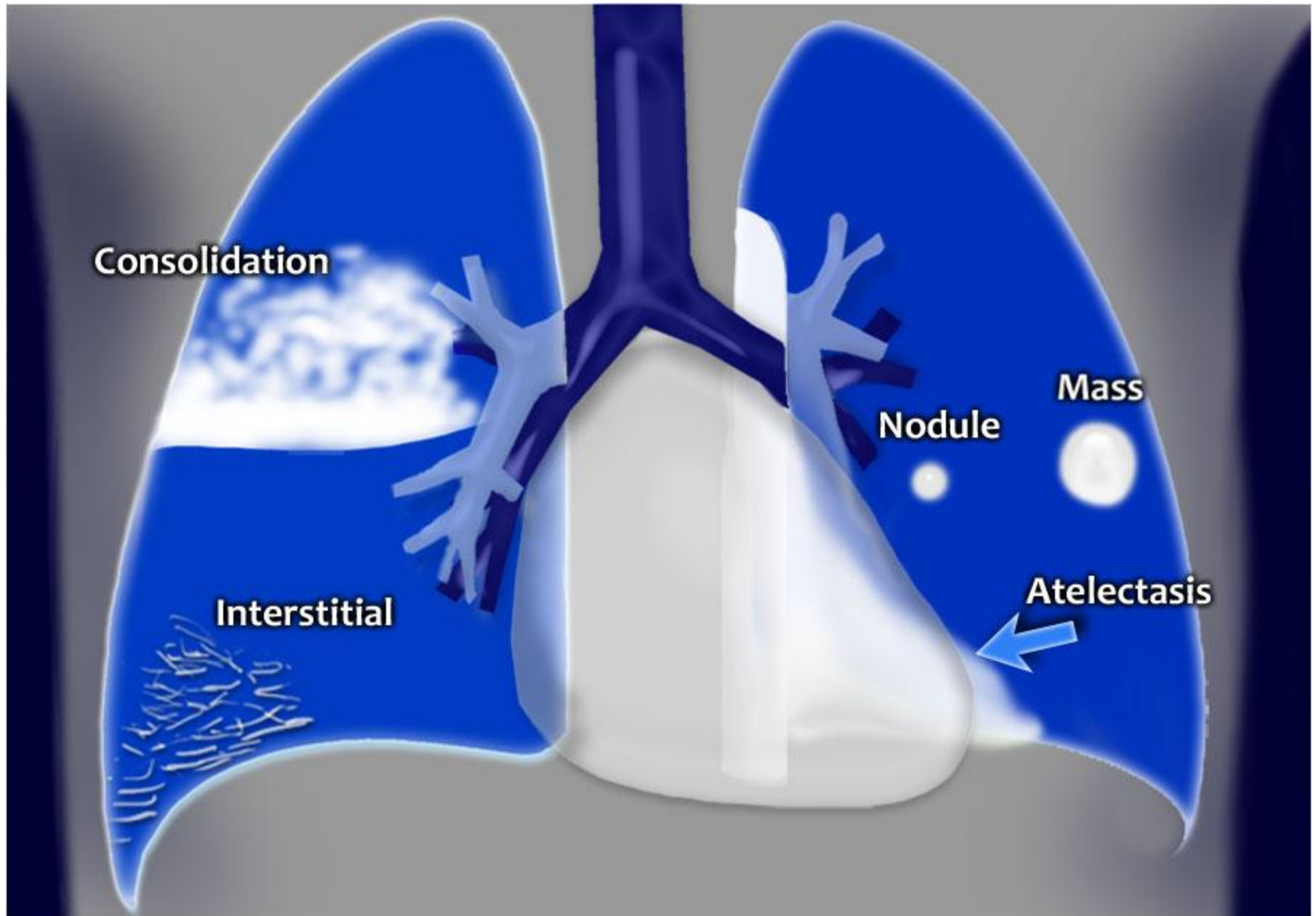
Atelectasis ??

Mass ??

Consolidation ??

Interstitial ??

4-Pattern approach



Whenever you see an area of increased density within the lung, it must be the result of one of these four patterns.

Consolidation - any pathologic process that fills the alveoli with fluid, pus, blood, cells (including tumor cells) or other substances resulting in lobar, diffuse or multifocal ill-defined opacities.

Interstitial - involvement of the supporting tissue of the lung parenchyma resulting in fine or coarse reticular opacities or small nodules.

Nodule or mass - any space occupying lesion either solitary or multiple.

Atelectasis - collapse of a part of the lung due to a decrease in the amount of air in the alveoli resulting in volume loss and increased density.

Here are the most common examples of these four patterns on a chest x-ray (click image to enlarge).

- Consolidation**

- Lobar consolidation

- Diffuse consolidation

- Multifocal ill-defined consolidations

- Interstitial**

- Reticular interstitial opacities

- Fine Nodular interstitial opacities

- Nodule or mass**

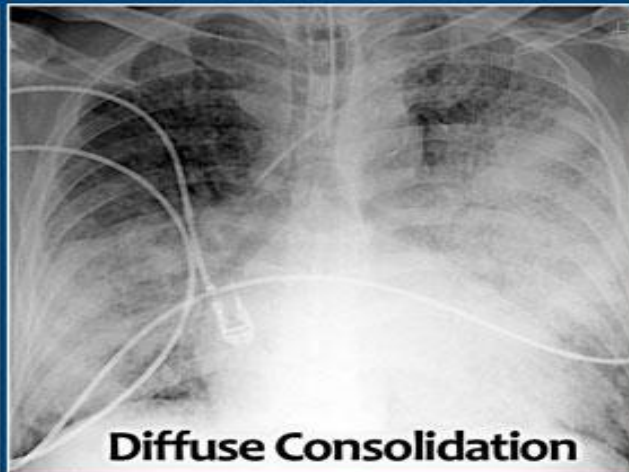
- Solitary Pulmonary Nodule

- Multiple Masses

- Atelectasis**



Lobar Consolidation



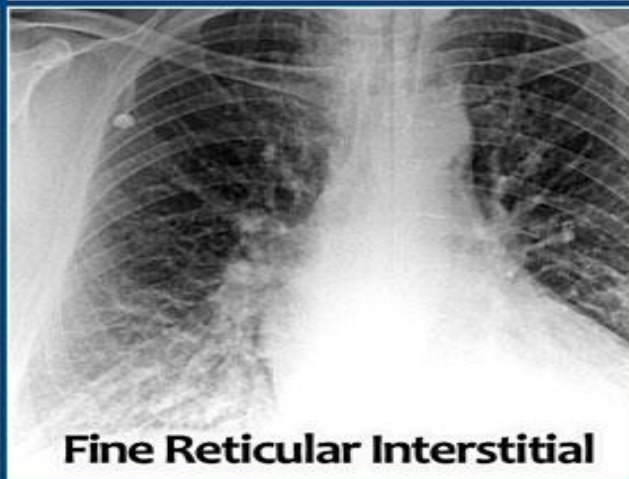
Diffuse Consolidation



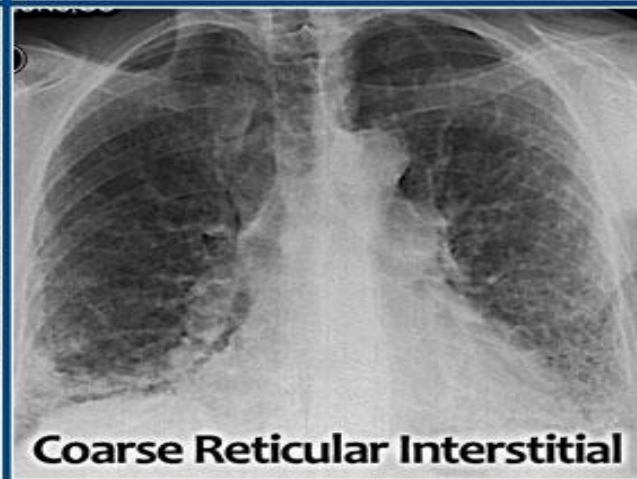
Multifocal ill-defined



Atelectasis



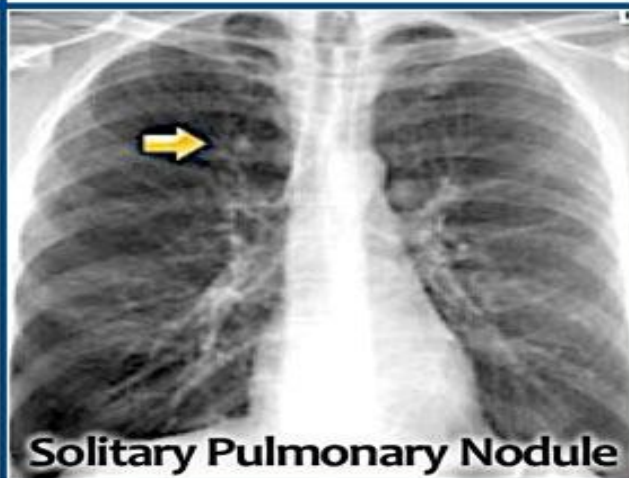
Fine Reticular Interstitial



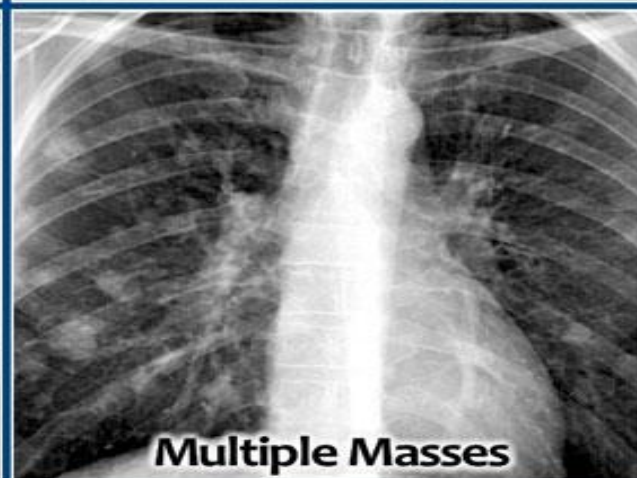
Coarse Reticular Interstitial



Fine Nodular



Solitary Pulmonary Nodule



Multiple Masses

You have to realize that ***it is not always possible to divide lung abnormalities into one of these four patterns***, but that should not be a problem.

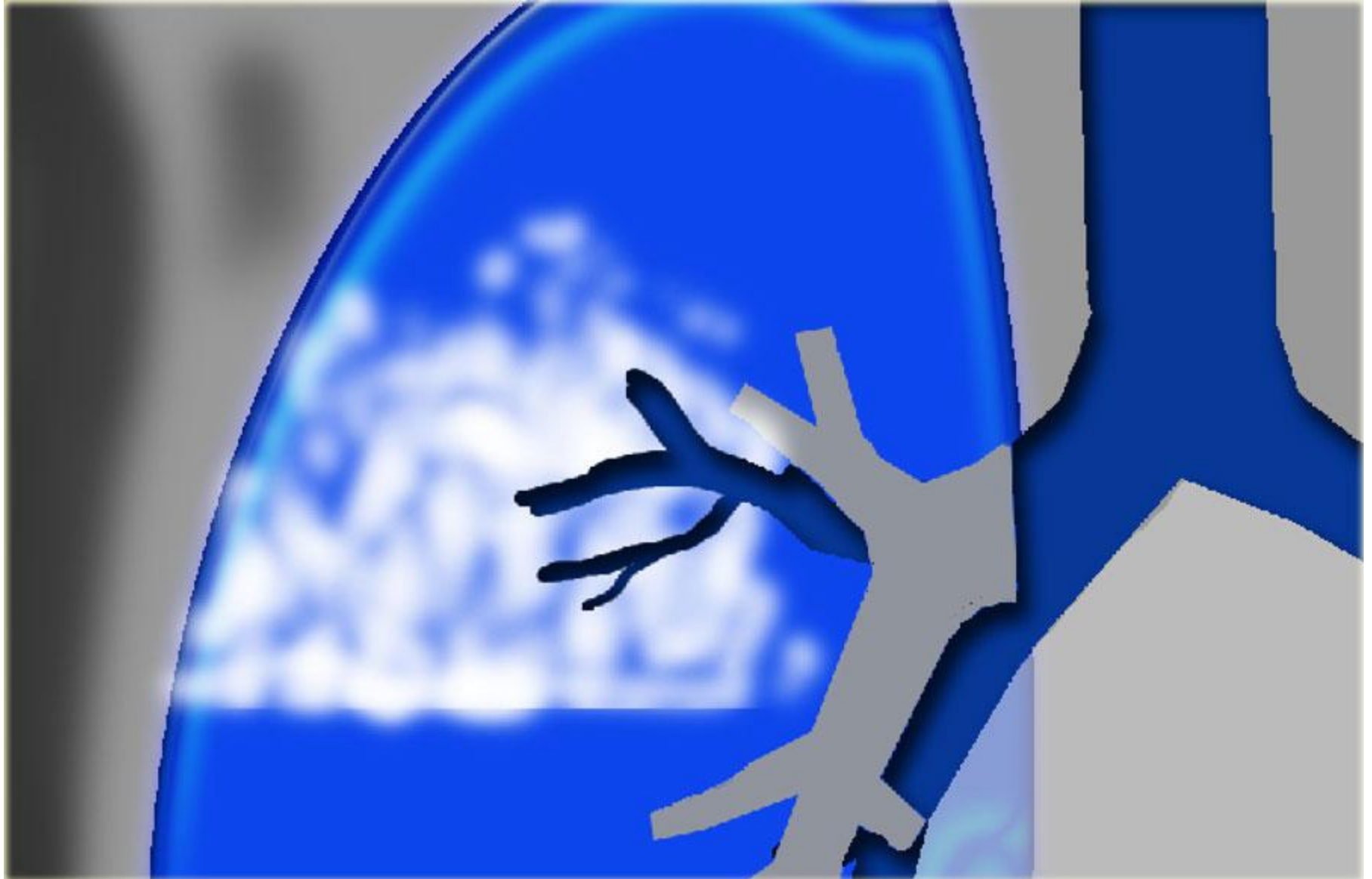
- Sometimes you are confronted with an abnormality that looks like a mass, but it could also be a consolidation.

Just do the work-up of both the differential diagnosis of masses and consolidation.

- In such a case information from clinical data, old films or follow-up films and CT-scan will usually solve the problem.

- Finally in some cases only biopsy will provide a diagnosis.

Consolidation



Consolidation is the result of replacement of air in the alveoli by transudate, pus, blood, cells or other substances.

Pneumonia is by far the most common cause of consolidation.

The disease usually starts within the alveoli and spreads from one alveolus to another.

When it reaches a fissure the spread stops there.

The key-findings on the X-ray are:

- ill-defined homogeneous opacity obscuring vessels
- [Silhouette sign](#): loss of lung/soft tissue interface
- Air-bronchogram
- Extension to the pleura or fissure, but not crossing it
- No volume loss

Differential diagnosis

The table summarizes the most common diseases, that present with consolidation.

Click to enlarge.

Chronic diseases are indicated in red.

A way to think of the differential diagnosis is to think of the possible content of the alveoli:

1. Water - transudate.
2. Pus - exsudate.
3. Blood - hemorrhage.
4. Cells - tumor, chronic inflammation.

Another way to think of consolidation, is to look at the pattern of distribution:

- Diffuse - perihilar (batwing) or peripheral (reversed batwing).
- Lobar or focal.
- Multiple - usually multiple ill-defined densities.

Content			
Water	Pus	Blood	Cells
Heart failure ARDS Low albumin Renal failure	Pneumonia	Trauma Goodpasture Henoch Schonlein SLE	BAC Organizing pneum Chron eosin pneum Sarcoid
Pattern			
Lobar	Diffuse	Multi-focal	
Lobar pneumonia - Streptococ pneum. - Klebsiella - TB - viral - fungal - Aspiration Neoplasm - Lungca with obstructive pneumonia - BAC - Lymphoma Hemorrhage - Contusion - Infarction Others - Organizing pneumonia - Eosinophilic pneum - Sarcoidosis - Sequestration - Mitral regurgitation with RUL edema	Edema - Heart failure - Volume overload - ARDS - Low albumin - Renal failure - Transfusion reaction Bronchopneumonia - Staph Aureus - Gram negative - PCP - Viral - fungal Neoplasm - BAC - Lymphoma Hemorrhage - SLE - Henoch-Schönlein - Wegener - Goodpasture Others - Organizing pneumonia - Eosinophilic pneum - Hypersensitivity pneum - Alveolar proteinosis	Bronchopneumonia - Staph Aureus - Legionella - Gram negative - Streptococcus pneum - Klebsiella - Pseudomonas - Anaerobe - PCP - TB - Aspiration Vascular - Septic emboli - Wegener's Neoplasm - BAC - Lymphoma - Metastases Others - Organizing pneumonia - Eosinophilic pneum - Lipoid pneumonia	

Now it is obvious that some diseases can have more than one pattern.

For instance a **lobar pneumonia** caused by **streptococcus pneumoniae** :

- may become diffuse if the patient does not respond to the treatment.

Other examples are organizing pneumonia (OP) and chronic eosinophilic pneumonia.

These diseases typically present as multifocal consolidations, but sometimes they may become diffuse.

OP is organizing pneumonia. When it is idiopathic it is called cryptogenic (COP). The old name is BOOP - Bronchiolitis Obliterans Organizing Pneumonia.

The new name for BAC - bronchoalveolar carcinoma is adenocarcinoma in situ.

Acute Consolidation

Pulmonary edema
Pneumonia
Aspiration
Hemorrhage / Contusion
Infarction

Chronic Consolidation

Neoplasm • post-obstruction
• Bronch-alveolar ca
• Lymphoma

Sarcoidosis
Organizing Pneumonia
Eosinophilic pneumonia
Alveolar proteinosis

It is very important to differentiate between **acute** consolidation and **chronic** consolidation, because it will limit the differential diagnosis.

In chronic disease we think of:

- Neoplasm with lobar or segmental post-obstructive pneumonia.
- Lung neoplasms like bronchoalveolar carcinoma and lymphoma.
- Chronic post-infection diseases like organizing pneumonia (OP) or chronic eosinophilic pneumonia, which both present with multiple peripheral consolidations.
- Sarcoidosis** is the great mimicker and sometimes the granulomatous noduli are so small and diffuse that they can present as consolidation.

This is known as *alveolar* sarcoidosis.

- Alveolar proteinosis** is a rare chronic disease that is characterized by filling of the alveoli with proteinaceous material.

Lobar consolidation

The most common presentation of consolidation is lobar or segmental.

The most common diagnosis is lobar pneumonia.

The table lists the differential diagnosis.

Lobar consolidation

Lobar pneumonia

- Streptococ pneum.
- Klebsiella
- TB
- Aspiration

Neoplasm

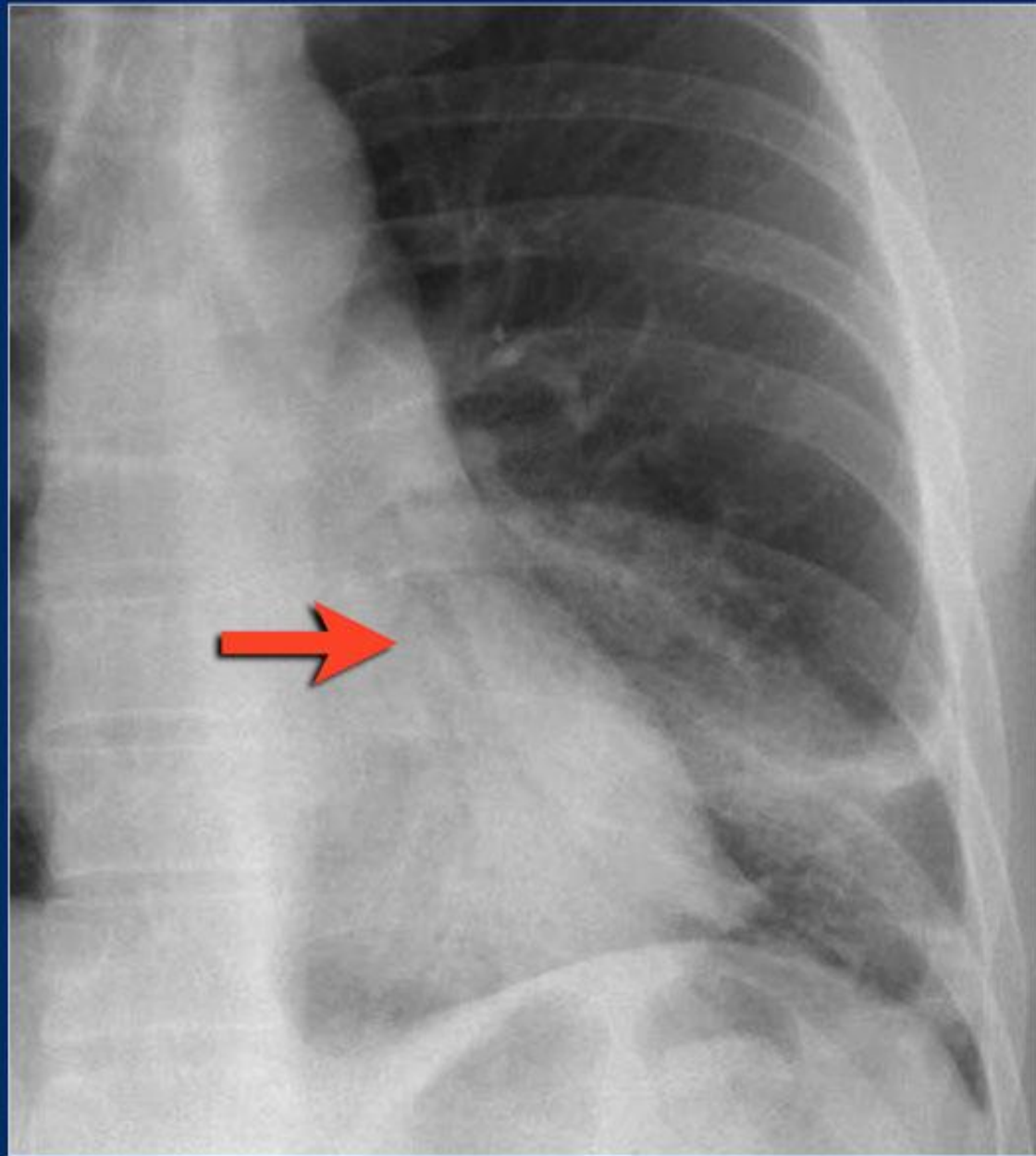
- Lungca with obstructive pneumonia
- BAC
- Lymphoma

Hemorrhage

- Contusion
- Infarction

Others

- Organizing pneumonia
- Eosinophilic pneum.
- Sarcoidosis
- Sequestration
- Mitral regurgitation with RUL edema



- Lobar pneumonia

Here a typical lobar consolidation.

First study the images, then continue reading.

The findings are:

- increased density with ill-defined borders in the left lung
 - the heart silhouette is still visible, which means that the density is in the lower lobe
- air-bronchogram
- Lobar consolidation is the result of disease that starts in the periphery and spreads from one alveolus to another through the pores of Kohn.

At the borders of the disease some alveoli will be involved, while others are not, thus creating ill-defined borders.

As the disease reaches a fissure, this will result in a sharp delineation, since consolidation will not cross a fissure.

As the alveoli that surround the bronchi become more dense, the bronchi will become more visible, resulting in an air-bronchogram (arrow).

In consolidation there should be no or only minimal volume loss, which differentiates consolidation from atelectasis.

Expansion of a consolidated lobe is not so common and is seen in *Klebsiella pneumoniae* and sometimes in *Streptococcus pneumoniae*, TB and lung cancer with obstructive pneumonia.

Lobar pneumonia

On the chest x-ray:

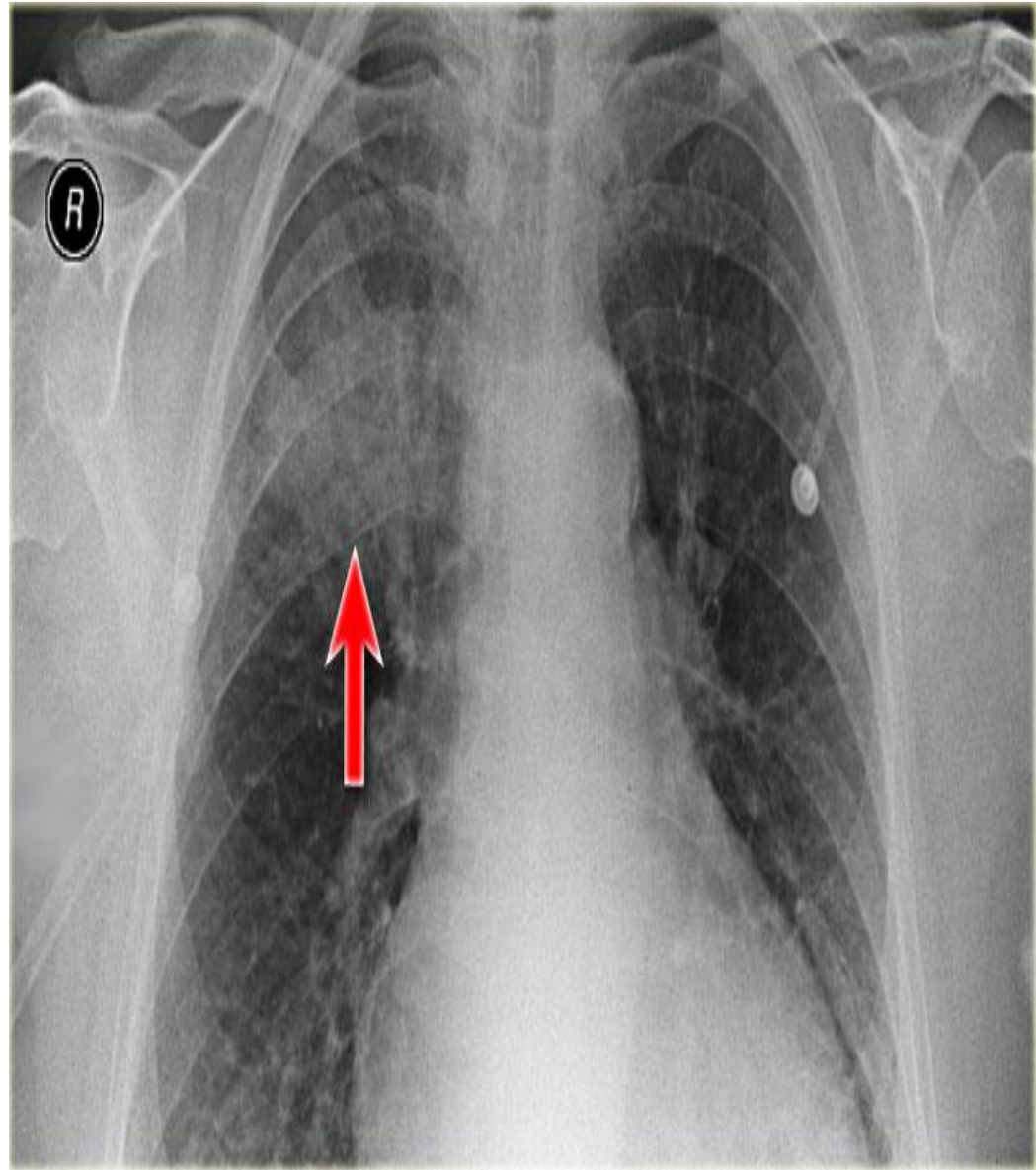
there is an ill-defined area of increased density in the right upper lobe without volume loss.

- The right hilus is in a normal position.
- Notice the air-bronchogram (arrow).

In the proper clinical setting this is most likely a lobar or segmental pneumonia.

* However if this patient had weight loss or long standing symptoms, we would include the list of causes of chronic consolidation.

This was an acute lobar pneumonia caused by *Streptococcus pneumoniae*.



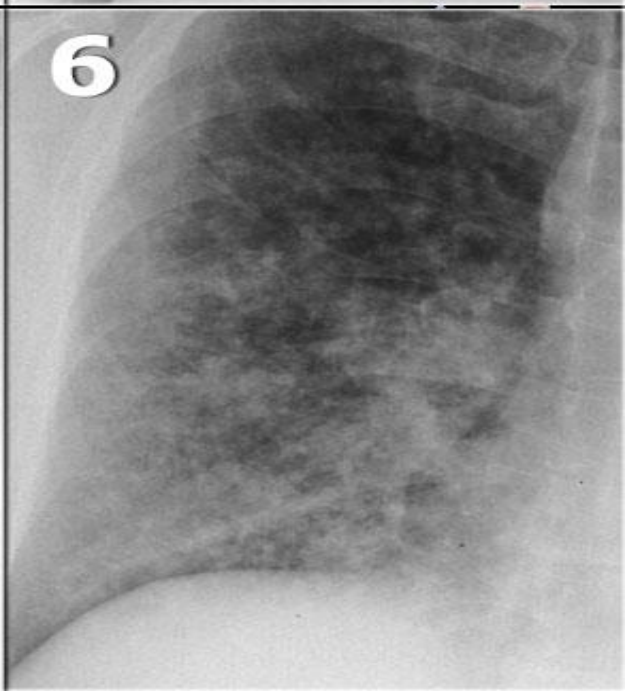
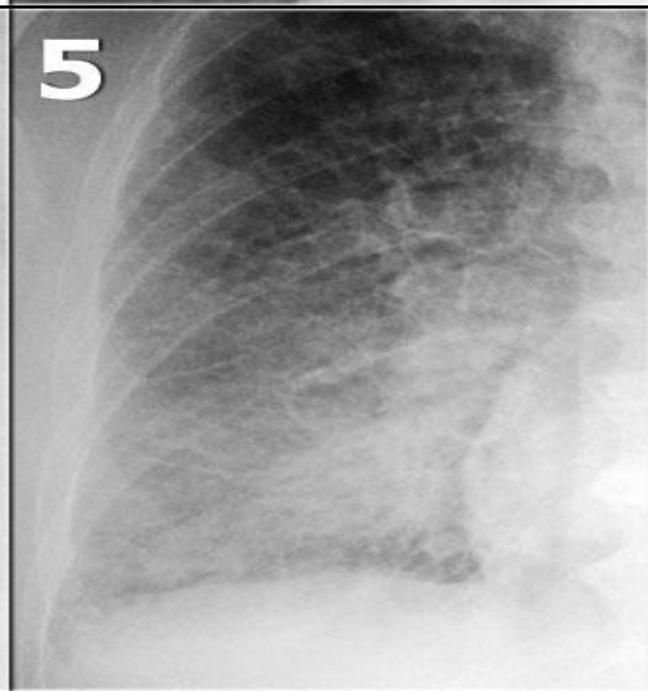
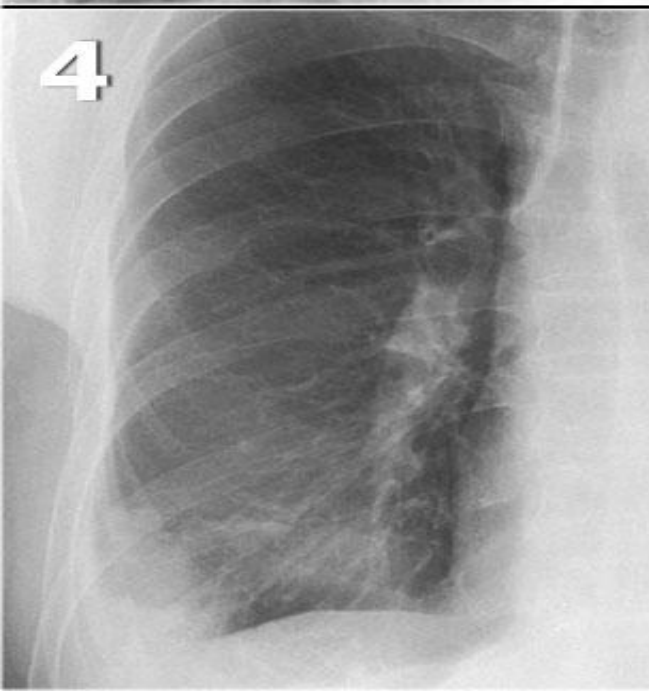
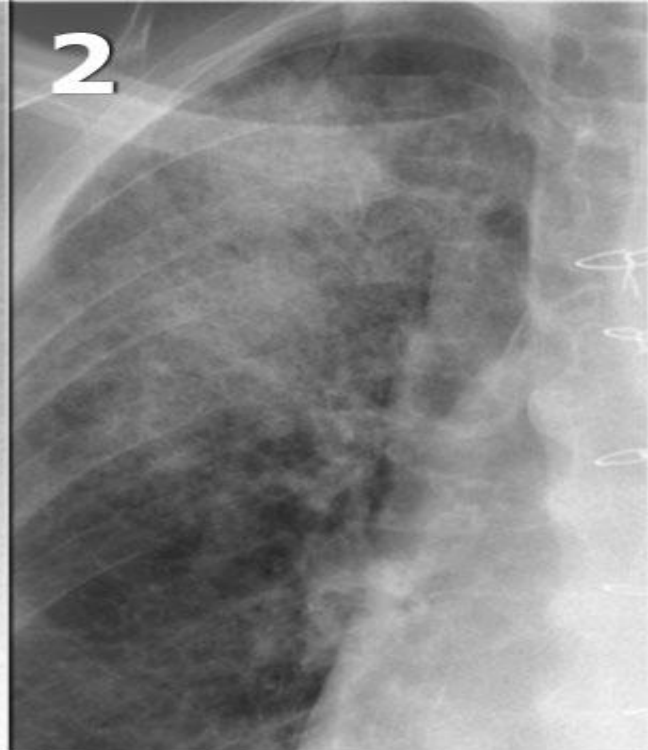
Based on the images alone, it is usually not possible to determine the cause of the consolidation.

Other things need to be considered, like acute or chronic illness, clinical data and other non-pulmonary findings.

Here we have a number of x-rays with consolidation.

Notice the similarity between these chest x-rays.

1. **Lobar pneumonia** - in a patient with cough and fever.
2. **Pulmonary hemorrhage** - in a patient with hemoptoe.
3. **Organizing pneumonia (OP)** - multiple chronic consolidations.
4. Infarction - peripheral consolidation in a patient with acute shortness of breath with low oxygen level and high D-dimer.
5. **Pulmonary cardiogenic edema** - filling of the alveoli with transudate in a patient with congestive heart failure. This would be more obvious if you were shown the whole image.
6. **Sarcoidosis** - at first glance this looks like consolidation, but in fact this is nodular interstitial lung disease, that is so wide-spread that it looks like consolidation.



Hemorrhage

In this case there was a solitary nodule in the right upper lobe and a biopsy was performed.

The lobar consolidation is the result of hemorrhage as a complication of the procedure.

Hemorrhage is seen in:

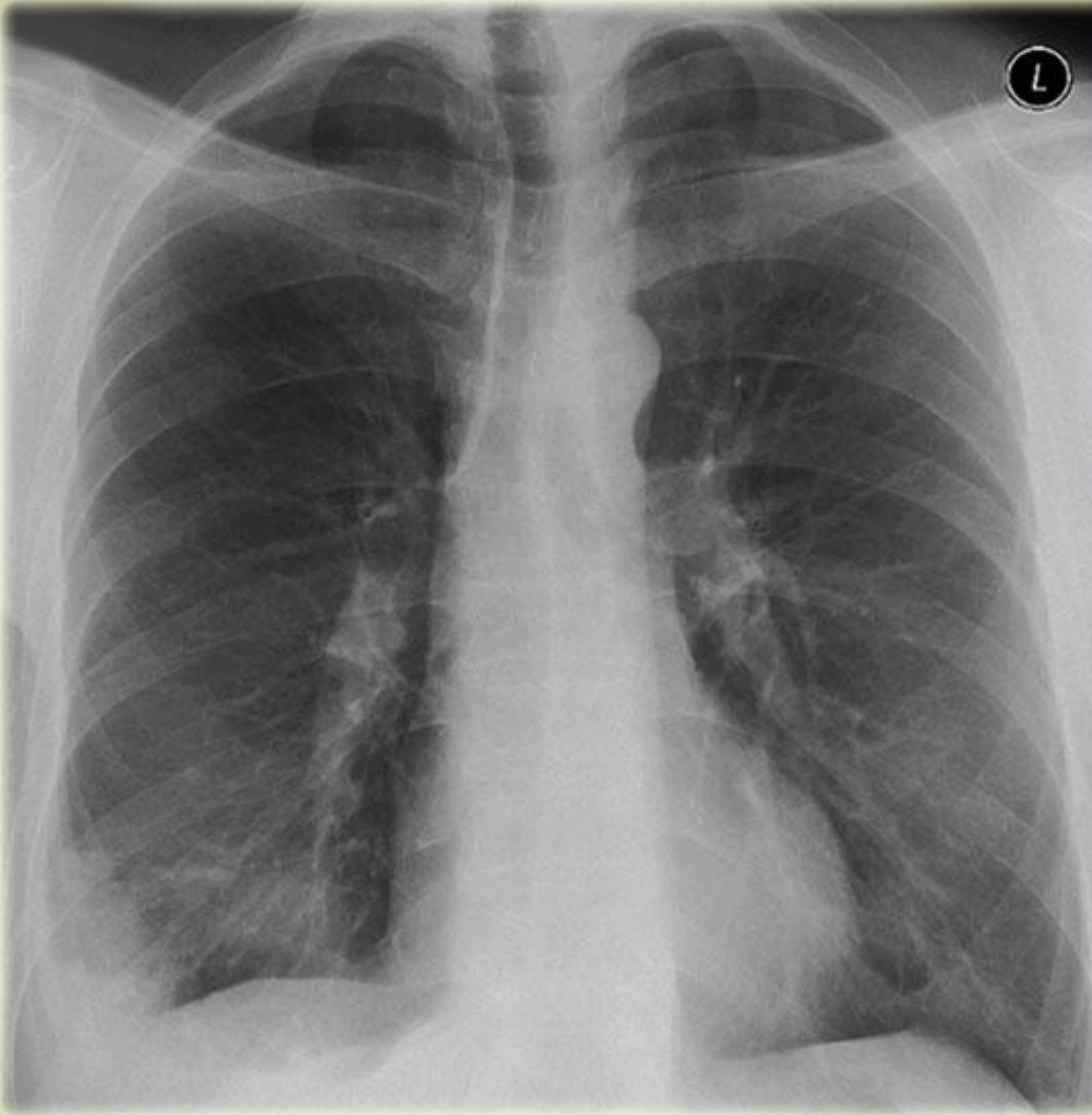
- Pulmonary contusion
- Pulmonary infarction
- Bleeding disorders: leukemia, anticoagulation therapy, diffuse intravascular coagulation.
- Vasculitis: SLE, Goodpasture's, Wegener's



- Hemorrhage post-biopsy

Lung infarction

- The radiographic features of **acute pulmonary thromboembolism** are insensitive and nonspecific.
- The most common radiographic findings in the Prospective Investigation of Pulmonary Embolism Diagnosis (PIOPED) study were atelectasis and patchy pulmonary opacity.
- In most cases of pulmonary emboli the chest x-ray is normal.
- In next cxr The peripheral consolidation is seen in the region of the emboli and can be attributed to hemorrhage in the infarcted area.



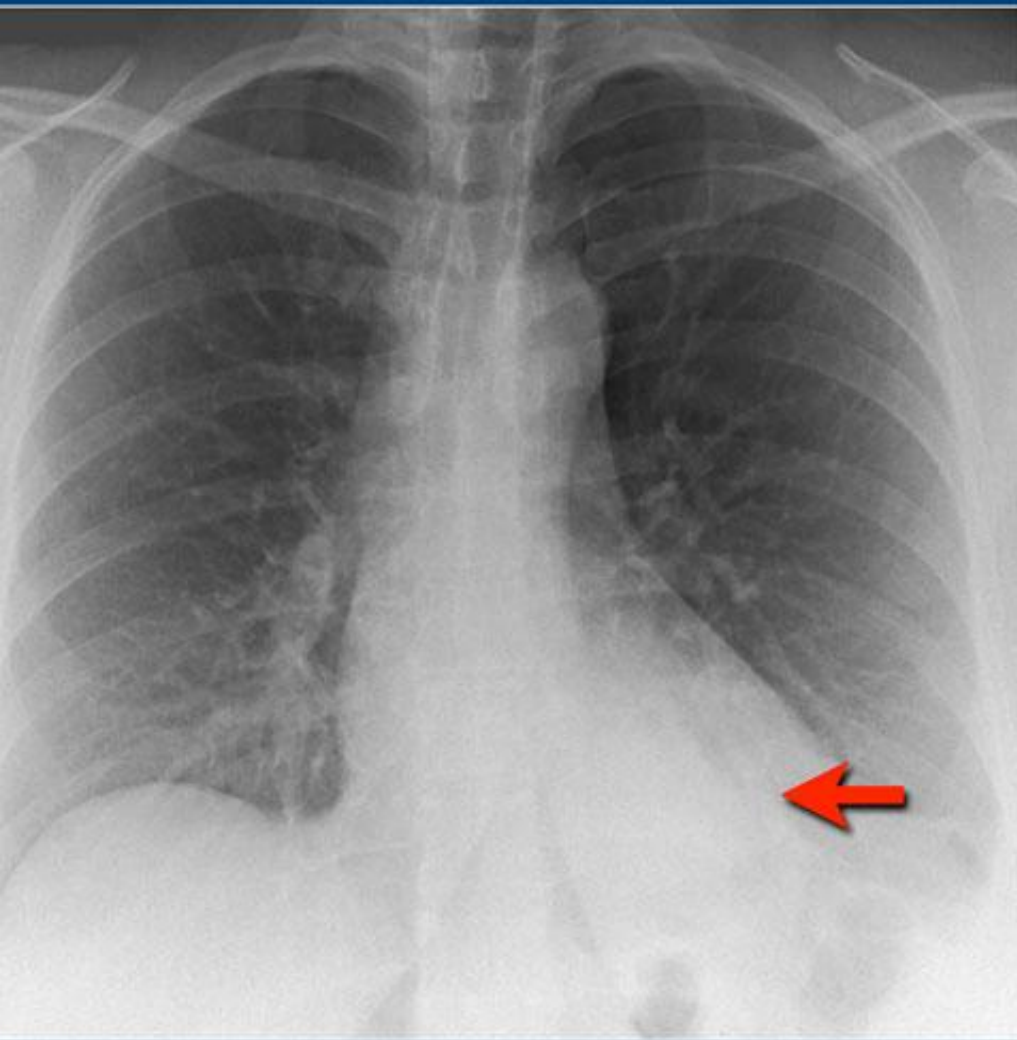
Lung infarction due to pulmonary emboli

Pulmonary sequestration

- This is an uncommon cause of lobar consolidation.
- It is a congenital abnormality.
- A nonfunctioning part of the lung lacks communication with the bronchial tree and receives arterial blood supply from the systemic circulation.

Patients present with recurrent infection when bacteria migrate through the **pores of Kohn**.

Notice the **feeding artery**, that branches off from the aorta (blue arrow).



- Pulmonary sequestration

Diffuse consolidation

- The most common cause of diffuse consolidation is pulmonary edema due to heart failure.
- This is also called cardiogenic edema, to differentiate it from the various causes of non-cardiogenic edema.
- The increased heart size is usually what distinguishes between cardiogenic and non-cardiogenic.
- Look for other signs of heart failure like redistribution of pulmonary blood flow, Kerley B-lines and pleural fluid.

However some patients, who have an acute cardiac infarction, may still have a normal heart size, while other patients who have a large heart due to a chronic heart disease, may have non-cardiac pulmonary edema due to a superimposed pulmonary infection, ARDS, near-drowning etc.

Diffuse consolidation

Edema

- Heart failure
- Volume overload
- ARDS
- Low albumin
- Renal failure
- Transfusion reaction

Bronchopneumonia

- Staph Aureus
- Gram negative
- PCP
- Viral - fungal

Hemorrhage

- SLE
- Henoch-Schönlein
- Wegener
- Goodpasture

-Others

- BAC
- Lymphoma
- Organizing pneumonia
- Eosinophilic pneumonia
- Hypersensitivity pneum

Congestive heart failure

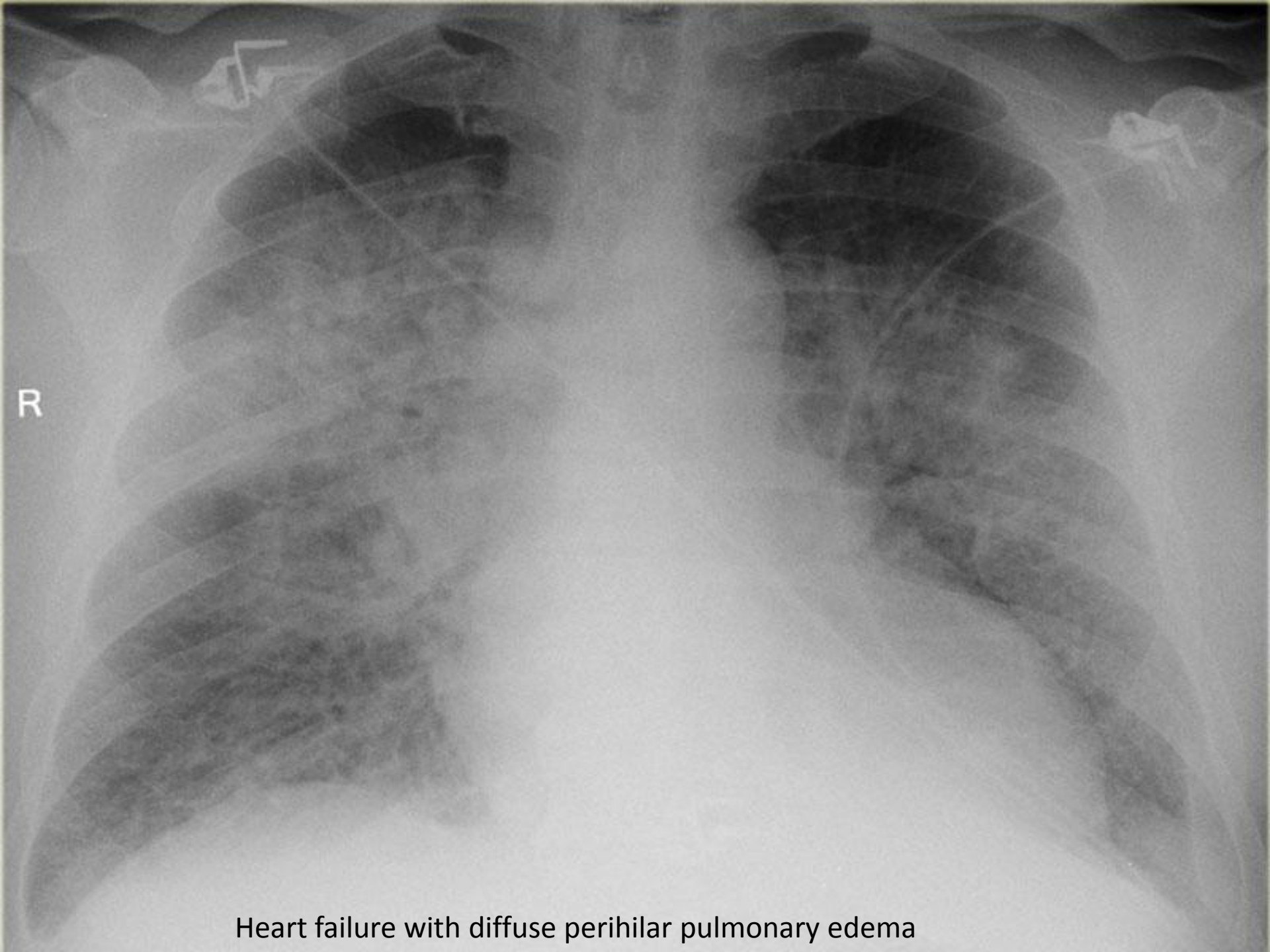
First study the images, then continue reading.

The findings are:

- **bilateral perihilar consolidation** with air bronchograms and ill-defined borders
- an increased heart size
- subtle interstitial markings
- probably a large [vascular pedicle](#)

All these findings indicate, that we are dealing with pulmonary edema due to heart failure.

You probably would like to look at old films to see if there are any changes.



Heart failure with diffuse perihilar pulmonary edema

Diffuse consolidation in bronchopneumonia

Here another case of diffuse consolidation.

This patient had fever and cough.

This was thought to be a diffuse bronchopneumonia.

Unlike lobar pneumonia, which starts in the alveoli, bronchopneumonia starts in the airways as acute bronchitis.

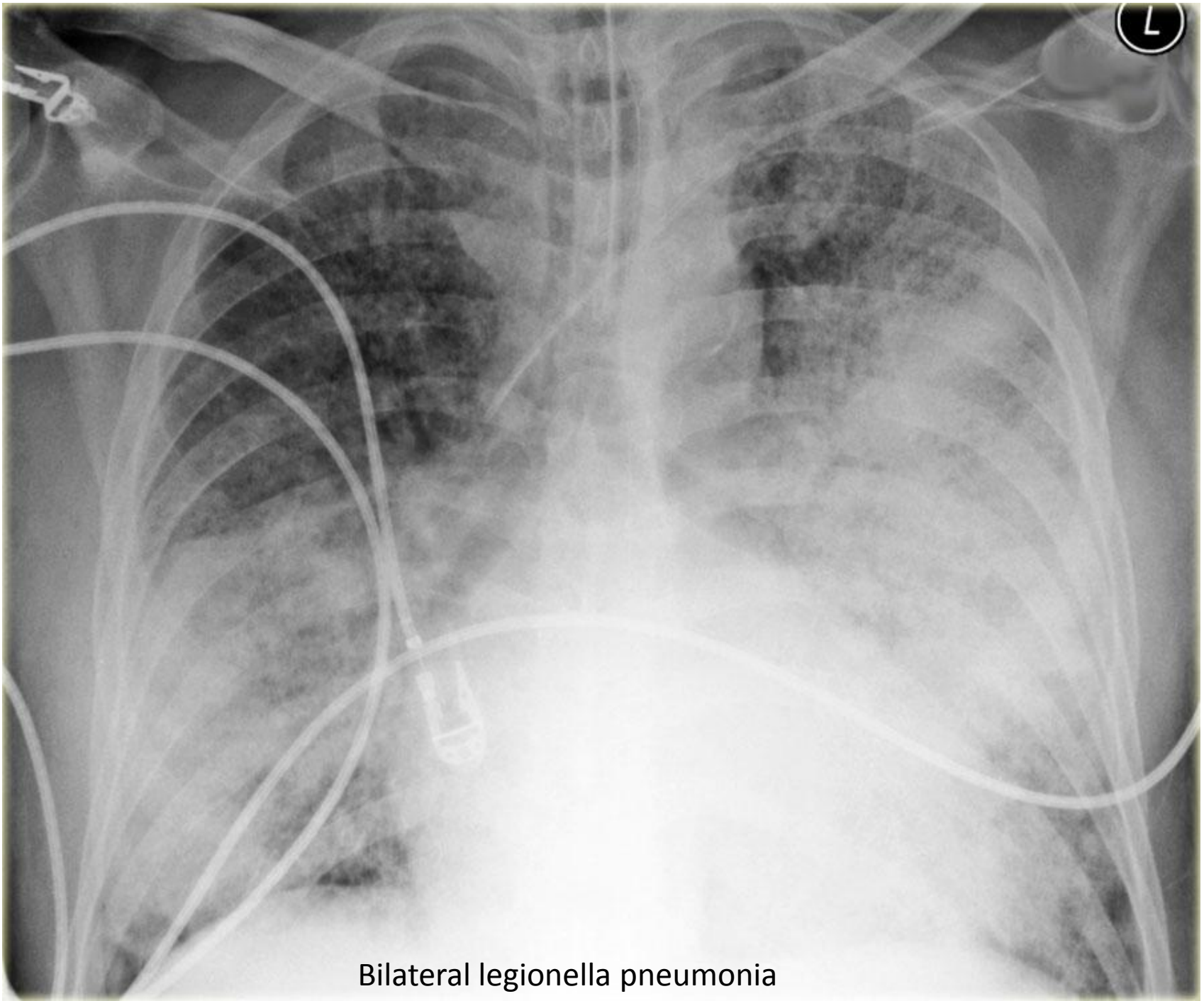
It will lead to multifocal ill-defined densities.

When it progresses it can produce diffuse consolidation.

The disease does not cross the fissures, but usually starts in multiple segments.

Bronchopneumonia can be caused by many micro-organisms.

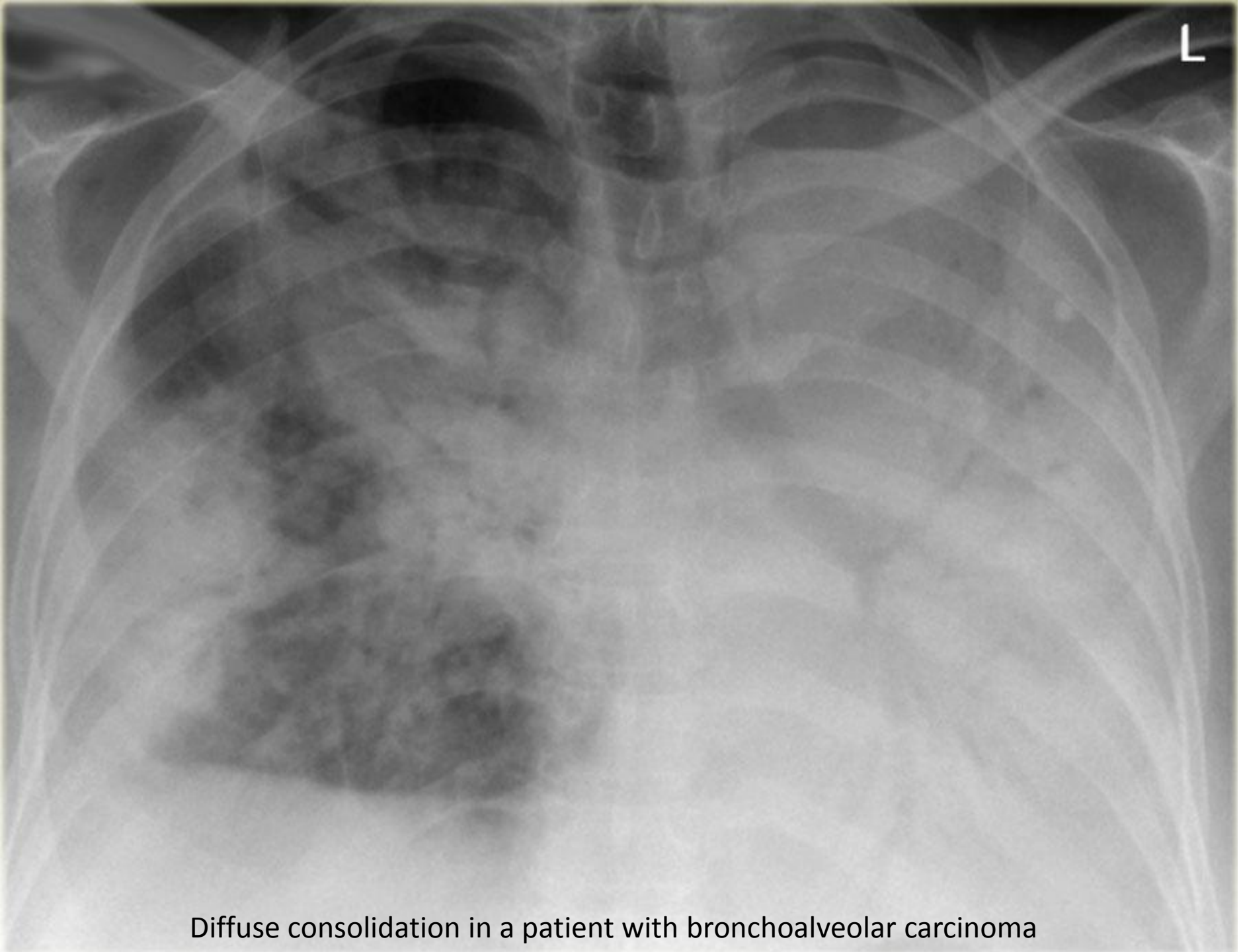
This proved to be legionella pneumonia.



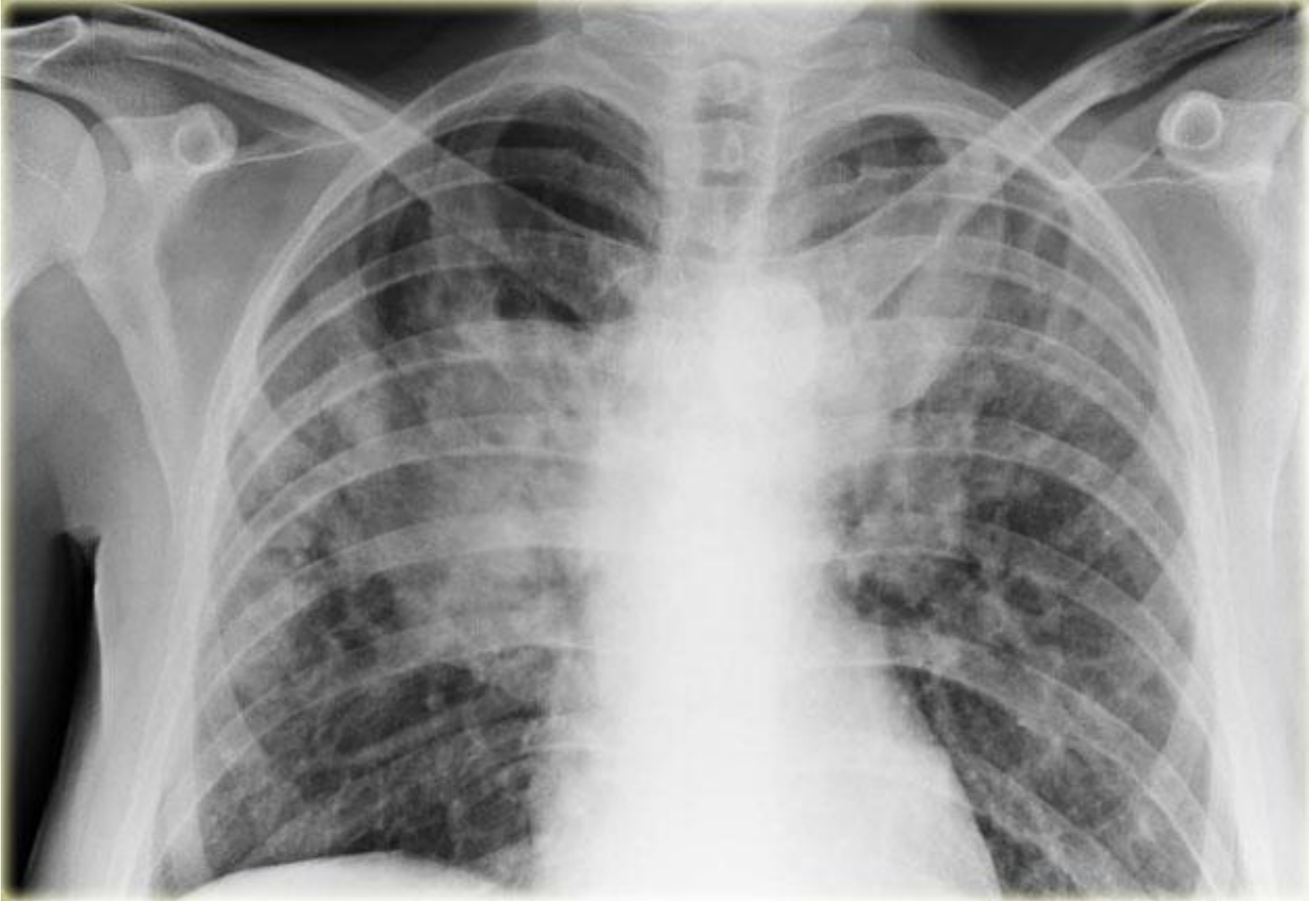
Bilateral legionella pneumonia

Diffuse consolidation in bronchoalveolar carcinoma

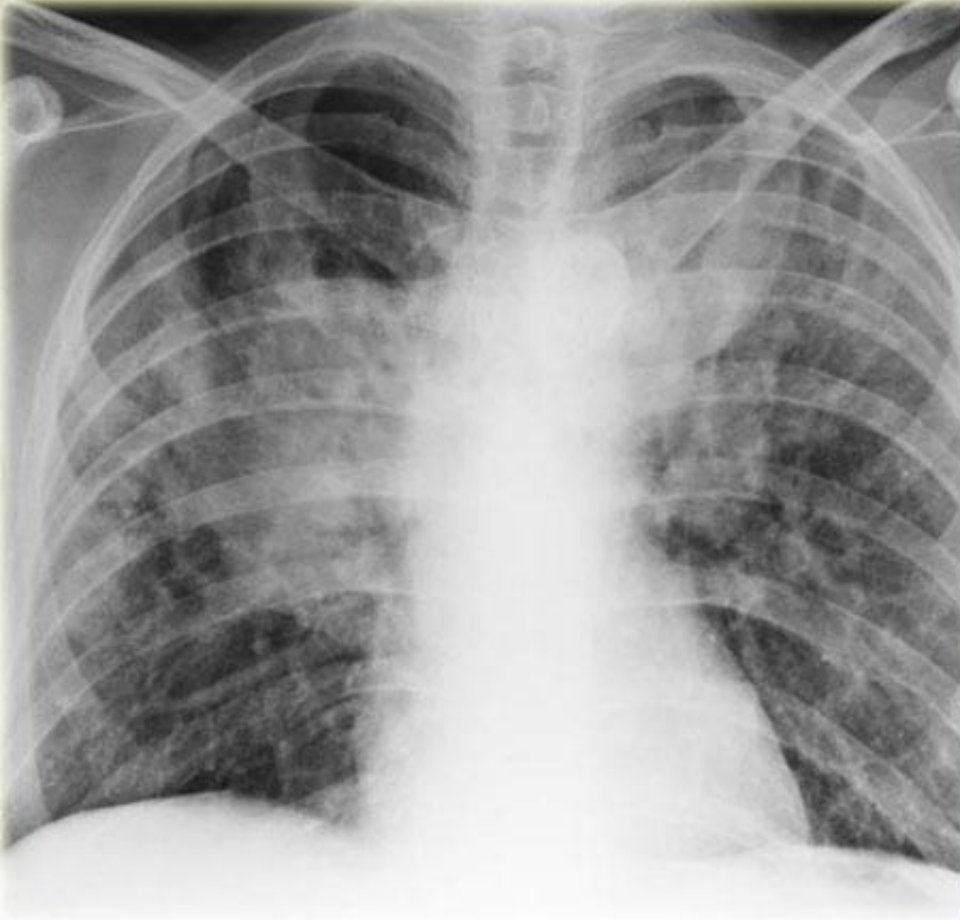
- The chest x-ray shows diffuse consolidation with 'white out' of the left lung with an air-bronchogram.
- This patient had a chronic disease with progressive consolidation. The disease started as a persistent consolidation in the left lung and finally spread to the right lung.
- Final diagnosis: **bronchoalveolar carcinoma.**



Diffuse consolidation in a patient with bronchoalveolar carcinoma



- This is a difficult case.
It demonstrates, that based on the x-ray alone, it is not certain which pattern we are looking at.
Are these densities masses or consolidation?
- Continue with the CT.



- The CT-image is not very helpful in the differentiation.
There are hypodense areas, which could be masses.
On the other hand this also could be areas of consolidation with hypodense areas due to necrosis.
- Finally the **diagnosis non Hodgkin's disease** was made based on biopsy.

Batwing

Pulmonary edema
Bronchopneumonia
PCP
Viral pneumonia

Reversed Batwing

Bronchoalveolar carcinoma
Sarcoidosis
Organizing Pneumonia
Eosinophilic pneumonia

Batwing

A bilateral perihilar distribution of consolidation is also called a Batwing distribution. The sparing of the periphery of the lung is attributed to a better lymphatic drainage in this area.

It is most typical of pulmonary edema, both cardiogenic and non-cardiogenic. Sometimes it is seen in pneumonias.

Reverse Batwing

Peripheral or subpleural consolidation is called reverse Batwing distribution. It is frequently seen in **chronic lung disease**.

Multifocal

- Multifocal consolidations are also described as multifocal ill-defined opacities or densities.
- In most cases these are the result of airspace-consolidations due to bronchopneumonia.
- As mentioned before **bronchopneumonia** starts in the bronchi and then spreads into the lungparenchyma.

This can lead to segmental, diffuse or multifocal ill-defined densities. In some cases however the underlying pathology of multiple ill-defined densities is interstitial disease, like in the alveolar form of sarcoidosis in which the granulomas are very small and fill up the alveoli.

Multiple ill-defined

Bronchopneumonia

- Staph Aureus
- Legionella
- Gram negative
- Streptococcus pneumonia
- Klebsiella
- Pseudomonas
- Anaerobe
- PCP
- TB

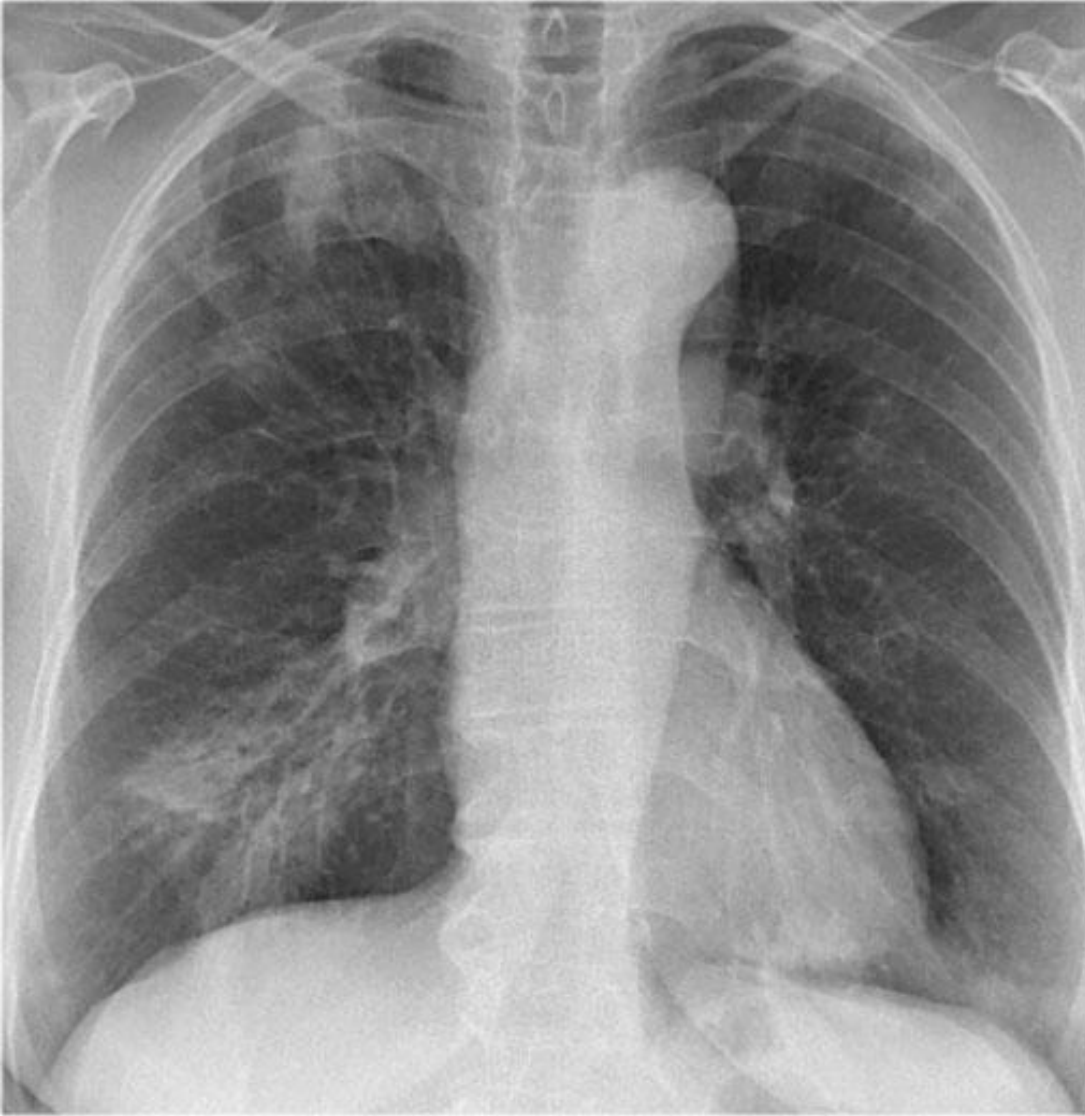
Vascular

- Septic emboli
- Wegener's

Neoplasm

- BAC
- Lymphoma
- Metastases

Sarcoidosis



- What do you see ?

- Notice that there are multiple densities in both lungs.

The larger ones are ill-defined and maybe there is an air-bronchogram in the right lower lobe.

- Probably we are dealing with multifocal consolidations, but one might also consider the possibility of multiple ill-defined masses.

There is a peripheral distribution.

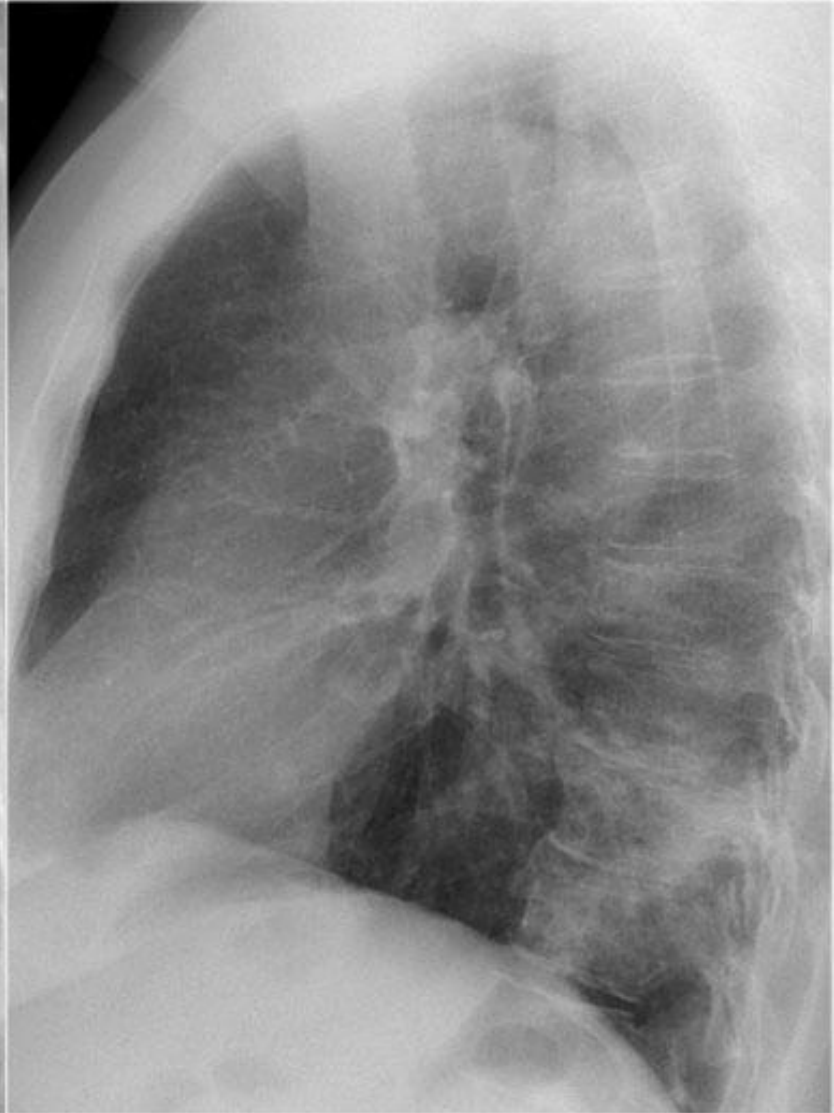
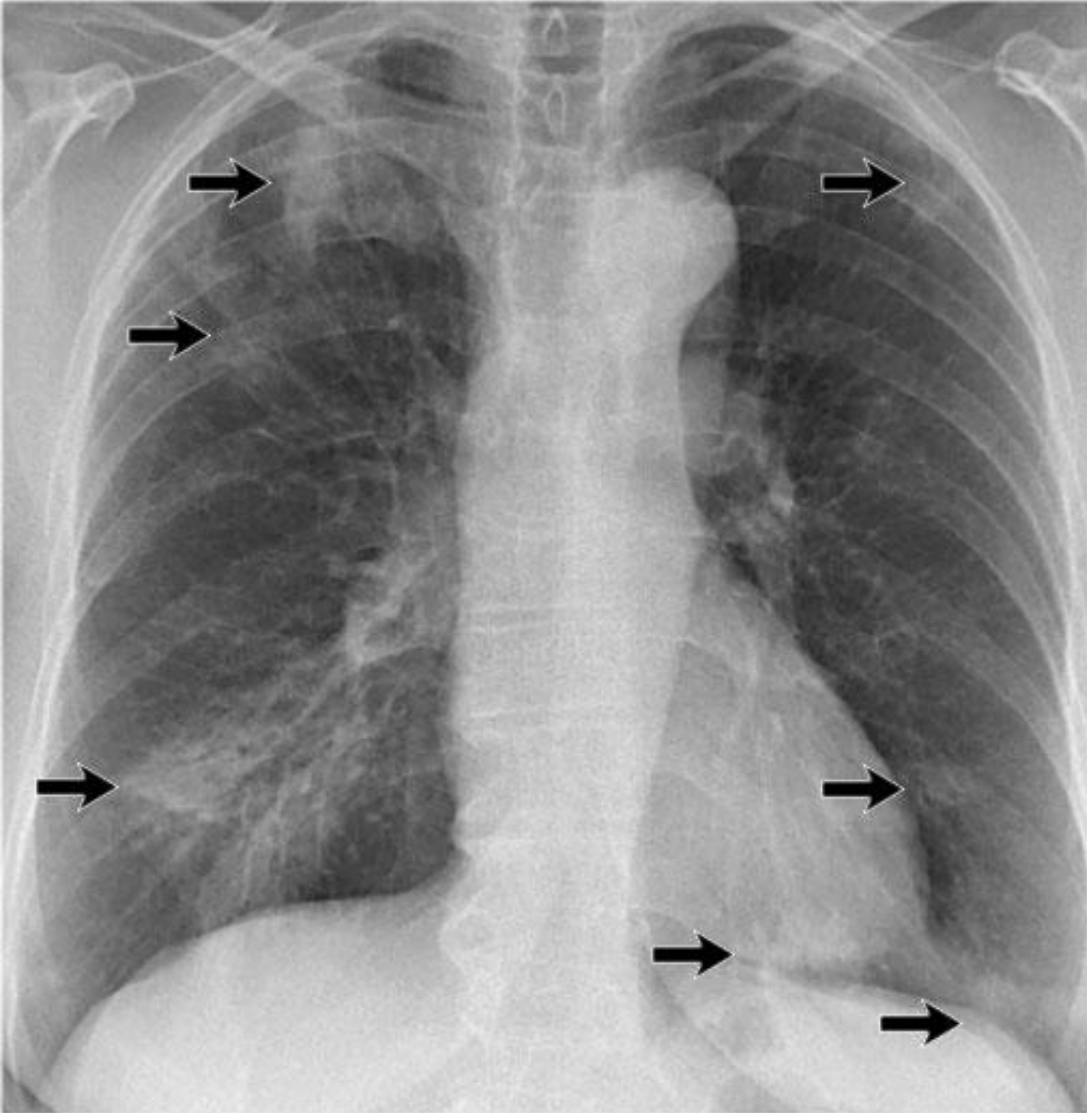
- This patient had a several month history of chronic non-productive cough, that did not respond to antibiotics.

So we are dealing with the differential diagnosis of chronic consolidation.

The lab-findings were normal which makes bronchoalveolar carcinoma and lymphoma less likely.

There was no eosinophilia, which excludes eosinophilic pneumonia.

Biopsy revealed the diagnosis of organizing pneumonia (OP) also known as BOOP.



Wegener's granulomatosis

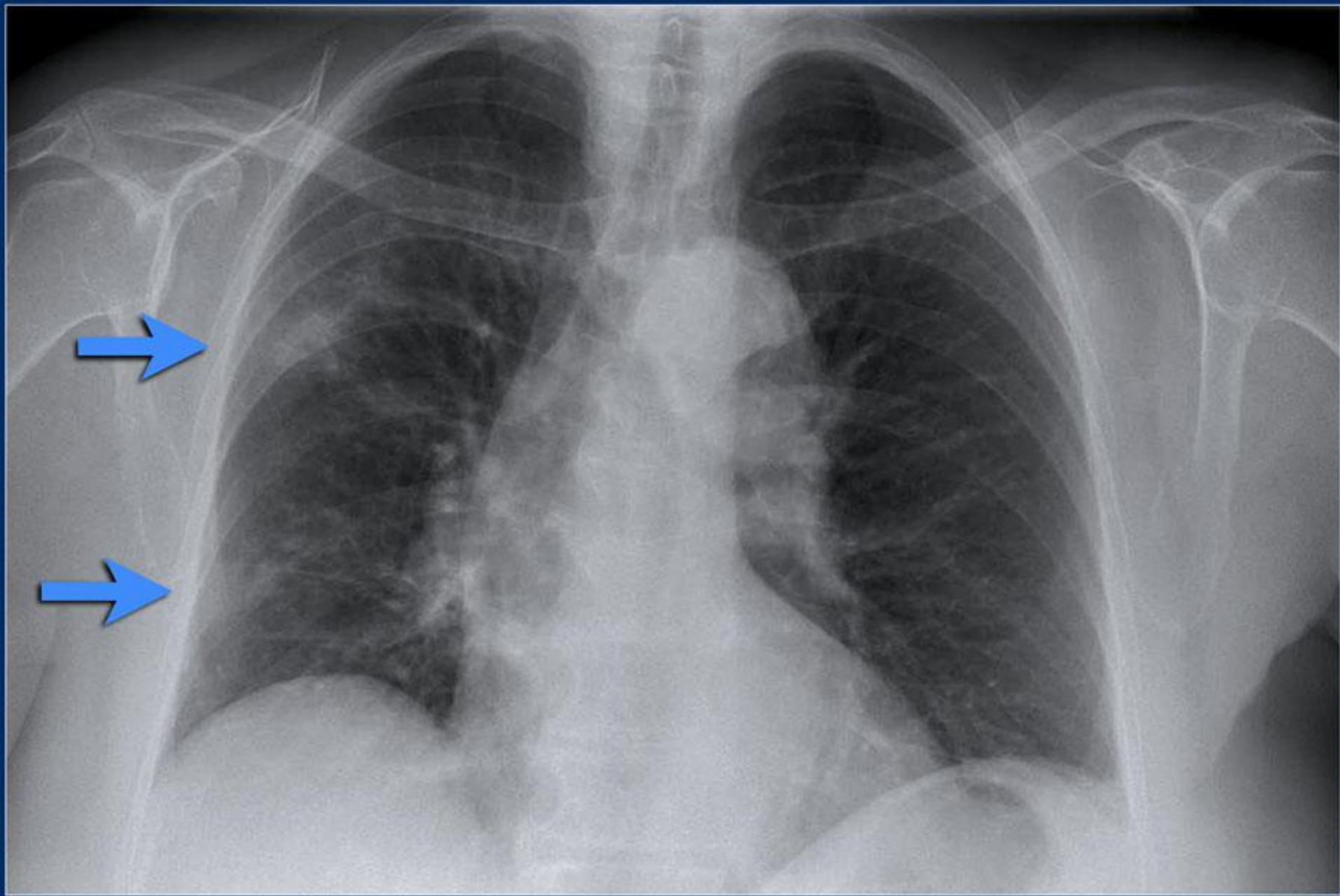
Wegener's is a collagen vascular disease with vasculitis involving the lung, kidney and sinuses.

- In the lung the vasculitis causes infarcts which first present as ill-defined areas of consolidation.

In a later stage these infarcts become more circumscribed and can be seen as multiple nodules or masses, sometimes with cavitation.

Here a patient with non-specific findings.

There are ill-defined densities in the right lung, which proved to be a manifestation of Wegener's.



Interstitial disease

Reticular pattern

- **Edema**
heart failure - fluid overload
- **Interstitial pneumonia**
Viral - PCP - Mycoplasma
- **Lymphangitis ca**
- **UIP** - *honeycombing*
IPF - RA - Systemic sclerosis
Dermatomyositis
- **Drugs**
Nitrofurantoin - Methotrexate -
Amiodarone
Busulfan - Bleomycin - Cytosine

Cystic disease

- Langerhans cell histiocytosis
- Lymphangioleiomyomatosis
- Pneumatoceles
- Lymphocytic Interstitial Pneumonia

Fine nodular

- Sarcoidosis
- Metastases
- TB

Differential diagnosis on HRCT

Most of our knowledge about imaging findings in interstitial lung disease comes from **HRCT**.

On HRCT there are four patterns: reticular, nodular, high and low attenuation (table).

On a Chest X-Ray it can be very difficult to determine whether there is interstitial lung disease and what kind of pattern we are dealing with.

On a CXR the most common pattern is *reticular*.

The *ground-glass pattern* is frequently not detected on a chest x-ray.

The *cystic pattern* is also difficult to appreciate on a chest x-ray.

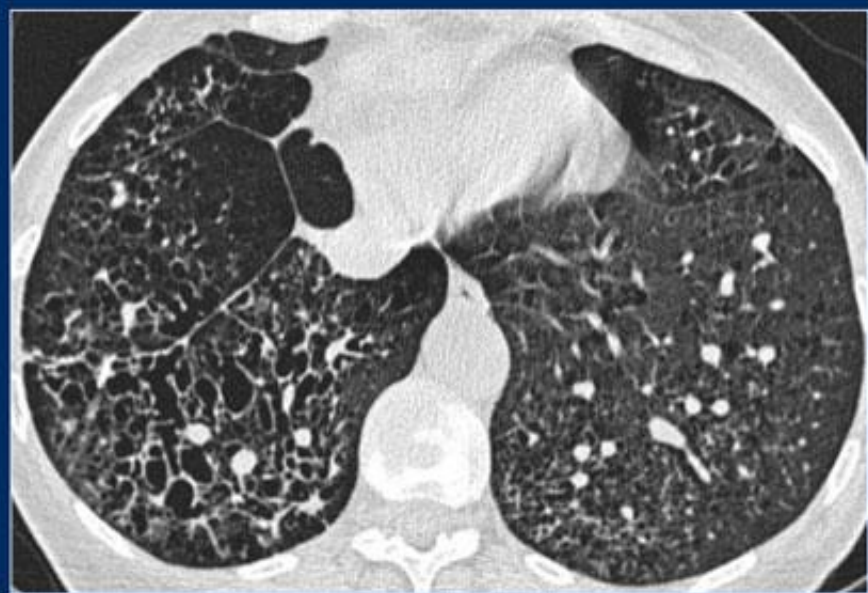
When the cysts have thick walls like in Langerhans cell histiocytosis or honeycombing, it frequently presents as a reticular pattern on a CXR.

However sometimes an interstitial pattern can be seen and in many cases UIP can be suspected based on the x-ray findings.

Cystic versus Reticular

- It can be difficult to determine whether we are dealing with a reticular pattern or a cystic pattern.
- The CXR is of a patient with Langerhans cell histiocytosis (LCH).
LCH is called a cystic disease.
- On the CXR it is difficult to see if this is a cystic or a reticular pattern.
In many of such cases a HRCT will give you more information.
This problem is also seen in patients with UIP.
- One of the prominent findings in UIP is honeycombing.

This creates a reticular pattern on the chest x-ray, because the cysts in honeycombing have thick walls.
We will show a case in a moment.



Reticular pattern in Congestive heart failure

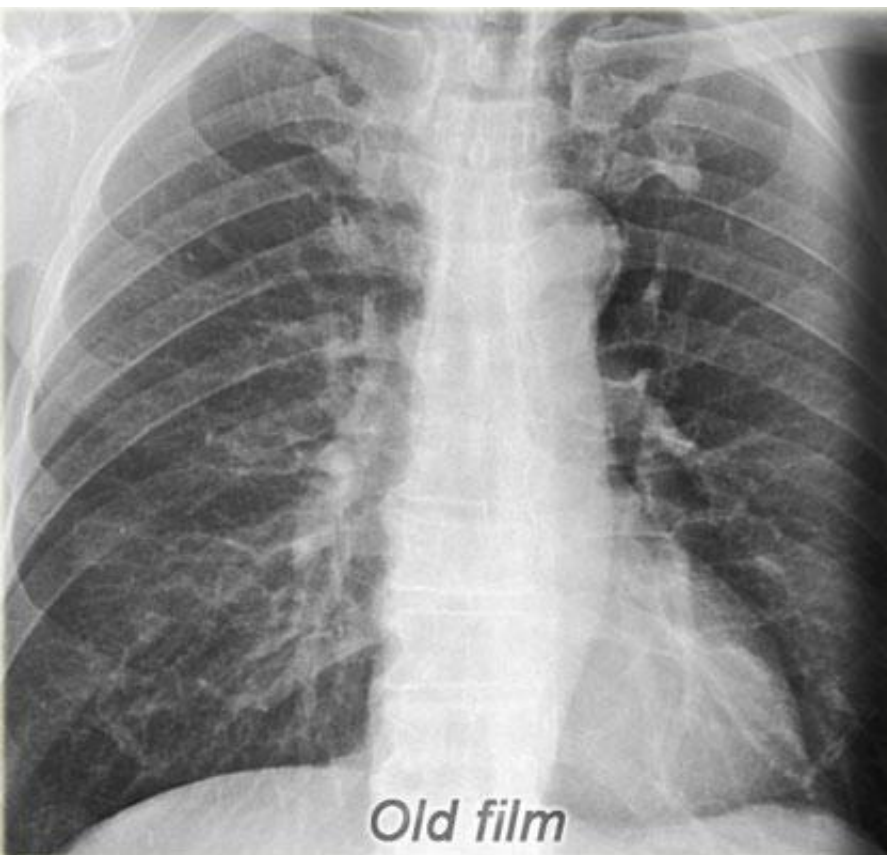
Study the images and then continue reading.

The findings are:

- Normal old film on the left.
- Reticular pattern especially in the basal parts of the lung. Some Kerley B lines are seen.
- Increased heart size.
- Pleural fluid seen on the left side.
- Pulmonary vessels are somewhat more prominent compared to the old film.

Based on these findings we can conclude that we are dealing with congestive heart failure.

This is the most common interstitial pattern on a CXR.



- Interstitial edema usually presents as reticulation.

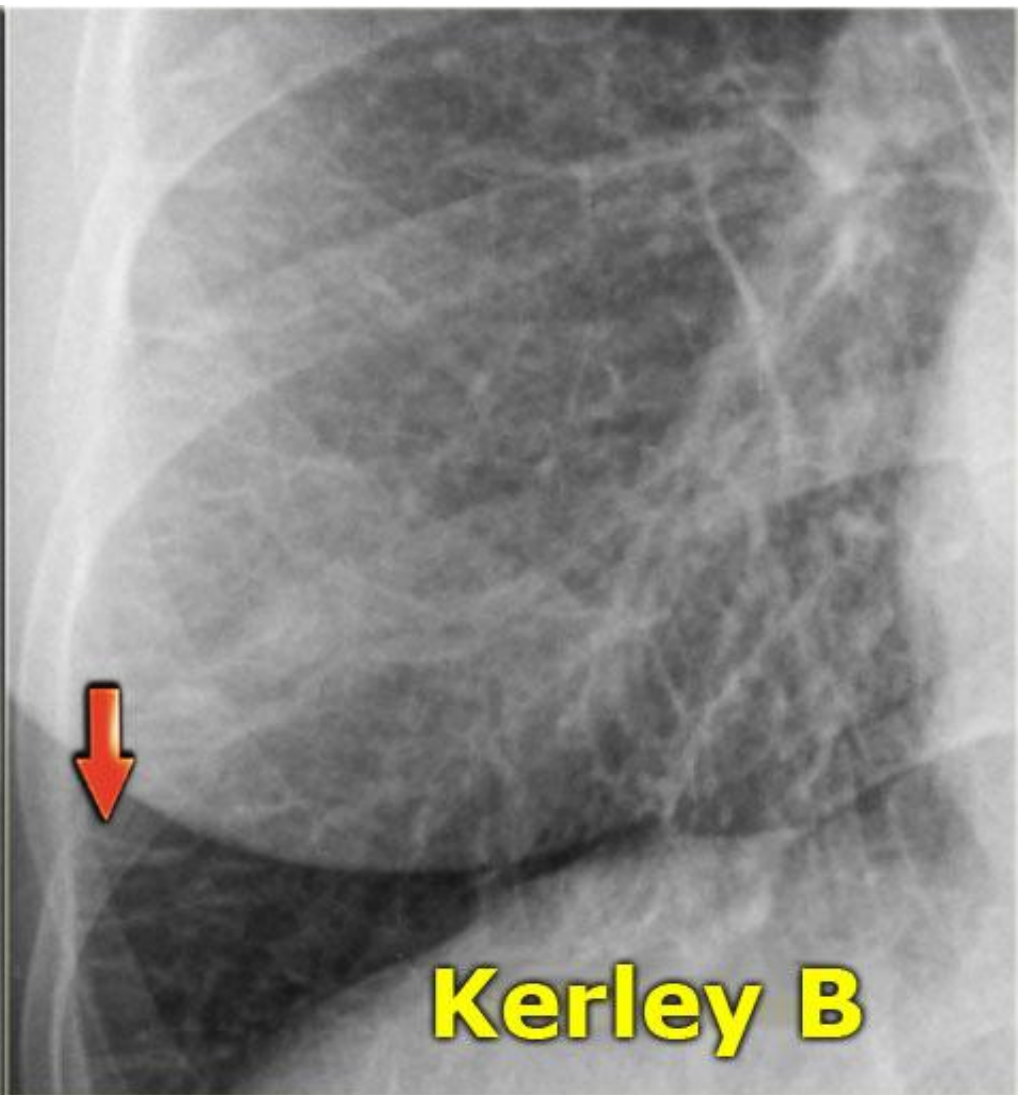
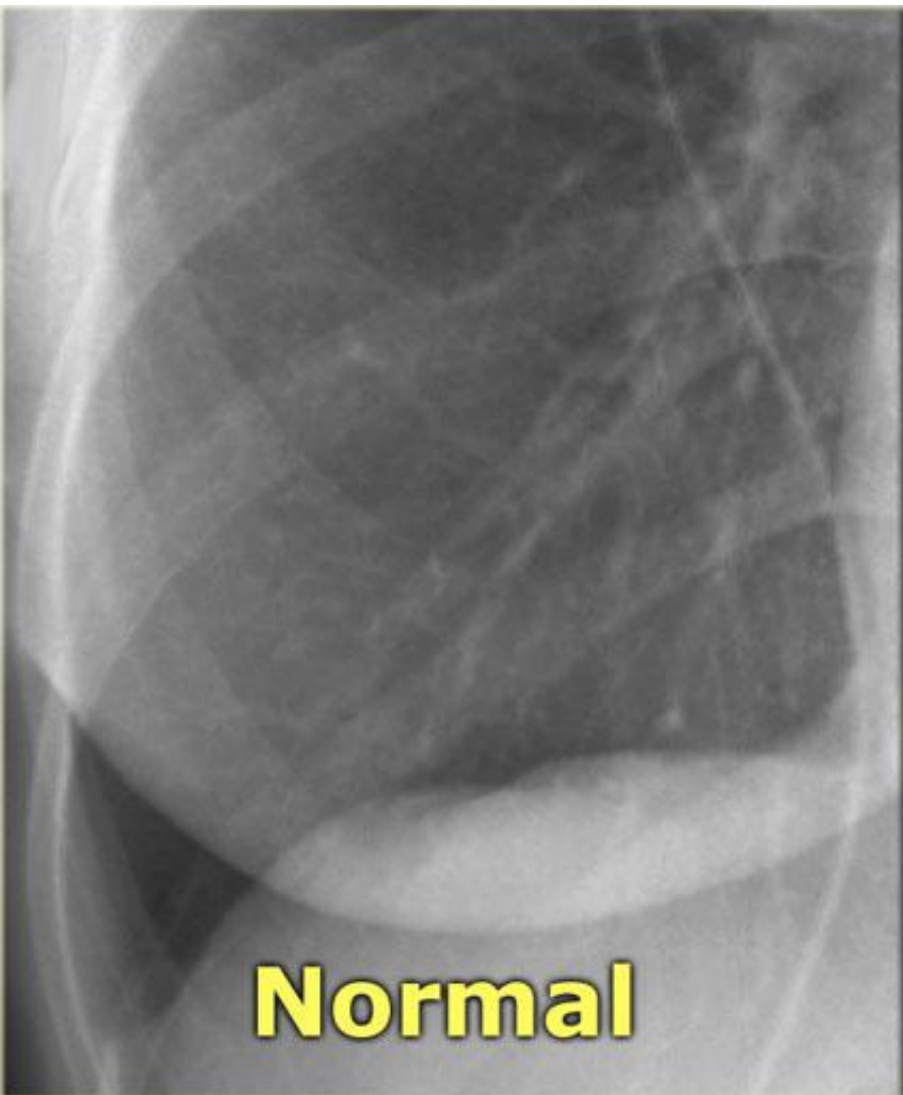
Sometimes **Kerley B** lines are visible.

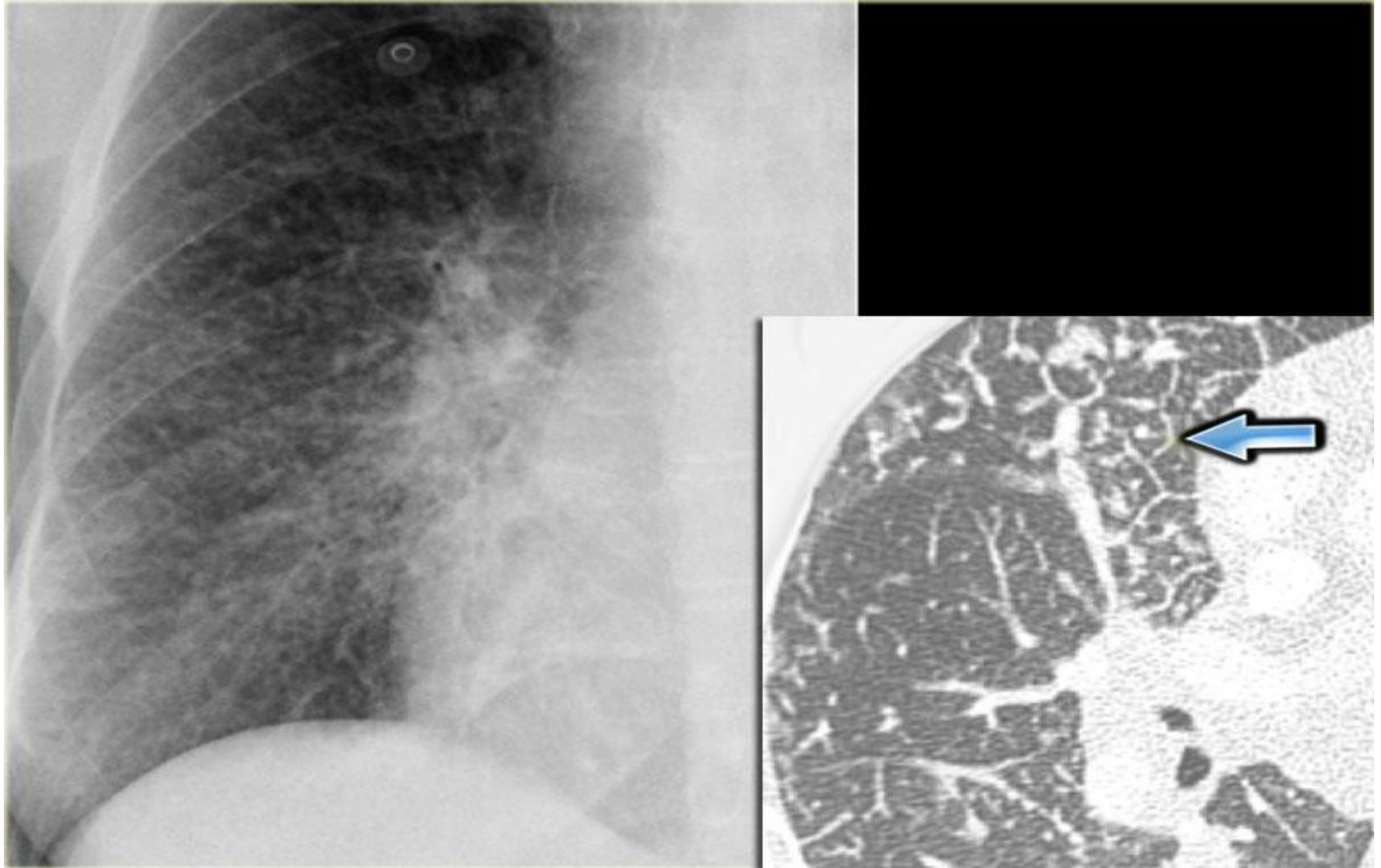
Here another example.

Kerley B lines are 1-2 cm long horizontal lines near the lateral pleura.

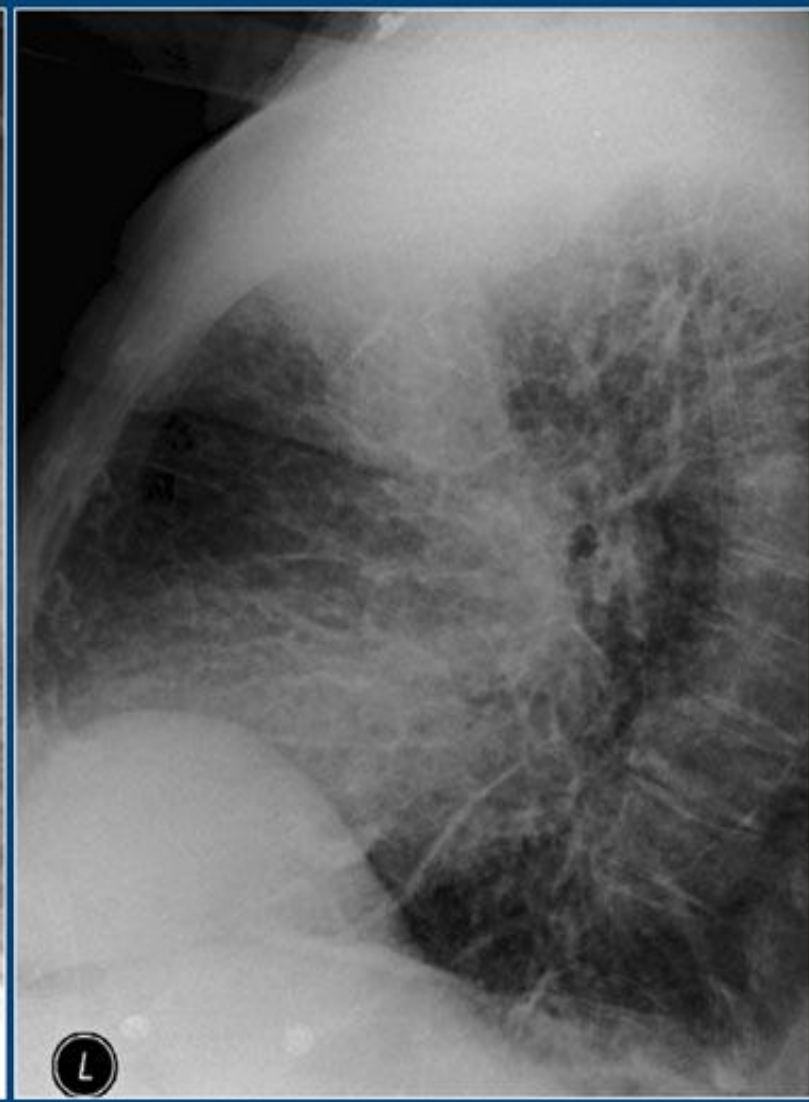
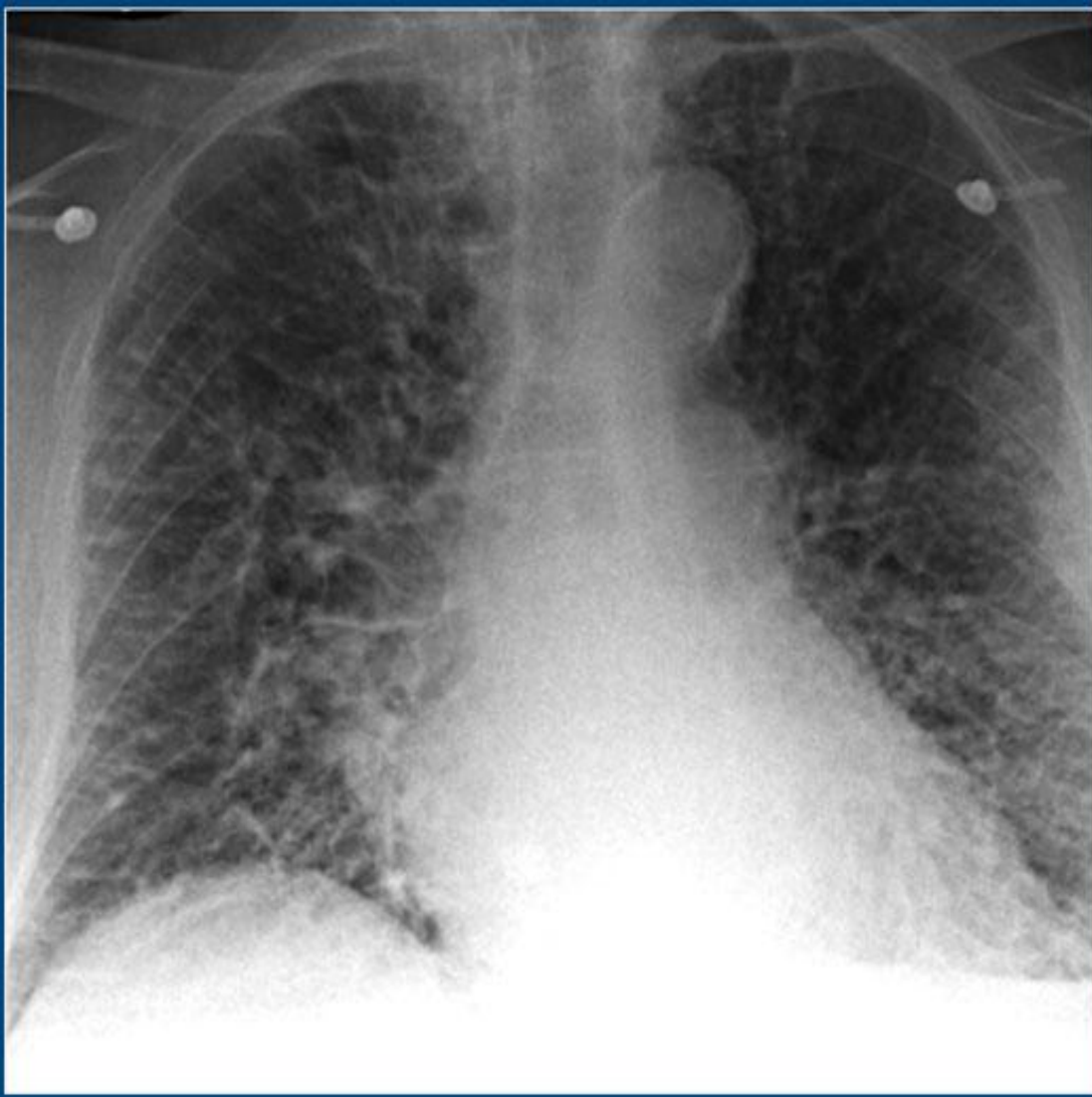
* The main differential diagnosis of Kerley B lines is:

1. **interstitial edema in heart failure**
2. **lymphangitis carcinomatosa**

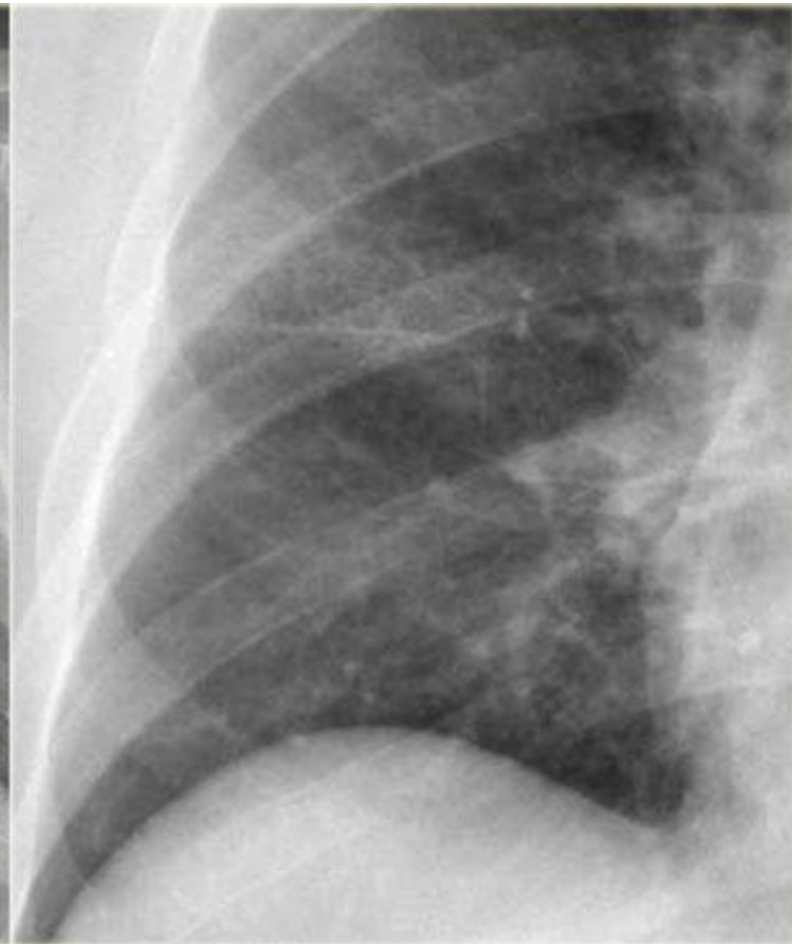




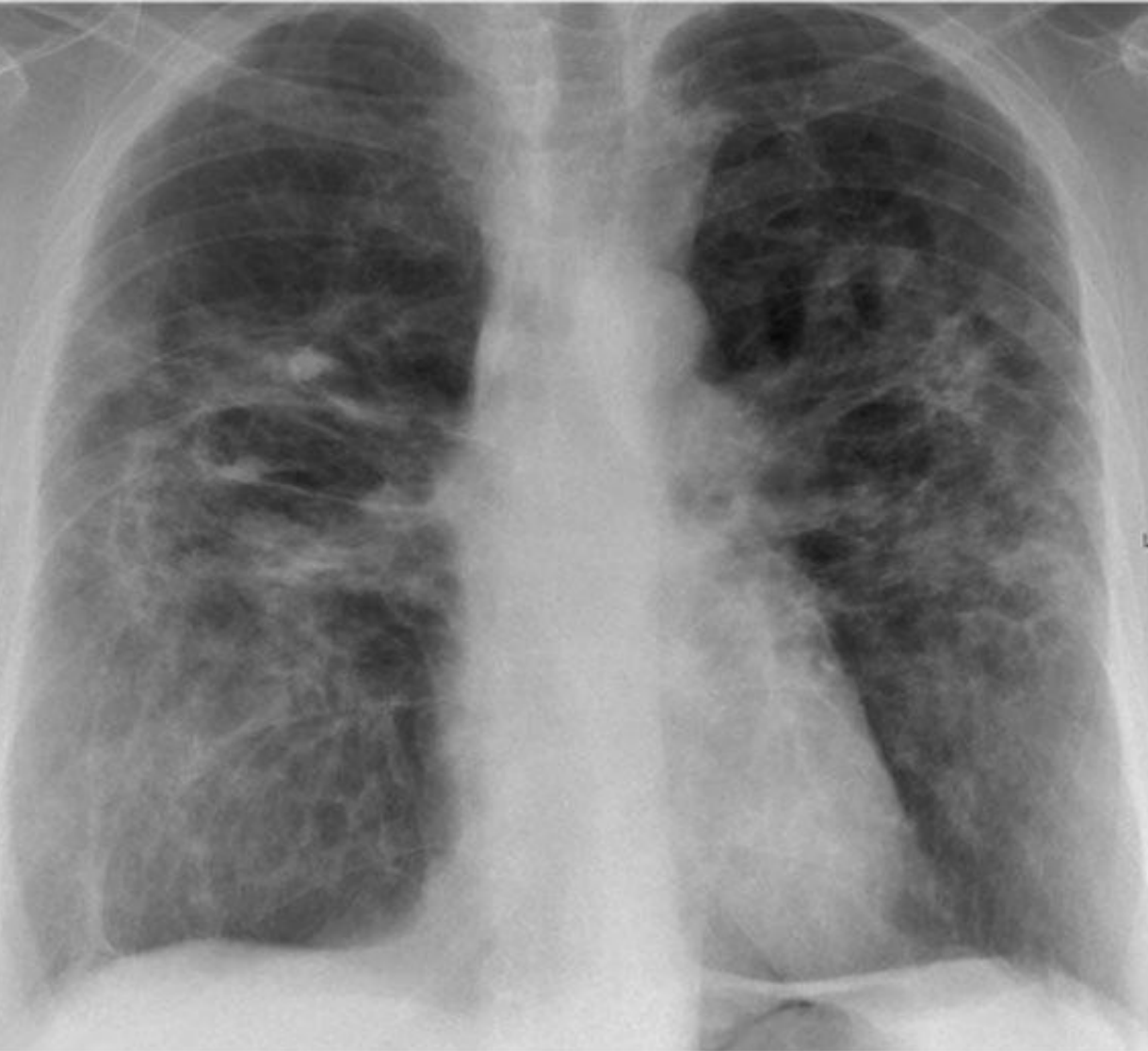
- Here another chest x-ray with interstitial edema and Kerley B lines in a patient with congestive heart failure. The CT shows the septal thickening.



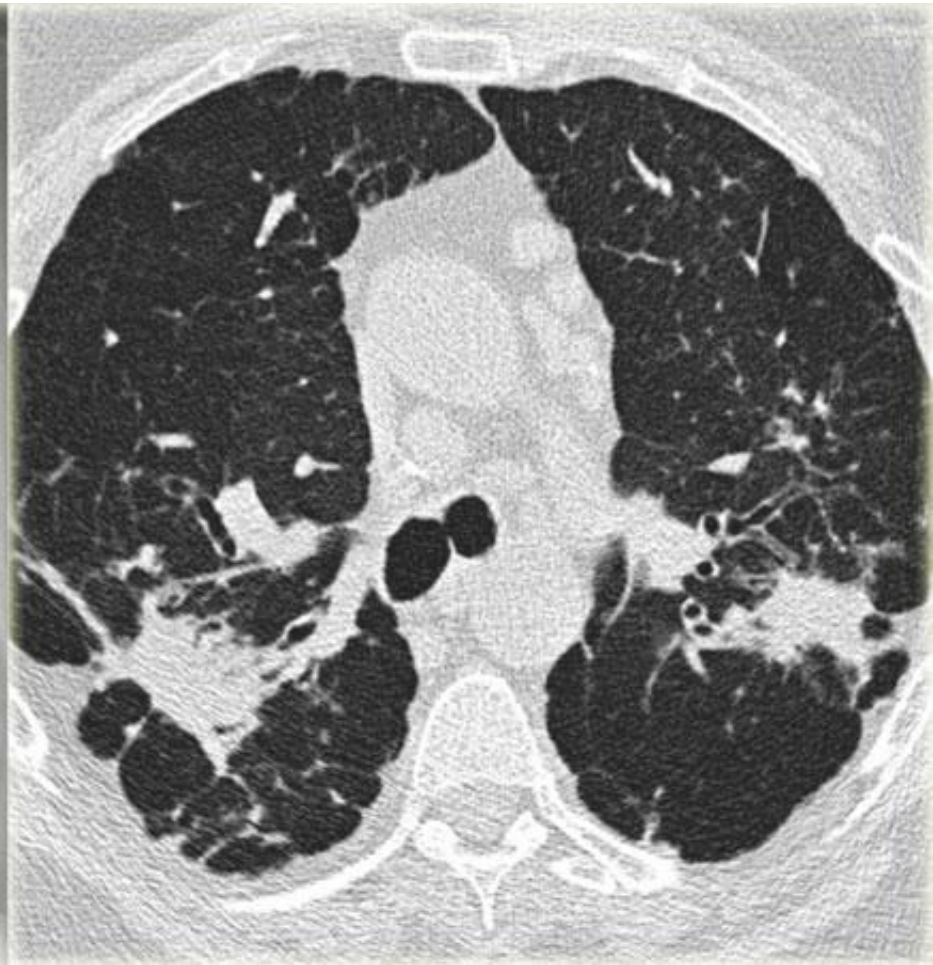
- Sometimes the reticulation is more coarse like in this case of congestive heart failure.



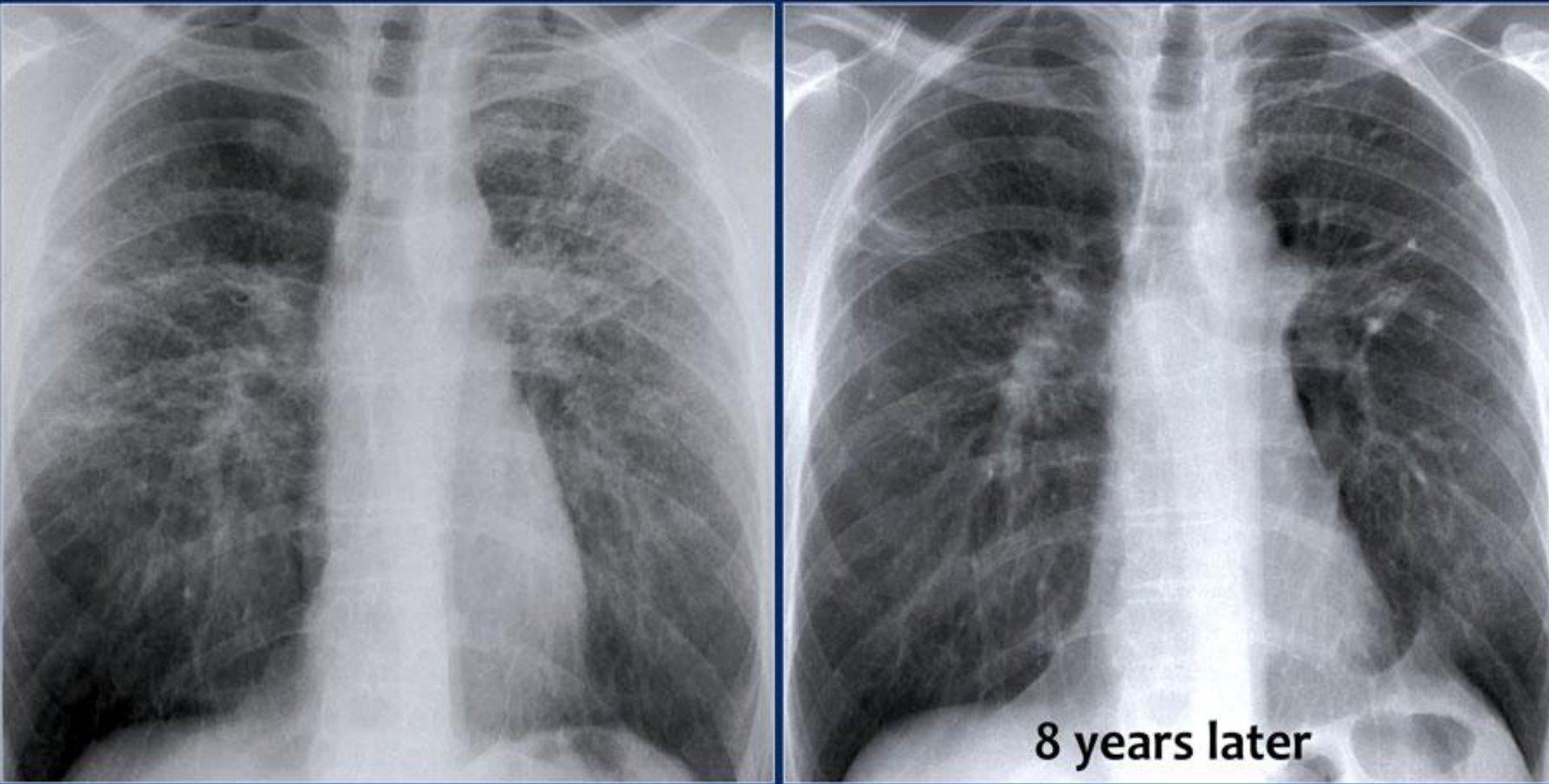
- **Sarcoidosis**
- In this case the chest x-ray shows subtle findings that could be described as fine reticulation.
- In many cases a HRCT is needed to determine the exact nature of the findings.
The HRCT - not shown - demonstrated a fine nodular appearance as a result of sarcoidosis.
- Notice the subtle irregular thickening of the minor fissure.
This is quite specific for sarcoidosis.



- **Longstanding Sarcoidosis**
- Here a typical chest film in a patient with long standing Sarcoidosis (stage IV). There is fibrosis in the upper zones.
- The differential diagnosis includes chronic hypersensitivity pneumonitis, which also results in fibrosis with upper lobe predominance.



- The HRCT demonstrates densities in both upper lobes. These are called conglomerate masses, which are the result of conglomerates of nodules.



- another patient with sarcoidosis.
- There is volume loss in the upper lobes as a result of fibrosis. The image on the left also shows densities in the lung. On a HRCT fine nodules were seen. A follow-up CXR shows resorption of most of the lung abnormalities. The fibrosis persists.

UIP

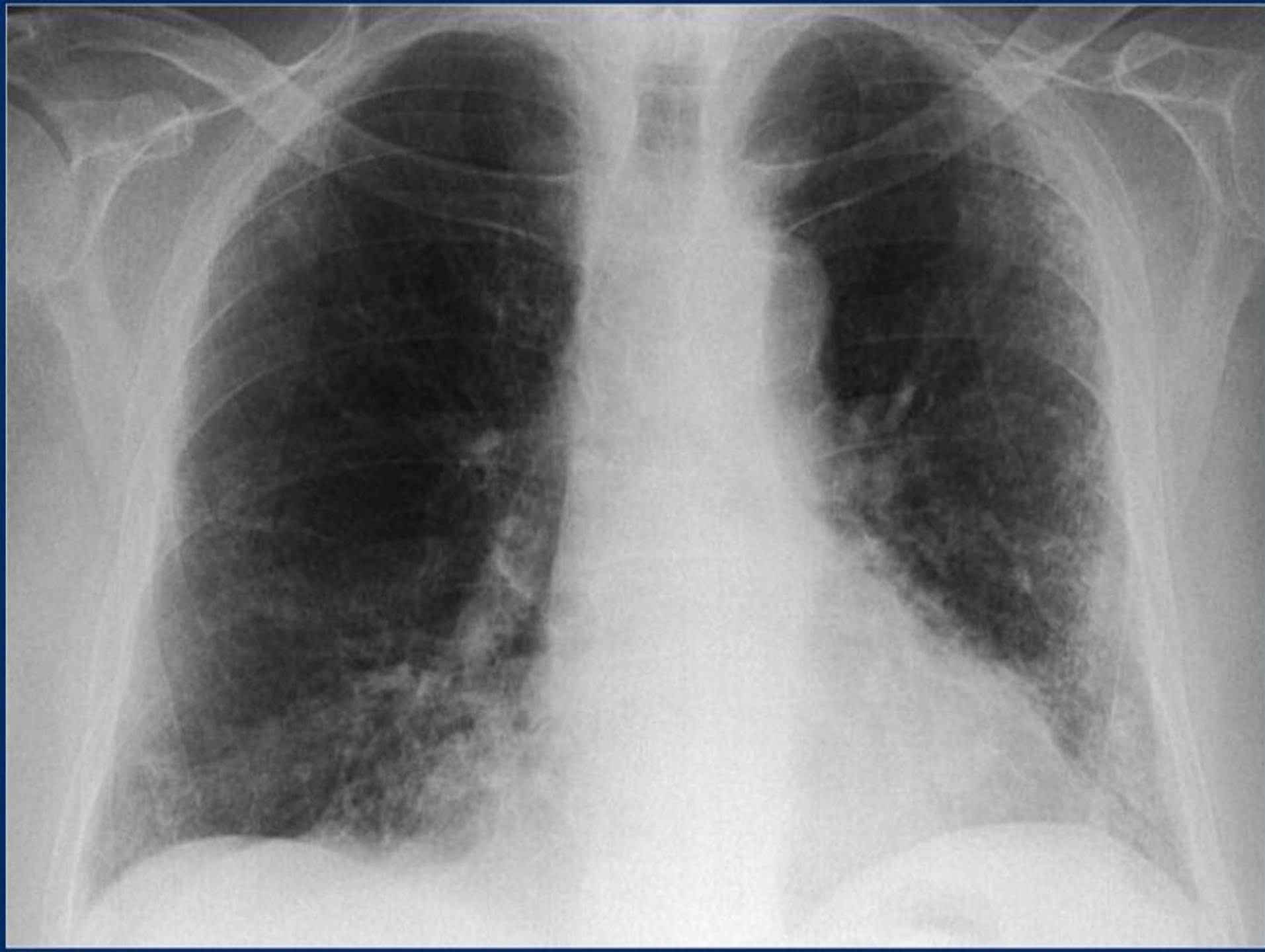
UIP is a histologic pattern of pulmonary fibrosis.

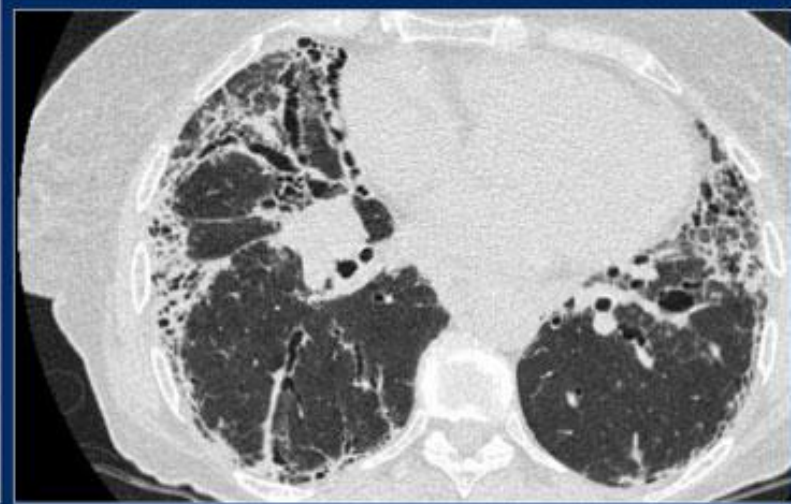
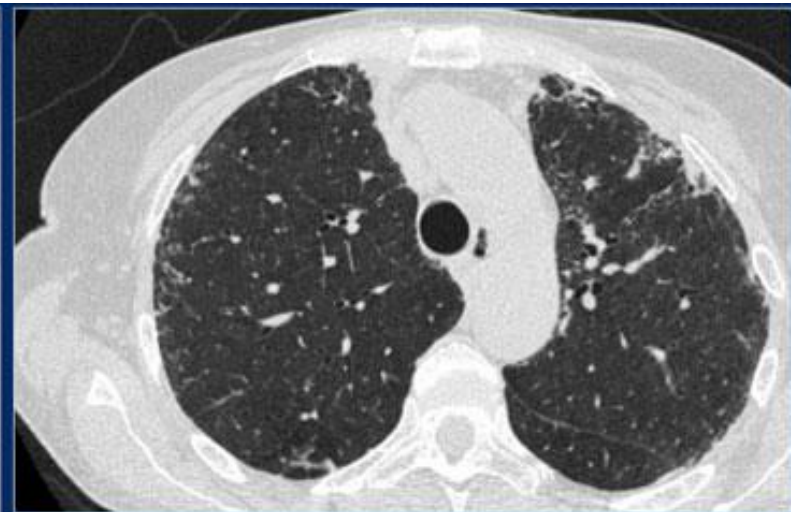
- On a chest X-ray UIP manifests as a reticular pattern particularly at the lung bases.
- In many cases you can suspect UIP on the CXR.
- A HRCT is needed to confirm the diagnosis by demonstrating honeycombing.

Here a CXR with a reticular pattern at the lung bases.

This pattern was first attributed to chronic congestive heart failure, but persisted on follow-up CXR's despite therapy.

HRCT demonstrated honeycombing.





- Here another case.
The CXR demonstrates a reticular interstitial pattern with a preference at the lung bases.
- The HRCT demonstrates honeycombing and traction bronchiectasis.

Interstitial pneumonias

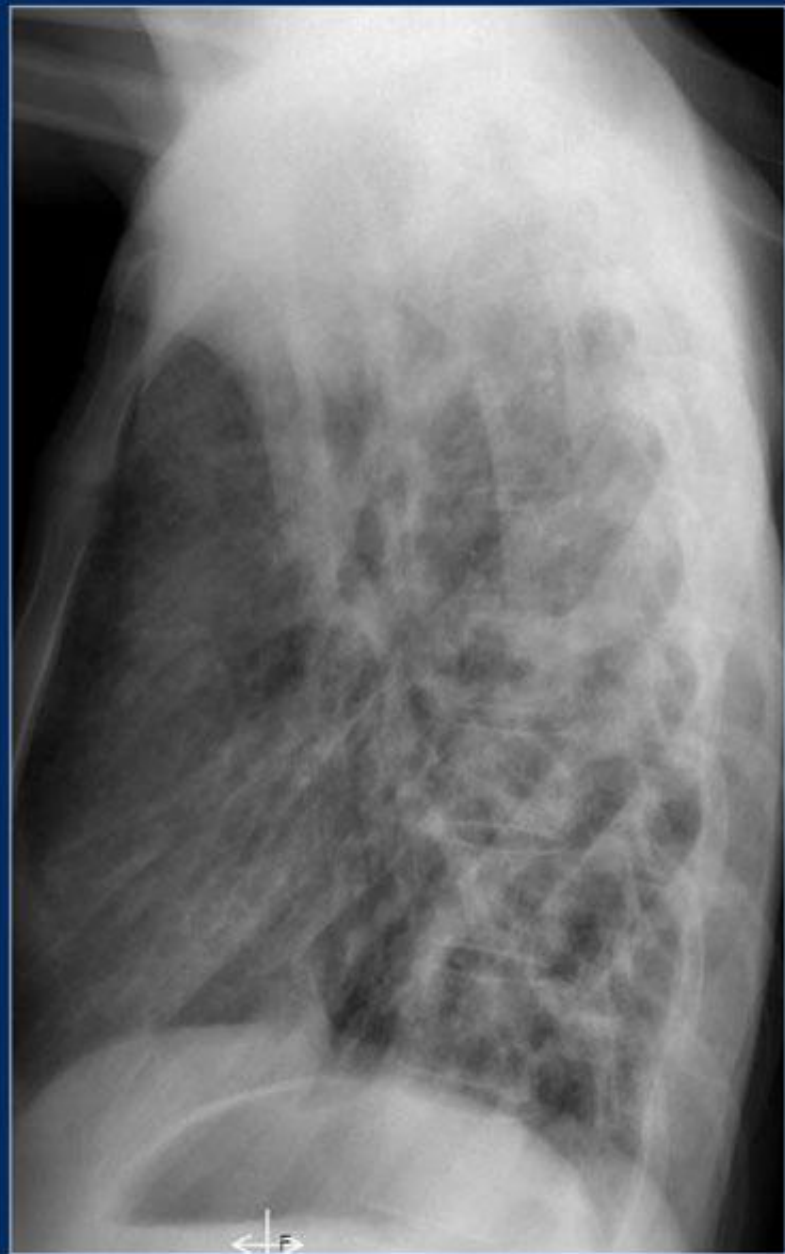
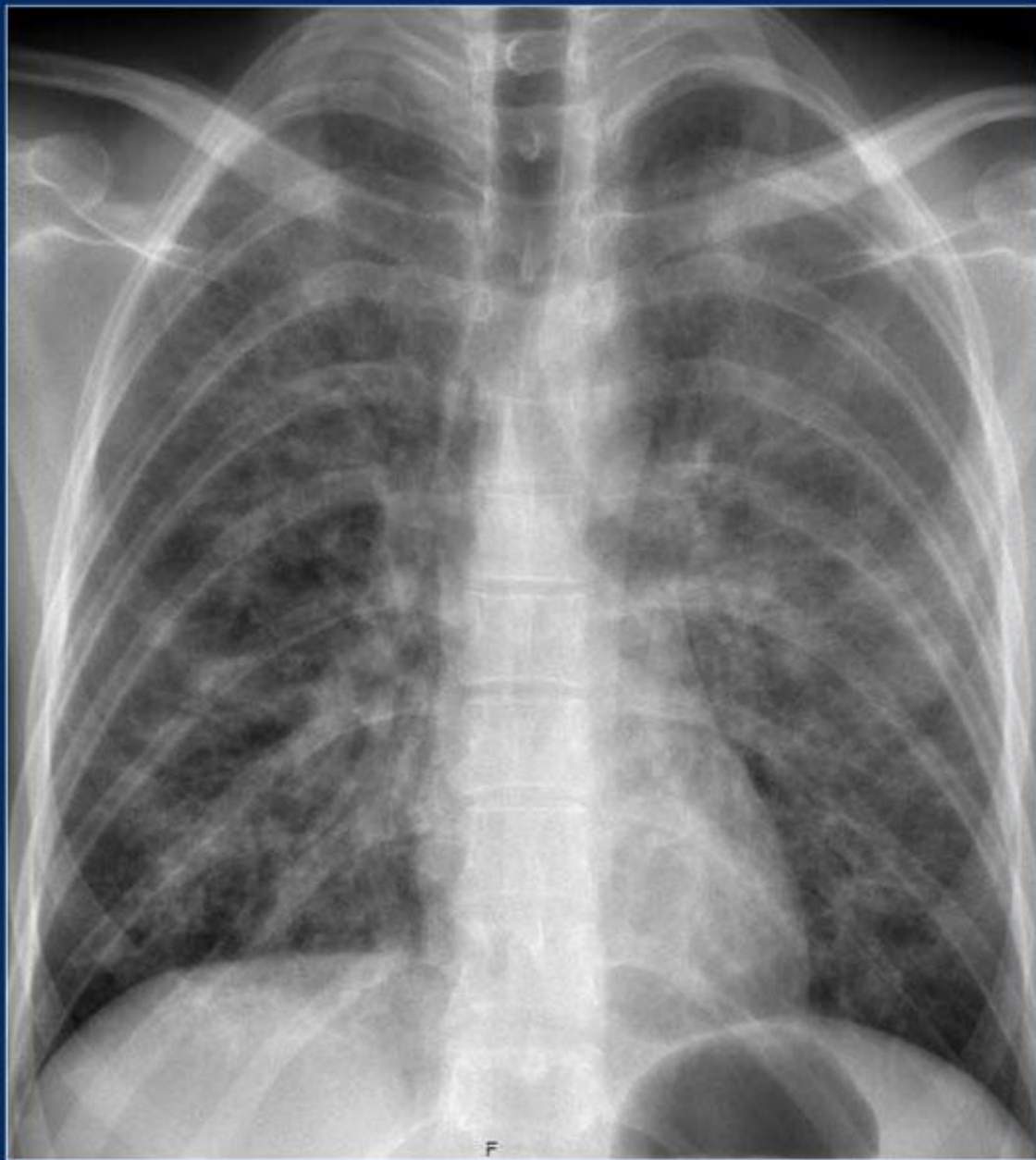
An acute reticular pattern is most frequently caused by interstitial edema due to cardiac heart failure.

The other cause is interstitial pneumonia:

- Viral
- PCP
- Mycoplasma pneumonia.

This patient presented with a nonproductive cough and some fever.

This was a PCP-infection as a first manifestation of AIDS.



- PCP

Sarcoidosis

On a CXR sarcoidosis usually first presents with hilar and mediastinal lymphadenopathy ([example](#)).

Parenchymal disease can present as consolidation or even as masses, but the most common presentation is a fine nodules.

Here a typical case.

When these small nodules coalesce, they may resemble consolidation.

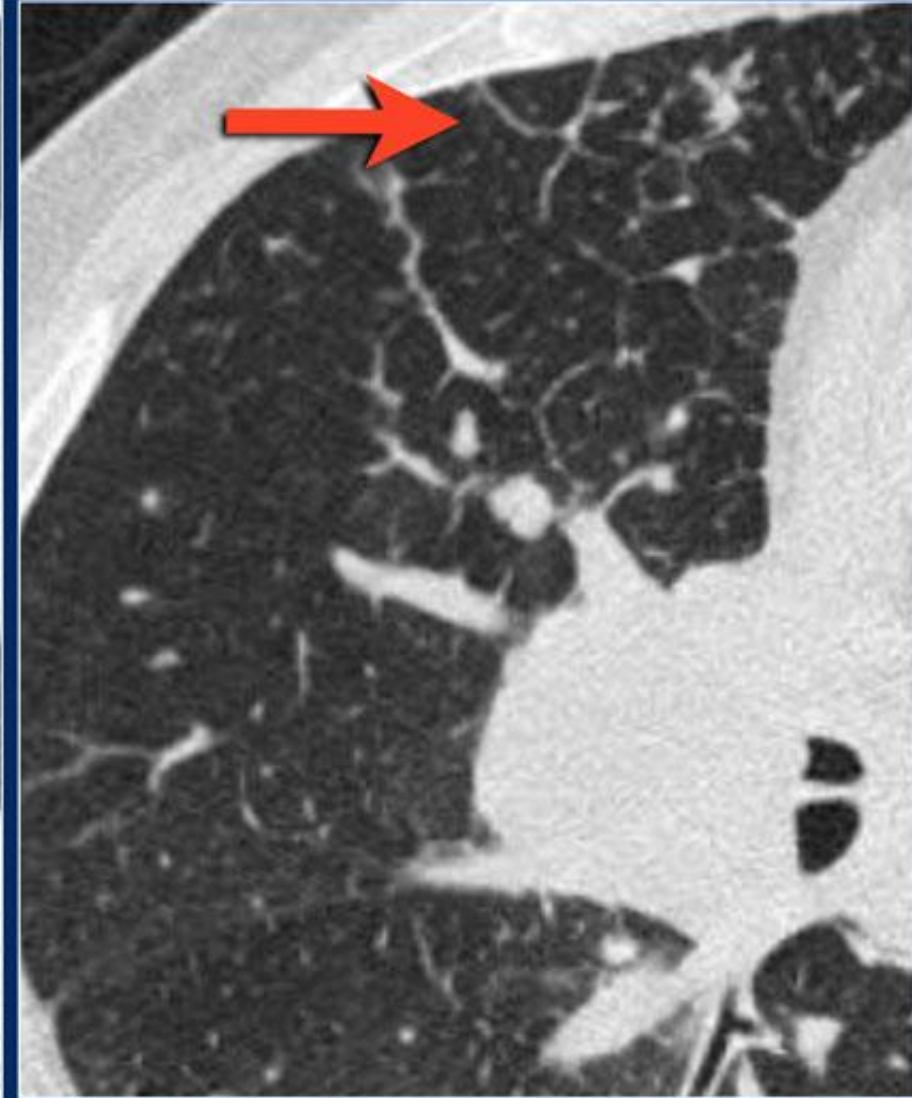


Sarcoidosis

Key Findings: Hilar and mediastinal lymphadenopathy and small nodules along bronchovascular bundle and fissures in a patient with fever, weight loss, fatigue and erythema nodosum

Lymphangitis carcinomatosa

Lymphangitis carcinomatosa also produces a reticular pattern. It is best appreciated on HRCT-images.



Key Findings: Focal smooth thickening of interlobular septa in a patient with known malignancy

Atelectasis

Atelectasis or lung-collapse:

- is the result of loss of air in a lung or part of the lung with subsequent volume loss
- due to airway **obstruction** or **compression** of the lung by pleural fluid or a pneumothorax.
- In many cases atelectasis is the first sign of a lung cancer.
- Evidently it is very important to recognize the various presentations of atelectasis, since some of them can be easily misinterpreted.

The key-findings on the X-ray are:

- Sharply-defined opacity obscuring vessels without air-bronchogram
- Volume loss resulting in displacement of diaphragm, fissures, hili or mediastinum

Atelectasis

Resorption

- Mucus plugging
- Tumor
- Foreign body

Relaxation

- Pleural effusion
- Pneumothorax
- Rounded atelectasis

Lobar atelectasis

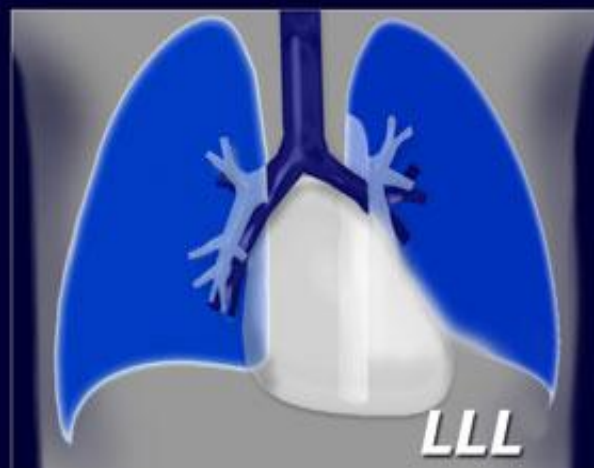
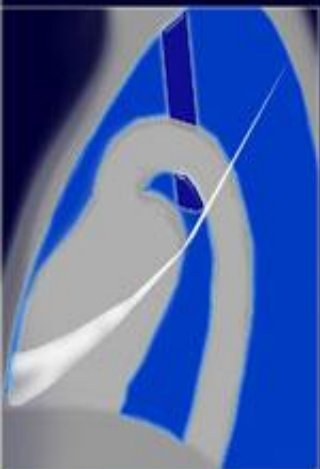
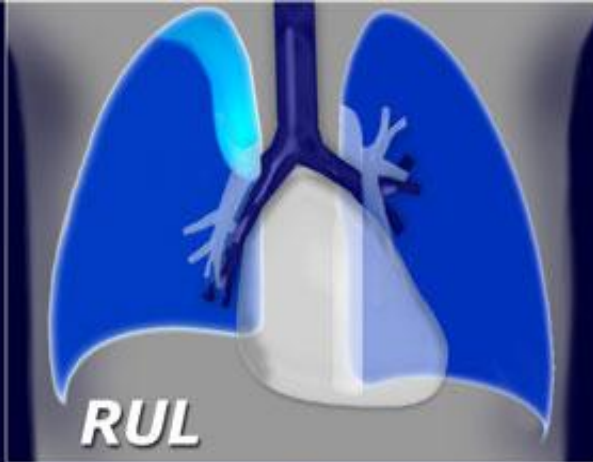
Lobar atelectasis or lobar collapse is an important finding on a chest x-ray and has a limited differential diagnosis.

The most common causes of atelectasis are:

- Bronchial carcinoma in smokers
- Mucus plug in patients on mechanical ventilation or asthmatics (ABPA)
- Malpositioned endotracheal tube
- Foreign body in children

***Sometimes lobar atelectasis produces only mild volume loss due to overinflation of the other lungparts.

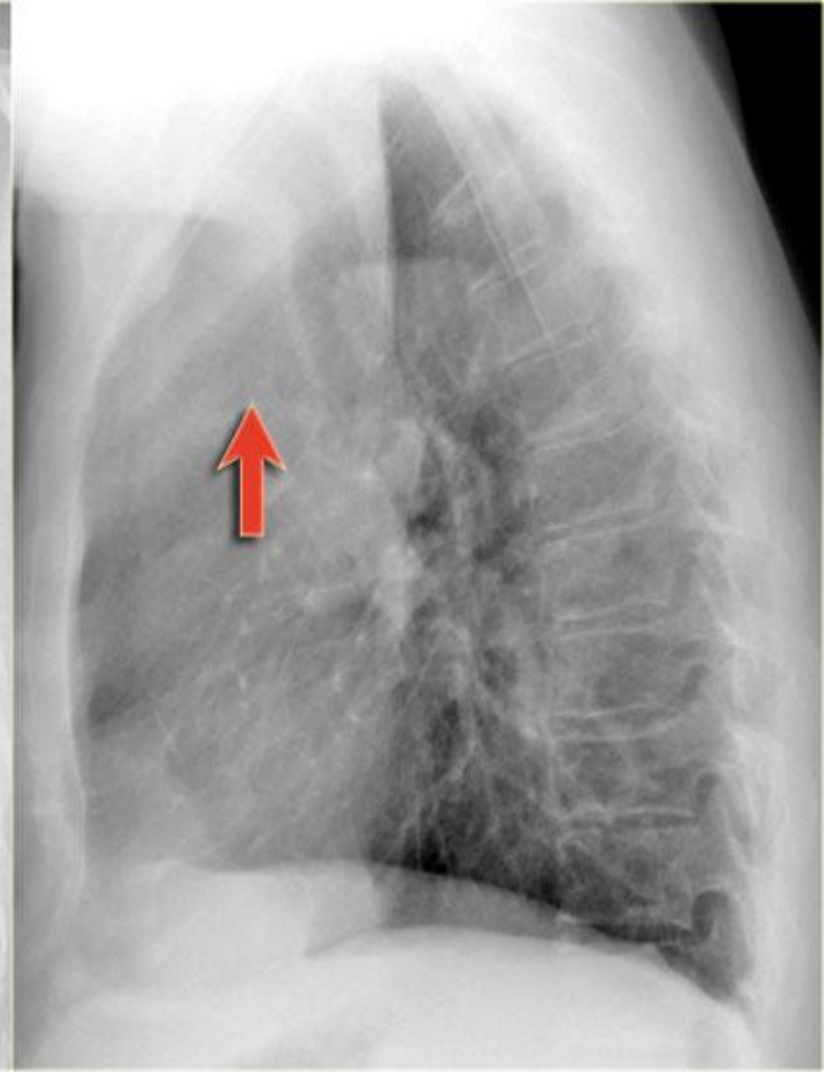
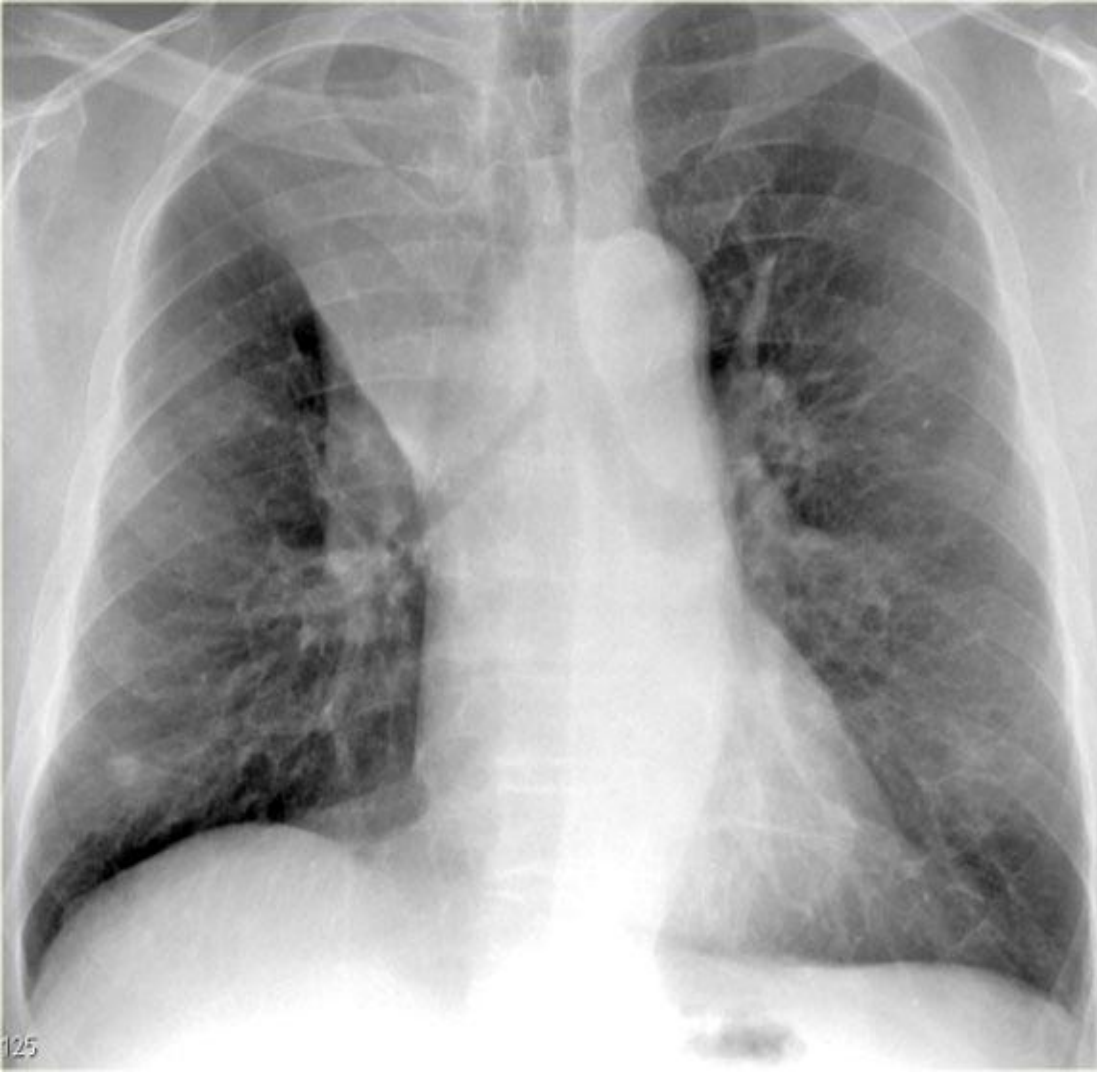
The illustration summarizes the findings of the different types of lobar atelectasis.

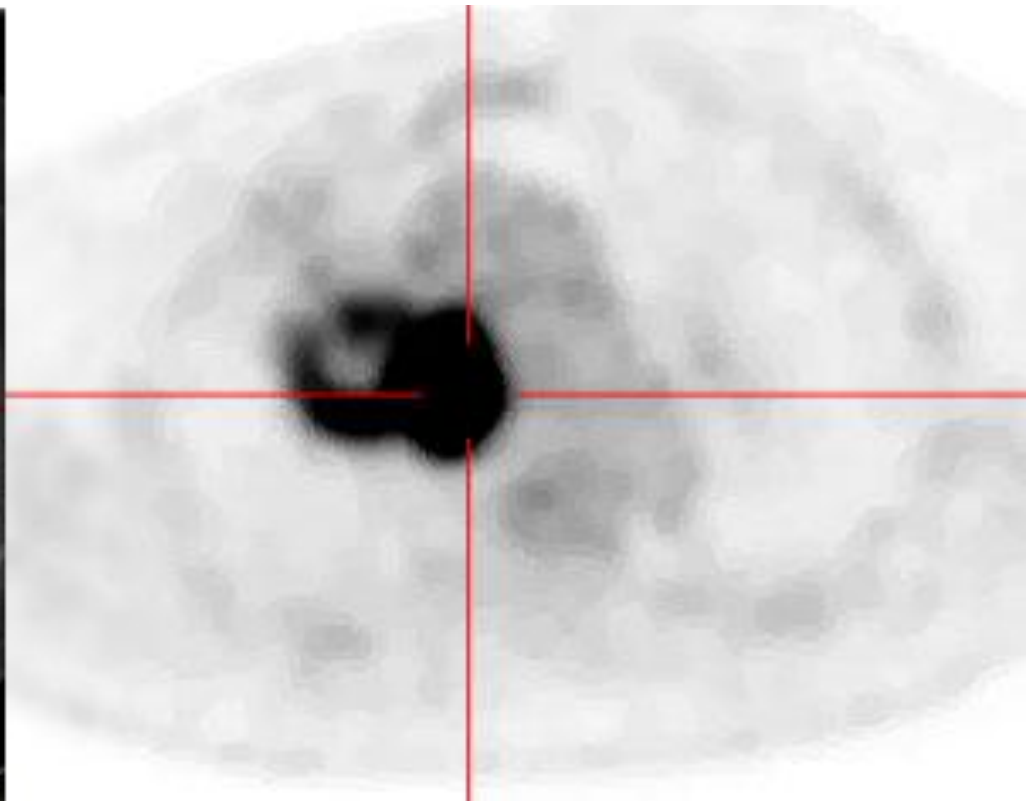
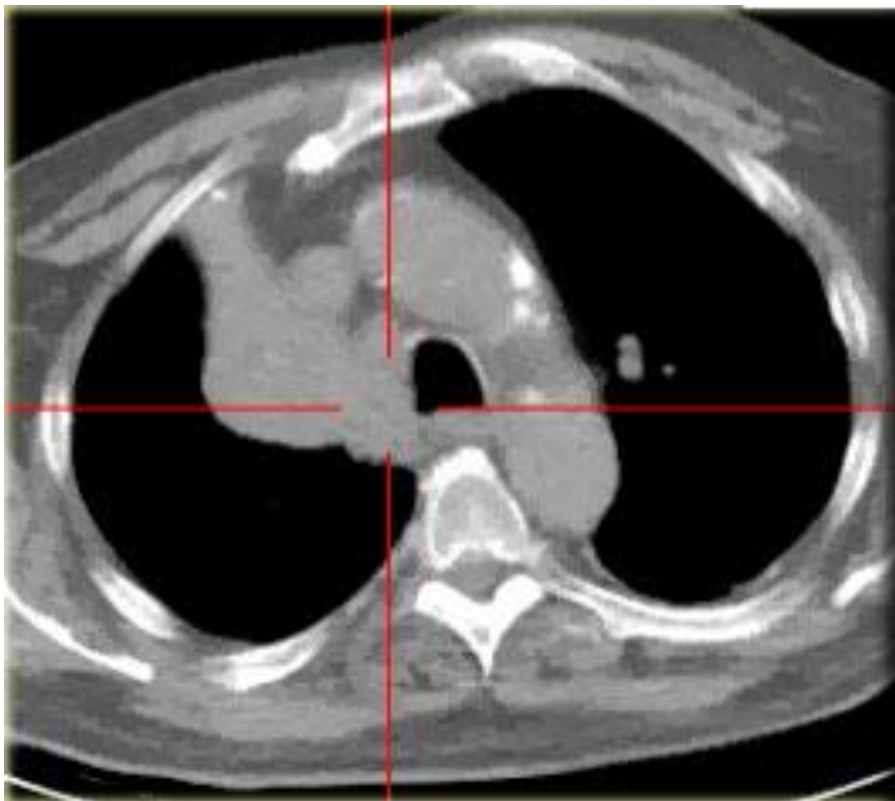


- **Right upper lobe atelectasis**
- First study the images, then continue reading.

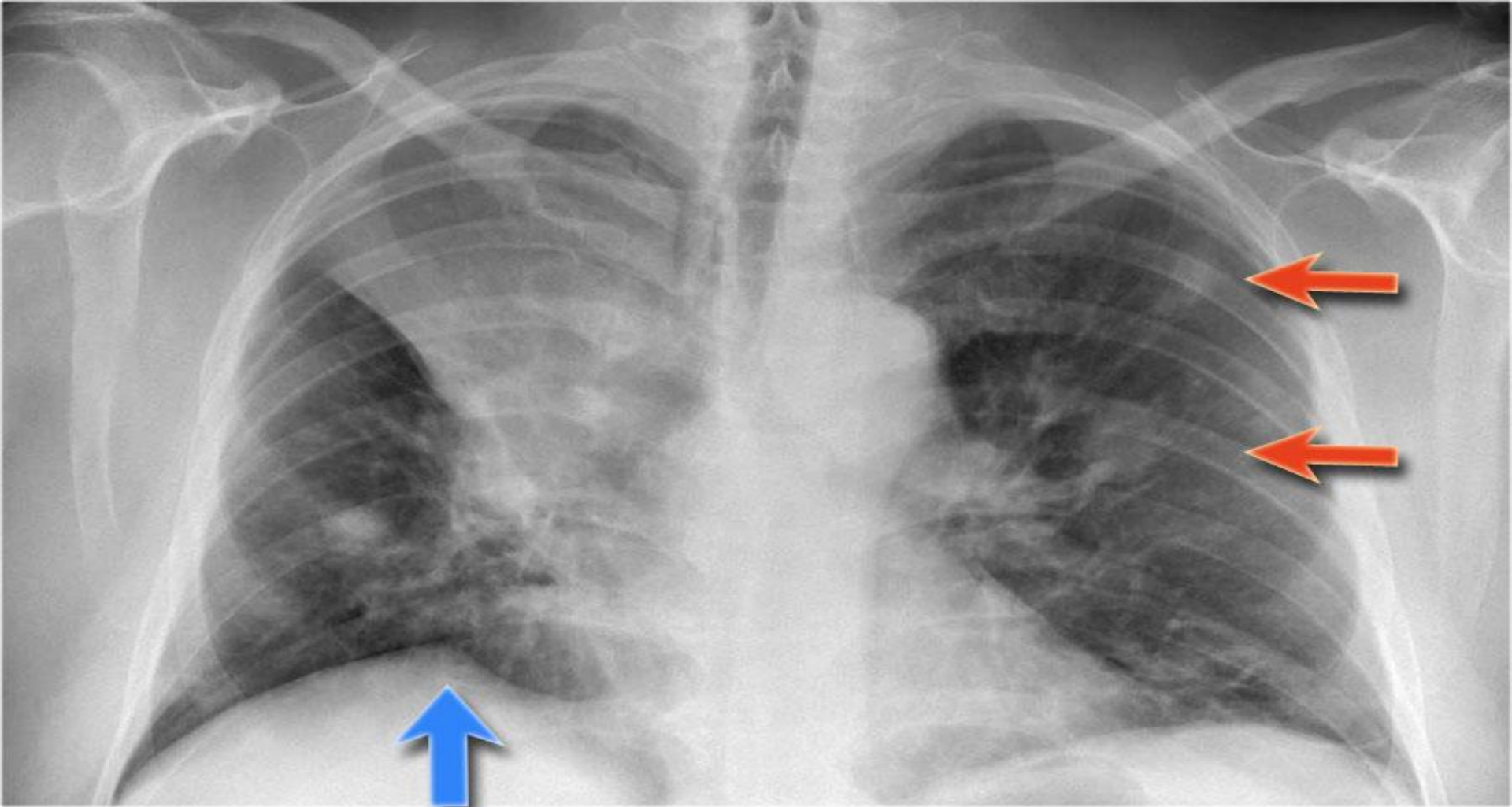
Findings:

- triangular density
- elevated right hilus
- obliteration of the retrosternal clear space (arrow)





- On the PET-CT a lungneoplasm is seen with subsequent atelectasis of the right upper lobe due to obstruction of the upper lobe bronchus.



- A common finding in atelectasis of the right upper lobe is 'tenting' of the diaphragm (blue arrow).
- This patient had a centrally located lung carcinoma with metastases in both lungs (red arrows).

Right middle lobe atelectasis

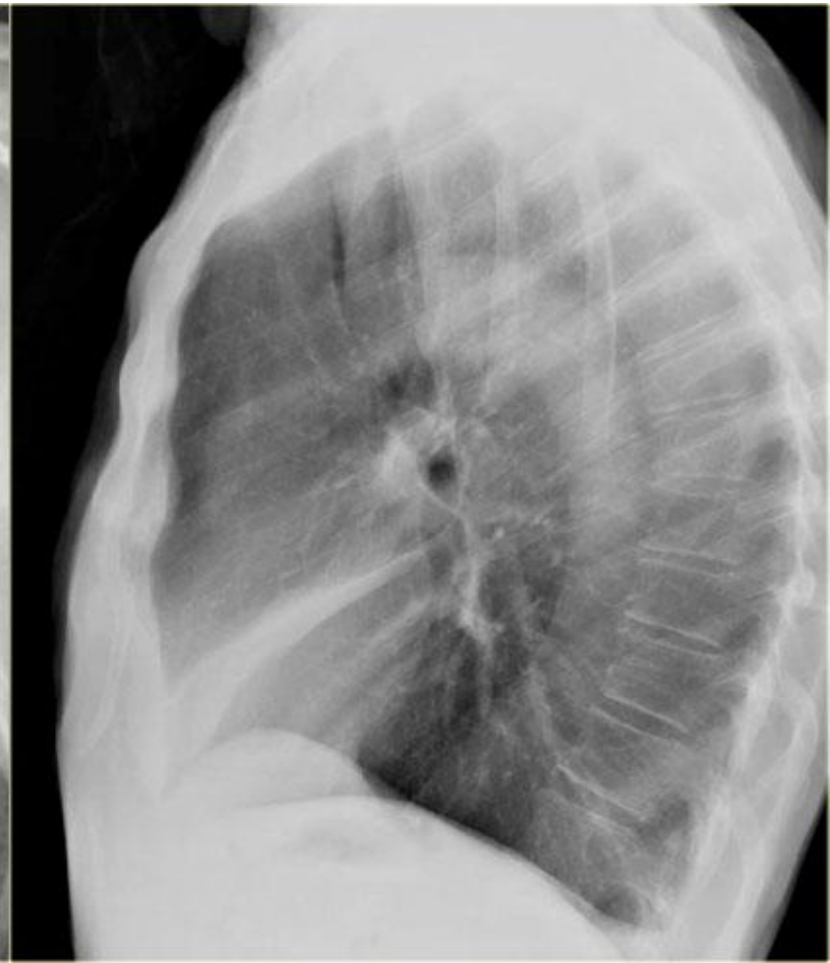
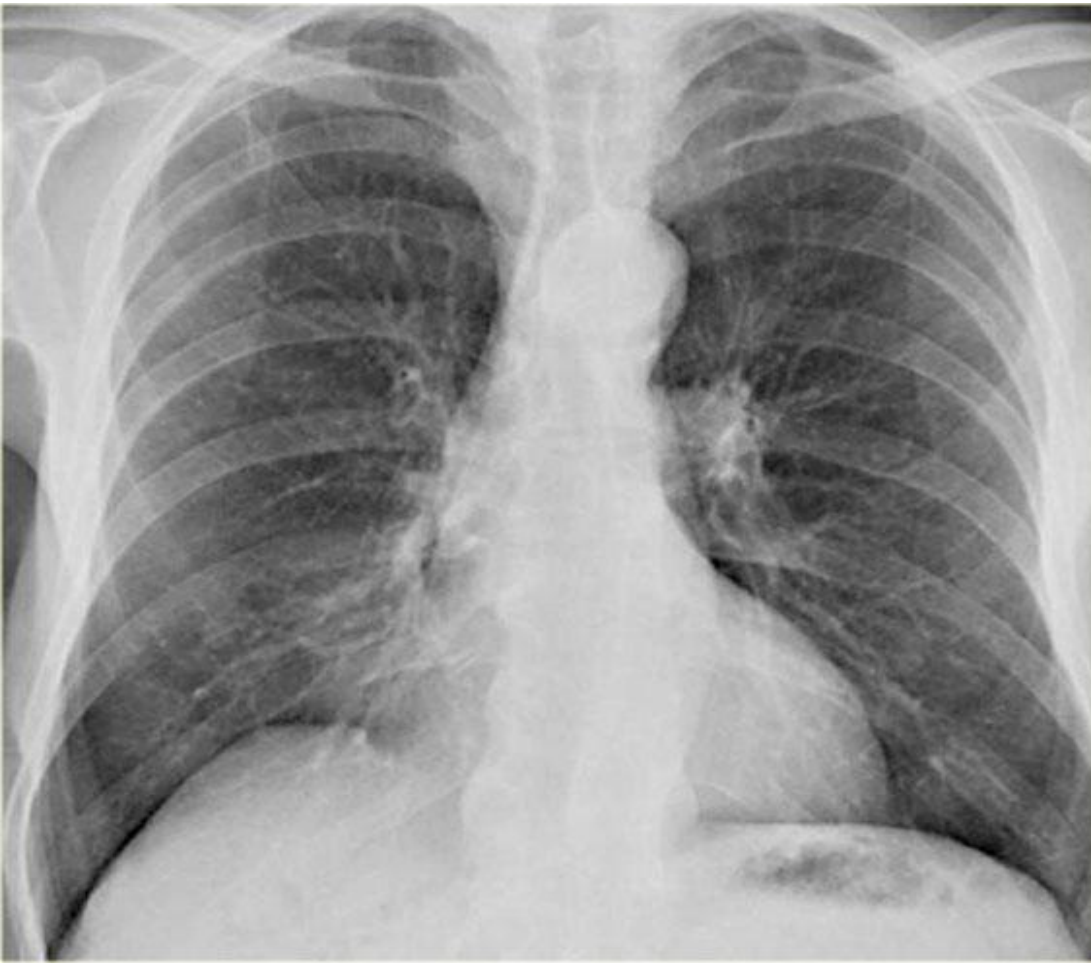
First study the x-rays and then continue reading. What are the findings?

Blurring of the right heart border (silhouette sign)

Triangular density on the lateral view as a result of collapse of the middle lobe

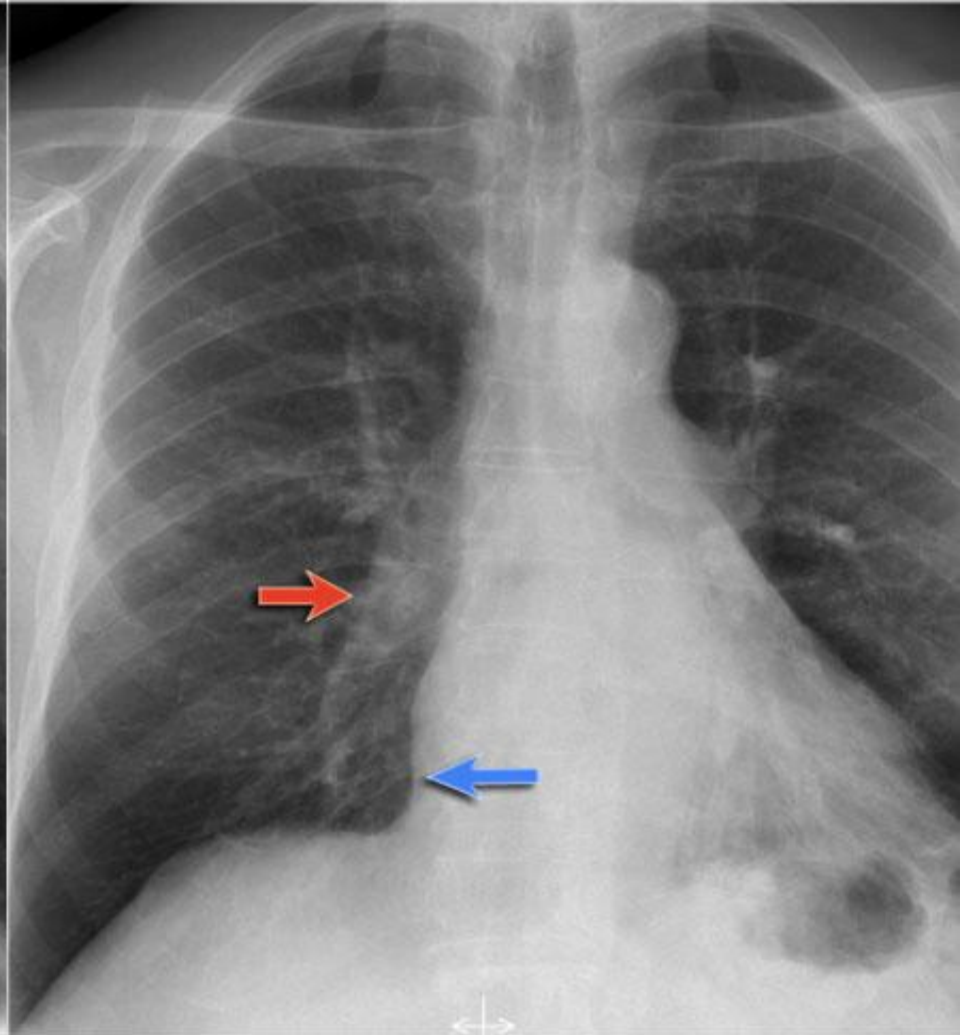
Usually right middle lobe atelectasis does not result in noticeable elevation of the right diaphragm.

A [pectus excavatum](#) can mimick a middle lobe atelectasis on a frontal view, but the lateral view should solve this problem.



- **Right lower lobe atelectasis**
- Chest x-rays of a 70-year old male who fell from the stairs and has severe pain on the right flank.
- There is some loculated pleural fluid posterolateral as a result of hemothorax.
- What are the pulmonary findings?
First study the images, then continue reading.





- There is a **right lower lobe atelectasis**.
- Notice the abnormal right border of the heart. The right interlobar artery is not visible, because it is not surrounded by aerated lung but by the collapsed lower lobe, which is adjacent to the right atrium.
- On a follow-up chest film the atelectasis has resolved. We assume that the atelectasis was a result of post-traumatic poor ventilation with mucus plugging.
- Notice the reappearance of the right interlobar artery (red arrow) and the normal right heart border (blue arrow).

Left upper lobe atelectasis

First study the x-rays, then continue reading. What are the findings?

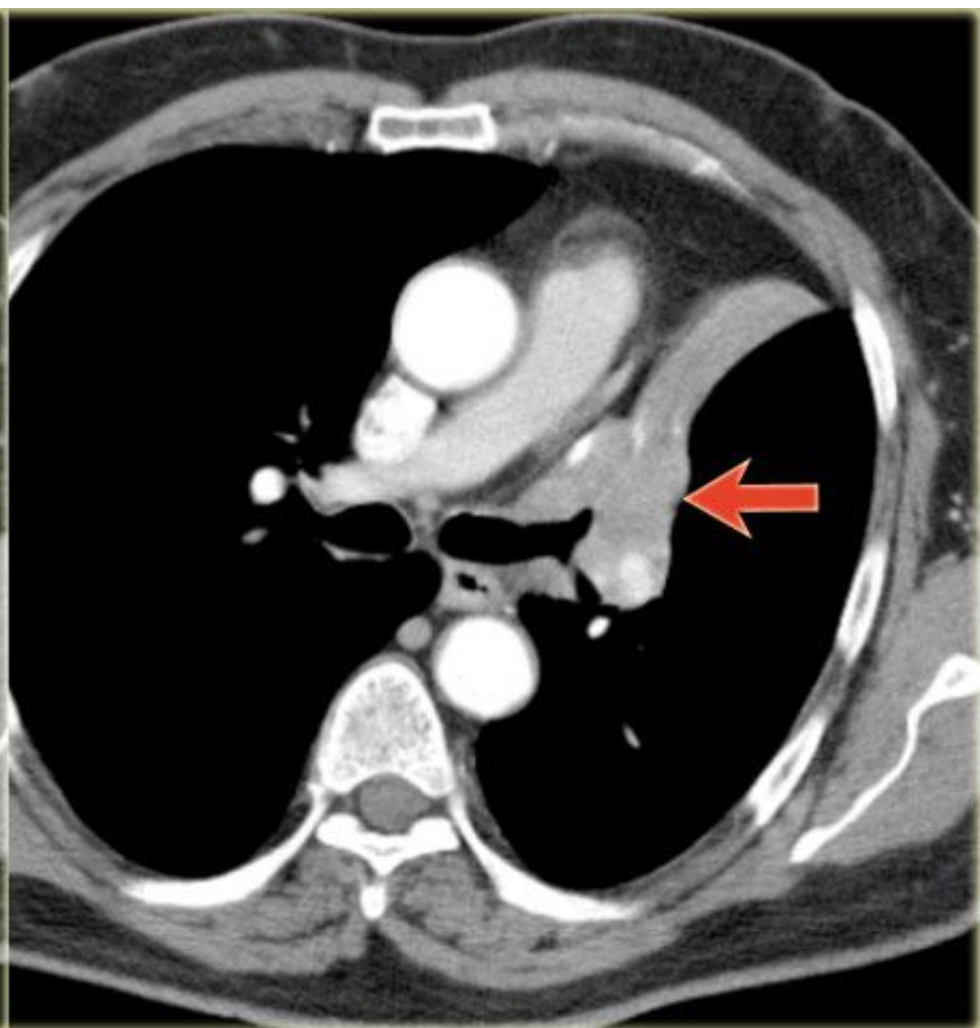
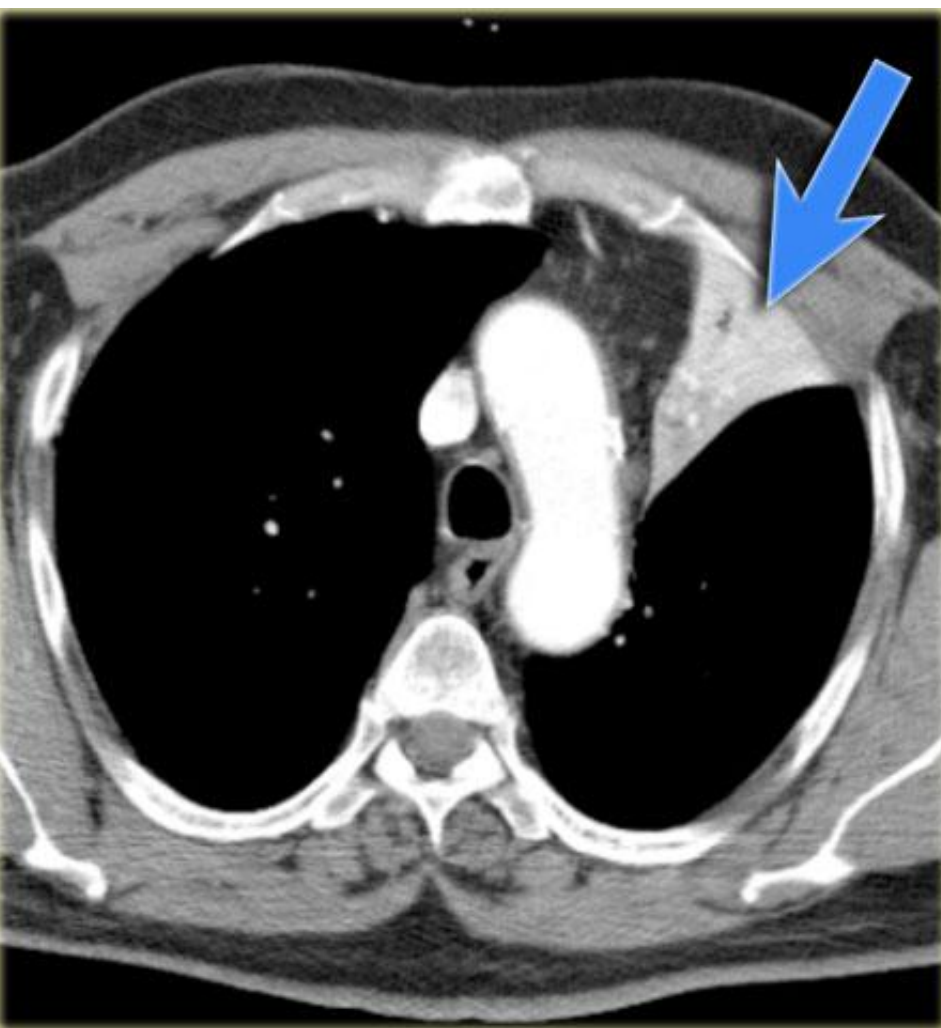
Minimal volume loss with elevation of the left diaphragm

Band of increased density in the retrosternal space, which is the collapsed left upper lobe

Abnormal left hilus, i.e. possible obstructing mass

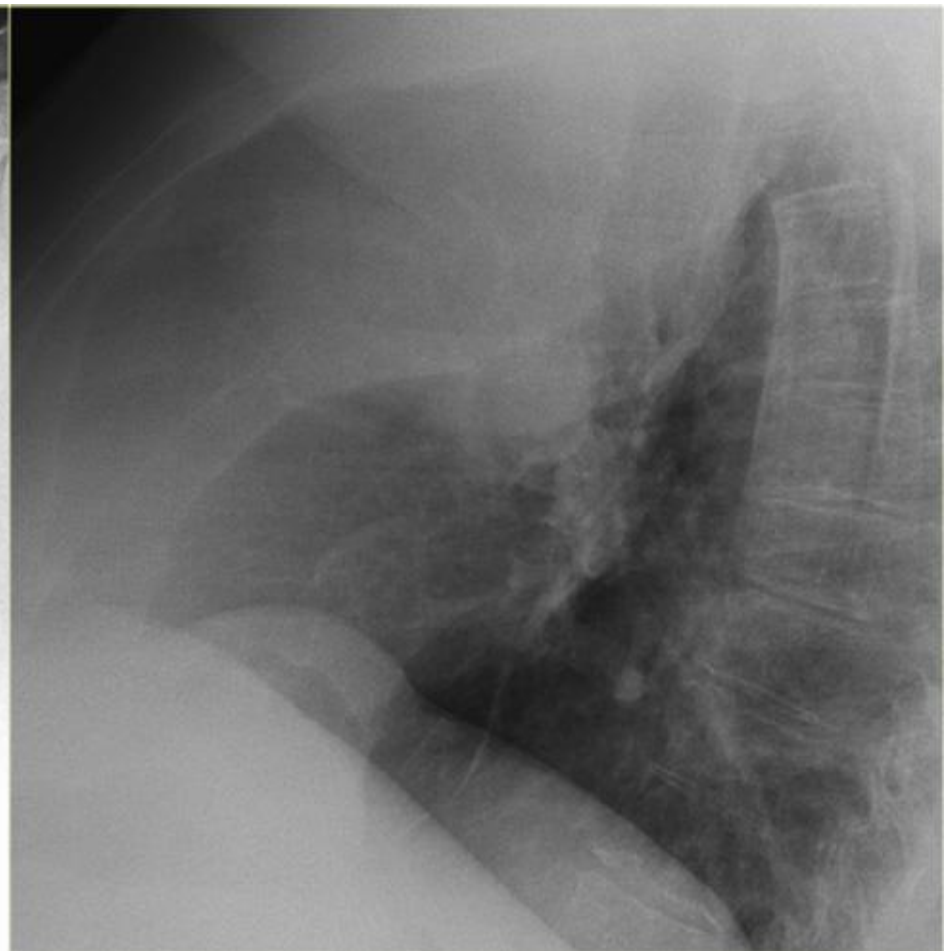
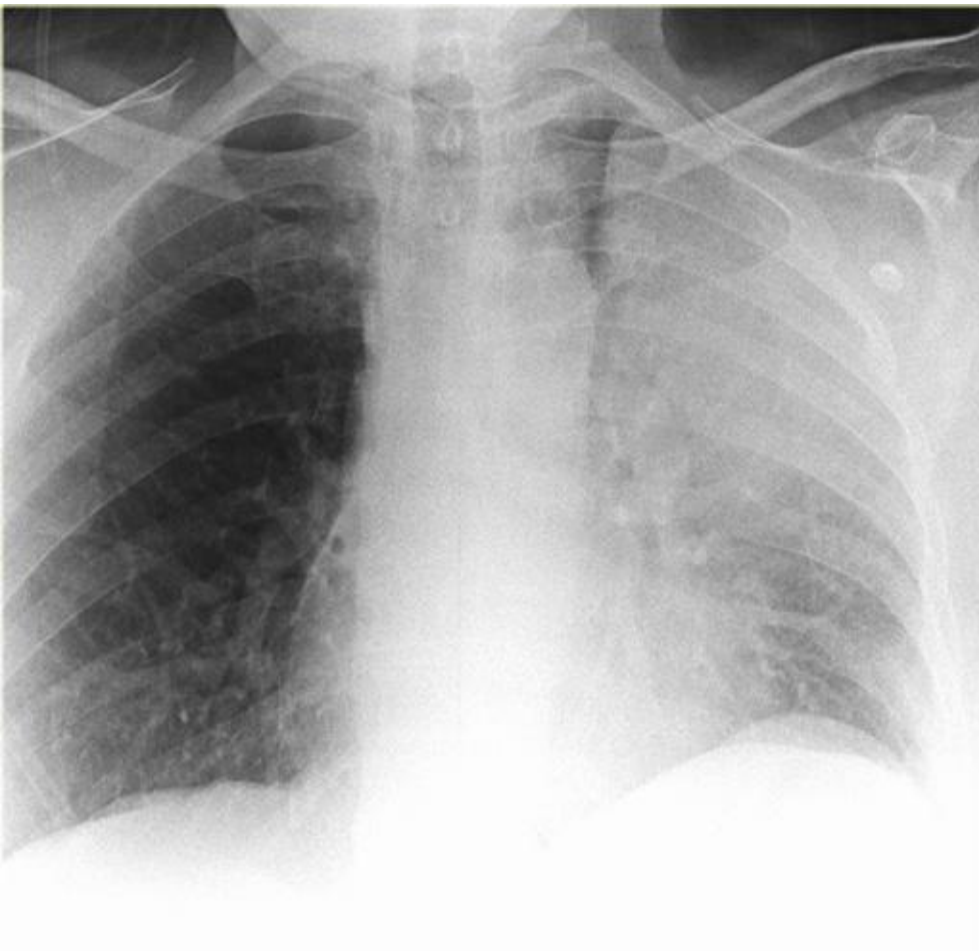
These findings indicate an atelectasis of the left upper lobe





The CT-images demonstrate the atelectasis of the left upper lobe (blue arrow).

There is a centrally located mass which obstructs the left upper lobe bronchus (red arrow).



- First study the x-rays then continue reading.
What are the findings and what sign is seen here?
- There is an ***atelectasis of the left upper lobe***.
You would not expect the apical region to be this dark, but in fact this is caused by overinflation of the lower lobe, which causes the superior segment to creep all the way up to the apical region.
- This is called the ***Luftschiel sign***.

First study the x-rays, then continue reading.

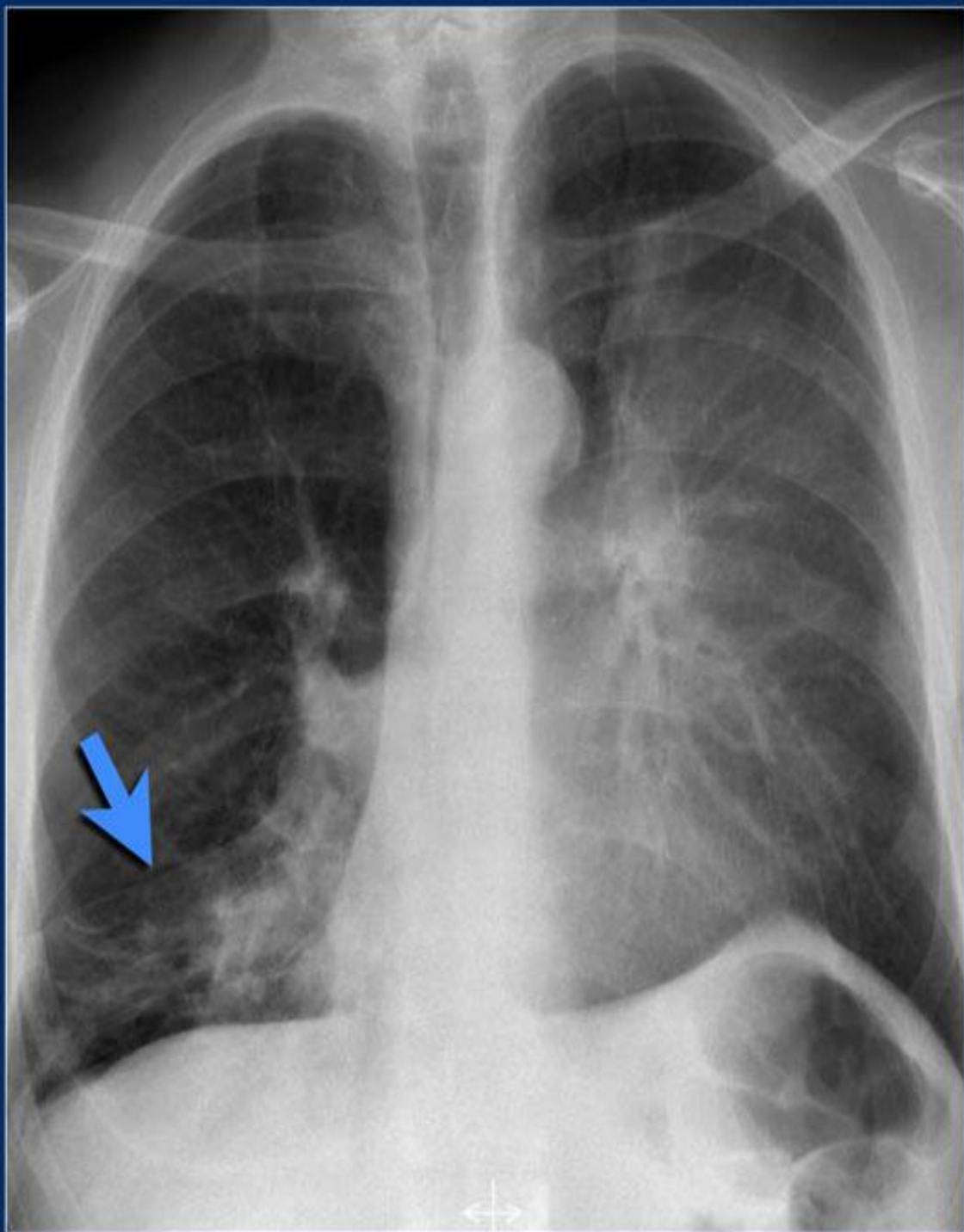
The findings are:

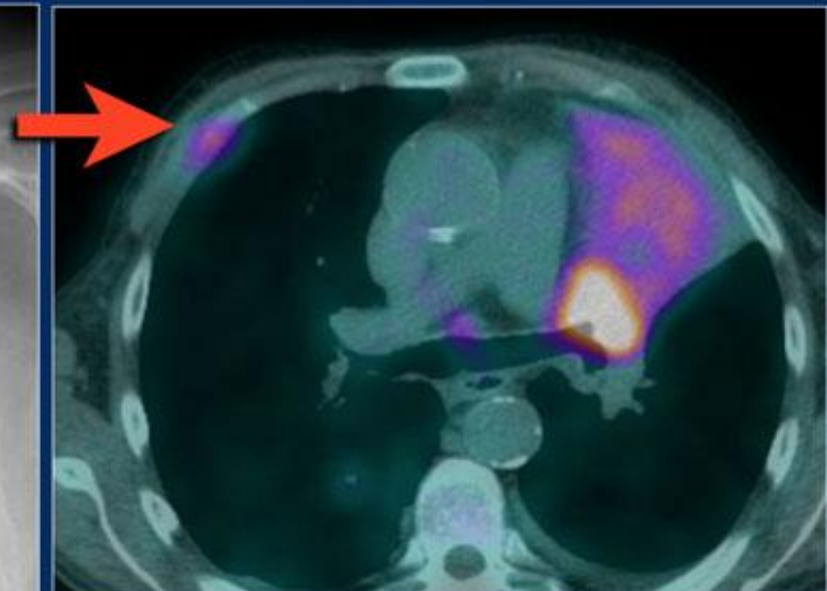
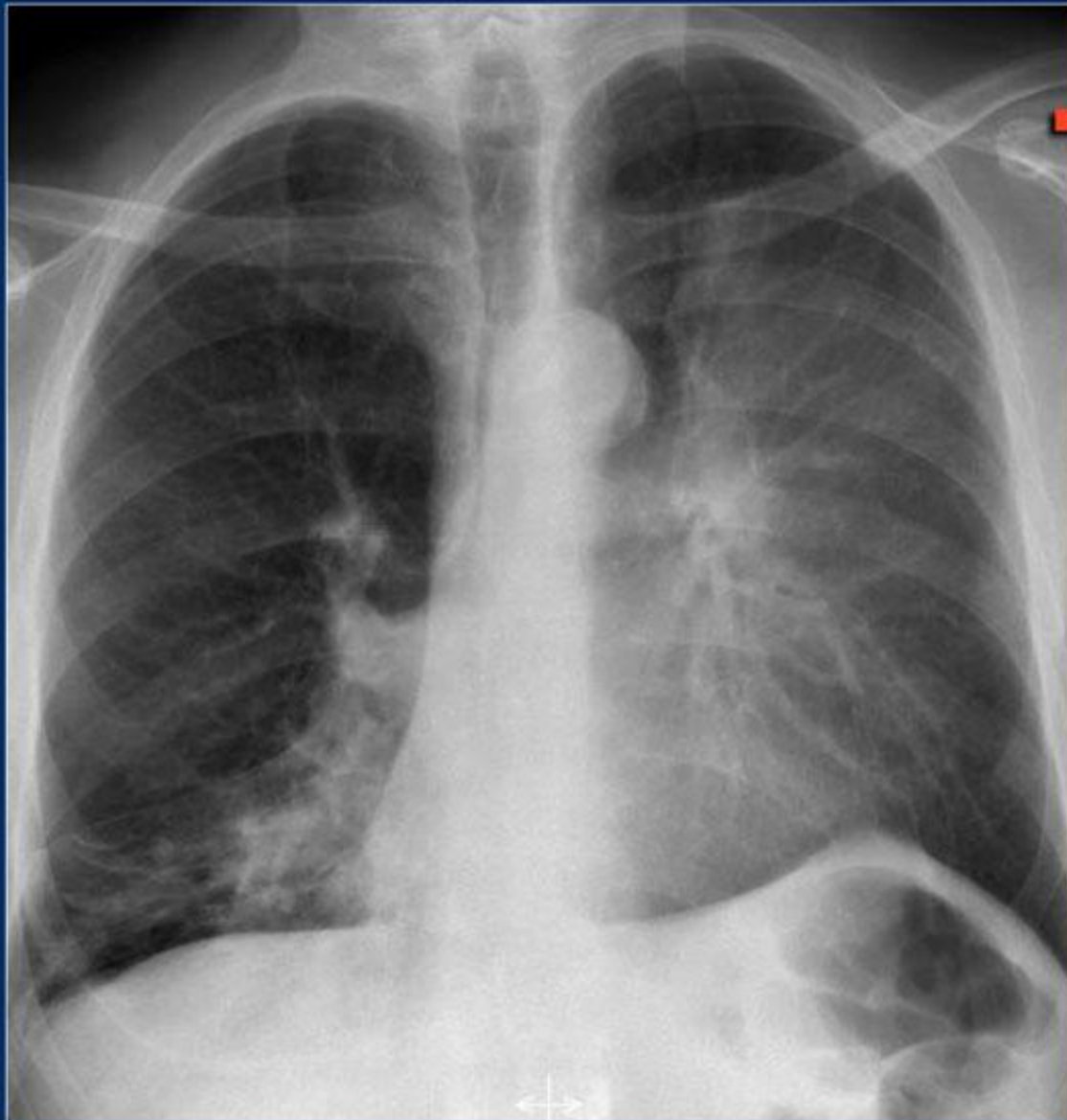
- Large density on the left with loss of cardiac silhouette.
- High position left diaphragm with tenting.
- Low position minor fissure
- Low position right hilum

These findings indicate a total atelectasis of the left upper lobe and possibly also partial atelectasis on the right.

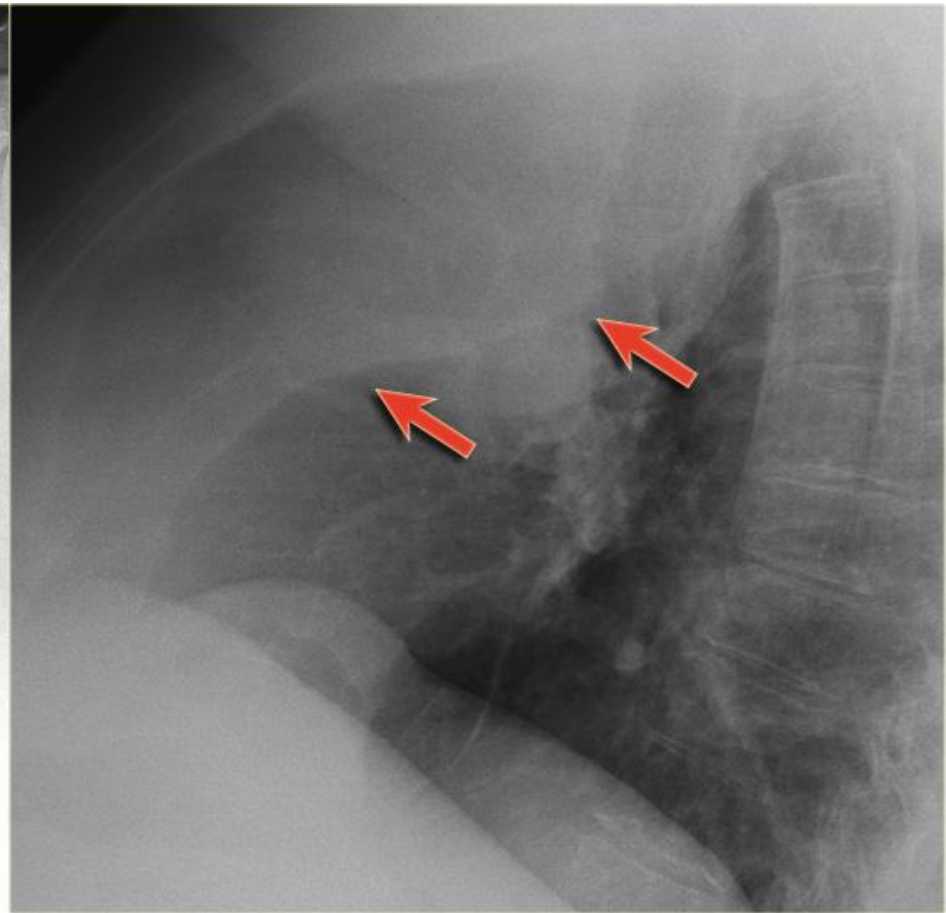
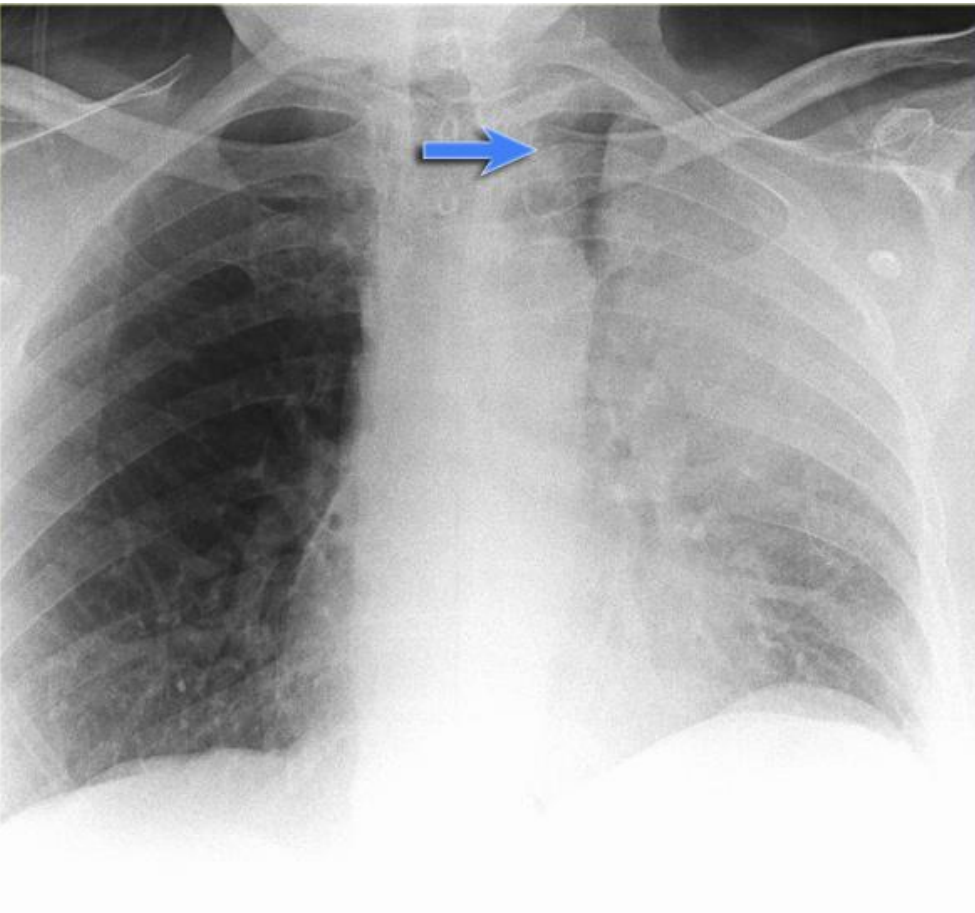
Since the silhouette of the right heart border is still visible, there is probably partial atelectasis of the lower lobe and not of the middle lobe.

Continue with the PET-CT...





- There were multiple bone metastases. One rib metastasis is indicated by the arrow.



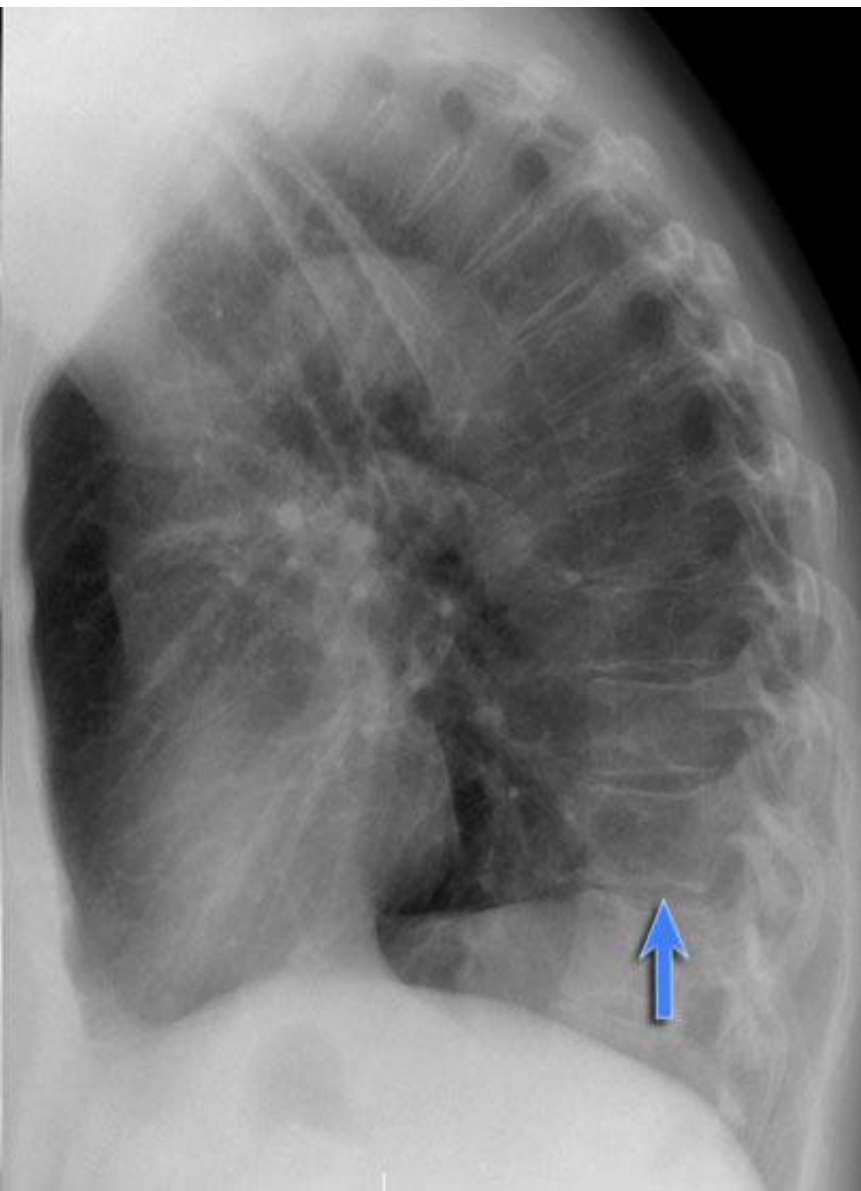
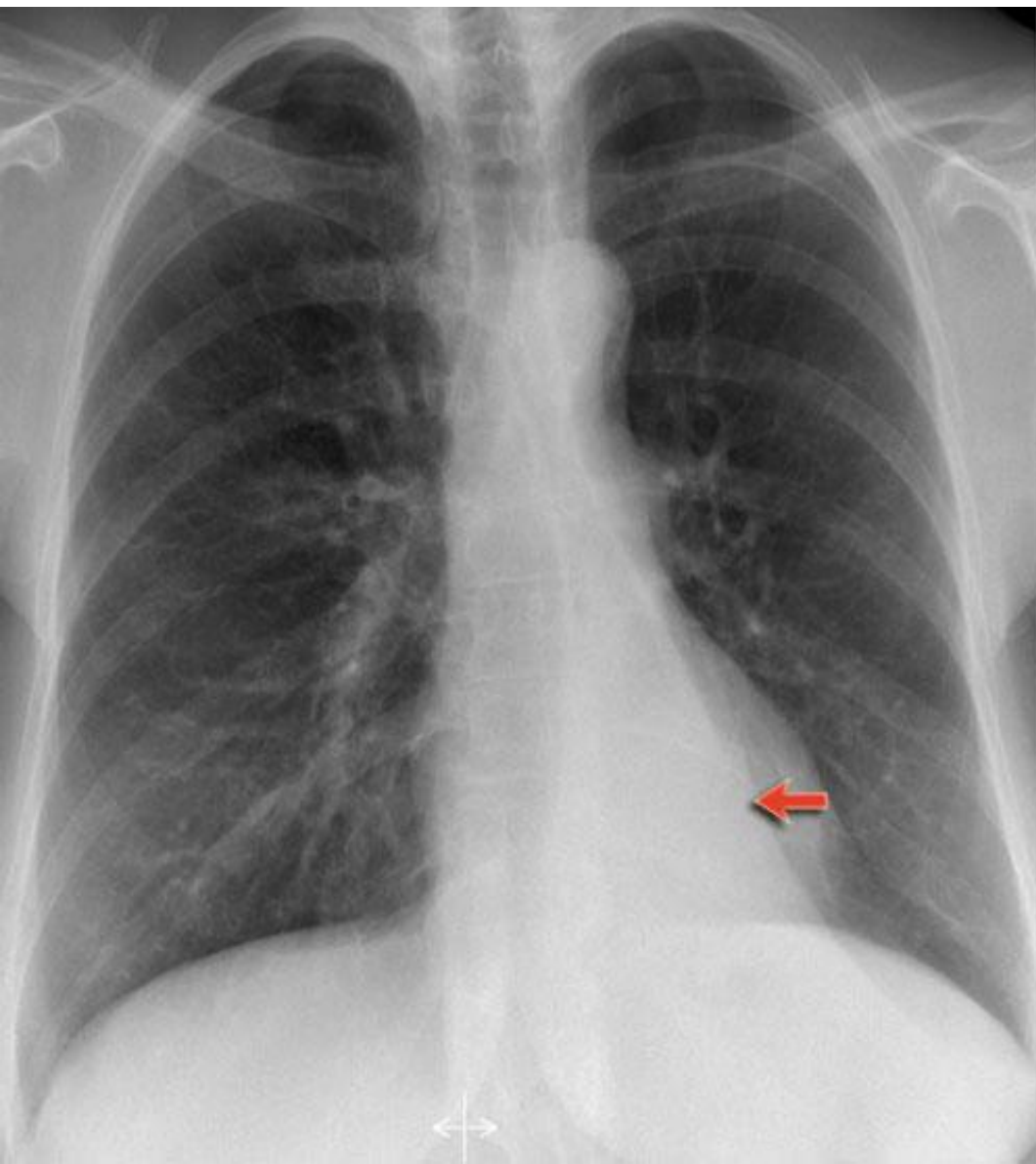
- **Luft sichel** means a **sickle of air** (blue arrow).
- Notice the bulging of the fissure on the lateral view. This is comparable to the *golden-S sign* in right upper lobe atelectasis and is suggestive of a centrally obstructing mass.

- Study the images and then continue reading.
- There is a total collapse of the left upper lobe. Notice the high position of the left hilum. There is only a subtle band of density projecting behind the sternum. This is the collapsed upper lobe.
- In this case there is compensatory overinflation of the left lower lobe resulting in a normal position of the diaphragm and the mediastinum.



- **Left lower lobe atelectasis**
- First study the x-rays then continue reading.
- Where is the abnormality located?





There is a triangular density seen through the cardiac shadow.

This must be an abnormality located posterior to the heart.

This is confirmed on the lateral view.

The contour of the left diaphragm is lost when you go from anterior to posterior.

As the title suggests this is **lower lobe atelectasis**.

- We cannot see the lower lobe vessels, because they are surrounded by the atelectatic lobe.
- Normally when you follow the thoracic spine from top to bottom, the lower region becomes less opaque.

Here we have the opposite (blue arrow).

Total atelectasis

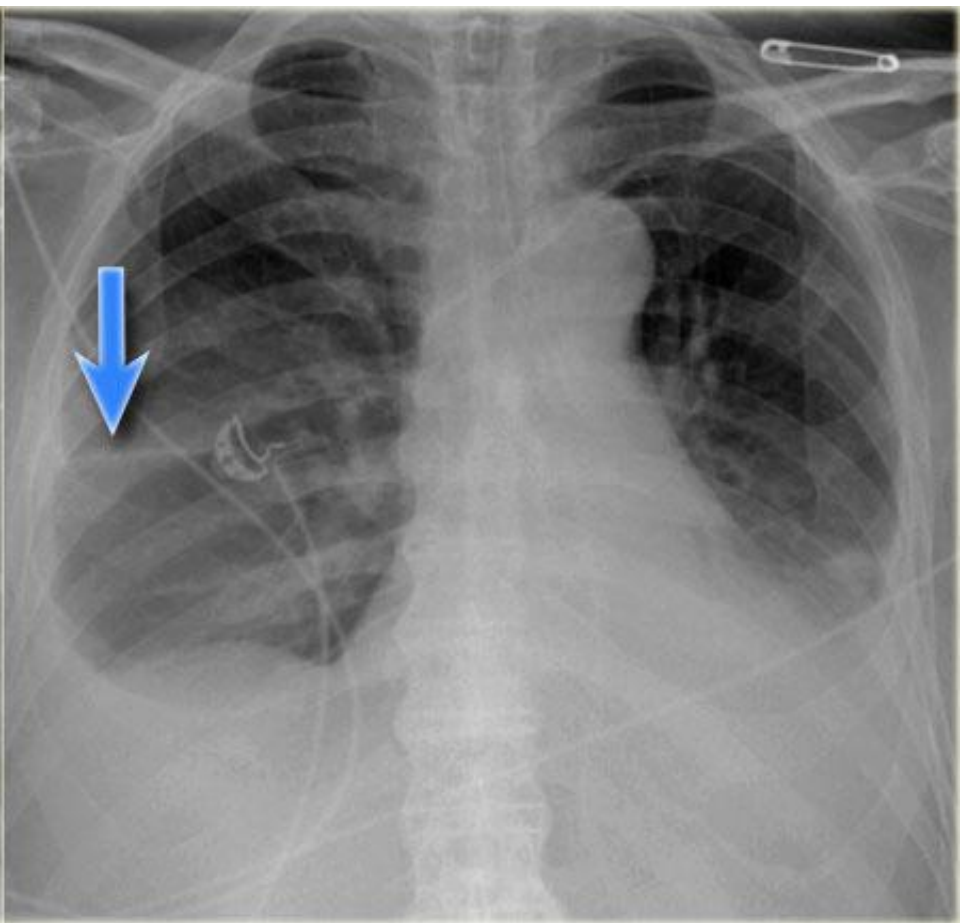
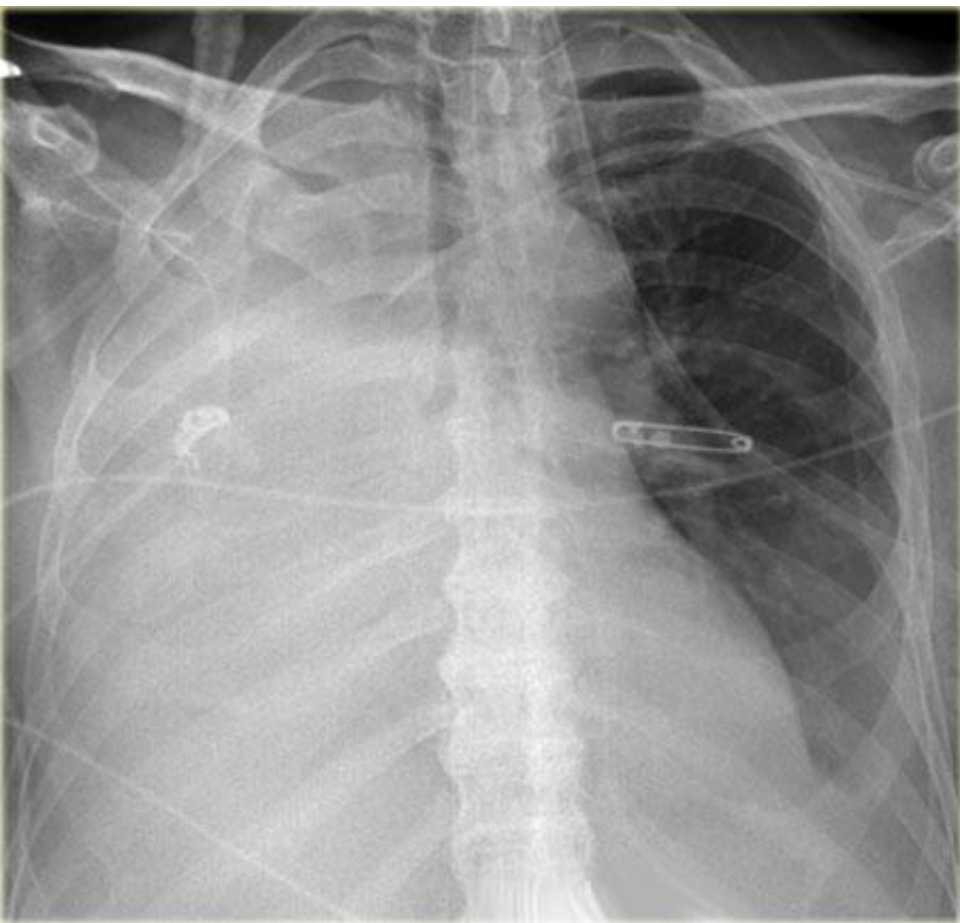
The chest x-ray shows total atelectasis of the right lung due to mucus plugging.

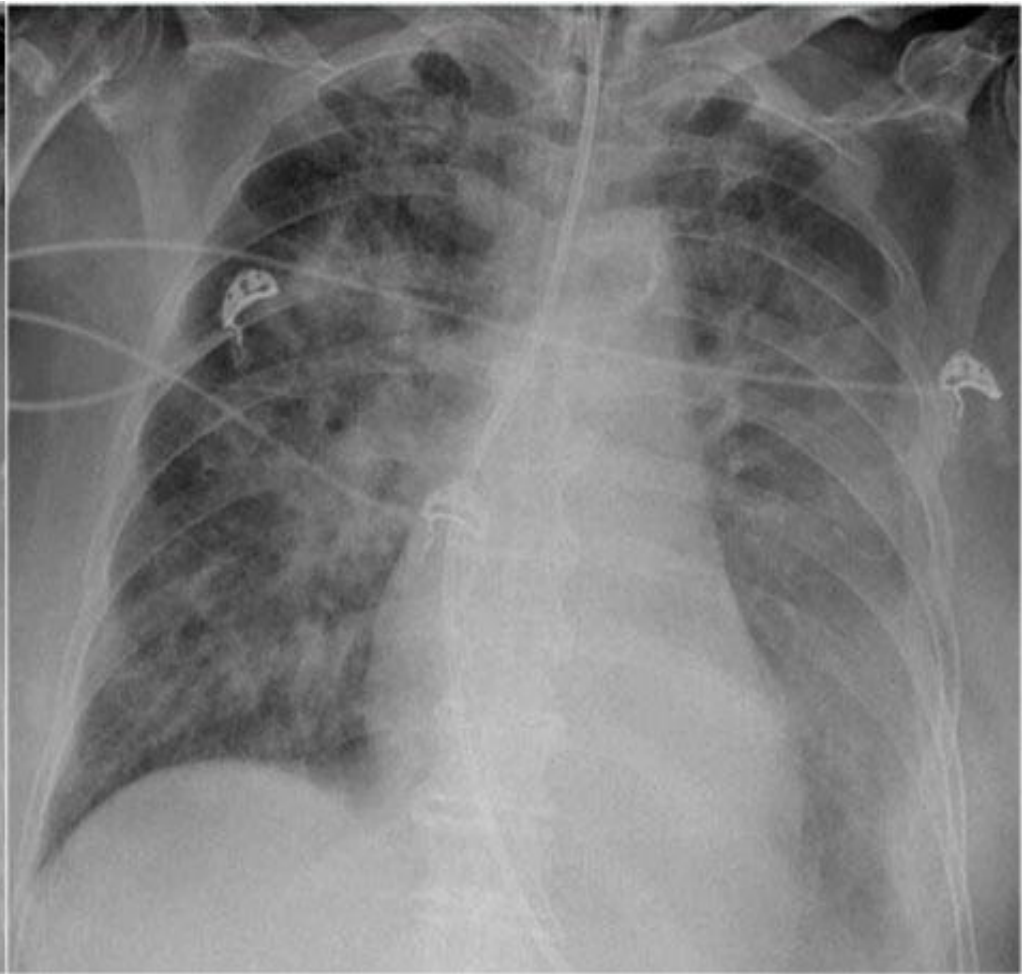
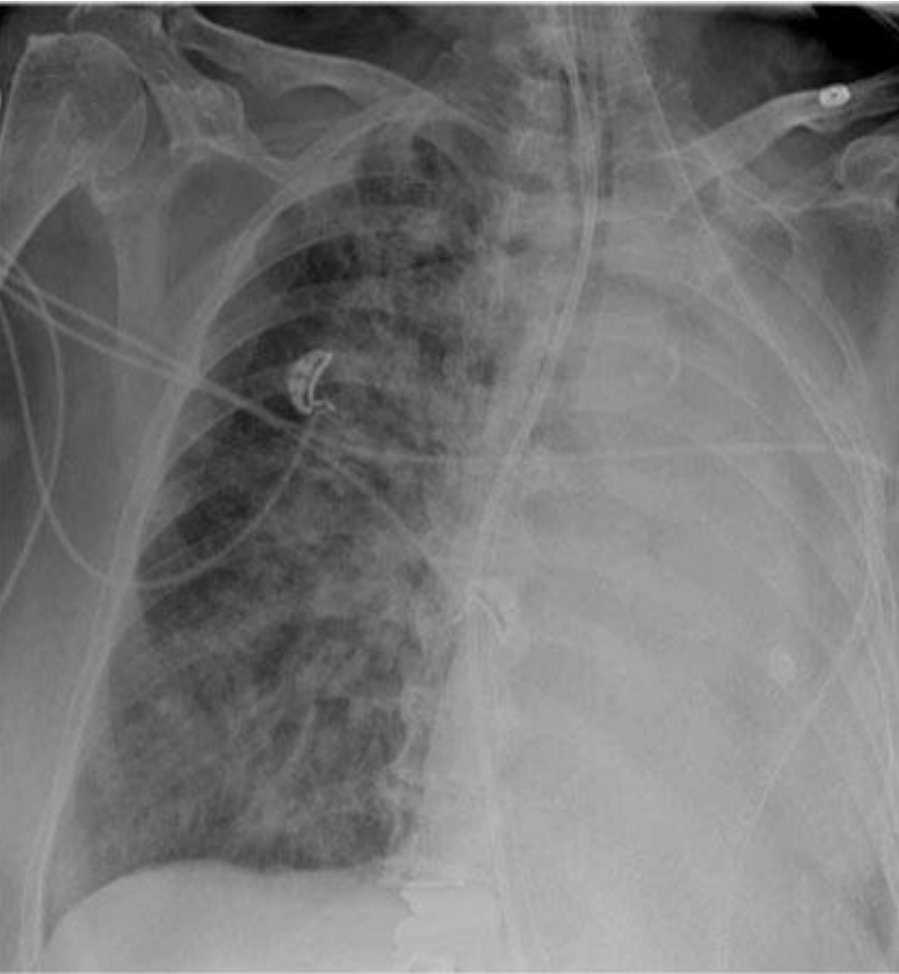
Notice the displacement of the mediastinum to the right.

Re-aeration on follow-up chest film after treatment with a suction catheter.

The mediastinum has regained its normal position.

A common cause of total atelectasis of a lung is a ventilation tube that is positioned too deep and thus obstructing one of the main bronchi.





- These images are of a patient who had widespread bronchopneumonia and was on ventilation.
- During follow up a white out on the left was seen. This was caused by a large mucus plug. After suction of the mucus plug the left lung was re-aerated.

- The chest x-ray shows a nearly total opacification of the left hemithorax.

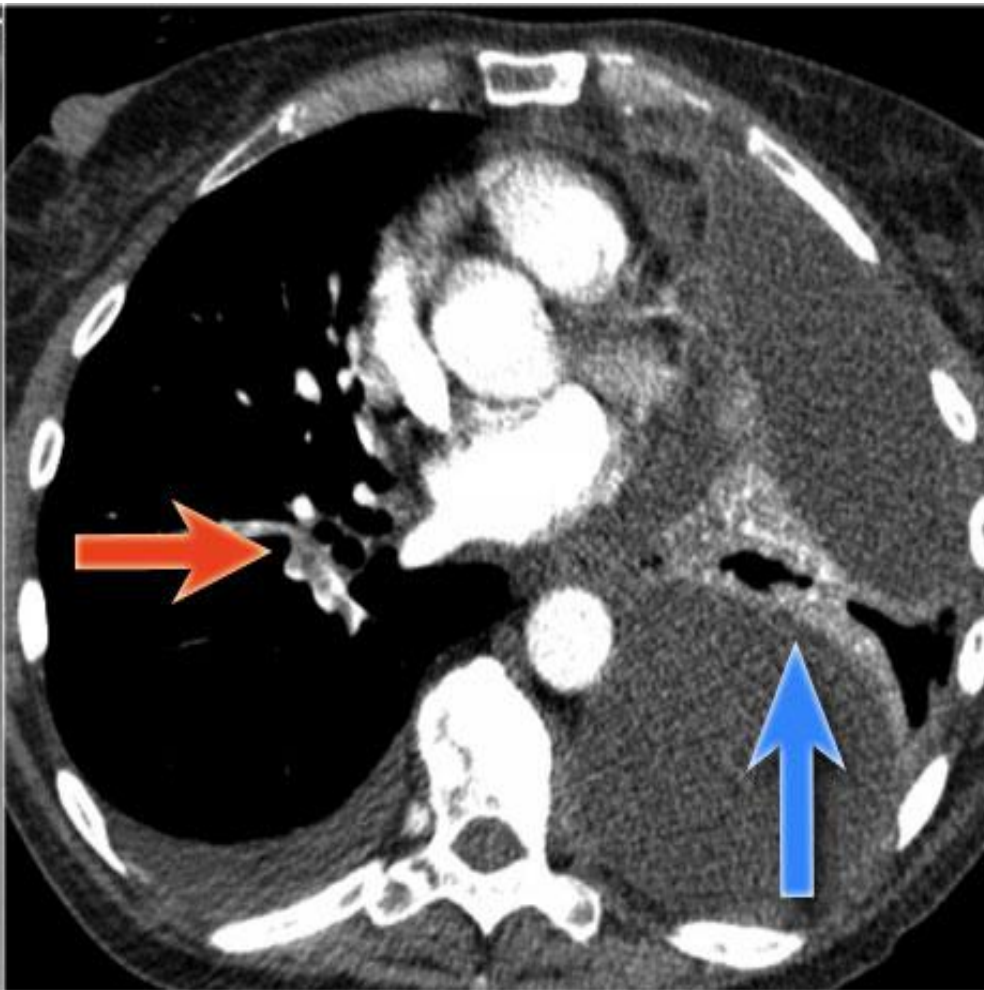
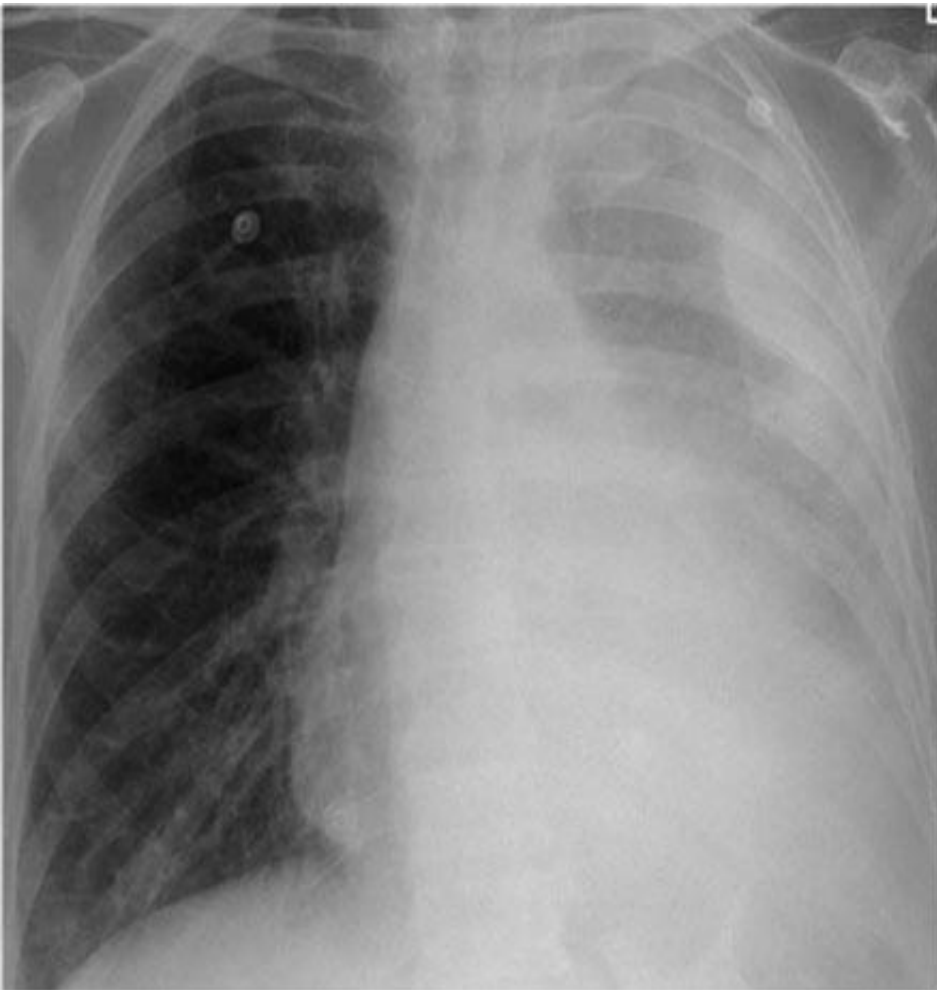
This patient was known to have pleuritic carcinomatosis.

The left lung is almost completely compressed by the pleural fluid.

Unlike most of the above cases, which were caused by obstruction, in this case the atelectasis is a result of compression.

The compression of the lung by the loculated fluid collections is best seen on the CT-image (blue arrow).

- The CT-scan was performed, because the patient was suspected of having pulmonary emboli (red arrow).



Rounded Atelectasis

CT findings

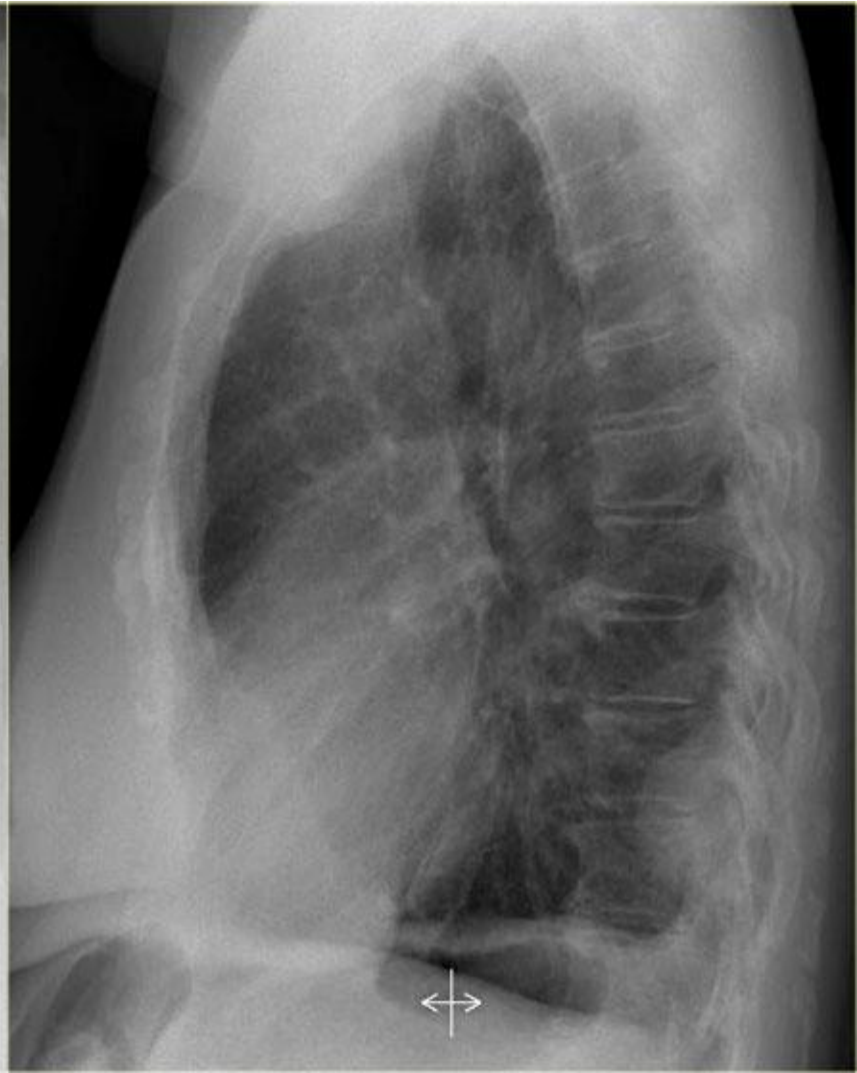
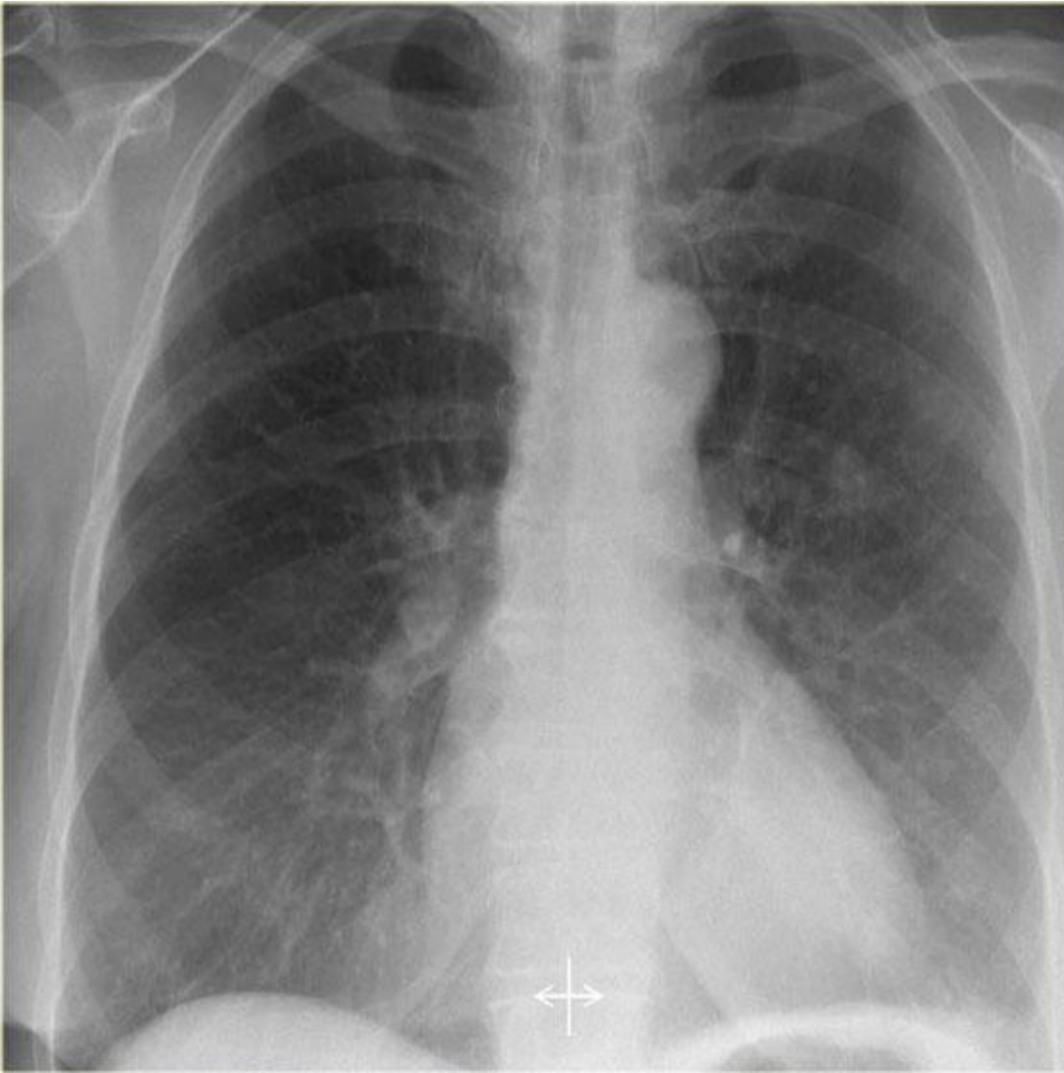
1. Pleural thickening
2. Round, oval or triangular shaped mass adjacent to the pleura
3. Comet tail sign - vessels converge into the mass in a curvilinear fashion

Rounded atelectasis

- The typical findings of rounded atelectasis on CT are pleural thickening, pleural-based mass and comet tail sign.
- The theory is that a local pleuritis causes the pleura to thicken and contract.
- The underlying lung shrinks and atelectasis develops in a round configuration.

The distorted vessels appear to be pulled into the mass and resemble a comet tail (4).

- First study the images and then continue reading.
- On the lateral view there is a mass-like lesion that is pleural-based.
The first impression is, that this is a pleural lesion.
- A CT was performed - see next images.



The CT shows a lesion that originates in the lung.

Many would have a lungcancer on the top of their differential diagnostic list.

However there is also some pleural thickening (red arrow) and vessels seem to swirl around the mass (blue arrows).

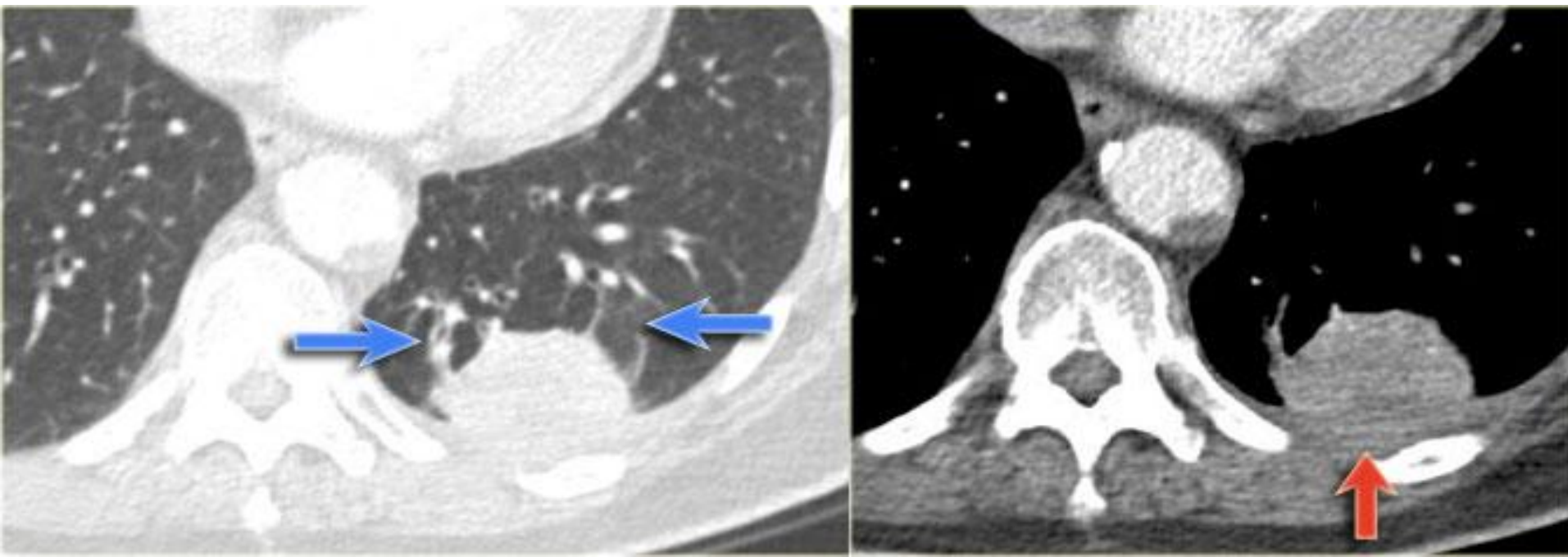
This is also described as the comet tail sign.

Whenever you see a pleural-based lesion that looks like a lungcancer, also consider the possibility of rounded atelectasis.

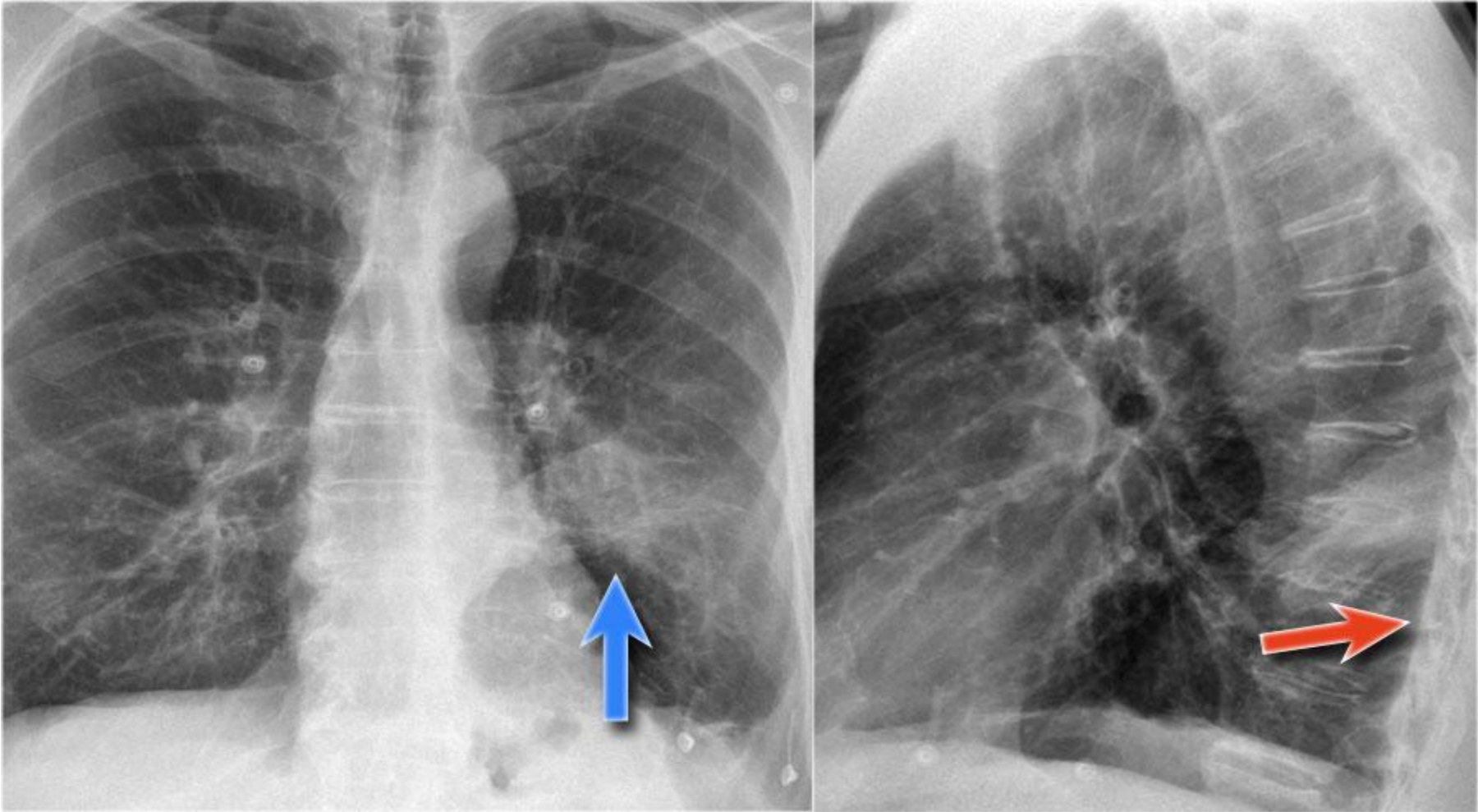
Rounded atelectasis is a benign lesion and when the findings are convincing, then biopsy is not needed.

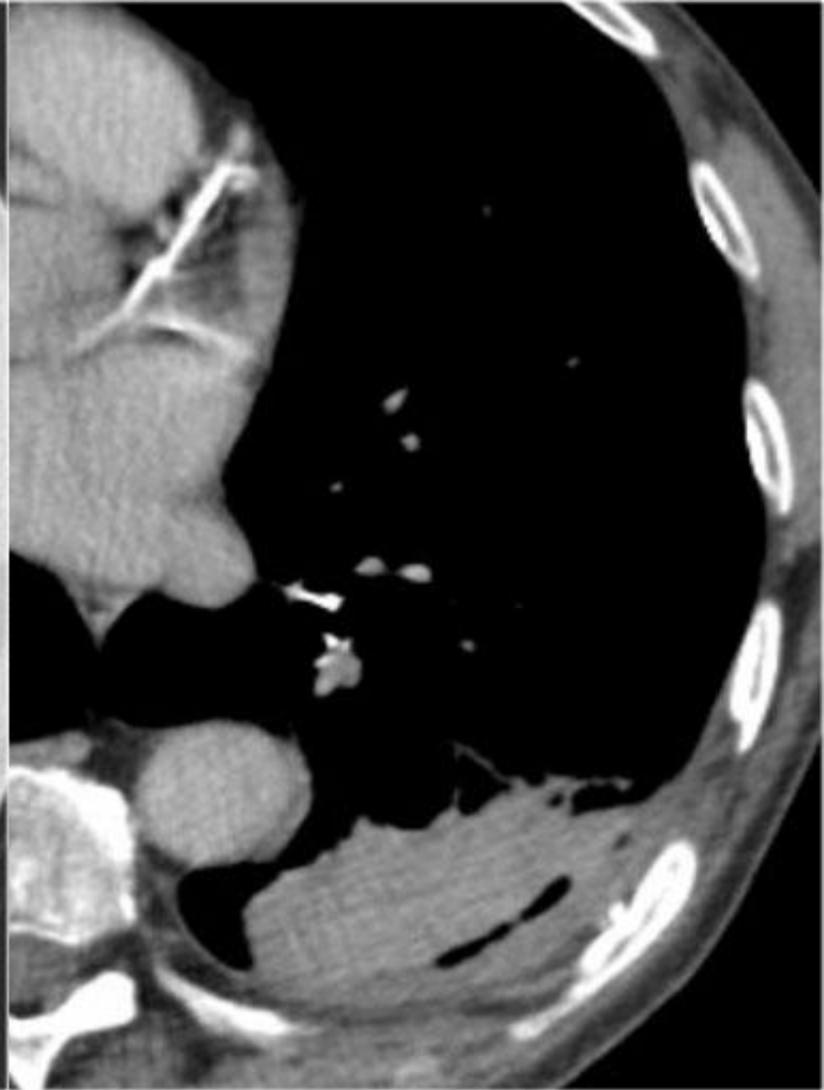
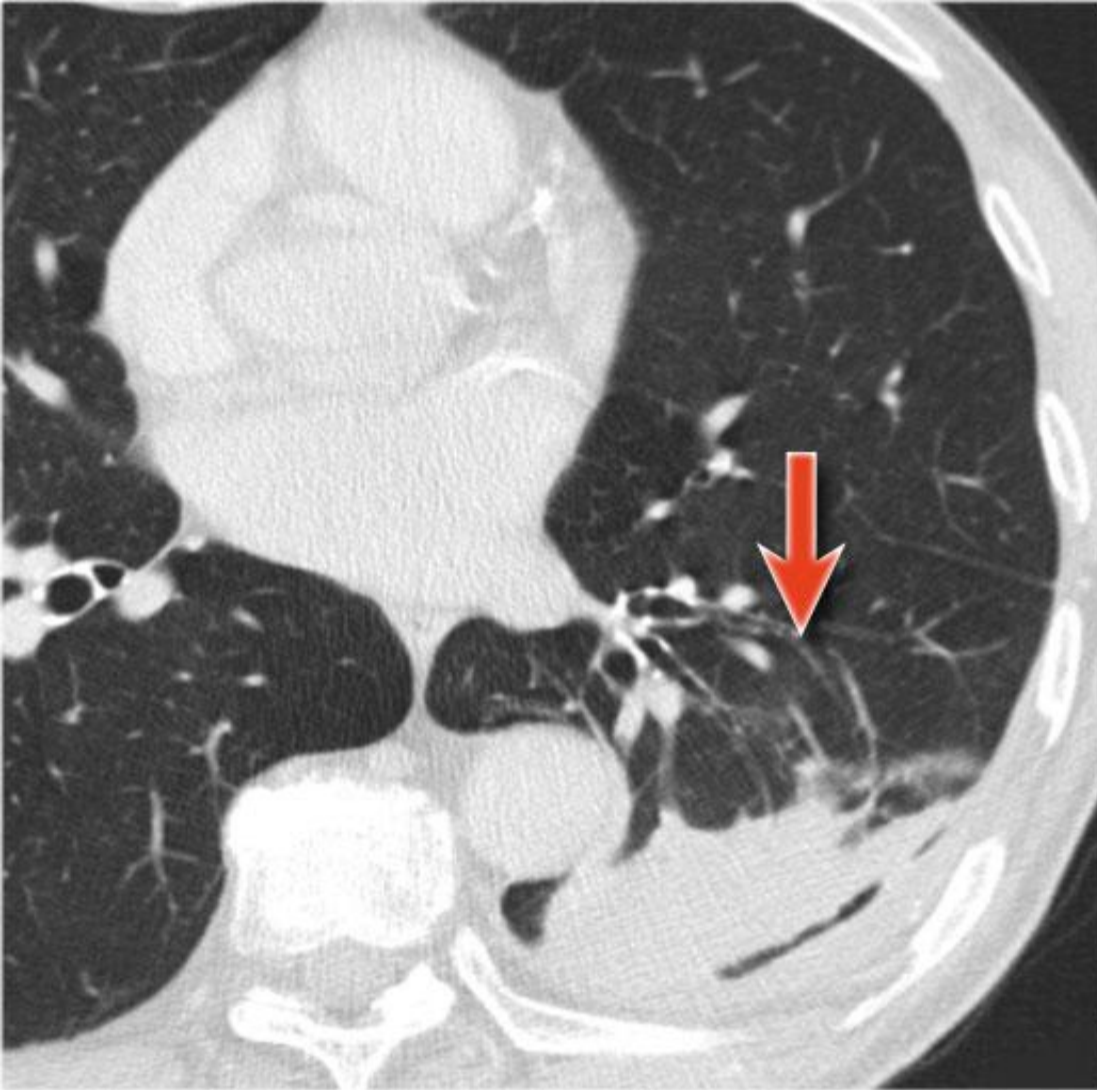
During follow up these lesions usually do not change in configuration.

Rounded atelectasis is frequently seen in patients with a history of asbest exposure.



- The images show a density posteriorly in the left lower lobe.
On the PA-film this looks like a mass or possibly a consolidation.
On the lateral film however the boundaries seem to be sharp, which is in favor of a mass.
Also notice that the pleura is thickened (red arrow).
- Although a peripheral lungcancer is on top of our list, we now also consider the possibility of rounded atelectasis.





- The CT-images show the typical features of a rounded atelectasis.
- There is an oval mass, pleural thickening and a comet tail sign (arrow).
- This lesion did not change in a two-year follow up.

Plate-like atelectasis

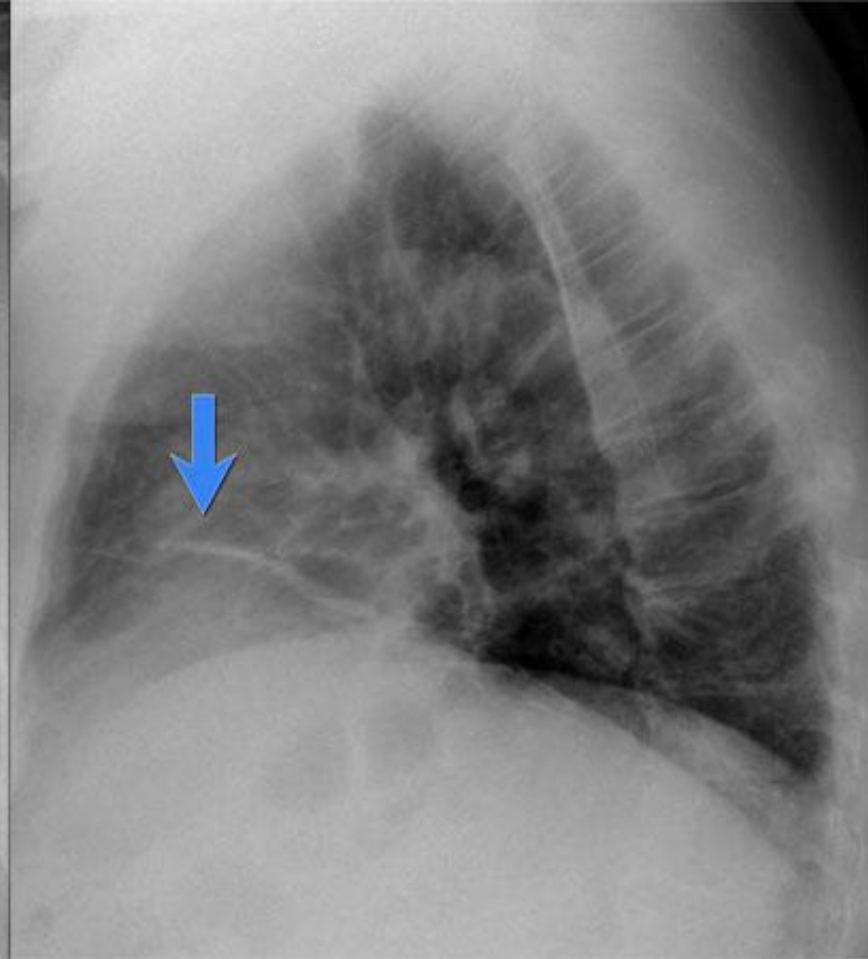
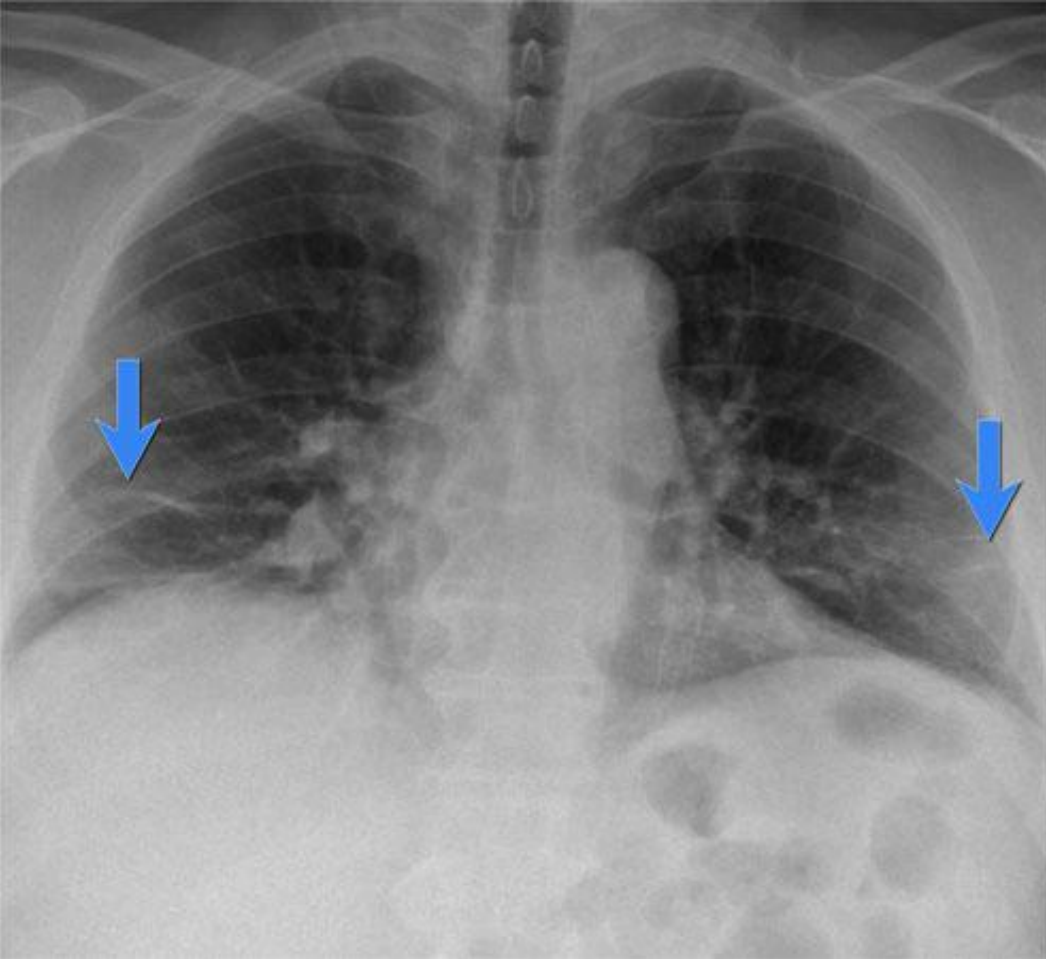
Plate-like atelectasis is a common finding on chest x-rays and detected almost every day.

They are characterized by linear shadows of increased density at the lung bases.

They are usually horizontal, measure 1-3 mm in thickness and are only a few cm long.

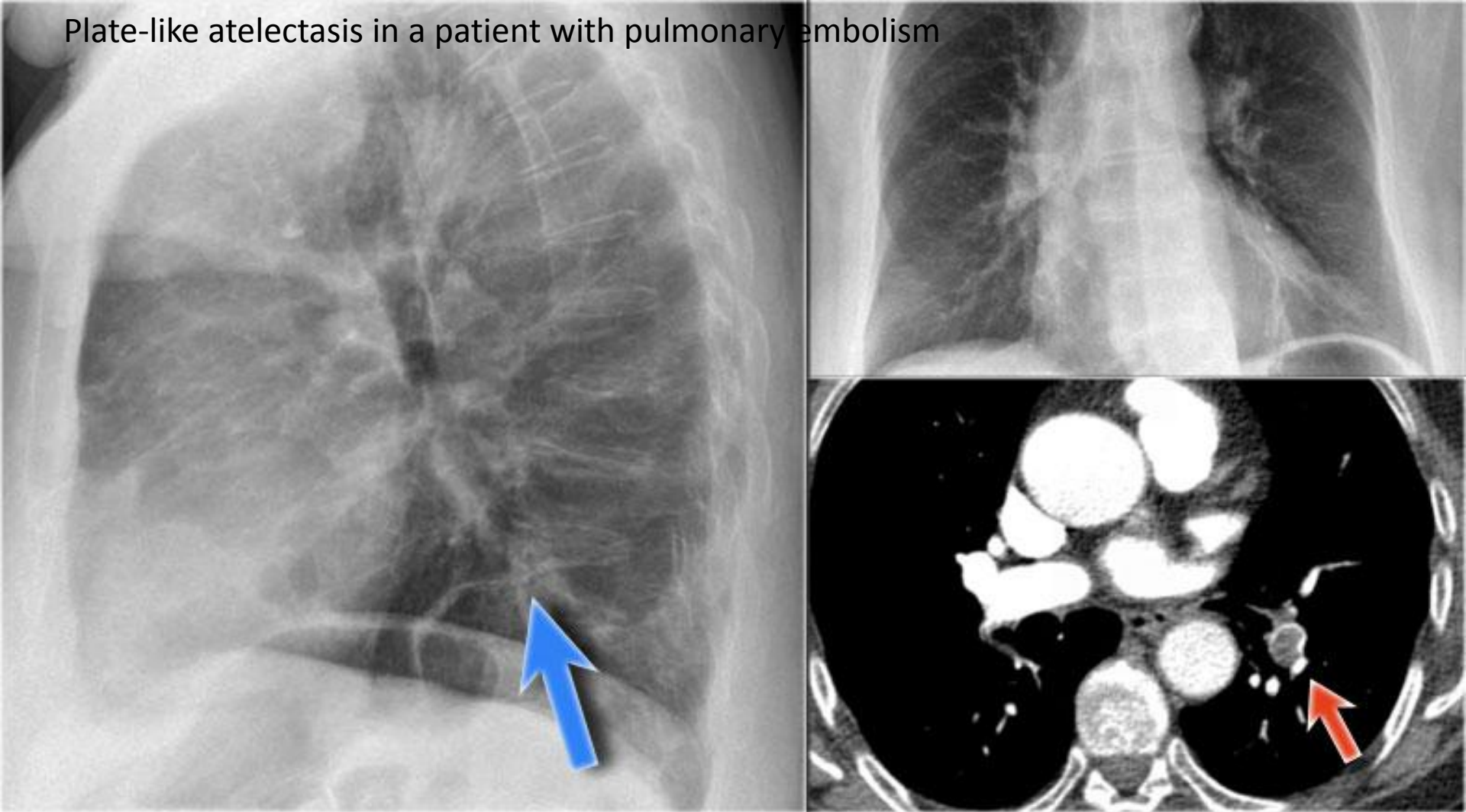
In most cases these findings have no clinical significance and are seen in smokers and elderly.

They are seen in patients, that are in a poor condition and who breathe superficially, for instance after abdominal surgery (figure).



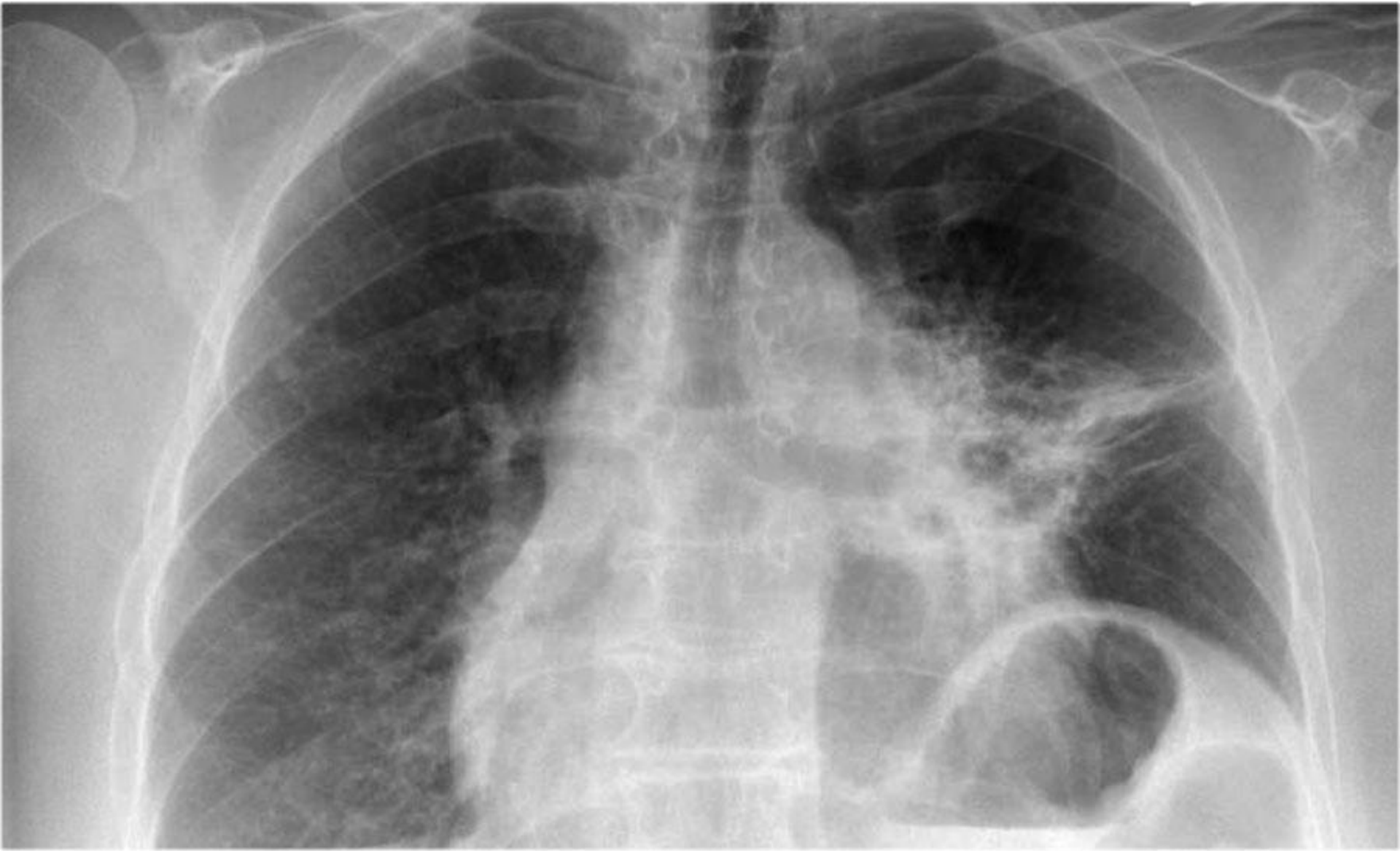
- Plate-like atelectasis due to poor inspiration in a patient who had abdominal surgery

Plate-like atelectasis in a patient with pulmonary embolism

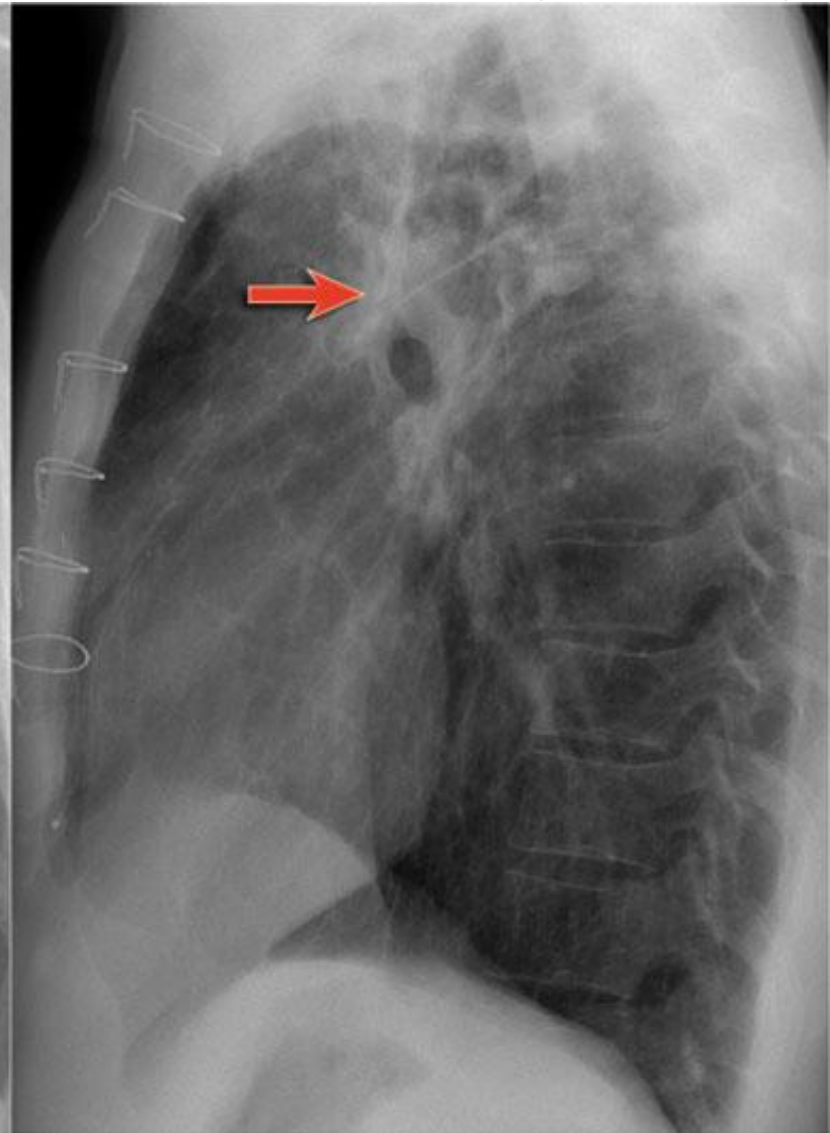
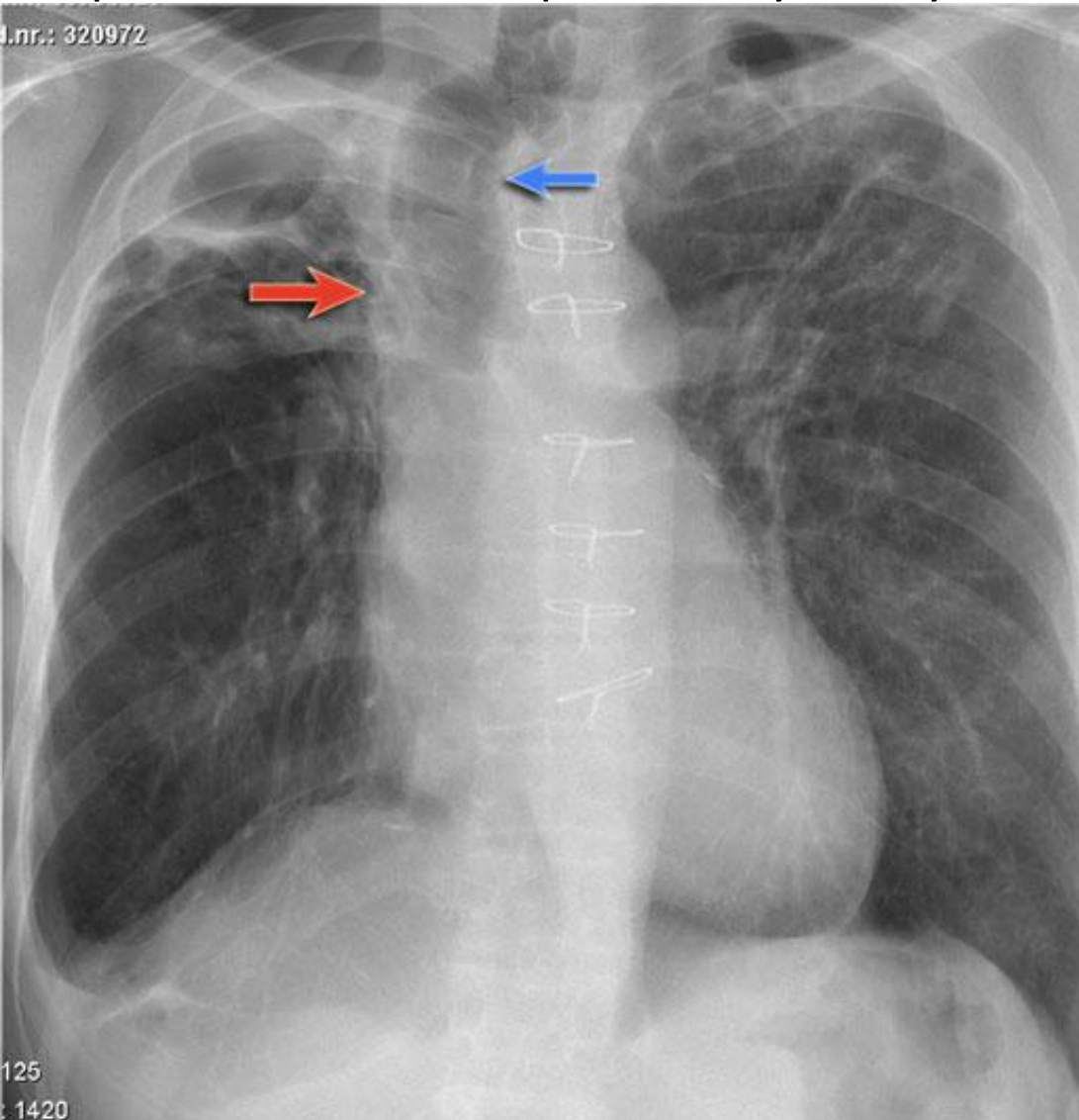


- Plate-like atelectasis is frequently seen in patients in the ICU due to poor ventilation.
- Plate-like atelectasis is also frequently seen in pulmonary embolism, but since it is non-specific, it is not a helpful sign in making the diagnosis of pulmonary embolism.

- **Cicatricial atelectasis**
- Atelectasis can be the result of fibrosis of lung tissue. This is seen after radiotherapy and in chronic infection, especially TB.
- Here we have a patient who was treated with radiotherapy for lung cancer. Notice the increased density of the lung tissue and the volume loss.



- Here we have a patient with atelectasis of the right upper lobe as a result of TB.
- Notice the deviation of the trachea. There is also some atelectasis of the left upper lobe, which results in a high position of the left pulmonary artery as seen on the lateral view (red arrow)



Nodules and Masses

Solitary Pulmonary Nodule

A solitary pulmonary nodule or SPN is defined as a discrete, well-marginated, rounded opacity less than or equal to 3 cm in diameter.

It has to be completely surrounded by lung parenchyma, does not touch the hilum or mediastinum and is not associated with adenopathy, atelectasis or pleural effusion.

The differential diagnosis of SPN is basically the same as of a mass except that the chance of malignancy increases with the size of the lesion.

Lesions smaller than 3 cm, i.e. SPN's are most commonly benign granulomas, while lesions larger than 3 cm are treated as malignancies until proven otherwise and are called masses.

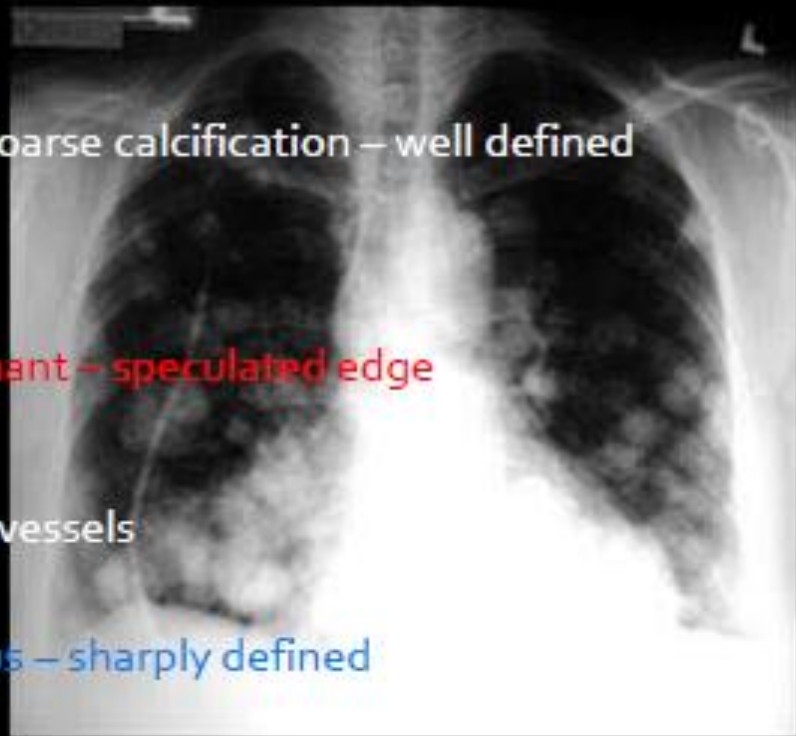
The tabel is adapted from [chest x-ray - a survival guide](#).

NODULES

3cm or less

6 COMMON

- ◆ Tuberculoma
 - ◆ Hamartoma
 - ◆ Bronchogenic ca.
 - ◆ Metastases
 - ◆ AVM + afferent & efferent vessels
 - ◆ Hydatid cyst
- Benign – Coarse calcification – well defined
- Malignant – speculated edge
- Homogeneous – sharply defined



SPN differential diagnosis

Most common

- Granuloma

Less common

- Bronchial carcinoma
- Metastasis
- Organizing pneumonia
- Hamartoma

Uncommon and rare

- Mucoid impaction
- Abscess
- Infected bulla
- Infarction
- Hematoma
- Rheumatoid nodule
- Wegener's granuloma
- Carcinoid
- Sarcoid granuloma
- AV-malformation
- Intrapulmonary lipoma or lymph node
- Hydatid cyst

Multiple nodules

pleural nodules

no pleural nodules

subpleural, septal,
peribronchovascular,

diffuse
no predominance

perilymphatic
distribution

random
distribution

centrilobular
distribution

Sarcoid
silicosis
lymphangitic ca

Miliary infect:
- TB, fungal
Metastases

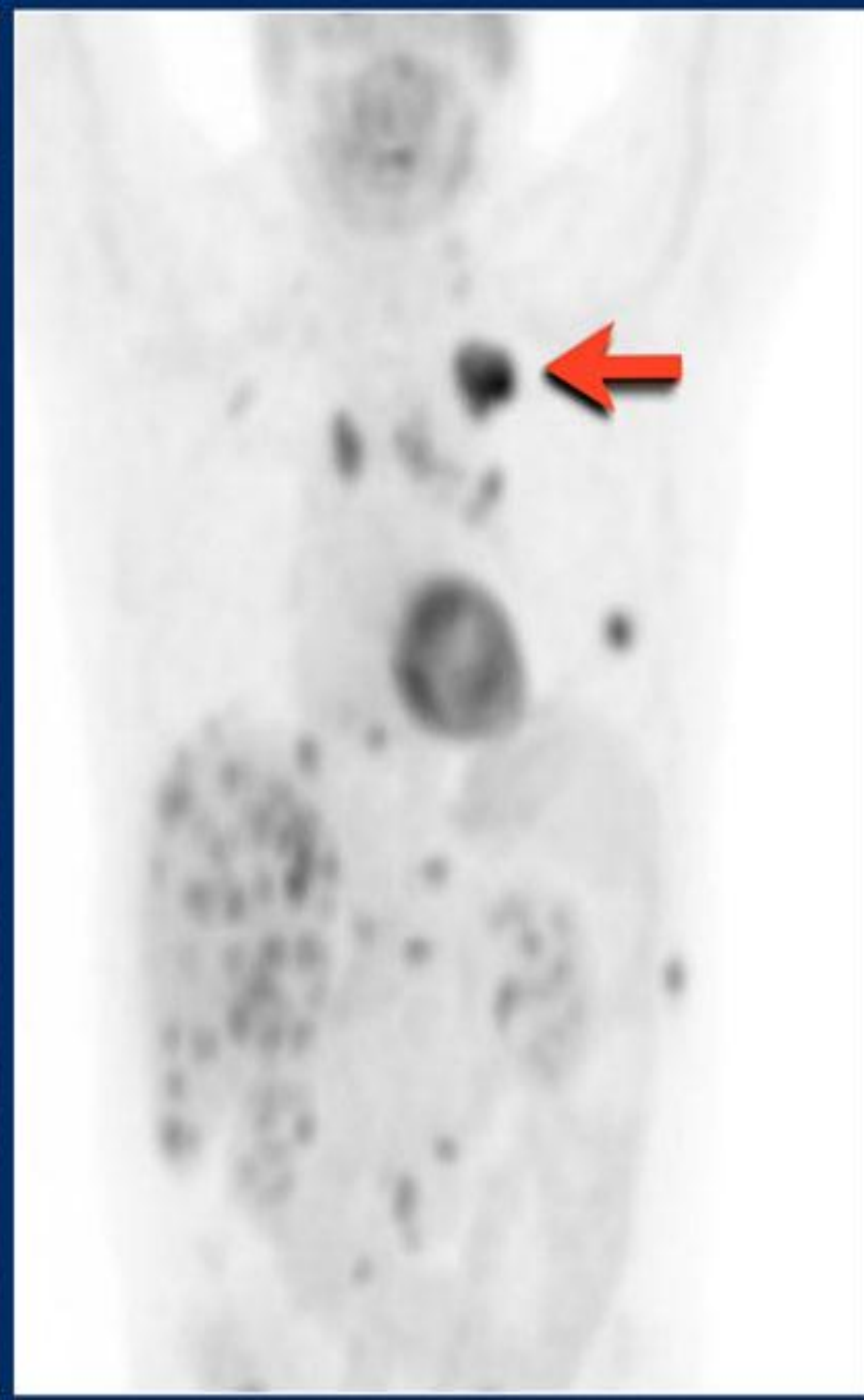
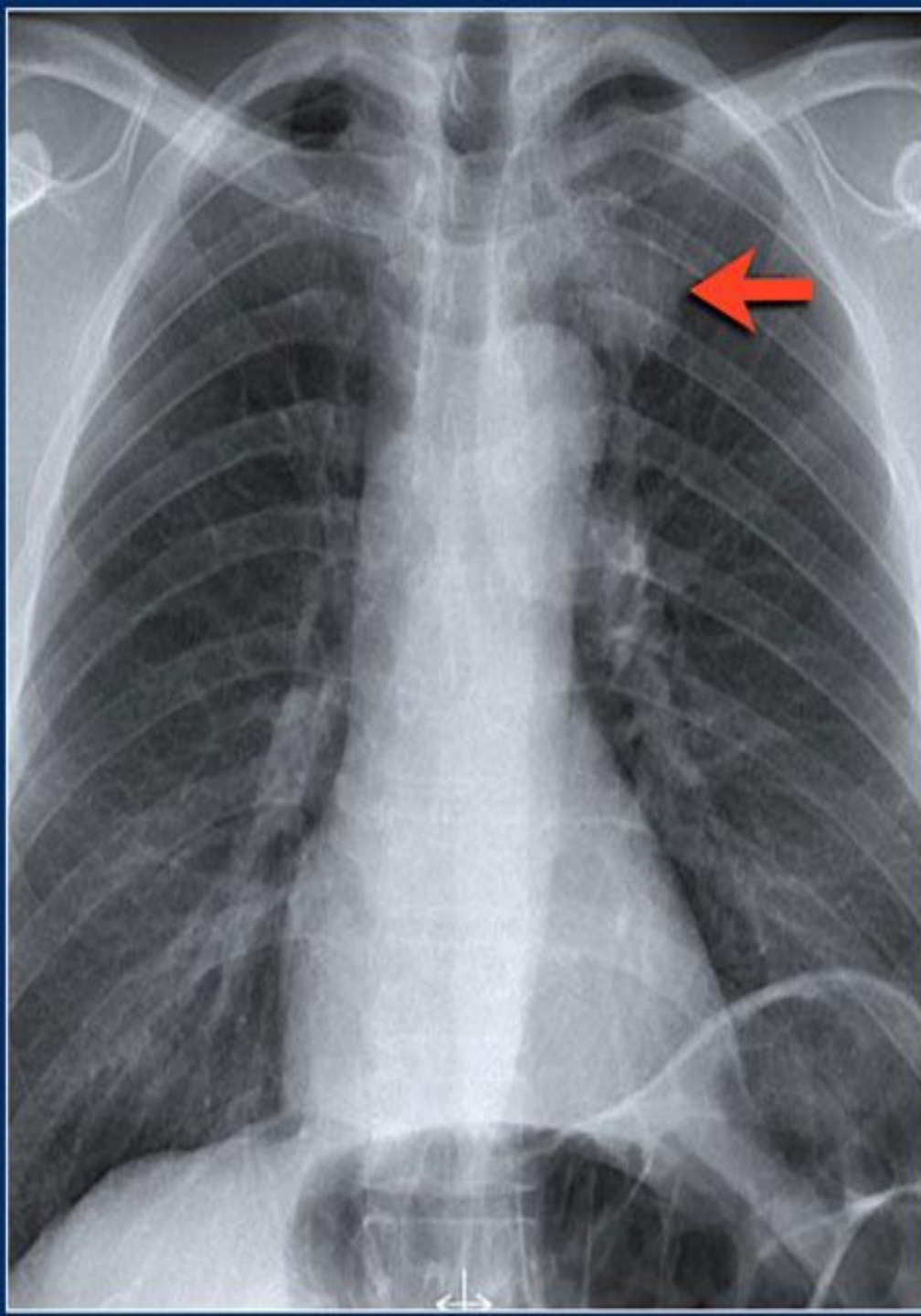
Hypersens. pneum
Resp bronchiolitis
Bronchioloalv ca
Infectious diseases

In lesions that do not respond to antibiotics, probably the most important non-invasive diagnostic tool is nowadays the PET-CT.

PET-CT can detect malignancy in focal pulmonary lesions of greater than 1 cm with a sensitivity of about 97% and a specificity of 78%.

False-positive findings in the lung are seen in granulomatous disease and rheumatoid disease.

False negatives are seen in low grade malignant tumors like carcinoid and alveolar cell carcinoma and lesions of less than 1 cm.



Fleischner Society recommendations for follow-up of nodules

Previous chest radiographs should be reviewed to determine if the lesion has been stable over 2 years.

If so, no further follow up is necessary, with the exception of *pure **ground-glass lesions on CT scans***, which can be slower growing.

For lesions with a benign pattern of calcification, further testing is not necessary.

Management of indeterminate lesions greater than 8-10 mm depends on clinical probability of malignancy, as follows:

- Low probability: Serial CT scanning at 3, 6, 12, and 24 months
- Intermediate probability: PET-CT, contrast-enhanced CT, transthoracic needle aspiration and/or transbronchial needle aspiration (TBNA)
- High probability: Surgical resection

Any unequivocal growth noted during follow up means that a definitive tissue diagnosis is needed.

Guidelines from the Fleischner Society

Low-risk patients

≤ 4 mm	No further investigation
4-6 mm	CT at 12 months
6-8 mm	CT at 6-12-18-24 months
> 8 mm	CT at 3-9-24 months contrast-enhanced CT PET and/or biopsy

High-risk patients

≤ 4 mm	CT at 12 months
4-6 mm	CT at 6-12-18-24 months
6-8 mm	CT at 3-6-9-12-24 months
> 8 mm	CT at 3-6-9-12-24 months contrast-enhanced CT PET and/or biopsy

Notes using Fleischner guidelines for Solid Pulmonary nodules

- Only apply in subjects 35 years or older.
- Subjects should be separated in low- and high-risk based on smoking characteristics.
- Do not apply in subjects with a known or suspected malignancy
- Measure the average of diameters in the axial plane
- Use thin-slice unenhanced (low-dose) scans to characterize and follow-up pulmonary nodules

Multiple masses

The differential diagnostic list of multiple masses is very long.

The most important diagnoses are listed in the table.

Sometimes it is difficult to differentiate multifocal consolidations from masses.

Multiple Masses - diff. diagn

Neoplastic

- Metastases
 - Renal - GI - melanoma - uterus
 - testis - sarcoma
- Uncommon:
 - Bronchoalveolar cell ca
 - Lymphoma - Kaposi

Infection

- TB
- Fungal
 - Histoplasmosis
 - Coccidioidomycosis
 - Invasive aspergillosis
- Septic emboli

Inflammation

- Sarcoid
- RA-nodules
- Wegener's granulomas
- Organizing pneumonia

Vascular

- Rendu-Osler - AV-malform.
- Pulmonary infarcts

Inhalation

- Mucoid impaction (ABPA)
- Silicosis

Metastases

-

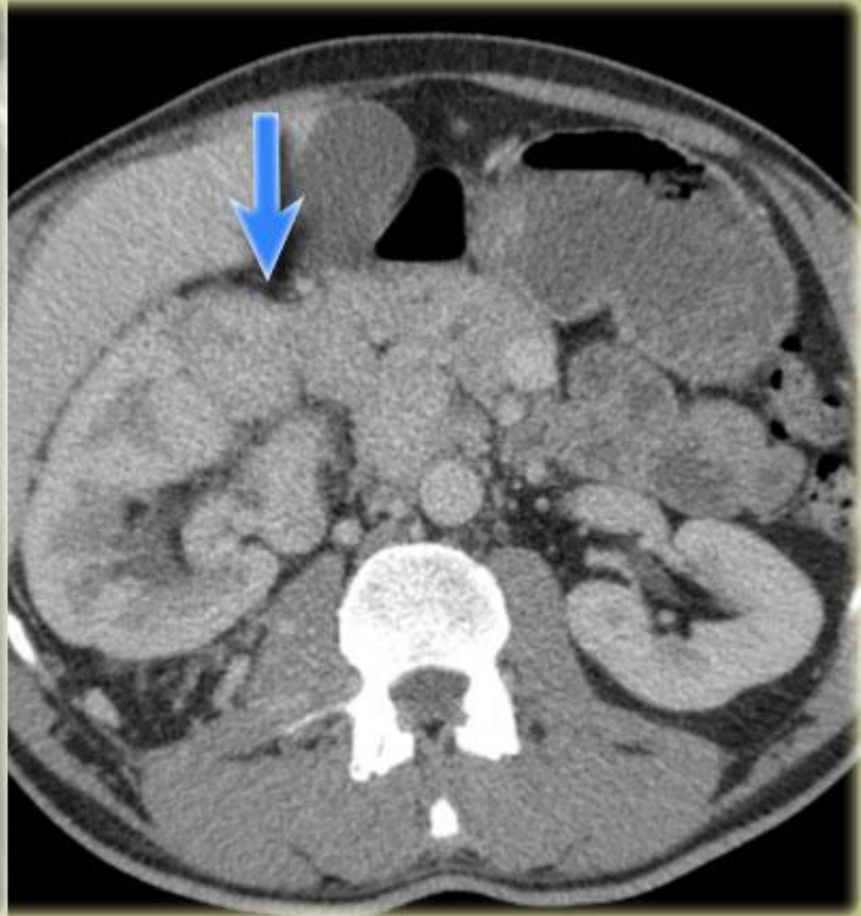
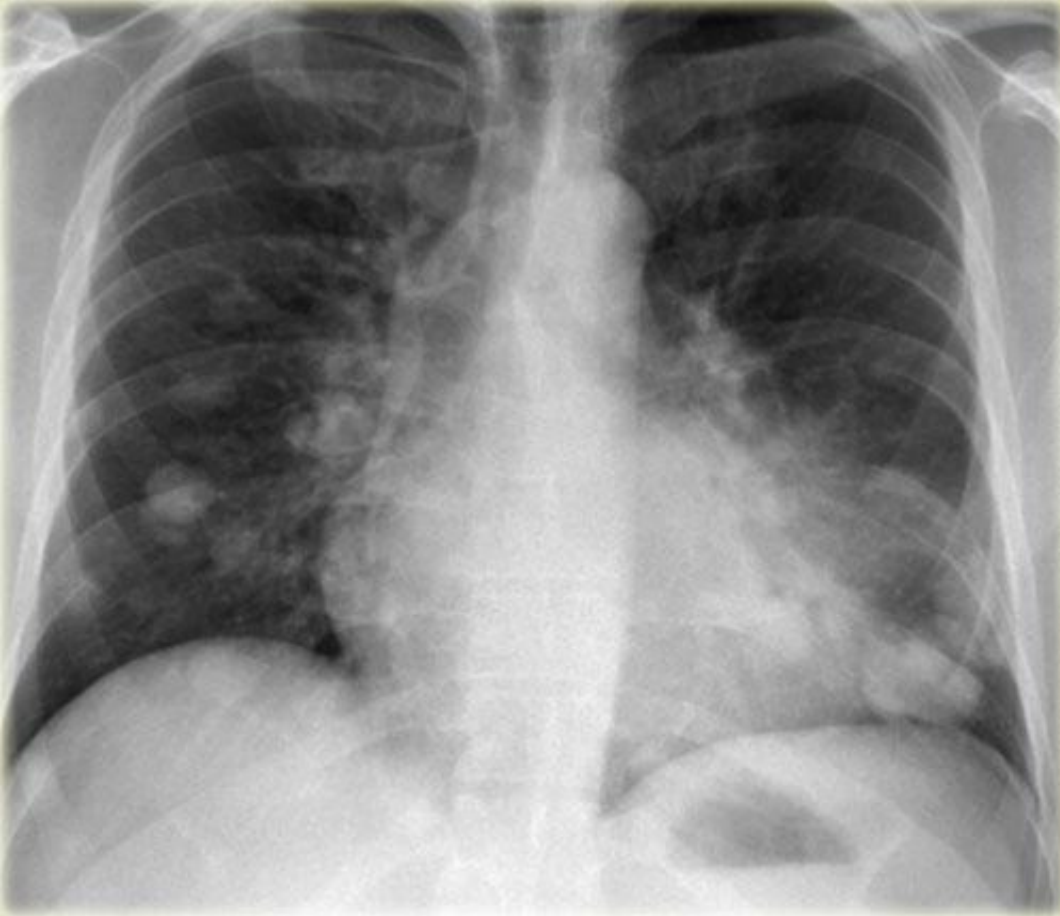
Metastases are the **most common cause of multiple pulmonary masses**.

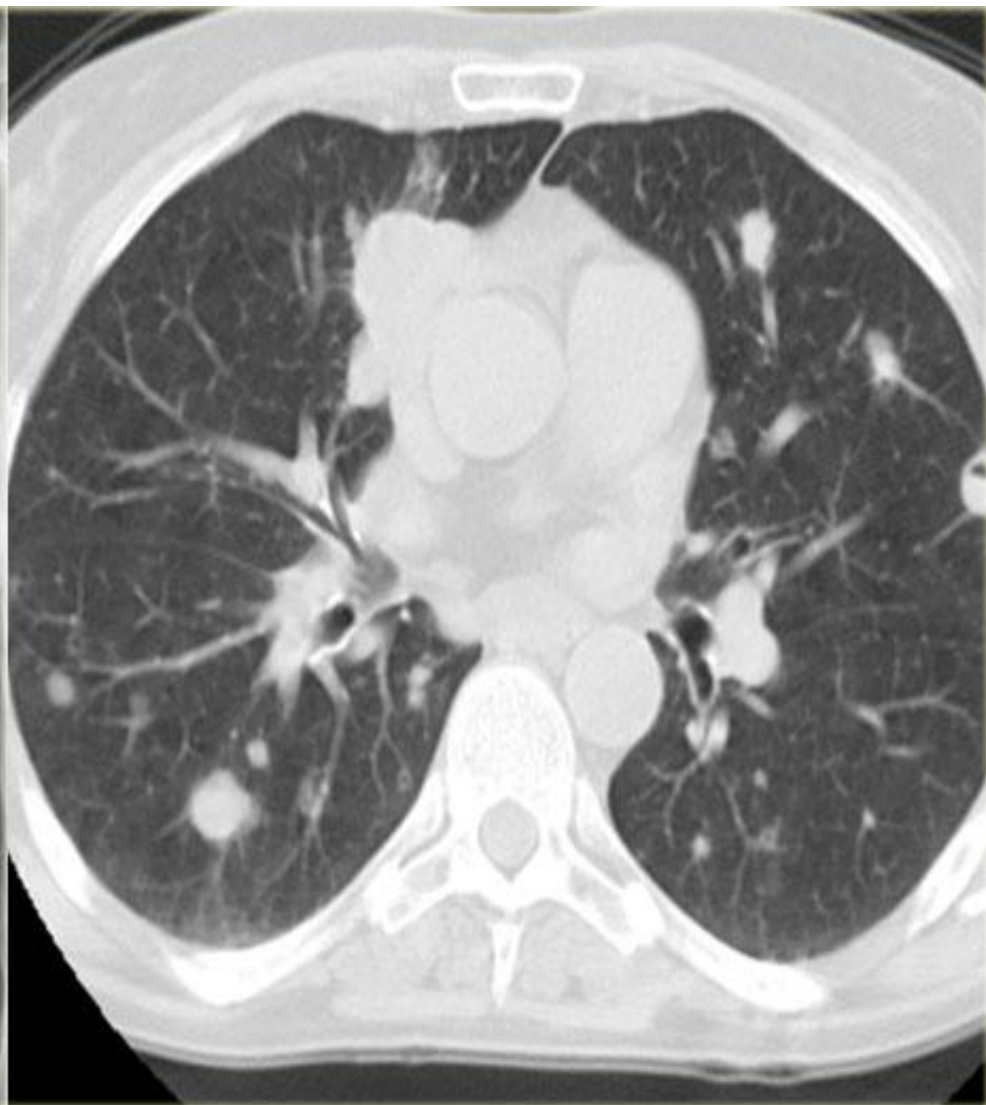
Usually they vary in size and are well-defined.

They predominate in the lower lobes and in the subpleural region.

- HRCT will demonstrate the [random distribution](#) unlike other diseases that have a perilymphatic or centrilobular distribution.

The images show a renal cell carcinoma that has invaded the inferior vena cava with subsequent spread of disease to the lungs.





- Here another patient with widespread pulmonary metastases of a cancer, that was located in the tongue.

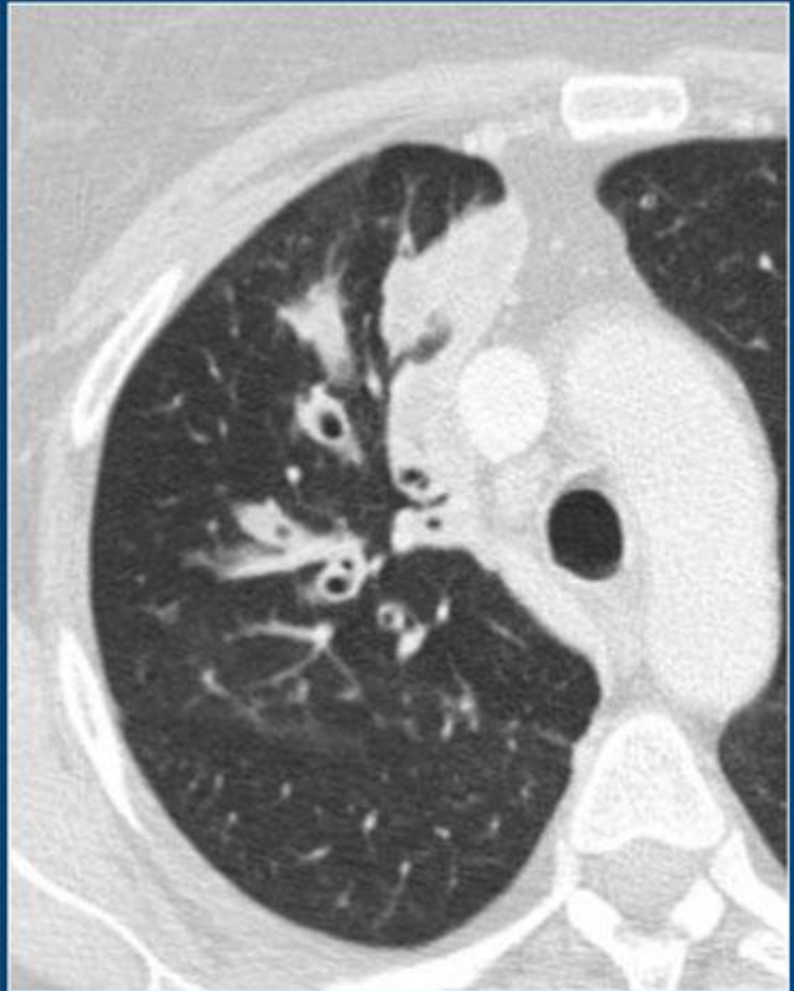
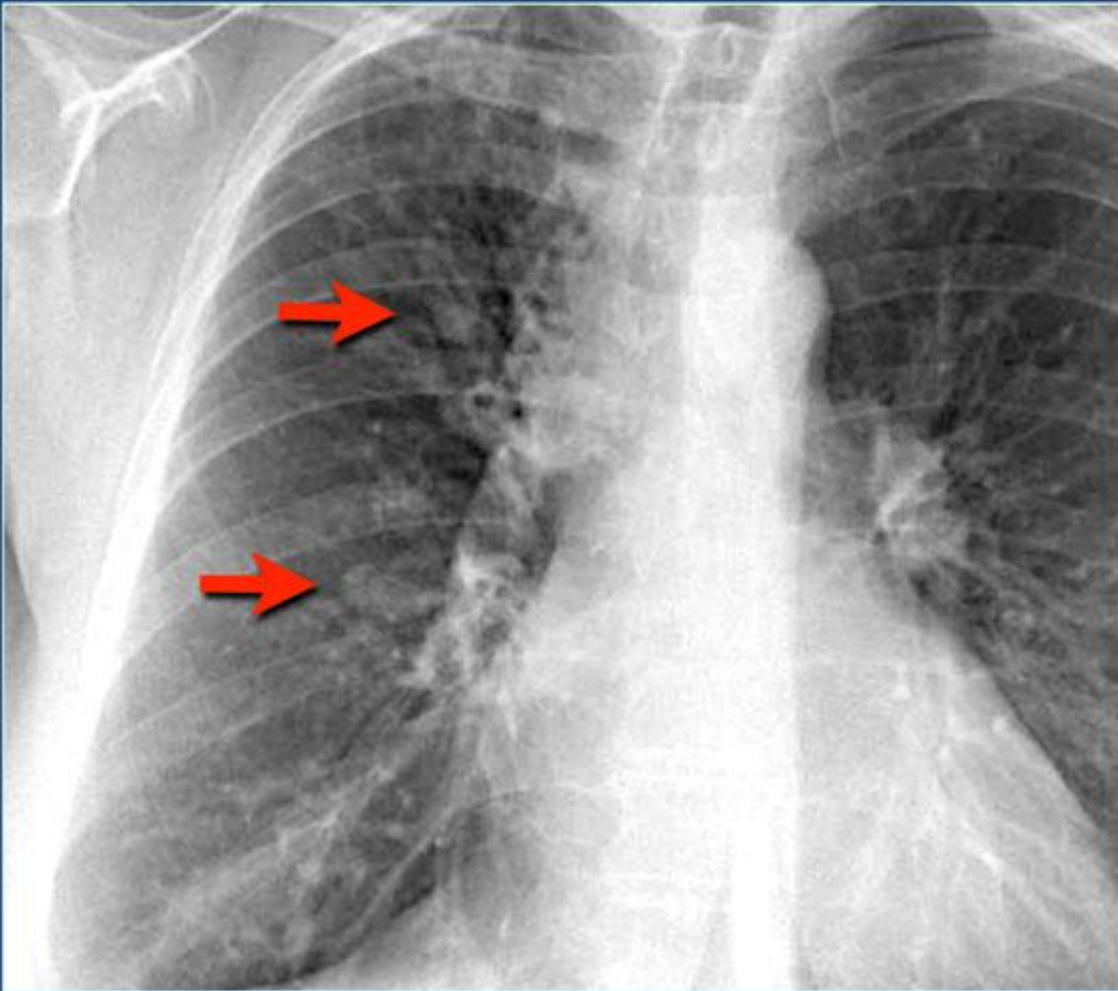
Mucoid impaction

- Mucus plugs or mucoid impaction can ***mimick*** the appearance of lung nodules or a mass.
- Sometimes differentiating mucus impaction from a lung cancer can be difficult.
- Mucoid impaction is commonly seen in patients with ***bronchiectasis***, as in cystic fibrosis (CF) and allergic bronchopulmonary aspergillosis (ABPA).
- ABPA is a hypersensitivity disorder induced by *Aspergillus*, that occurs in patients with asthma or CF.

It is also seen in bronchial obstruction caused by an obstructing tumor or bronchial atresia.

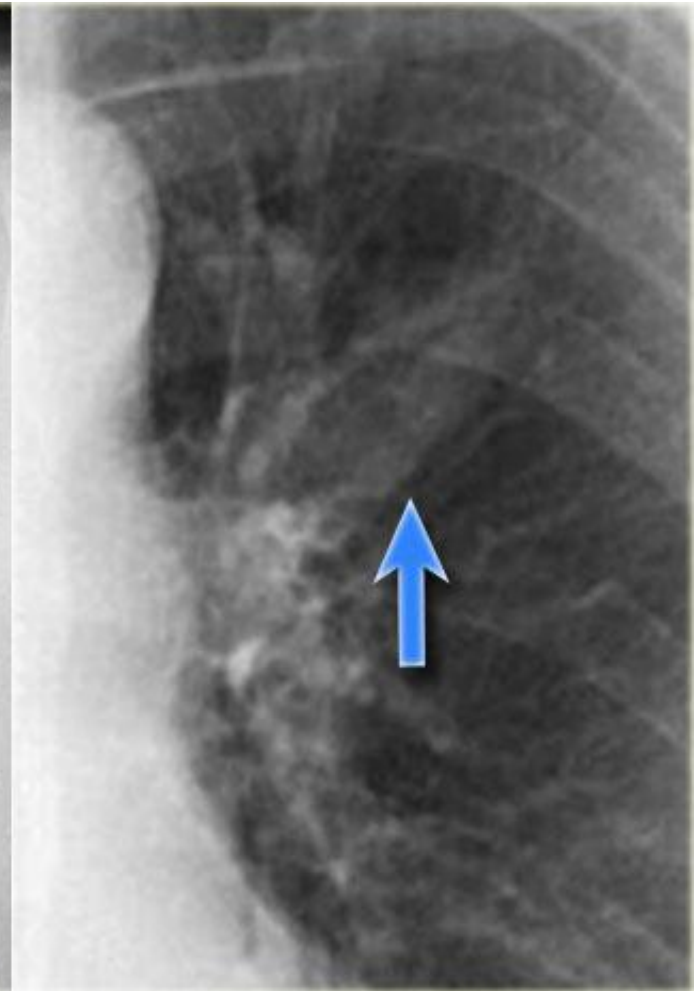
In this case there are some mass-like structures in the right lung.

CT demonstrated bronchiectasis with mucoid impaction.



Mucoid impaction

- A more common presentation of mucoid impaction is seen here.
- This is the typical 'finger-in-glove' appearance of mucoid impaction.
- The mucus in the dilated bronchi looks like the fingers in a glove.



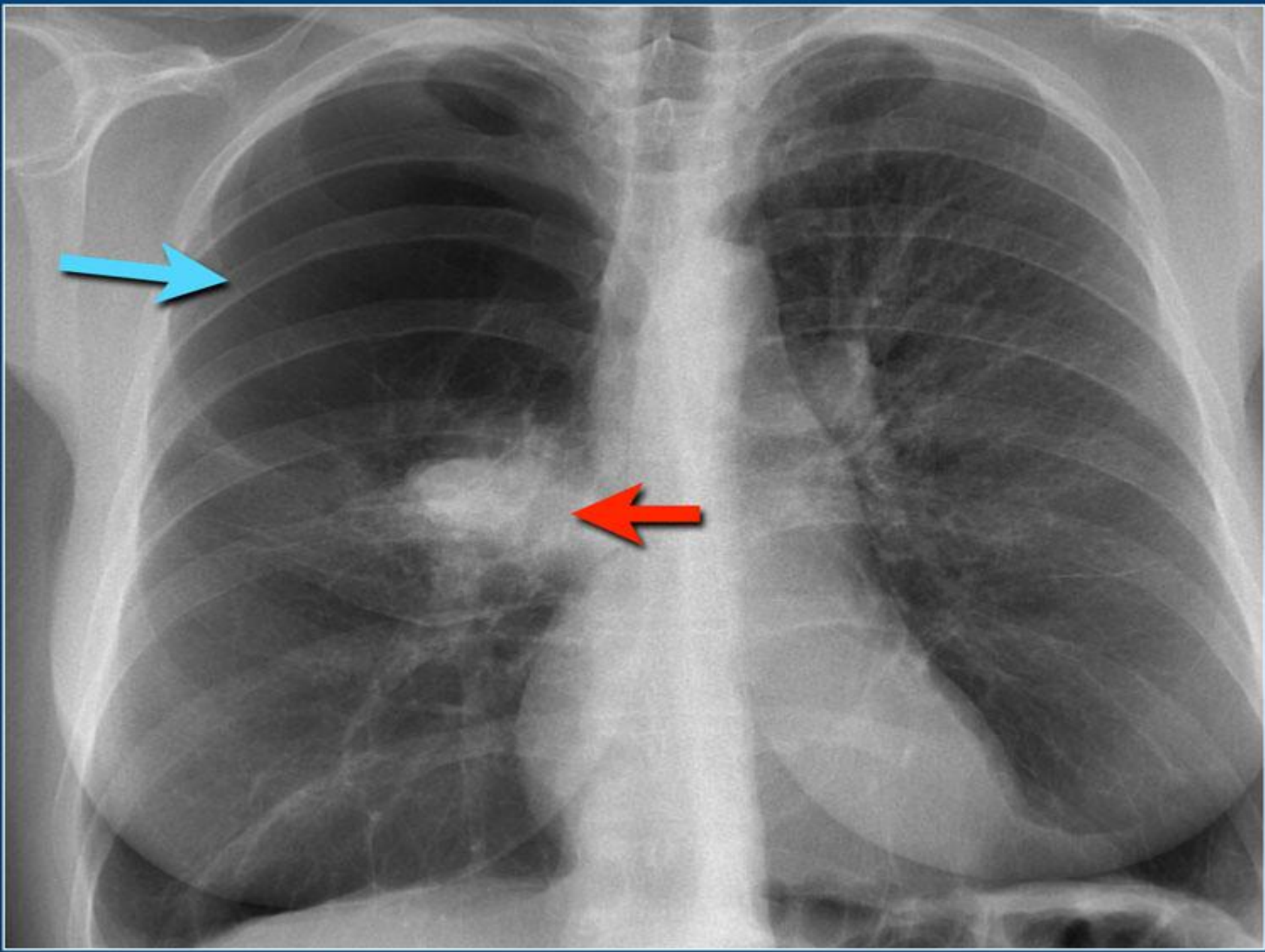
Bronchial atresia

Bronchial atresia is a congenital abnormality resulting from interruption of a bronchus with associated peripheral mucus impaction and associated hyperinflation of the obstructed lung (10).

The hyperinflation of the affected lungsegment is caused by collateral ventilation through the pores of Kohn.

The characteristic finding is a hyperlucent area of the lung surrounding a branching or nodular opacity that extends from the hilum.

Notice the central mass surrounded by hyperlucent lung (blue arrow).



**Decreased density
or
Lucencies**

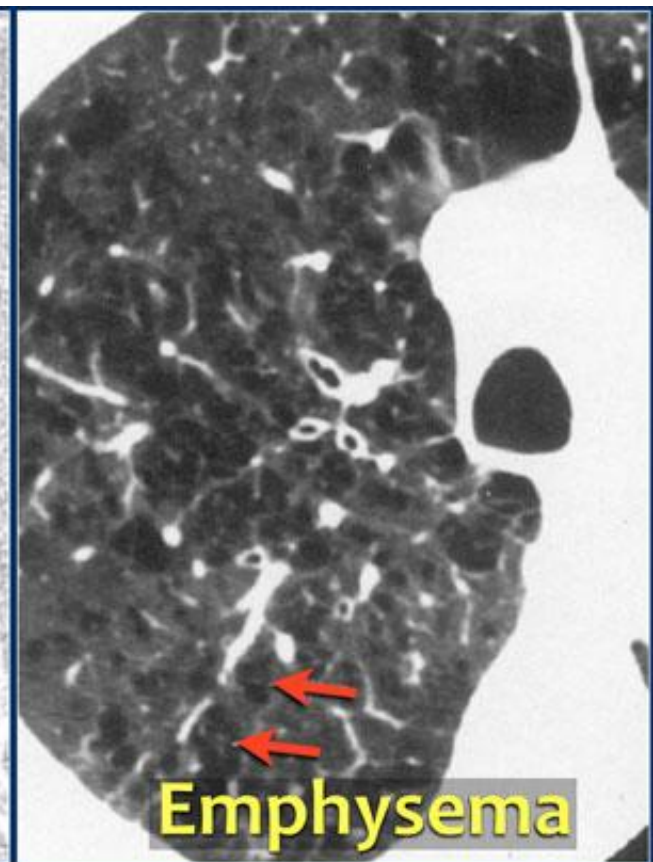
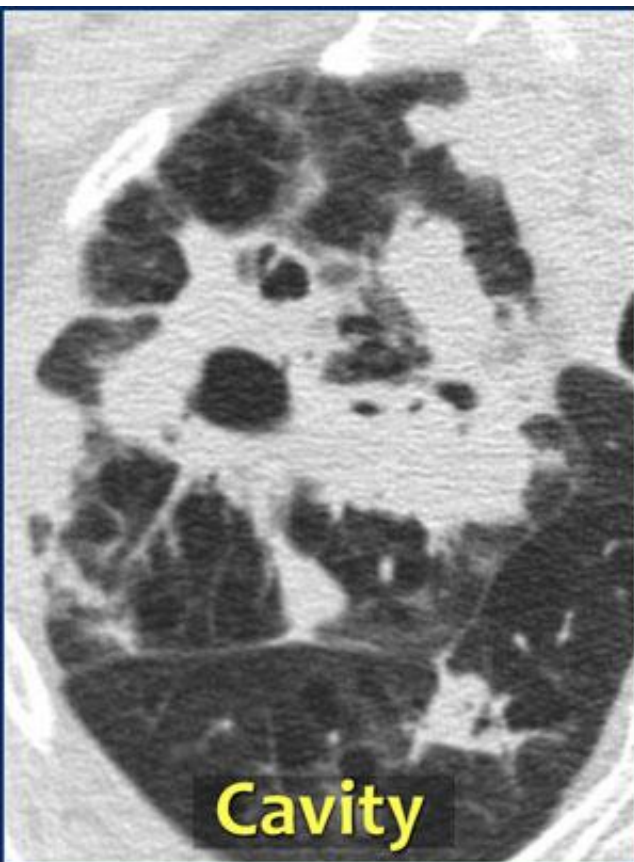
Radiologists use many terms to describe areas of decreased density or lucencies within the lung, like ***cyst, cavity, pneumatocele, emphysema, bulla, honeycombing, bleb etc.***

Many of these terms are based on the pathogenesis of the abnormality. This makes it difficult to use these terms, since in many cases when we describe a chest X-ray, we are trying to figure out what the pathology could be.

A more practical approach is to describe areas of decreased density in the lung as:

- Cavity** - lucency with a thick wall
- Cyst** - lucency with a thin wall
- Emphysema** - lucency without a visible wall

Cavity - wall > 3mm	Cyst - wall ≤ 3mm	Multiple
<u>Infection</u> • Staphylococcus - Klebsiella • Gram negative • Anaerobe in aspiration • TB - Fungal (aspergillus) <u>Neoplasm</u> • Lungca - squamous cell - BAC • Metastases <u>Vascular</u> • Lunginfarction	<u>Congenital</u> • Bronchogenic cyst • CAM - cystic adenomatoid malformation <u>Bulla</u> <u>Pneumatocele</u> • posttraumatic	<u>Infection</u> • Staphylococcus - Klebsiella • Gram negative • PCP • Anaerobe in aspiration • TB - Fungal (aspergillus) <u>Neoplasm</u> • Lungca - squamous cell - BAC • Metastases <u>Collagen-vascular</u> • thrombo- and septic emboli • RA - Wegener's <u>Emphysema</u> <u>Cystic Lung disease</u> • Langerhans cell histiocytosis • Lymphangioliomyomatosis • Honeycombing



CAVITARY LESIONS

Contents



Air fluid level

Surface of the fluid level

Straight

Abscess

Wavy

ruptured hydatid cyst

Thick

Regular

Chronic abscess

Irregular

Cavitating neoplasm

Air only

Wall thickness

Thin

Site

Central
Pneumatocele

Peripheral
emphysematous
bullae



Cavities frequently arise within a mass or an area of consolidation as a result of necrosis.

We will discuss them here, because the prominent feature is the lucency.

- In the differential diagnosis there is overlap between cavities and cysts.

Cavities can heal and end up as lungcysts and lungcysts can become infected and turn into thick walled cavities.

Sometimes emphysematous bullae have visible walls that measure less than 1 mm.

To differentiate them from cysts, is to look at the surrounding lung parenchyma.

- Cysts occur without associated pulmonary emphysema.

Cysts usually contain air, but occasionally contain fluid or solid material.

The term is mostly used to describe enlarged thin-walled airspaces in patients with lymphangioleiomyomatosis or Langerhans cell histiocytosis.

Thicker-walled honeycomb cysts are seen in patients with end-stage fibrosis ([11](#)).

Cavitation

Pneumonia

In virulent pyogenic infections an abscess may form within the consolidated lung as a result of necrosis due to vasculitis and thrombosis.

When some of the pus is coughed up, a cavity can be seen on the chest film.

These patients are usually very ill.

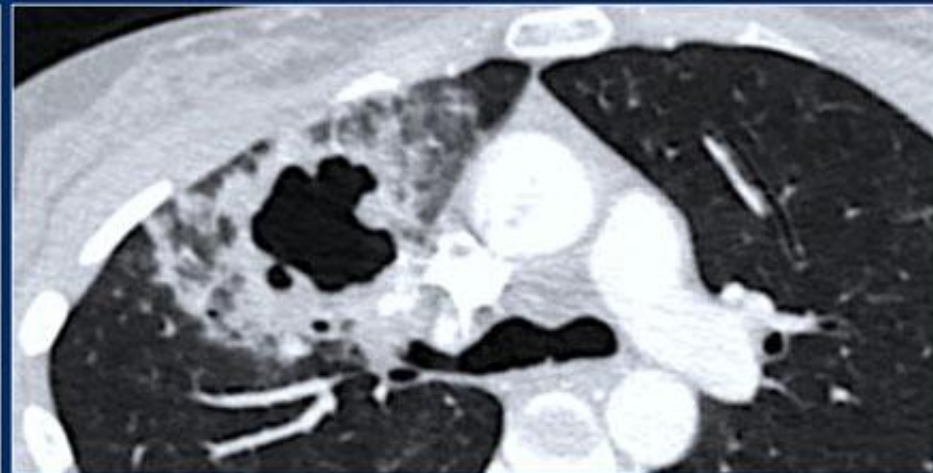
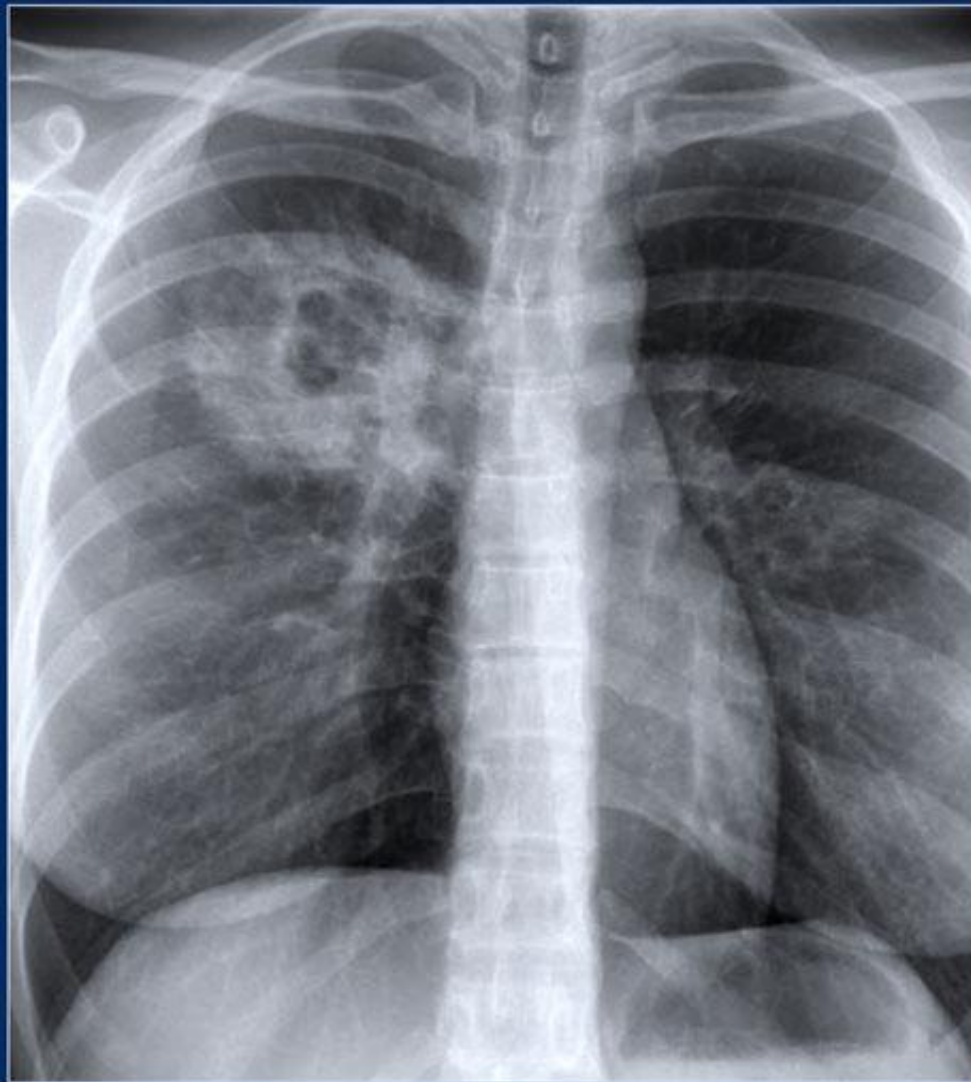
In granulomatous infection like TB, cavities may form, but these patients are usually not that ill.

Cavitation is not seen in viral pneumonia, mycoplasma and rarely in streptococcus pneumoniae.

These images are of a young patient with pneumonia.

No micro-organism could be isolated.

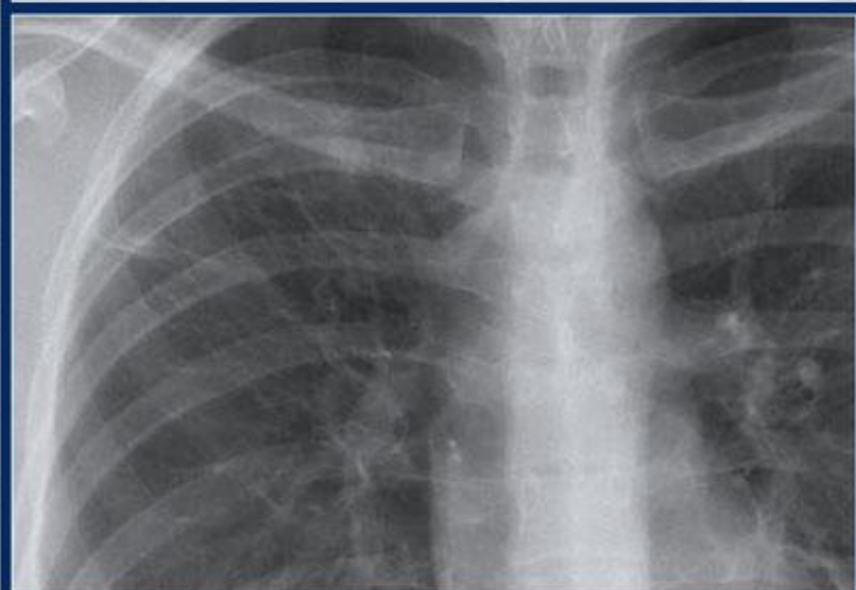
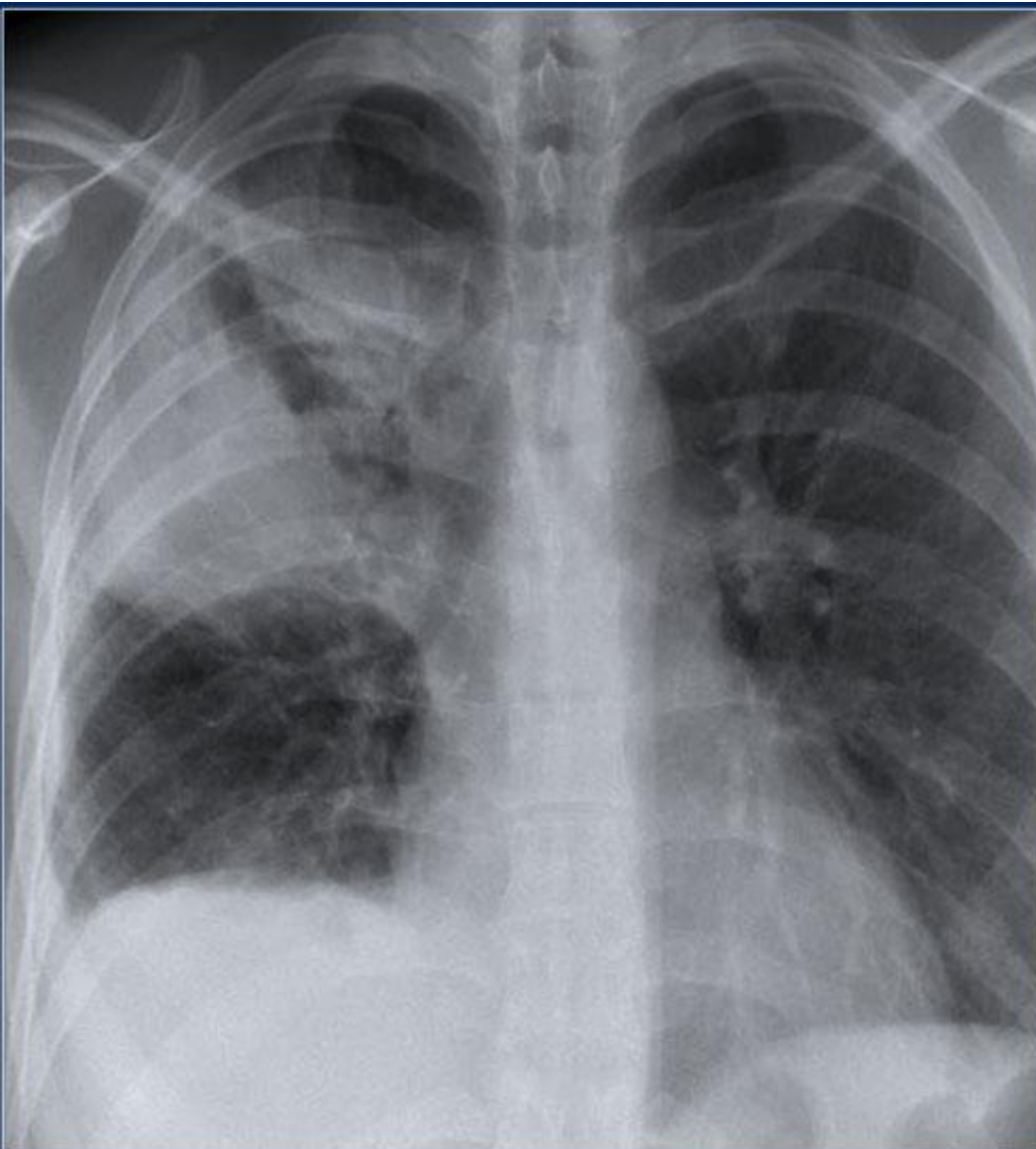
Within one month after treatment with antibiotics, there was almost complete resolution of the consolidation and the cavity



- Pneumonia with cavitation

Pneumonia

- Here another example of a pneumonia with cavitation.
- Notice the destruction of lung parenchyma as seen on the CT.
- At one year follow up only minimal changes are seen on the CXR.



TB

This patient presented first with the CXR on the left.

First study the images.

Then continue reading.

The findings are:

Widespread ill-defined densities, which are probably small consolidations.

Cavity in the right upper lobe.

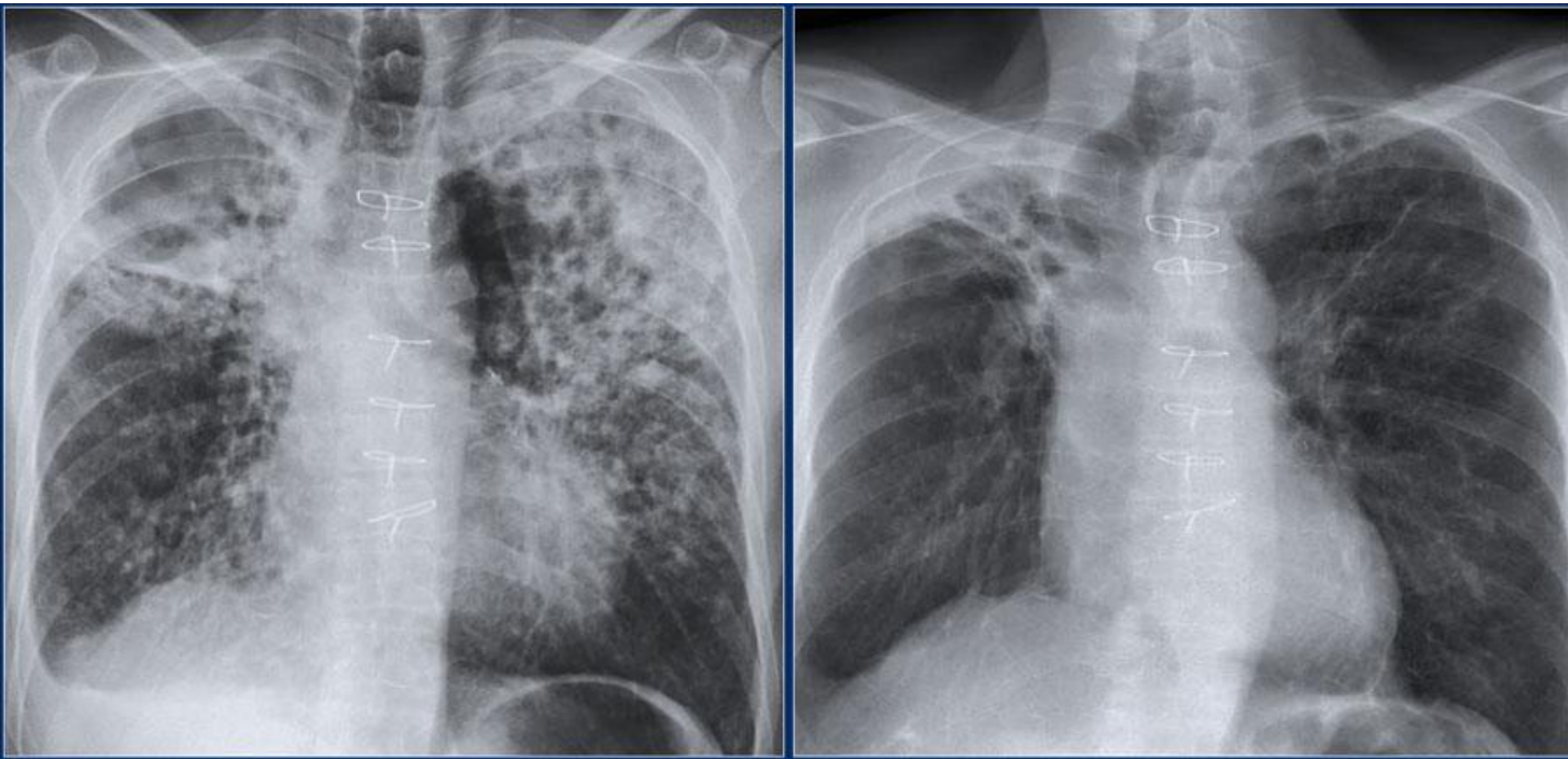
We can assume that this is reactivation of a latent TB.

Culture was positive for TB.

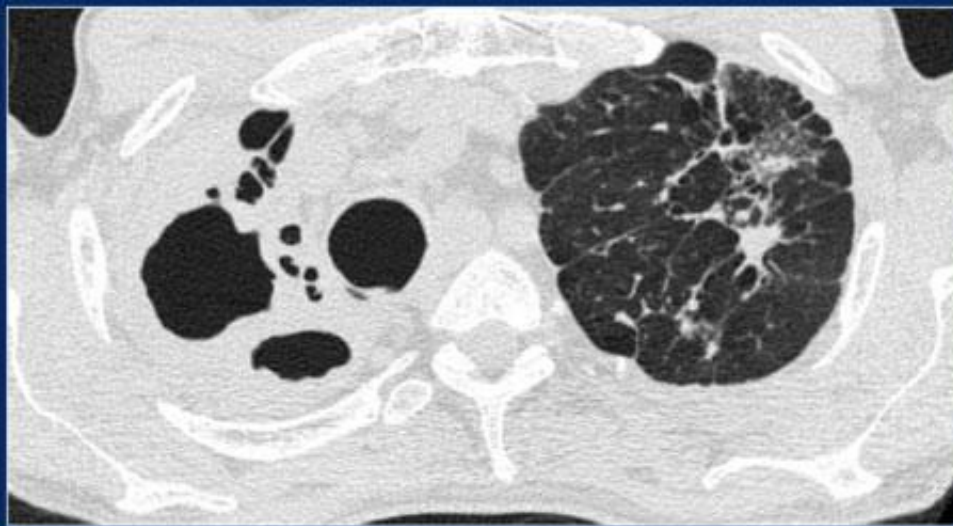
•A CXR some years later on the right shows:

- Right upper lobe atelectasis
- Deviation of the trachea
- Scarring and cavitation of the remnants of the upper lobe
- In left upper lobe minimal fibrosis and cavitation.

This is better appreciated on a CT.....Continue...



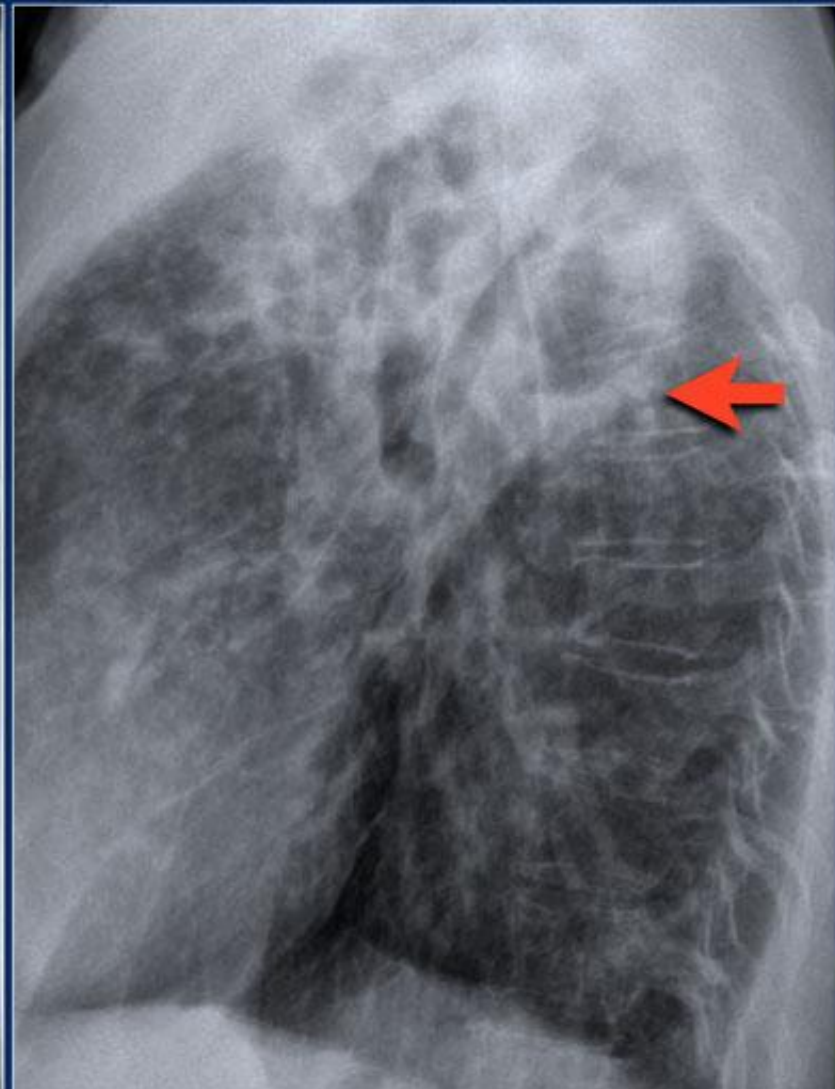
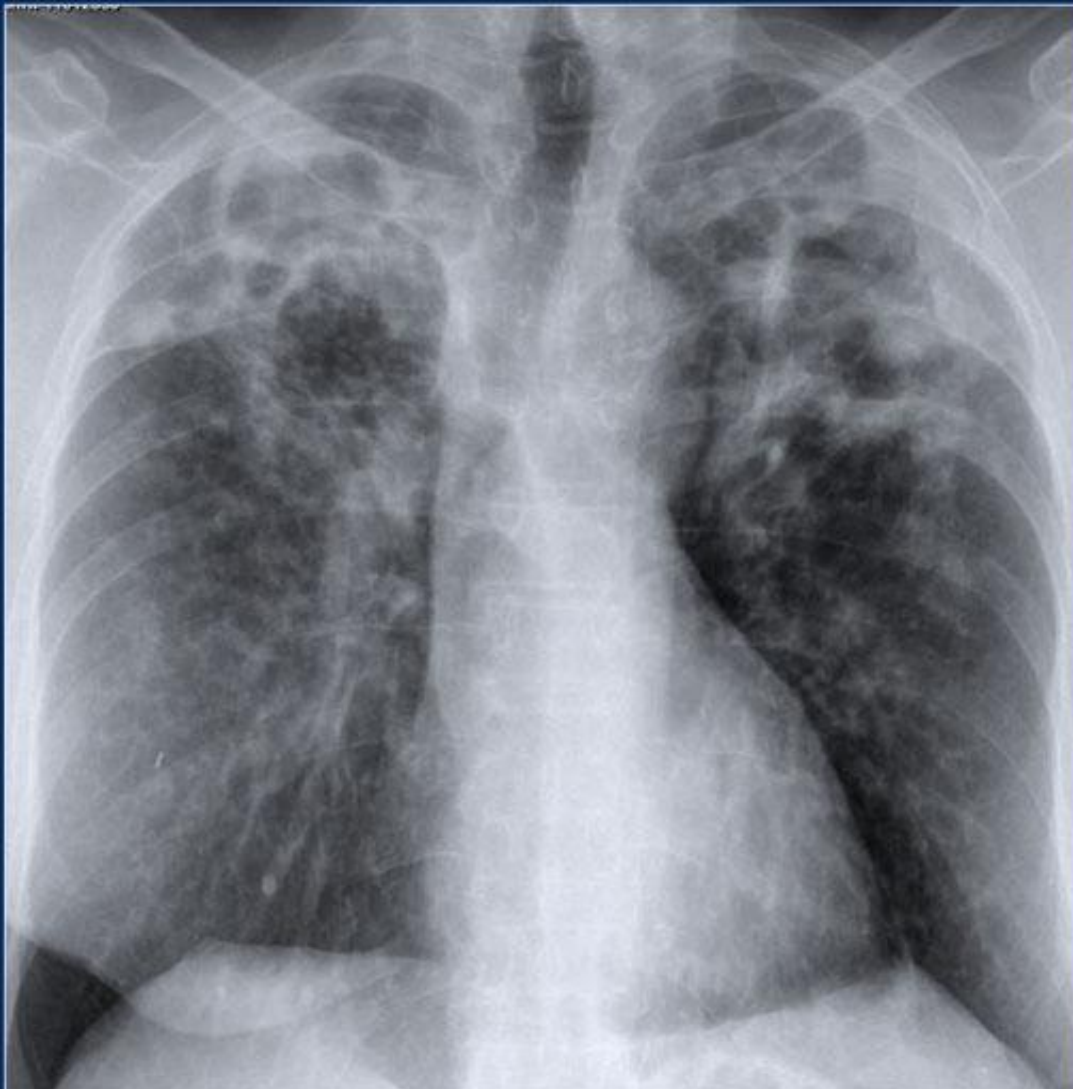
- Postprimary TB

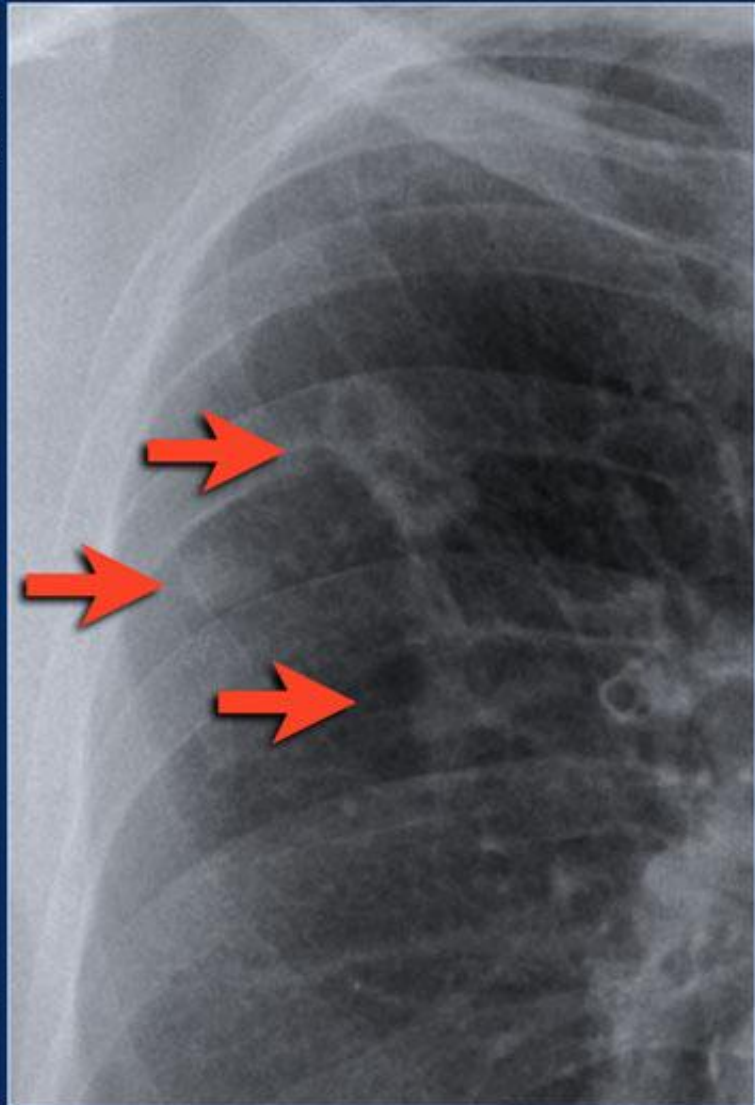
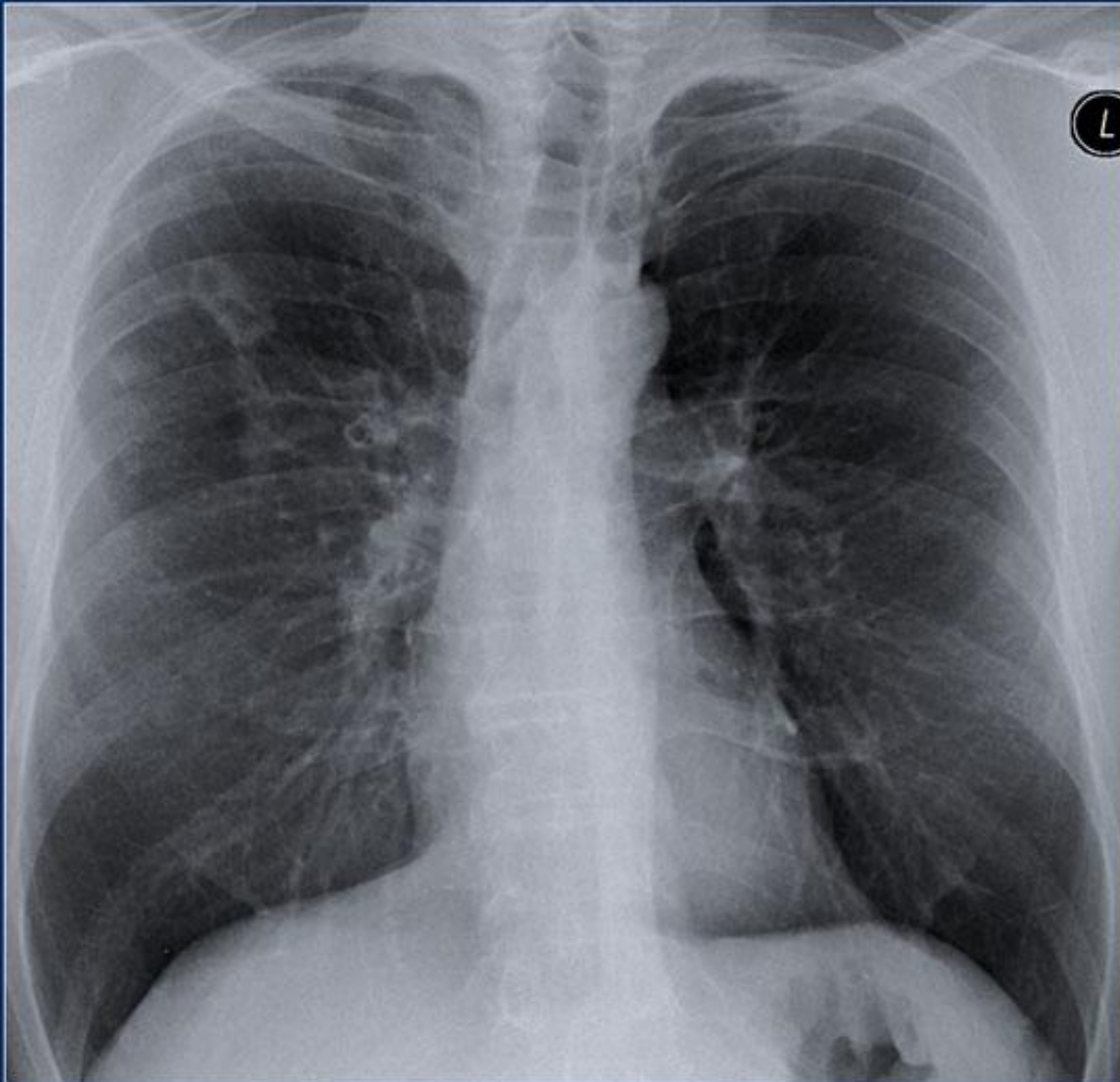


Same patient

- Notice the cavitation especially on the right. In the left upper lobe there is probably some traction-bronchiectasis due to the fibrosis.

- **Nontuberculous mycobacteria**
- Nontuberculous mycobacteria, also known as atypical mycobacteria, are all the other mycobacteria which can cause pulmonary disease resembling TB.
- Here a patient with active disease in both upper lobes due to infection with atypical mycobacterium.
Notice the air-fluid level indicating pus within the cavity (arrow).





- Here another patient with a mycobacterium infection.
- Notice the nodules with cavitation.
Continue with the CT-images.

- Same patient with nontuberculous mycobacteria infection.
Multiple small cavities are seen.



Septic emboli

Septic emboli usually present as multiple ill-defined nodules.

In about 50% cavitation is seen.

CT demonstrates more lesions than the chest film and can suggest the diagnosis in the proper clinical setting by demonstrating wedge-shaped peripheral lesions abutting the pleura, air-bronchograms within the ill-defined nodules and a feeding vessel sign ([7](#)).

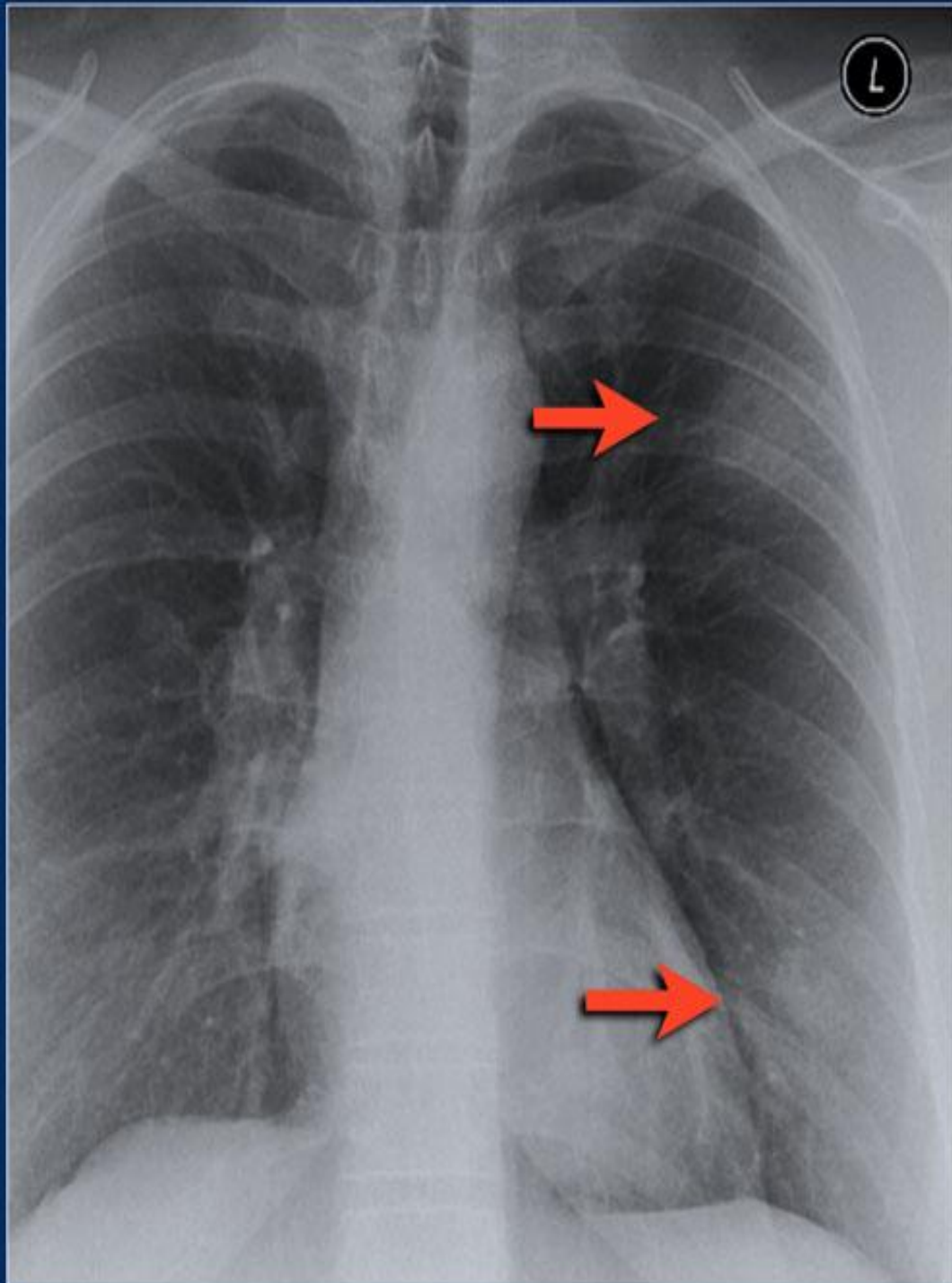
Some argue whether there is really something like a feeding vessel sign ([8](#)).

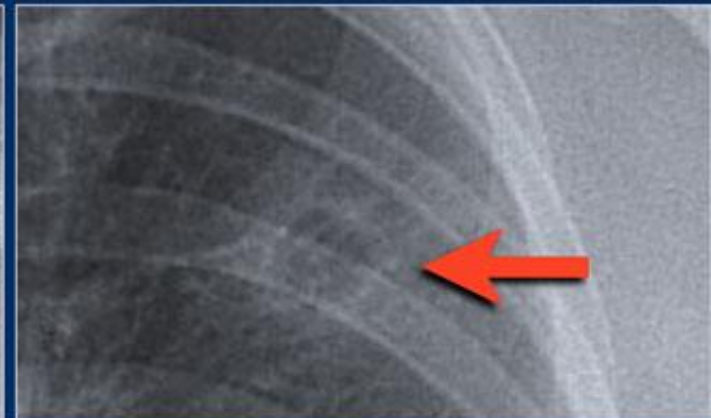
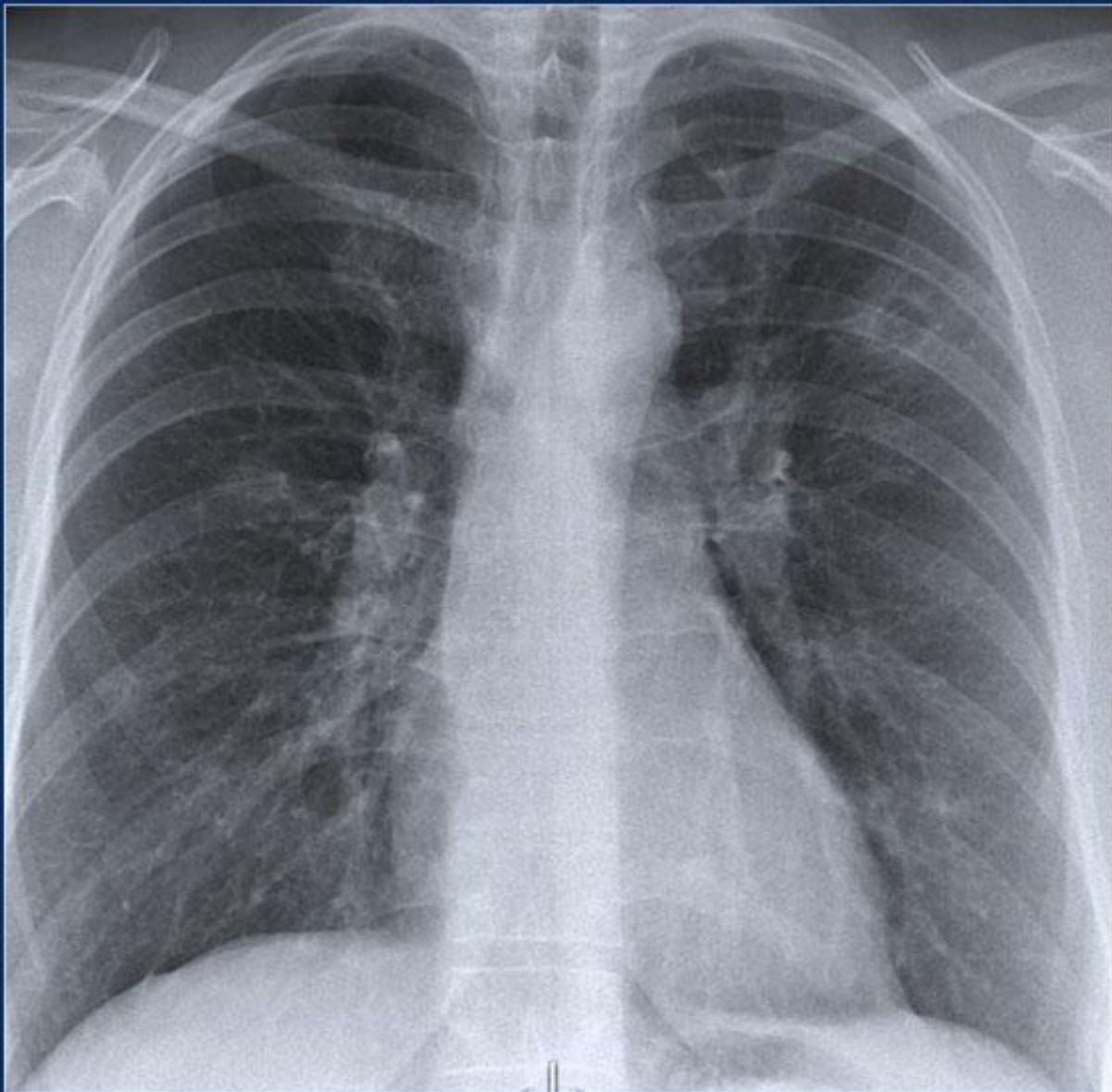
Here a patient with septic emboli.

The chest film shows two ill-defined densities in the left lung, which are probably consolidations.

On the CT cavitation is seen and another density with cavitation in the right lung.

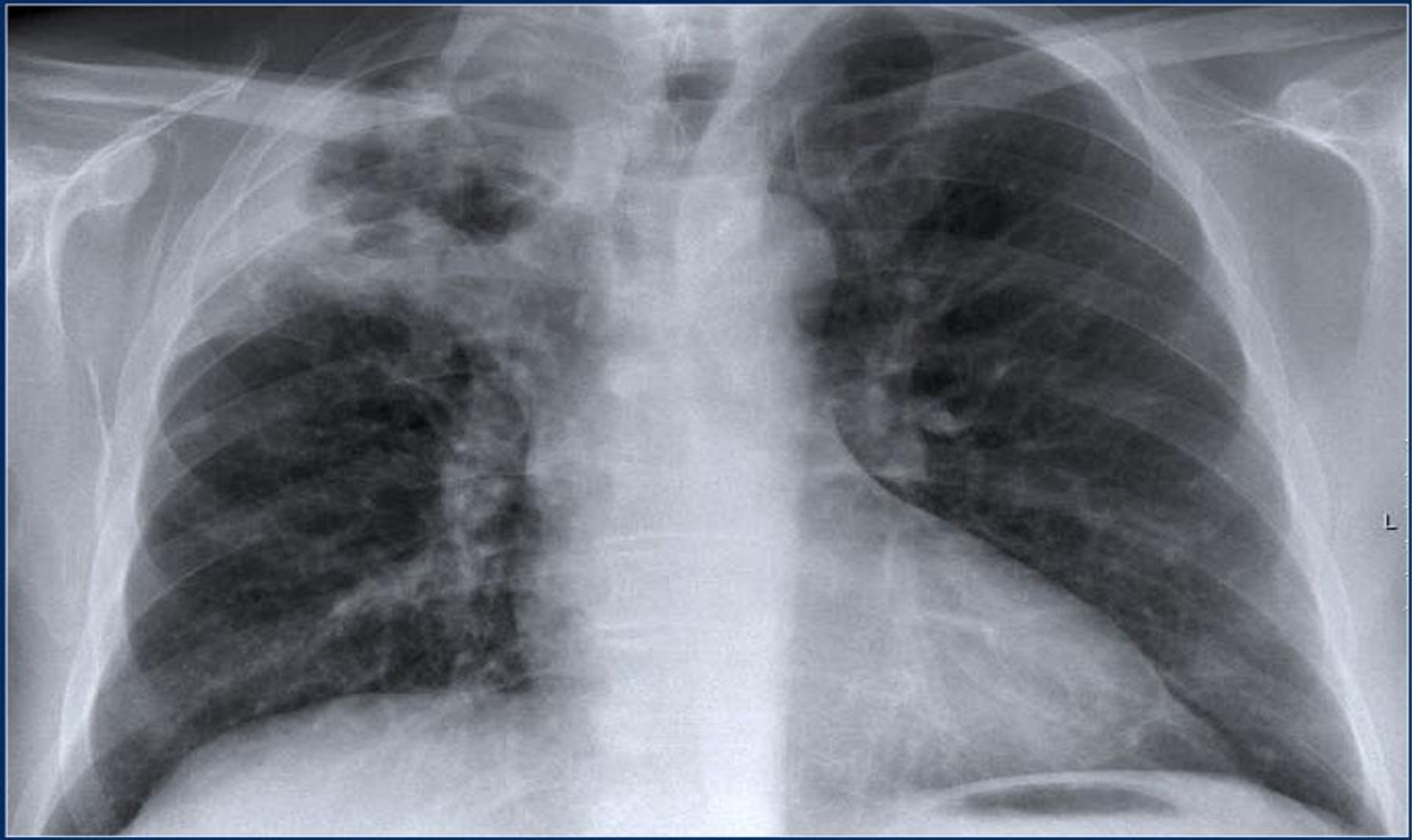
Continue with follow up film.





- **Septic emboli**
- Same patient.
- On a follow up CXR only a small lungcyst is seen.

- **Lungcancer**
 - 10% of lungcancers cavitate, most commonly squamous cell carcinoma. Small cell lungcancer does not cavitate.
Bronchoalveolar carcinoma, or now called adenocarcinoma in situ, may occasionally cavitate and sometimes present as multiple lesions.
- a **chest x-ray** of a large cavitating lung cancer, which started as a small mass.



Lung infarction

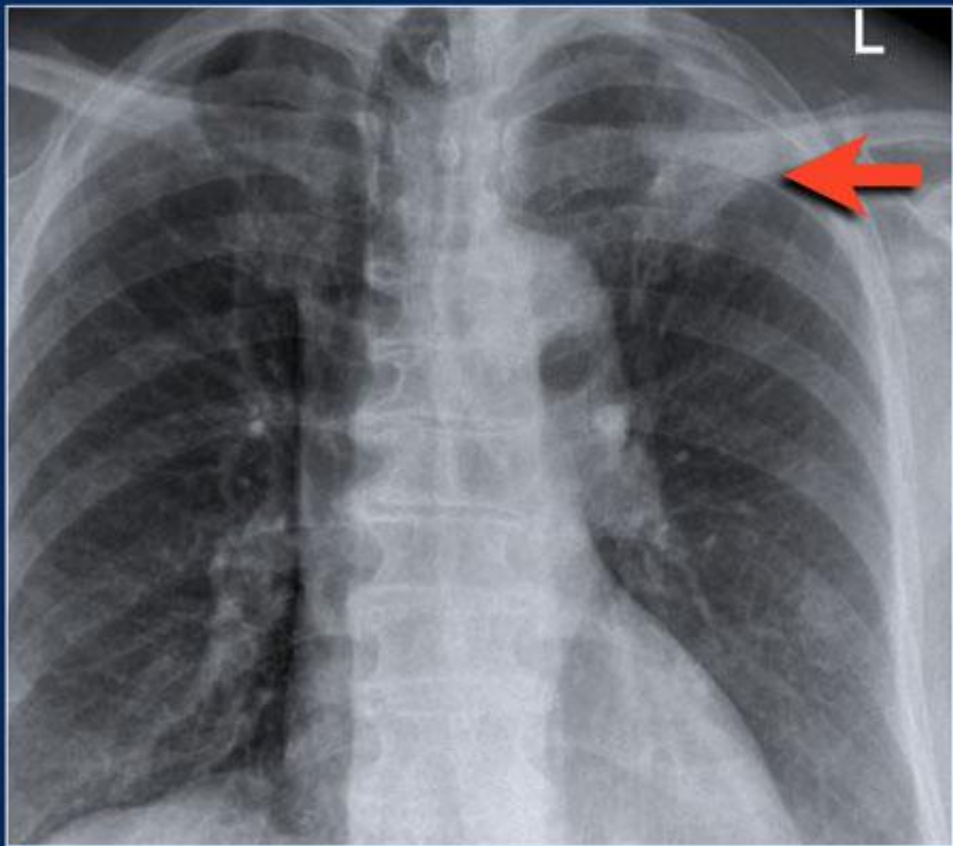
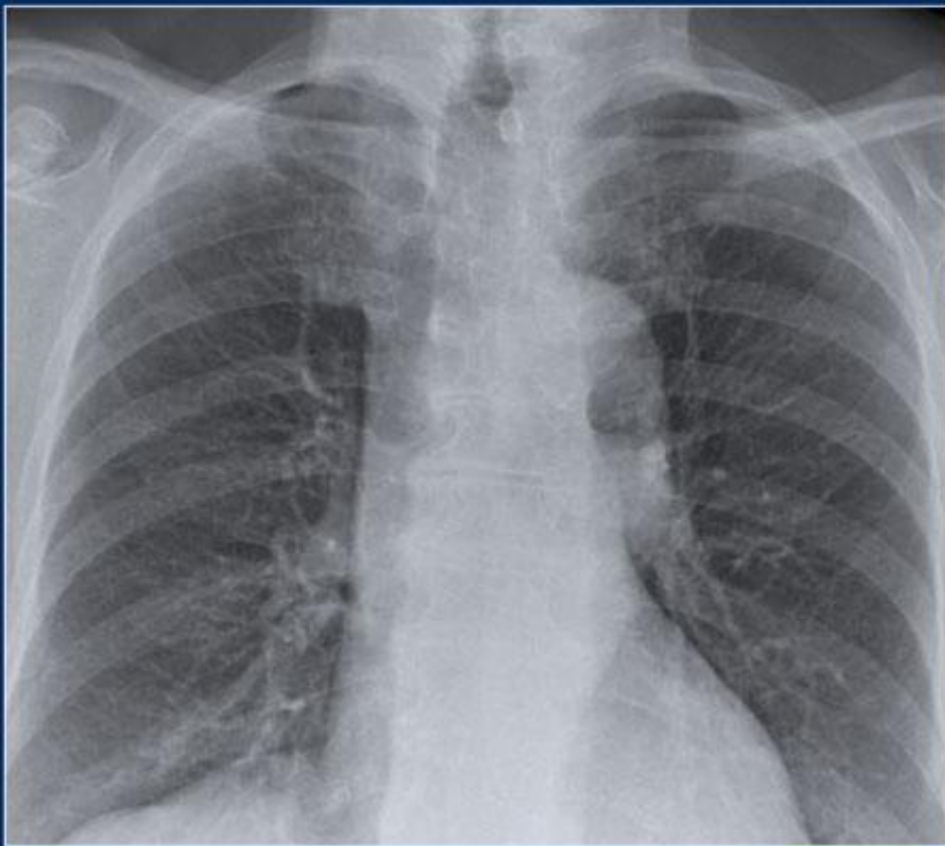
In pulmonar embolism it is not common to see consolidation. The consolidation is a result of lunginfarction and bleeding into the alveoli.

In this case a lung cyst has formed in the infarcted area.

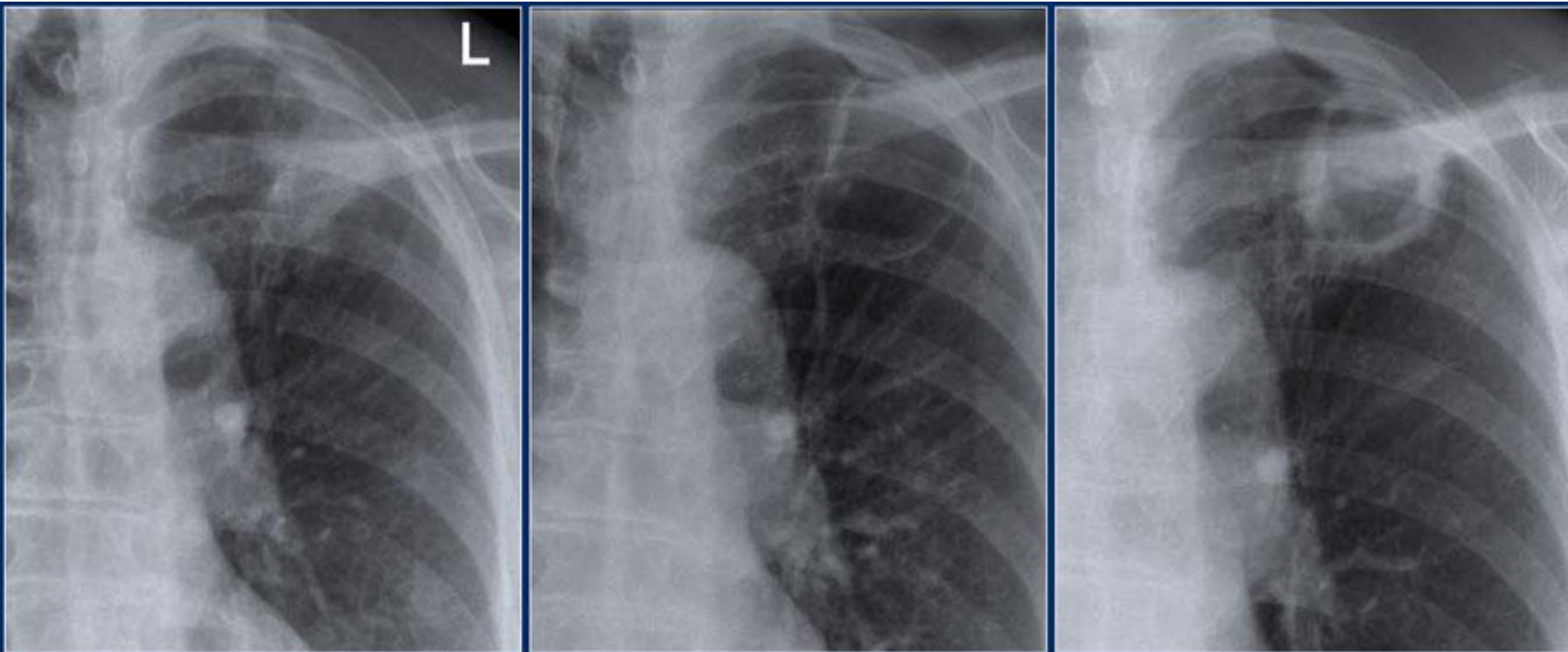
Here we see an old chest film, which is normal.

The pulmonary embolus has caused a **triangular density** on the chest film (arrow).

On the CT we can see, that it is a segmental consolidation. Continue with the follow up films.



- **Lung infarction**
- Same patient.
- On follow up films first a cyst is seen.
One year later there is a thick wall probably as a result of secondary infection.



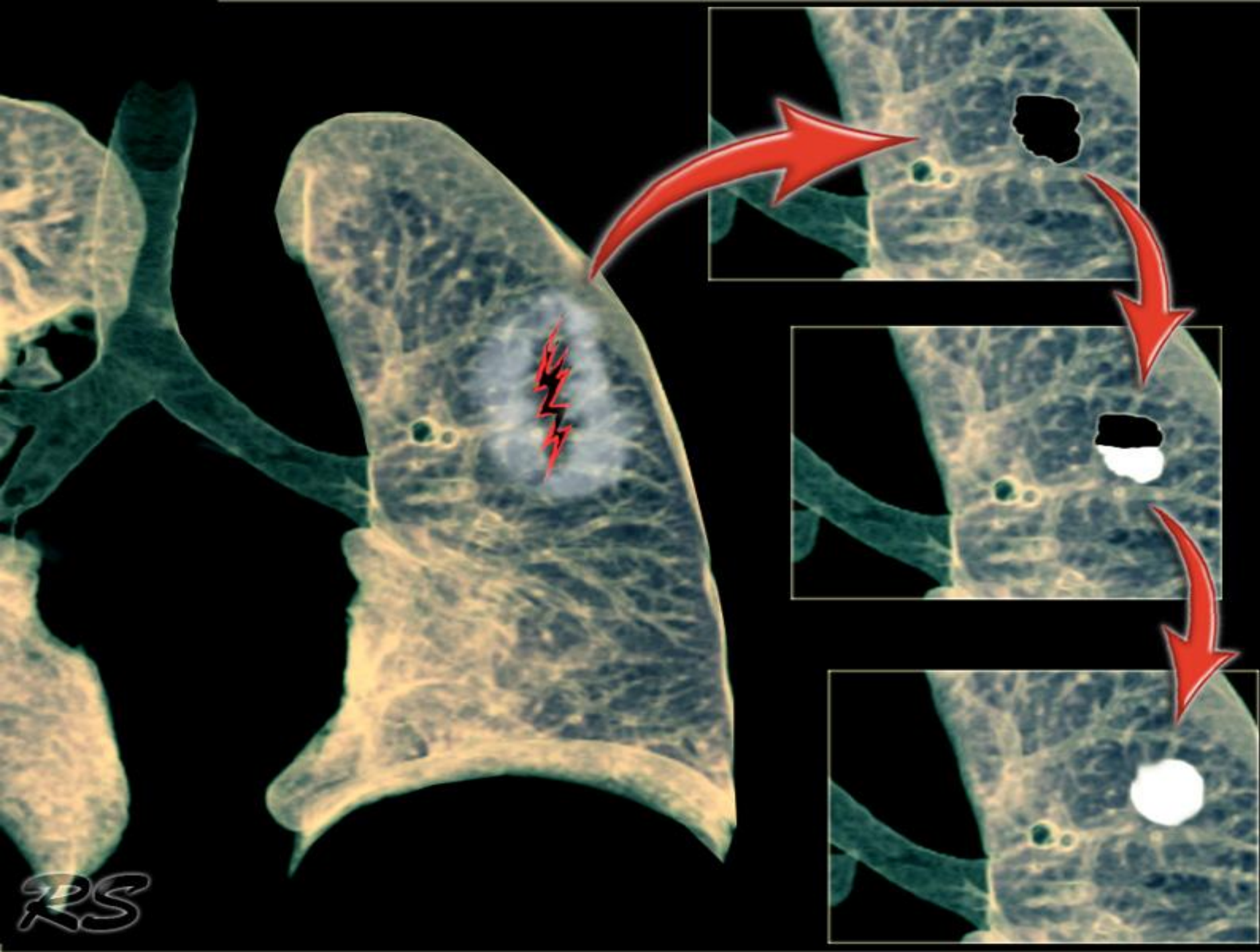
Pneumatocele

The term pneumatocele is used to describe a lungcyst, which is most frequently caused by acute pneumonia, trauma, or aspiration of hydrocarbon fluid and is usually transient.

The mechanism is believed to be a combination of parenchymal necrosis and check-valve airway obstruction ([11](#)).

The illustration shows a pneumatocele as a result of a trauma.

When it fills with fluid, it may resemble a solitary pulmonar nodule.



TABLES & FIGURES

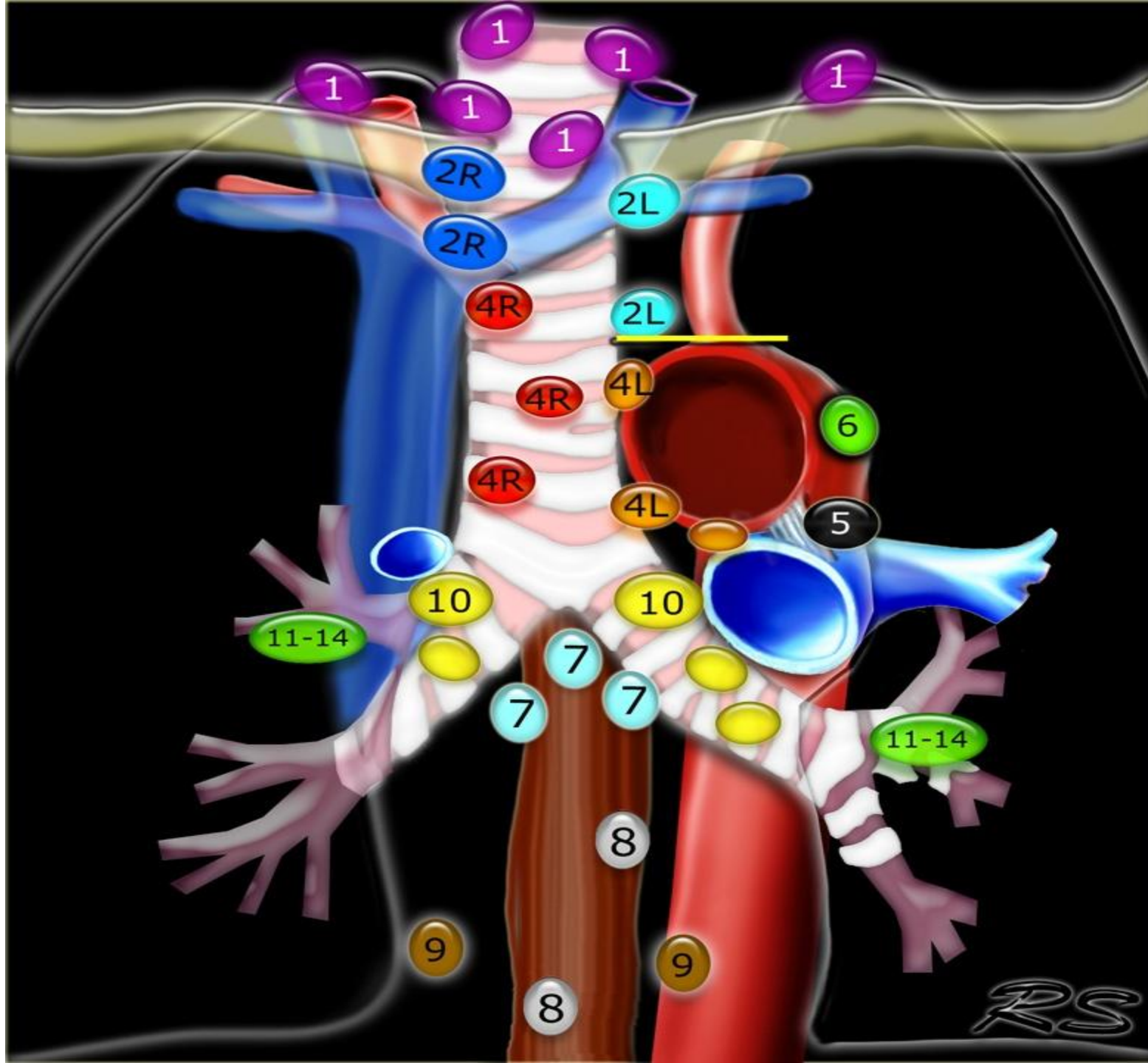
TABLES & FIGURES

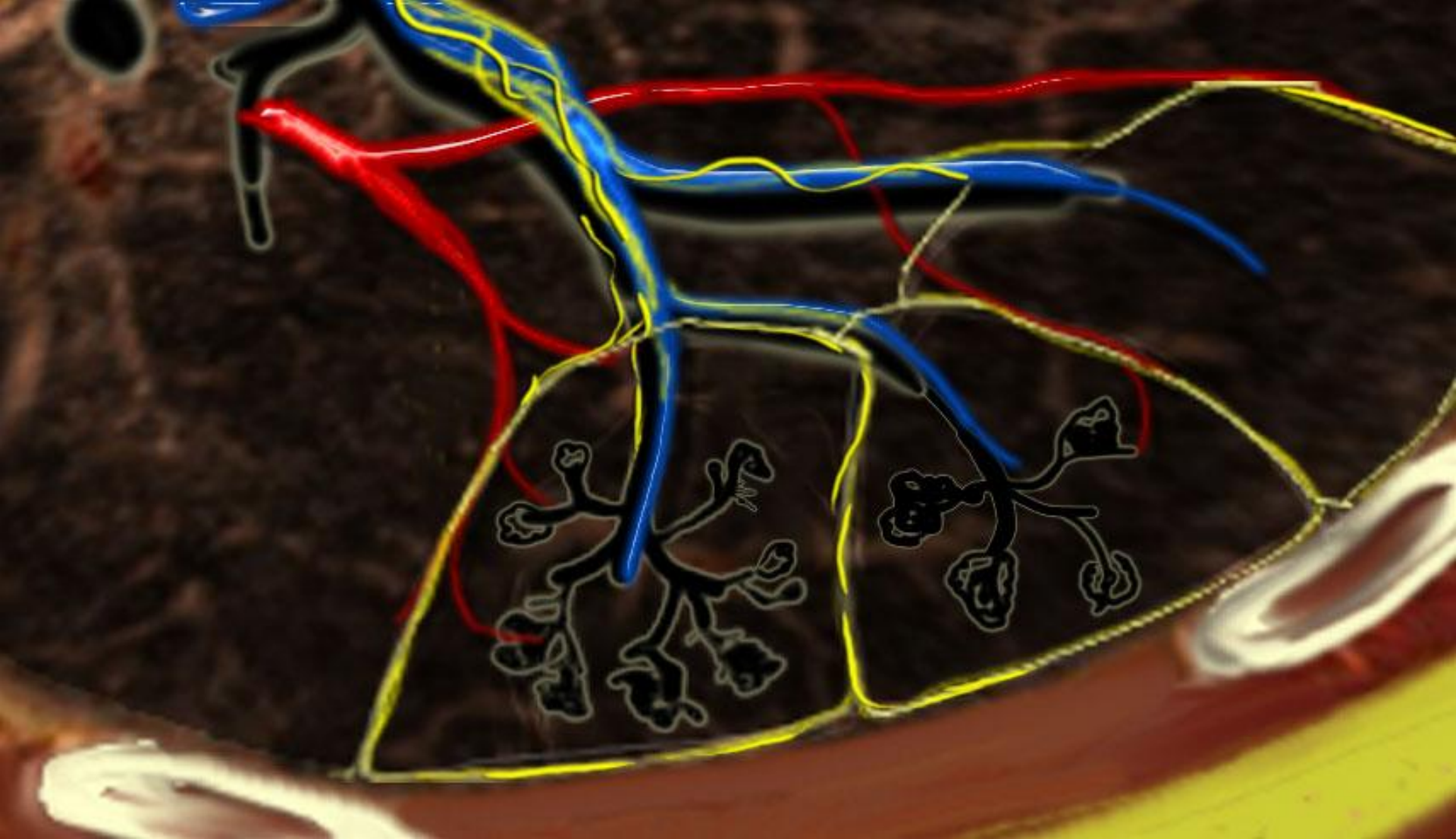
	Diam	Scopy	Atelectasis	Invasion	Nodules
T1	T1a ≤ 2cm T1b >2-3cm	No invasion lobar bronchus			
T2	T2a: >3-5cm T2b: >5-7cm or	main bronchus > 2cm to carina	Atelectasis or obstructive pneumonia to hilus not entire lung	visceral pleura	
T3	> 7cm or	< 2cm to carina	Whole lung	Chest wall diafragm phrenic nerve mediast pleura parietal pericard	Nodules in same lobe
T4		Tumor in carina		Heart great vessels trachea esophagus spine	Nodules in other ipsilateral lobes

- new T-staging according to the 7th edition of the TNM-staging of lungcancer

Regional lymph nodes (N)

- | | |
|-----------|--|
| N1 | In ipsilateral peribronchial and/or ipsilateral hilar lymph nodes and intrapulmonary nodes |
| N2 | In ipsilateral mediastinal and/or subcarinal lymph nodes |
| N3 | In contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene or supraclavicular lymph nodes |





- Secondary lobules. The centrilobular artery (in blue: oxygen-poor blood) and the terminal bronchiole run in the center. Lymphatics and veins (in red: oxygen-rich blood) run within the interlobular septa

Common HRCT diagnoses

Sarcoidosis
Histiocytosis
Idiop pulm fibrosis
Tumor
Failure (edema)
Aspiration
Collagen vascular
Environmental
Drug reaction

90%

UIP
Hypersensitivity P
Sarcoid
Asbestosis
BOOP
Langerhans hist

80%

UIP
Silicosis
Sarcoid
Lymphangitic
Hypersensitivity

83%

	Lesions	Fluid	Fat	Vascular
Anterior	Thymic Lymphoma Germ Cell Goiter	Thymic C Thymoma Pericardial C Germ Cell Lymphoma	Germ cell-b Thymolipoma Fat Pad	Thyroid Cardiac Coronary
Middle	Lymph nodes Duplication C Arch anomaly	Duplication C Necrotic nodes Pericard recess Retroperitoneal	Lipoma Esophageal FV polyp	Arch anomaly Azygous Vein Vascular nodes
Posterior	Neurogenic Bone and marrow	Neuroenteric C Schwannoma Meningocoele	Extramedullary Hematopoiesis	Desc Aorta
>1 comp	Infection Hemorrhage Lung Cancer	Lymphangioma Mediastinitis	Liposarcoma	Hemangioma

- Mediastinal Masses

Thank You

Good Luck

Ahmad Mokhtar Abodahab
Mar 2017

